TJA1104

MACsec enabled 100Base-T1 PHY

Rev. 1.0 — 17 April 2024

Product data sheet
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1 General description

The TJA1104 is a 100BASE-T1 compliant MacSec enabled Ethernet PHY. The device provides 100 Mbit/s transmit and receive capability over unshielded twisted-pair cables, supporting a cable length of up to 15 m. It is available in four variants. The TJA1104A/TJA1104C offer standard MII / RMII and RGMII(-ID) MAC interfaces. TJA1104A and TJA1104B have MACsec enabled. TJA1104C and TJA1104D are variants without MACsec. The variants TJA1104B and TJA1104D provide SGMII.

The product integrates an IEEE 802.1AE-2018 compliant MACsec subsystem, which is configurable by the host controller via MDIO. MACsec ensures the authenticity, integrity, and (optional) also confidentiality of any communication over the link. The TJA1104 supports both, point-to-point and end-to-end secure associations. The implementation of MACsec is fully AVB and TSN compliant and maintains predictable latency for both, controlled, and uncontrolled data path. Also the MACsec congestion avoidance techniques supported by TJA1104 retain full compliance with PTP. A rich set of advanced frame filtering options allow to bypass selected non-MACsec frame types.

TJA1104 is fully drop-in compatible to TJA1103, with extended register space and feature set. If MACsec is disabled, the TJA1104 behaves similar to TJA1103.

For sustainable system efficiency and flexibility, the device is optimized for lowest power consumption and highest robustness, while maintaining a small package footprint. The TJA1104, including the MACsec subsystem, is developed in compliance with ISO 26262, achieving ASIL-B.

Adequate safety features have been implemented on-chip, to support functional safety concepts on system level. Additional documentation, including a safety manual[10], is available on request. In combination with state of the art diagnostic features, the device is perfectly suited to support high availability of communication paths in upcoming automotive electrical/electronic (E/E) architectures.

The TJA1104 is designed to meet the harsh requirements of automotive applications. It is typically used to provide authenticity of communication of end nodes like radar, LiDAR, and camera sensors for driver assistance systems (ADAS), as well as a variety of electronic control units (ECUs) like zonal gateways, telematics and head units.

The device supports OPEN Alliance TC-10 sleep / wake over data-line. By eliminating the need for additional cabling, the feature saves cost and weight.

Due to its efficient IEEE1588v2 and 802.1AS-2020 compliant time stamping capability, the TJA1104 provides high-resolution accuracy for global time synchronization, which is required to support the timing critical data fusion functions of self-driving cars. MACsec can be applied to protect the time stamp.

2 Features and benefits

2.1 General

- IEEE 802.3 Clause 97 compliant 100BASE-T1 PHY
- AEC-Q100 Grade 1 (-40 °C to +125 °C ambient)
- Compact HVQFN 36-pin package (6 × 6 mm) for PCB space-constraint applications
- Standard MII / RMII / RGMII- (TJA1104A/TJA1104C)
  - Integrated delay (RGMII-ID) in receive and transmit clock path
  - 3.3 V, 2.5 V, or 1.8 V LVCMOS I/O standard for MII/RMII/RGMII/MDIO
- SGMII-compliant (TJA1104B/TJA1104D) MAC interface
- OPEN Alliance TC-10 compatible sleep / wake-forwarding over data line
  - Low-power Sleep mode
- ISO 26262, ASIL-B compliant
- IEEE1588v2 and 802.1AS-2020 2-step timestamping support[1]
  - Synchronization of timestamp clock with other switch or PHY devices
  - Timestamp delivery via in-band (RX-only) or SMI communication
  - RX and TX start of frame (SOF) indication via pins
- IEEE1149.1 and IEEE1149.6 compliant JTAG interface for boundary scan
- MDIO register access
  - IEEE 802.3 Clause 45
  - IEEE 802.3 Clause 22 (indirect access)

2.2 Secure communication

- IEEE 802.1AE-2018 MACsec at full wireline performance
- Support for integrity protection and confidentiality protection
- Support for two concurrent security associations per secure channel for re-keying
- Support for 4 end-to-end secure channels in TX and RX direction
- Cipher suites:
  - AES-GCM-128
  - AES-GCM-256
- Extended packet number (XPN) for reduced rekeying frequency
- AVB, TSN compatible flow-control:[2]
  - Cut-through, pre-gapped IPG[3]
  - Cut-through, standard IPG and vendor-specific MACsec TX/RX control tag in TX direction
- Uncontrolled / controlled port and secure channel selection:
  - Internal lookup table with ternary match-based frame filtering
    - 8 uncontrolled / controlled port filter rules
    - 4 secure-channel selection rules
  - In-band host-side selection through vendor-specific MACsec TX/RX control tag in TX and RX direction
- Replay protection

1 The PTP frame parser cannot recognize PTP frames if an optional, vendor specific MACsec TX/RX control tag is present.
2 PAUSE frame flow-control not supported. The PAUSE mechanism interferes with real-time traffic and introduce undesired packet delay variation.
3 IPG must be extended large enough to allow for adding SecTag and ICV.

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- Jumbo-frame support
- Clear-tag support
  - Up to two VLAN tags and one custom tag (for example 802.1CB R-tag)
- Diagnostics and statistics (MIB) counters

2.3 Optimized for automotive use cases

- Support for shielded and unshielded twisted-pair cable
- Adaptive receive equalizer optimized for automotive cable length of up to at least 15 m
  - Echo cancellation up to 25 m[4]
- Enhanced integrated pulse shaping for low RF emissions
- EMC-optimized output driver configuration for RGMII and SGMII
- MDI pins protected against ESD up to ±4 kV HBM and ±6 kV IEC 61000-4-2
- MDI pins protected against transients in automotive environment IEC 62228-5
- MDI pins do not need external filtering or ESD protection

2.4 Low-power mode

- Inhibit output for voltage regulator control
- Robust remote wake-up detection via bus lines
- OPEN Alliance TC-10 compliant sleep/wake-up
  - PHY-level wakeup-forwarding via dedicated pin
- Wake-up via SMI-access

2.5 Diagnosis

- OPEN Alliance TC-1 Advanced PHY Features
  - Signal Quality Indicator (SQI)
  - Signal MSE (Mean Square Error) indicating the signal error at the slicer
  - Diagnosis of cable errors (shorts and opens)
  - Link training time measurement
  - Link losses since reset/power on
  - Symbol errors since power on
- Optional Ethernet frame generator and checker for built-in self-test (BIST)
- Latent fault detection
- Real-time monitoring of link stability and data quality
- Gap-free supply undervoltage detection
- Temperature monitor with overtemperature detection
- Internal, external and remote loopback modes for diagnostics

2.6 Miscellaneous

- 3.3 V or 2.5 V single supply operation
- 3.3 V, 2.5 V, 1.1 V individual supply operation
- 3.3 V, 2.5 V I/O for RST_N, INH, and wake interface
- GPIO, LED, PTP_SYNC, and reference clock inputs/outputs

4 OPEN Whole communication channel (WCC) requirements as per OPEN Alliance Definitions for Communication Channel v1.0.

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- Support for crystal or digital clock input
- Pin-strapping to configure
  - Managed and autonomous operation modes
  - master / slave mode
  - Polarity correction
  - xMII / JTAG modes
  - PHY address
- On-chip termination resistors
- Jumbo frame support up to 16 kB

3 Ordering information

Table 1. Ordering information

Type number	Package		
	Name	Description	Version
TJA1104AHN	HVQFN36	plastic thermal enhanced very thin quad flat package; no leads; 36 terminals; body 6 × 6 × 0.85 mm, with MACsec	SOT1092-2
TJA1104BHN			
TJA1104CHN		plastic thermal enhanced very thin quad flat package; no leads; 36 terminals; body 6 × 6 × 0.85 mm, without MACsec	
TJA1104DHN			

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4 Block diagram

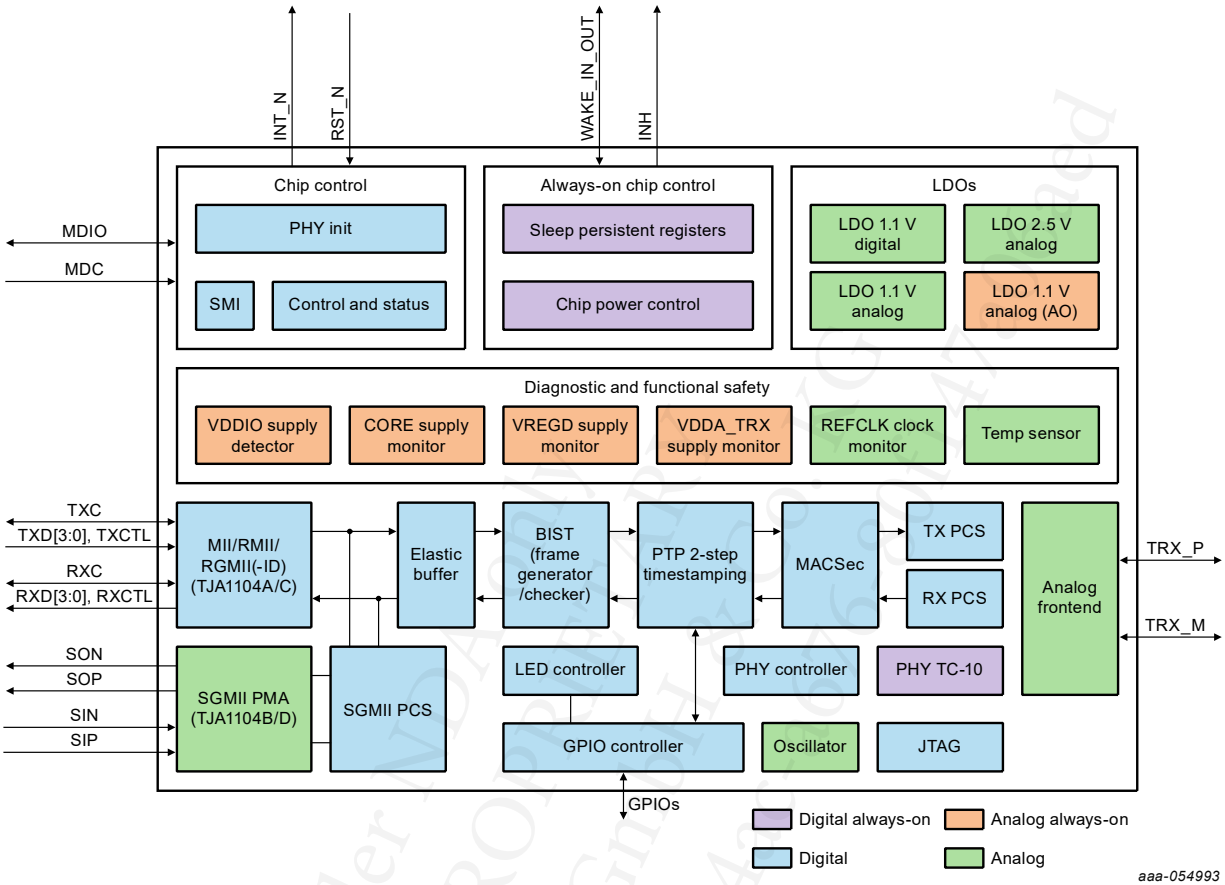


Figure 1. Block diagram

5 Pinning information

5.1 Pinning

The pin configuration of the TJA1104 is shown in Figure 2 and Figure 3.

The following standard interfaces are provided by the TJA1104: MII, RMII, RGMII (TJA1104A/TJA1104C variants), SGMII (TJA1104B/TJA1104D variants), MDIO, MDI.

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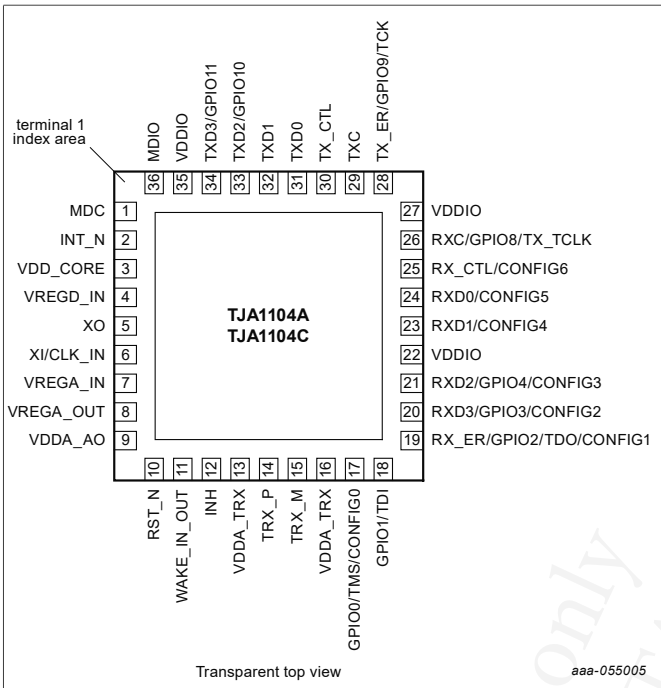


Figure 2. Pinning TJA1104A/TJA1104C

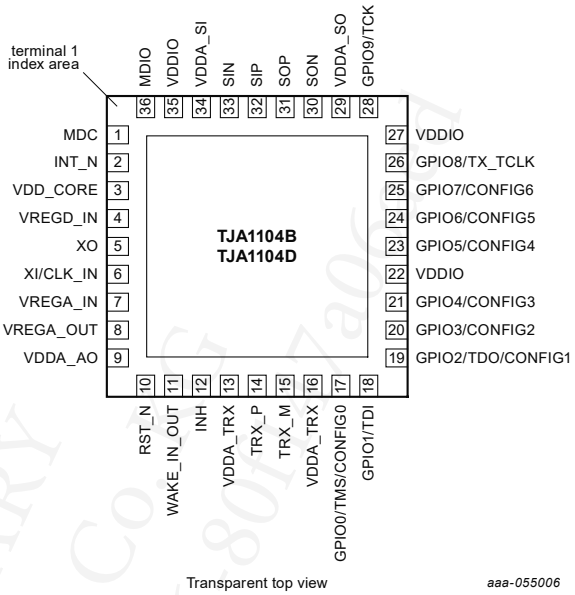


Figure 3. Pinning TJA1104B/TJA1104D

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5.2 Pin description

Table 2. Pinning TJA1104A and TJA1104C

Symbol	Pin	Type[1][2]	IO supply domain[3]	Description
MDC	1	I	VDDIO	SMI clock input
INT_N	2	O, PU	VDDIO	Interrupt low-side switch output
VDD_CORE	3	P	—	Digital supply (1.1 V)
VREGD_IN	4	P	—	Regulator Input (2.5 V, 3.3 V)
XO	5	AO	—	Crystal feedback
XI / CLK_IN	6	AI	—	Crystal input / Digital clock input
VREGA_IN	7	P	—	Analog Regulator input (2.5 V, 3.3 V)
VREGA_OUT	8	P	—	Analog Regulator Output (2.5 V)
VDDA_AO	9	P	—	Always on supply (2.5-3.3V)
RST_N	10	I, PU	VDDA_AO	Reset Input (VDDA_AO domain)
WAKE_IN_OUT	11	IO	VDDA_AO	Local/forwarding wake-up input/output (configurable)
INH	12	O	VDDA_AO	Inhibit high-side switch output for Vreg control
VDDA_TRX	13	P	—	MDI analog voltage supply (2.5 V)
TRX_P	14	AIO	—	Plus terminal for Tx/Rx signal
TRX_M	15	AIO	—	Minus terminal for Tx/Rx signal
VDDA_TRX	16	P	—	MDI analog voltage supply (2.5 V)
TMS / GPIO0 / CONFIG0	17	IO	VDDIO	JTAG Test mode select, GPIO0, CONFIG0
TDI / GPIO1	18	IO	VDDIO	JTAG Test data in. GPIO1
RX_ER / GPIO2 / TDO / CONFIG1	19	IO	VDDIO	RMII/MII RX_ER, GPIO2, JTAG Test data out, CONFIG1
RXD3 / GPIO3 / CONFIG2	20	IO	VDDIO	RGMII/RMII/MII RX Data 3 output, GPIO3, CONFIG2
RXD2 / GPIO4 / CONFIG3	21	IO	VDDIO	RGMII/RMII/MII RX Data 2 output, GPIO4, CONFIG3
VDDIO	22	P	—	Digital interface I/O voltage (1.8 V, 2.5 V, 3.3 V)
RXD1 / CONFIG4	23	IO	VDDIO	RGMII/RMII/MII RX Data 1 output, CONFIG4
RXD0 / CONFIG5	24	IO	VDDIO	RGMII/RMII/MII RX Data 0 output, CONFIG5
RX_CTL / CONFIG6	25	IO	VDDIO	RGMII/RMII/MII interface receive control input, CONFIG6
RXC / GPIO8 / TX_TCLK	26	IO	VDDIO	RGMII/MII interface receive clock, GPIO8, test clock TX_TCLK
VDDIO	27	P	—	Digital interface I/O voltage (1.8 V, 2.5 V, 3.3 V)
TX_ER / GPIO9 / TCK	28	IO	VDDIO	JTAG Test clock, MII TX_ER, GPIO9
TXC	29	IO	VDDIO	RGMII/RMII/MII Transmit Clock input
TX_CTL	30	IO	VDDIO	RGMII/RMII/MII interface transmit control input
TXD0	31	IO	VDDIO	RGMII TX Data input

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Table 2. Pinning TJA1104A and TJA1104C*...continued*

Symbol	Pin	Type[1][2]	IO supply domain[3]	Description
TXD1	32	IO	VDDIO	RGMII/RMII/MII TX Data input
TXD2 / GPIO10	33	IO	VDDIO	RGMII/RMII/MII TX Data input, GPIO10
TXD3 / GPIO11	34	IO	VDDIO	RGMII/RMII/MII TX Data input, GPIO11
VDDIO	35	P	—	Digital interface I/O voltage (1.8 V, 2.5 V, 3.3 V)
MDIO	36	IO	VDDIO	SMI Data I/O
GNDPAD	37	G	—	Exposed die pad

[1] I: digital input; O: digital output; IO: digital input/output; AO: Analog output; AI: Analog Input; AIO: Analog input/output; PU: weak pull-up; P: power supply. G: ground
[2] Pin type depends on functional mode; consult the functional description for details.
[3] Only applicable to CMOS input/output pins

Table 3. Pinning TJA1104B and TJA1104D

Symbol	Pin	Type[1][2]	IO supply domain[3]	Description
MDC	1	I	VDDIO	SMI Clock Input
INT_N	2	O, PU	VDDIO	Interrupt low-side switch output
VDD_CORE	3	P	—	Digital supply (1.1 V)
VREGD_IN	4	P	—	Regulator Input (2.5 V, 3.3 V)
XO	5	AI	—	Crystal feedback
XI / CLK_IN	6	AI	—	Crystal input / Digital clock input
VREGA_IN	7	P	—	Analog Regulator input (2.5 V, 3.3 V)
VREGA_OUT	8	P	—	Analog Regulator Output (2.5 V)
VDDA_AO	9	P	—	Always on supply (2.5 V, 3.3 V)
RST_N	10	I, PU	VDDA_AO	Reset Input
WAKE_IN_OUT	11	IO	VDDA_AO	Local/forwarding wake-up input/output (configurable)
INH	12	O	VDDA_AO	Inhibit high-side switch output for Vreg control
VDDA_TRX	13	P	—	MDI analog voltage supply (2.5 V)
TRX_P	14	AIO	—	Plus terminal for Tx/Rx signal
TRX_M	15	AIO	—	Minus terminal for Tx/Rx signal
VDDA_TRX	16	P	—	MDI analog voltage supply (2.5 V)
GPIO0 / TMS / CONFIG0	17	IO	VDDIO	GPIO0, JTAG Test mode select, CONFIG0
GPIO1 / TDI	18	IO	VDDIO	GPIO1, JTAG Test data in
GPIO2 / TDO / CONFIG1	19	IO	VDDIO	JTAG Test data out / CONFIG1 / GPIO2
GPIO3 / CONFIG2	20	IO	VDDIO	CONFIG2, GPIO3
GPIO4 / CONFIG3	21	IO	VDDIO	CONFIG3, GPIO4
VDDIO	22	P		Digital interface I/O voltage (1.8 V, 2.5 V, 3.3 V)
GPIO5 / CONFIG4	23	IO	VDDIO	CONFIG4, GPIO5

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Table 3. Pinning TJA1104B and TJA1104D*...continued*

Symbol	Pin	Type[1][2]	IO supply domain[3]	Description
GPIO6 / CONFIG5	24	IO	VDDIO	CONFIG5, GPIO6
GPIO7 / CONFIG6	25	IO	VDDIO	CONFIG6, GPIO7
GPIO8 / TX_TCLK	26	IO	VDDIO	test clock TX_TCLK, GPIO8
VDDIO	27	P	—	Digital interface I/O voltage (1.8 V, 2.5 V, 3.3 V)
TCK / GPIO9	28	IO	VDDIO	JTAG Test clock, GPIO9
VDDA_SO	29	P	—	Analog SGMII transmitter supply (1.1 V)
SON	30	AO	—	Minus Terminal for SGMII output
SOP	31	AO	—	Plus Terminal for SGMII output
SIP	32	AI	—	Plus Terminal for SGMII input
SIN	33	AI	—	Minus Terminal for SGMII input
VDDA_SI	34	P	—	Analog SGMII receiver supply (1.1 V)
VDDIO	35	P	—	Digital interface I/O voltage (1.8 V, 2.5 V, 3.3 V)
MDIO	36	IO	VDDIO	SMI Data I/O
GNDPAD	37	G	—	Exposed die pad

[1] I: digital input; O: digital output; IO: digital input/output; AO: Analog output; AI: Analog Input; AIO: Analog input/output; PU: weak pull-up; P: power supply. G: ground
[2] Pin type depends on functional mode; consult the functional description for details.
[3] Only applicable to CMOS input/output pins

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6 Functional description

6.1 Operating modes

A mode transition diagram showing all internal device- and port-level state transitions is shown in Figure 4. It represents externally observable behavior.

Internally, the device architecture, including the register layout and interrupt tree, is scalable and extends to multiport PHY devices.

See the TJA1104 application notes [8] for a detailed description of application use-cases.

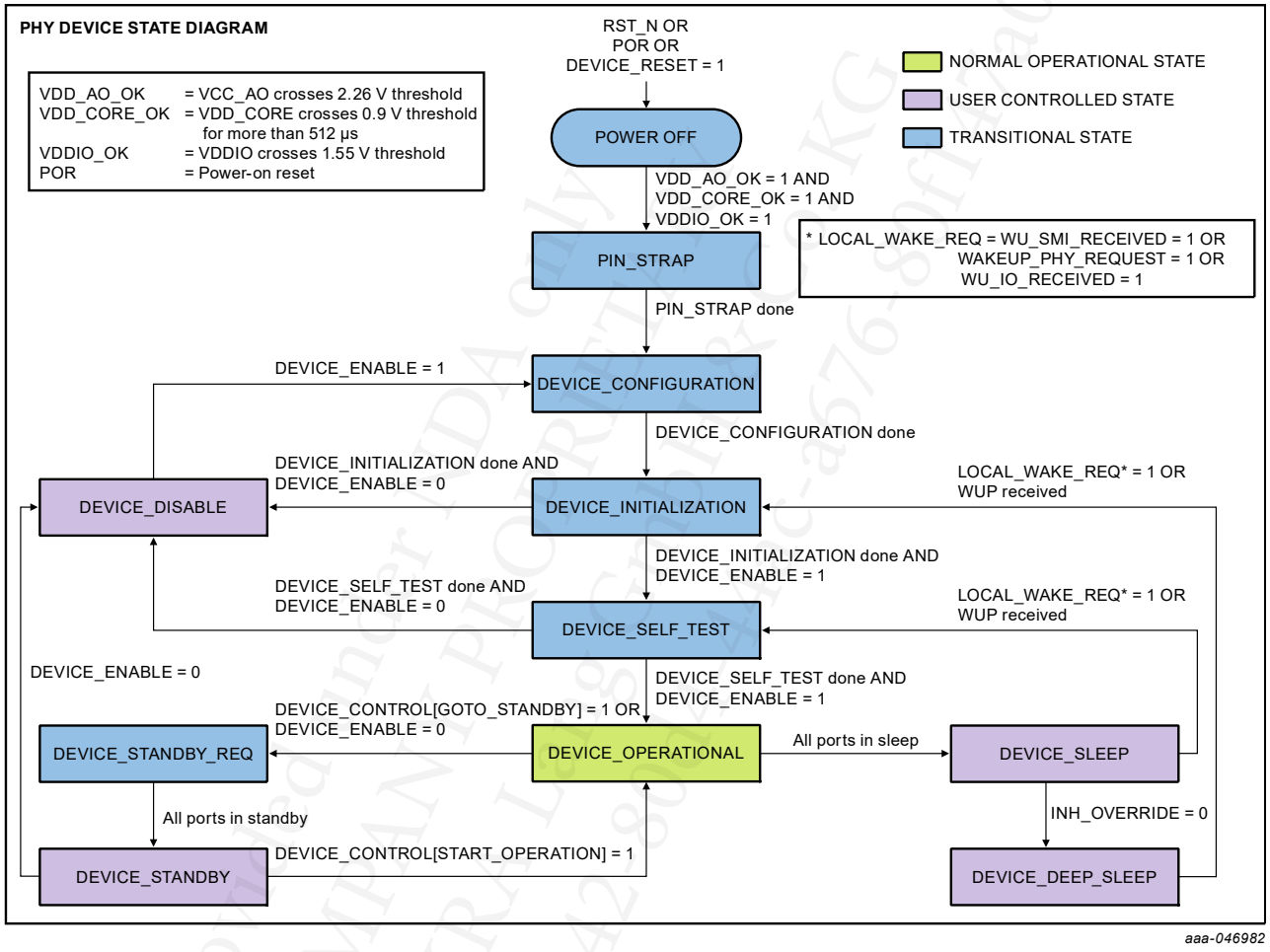


Figure 4. Device-level mode transitions

6.2 Managed/autonomous operation

The device is configured in either managed or autonomous mode, which determines the startup behavior. The mode can be set via pin strapping (see Table 18) or via bit PHY_CONFIG[AUTO_OPERATION] (Table 342).

In managed mode, the device transitions to STANDBY state after power-on and device initialization. This allows additional configuration options to be set before bringing the device into OPERATIONAL state by setting PHY_CONTROL[START_OPERATION]. SMI access is required to trigger this transition.

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In autonomous mode, the device proceeds to OPERATIONAL state without any interaction needed. In OPERATIONAL state, link-up can begin immediately. This enables very fast link-up after reset and allows parallel link-up and host controller boot-up. SMI access is optional in this mode if further configuration or status access is not needed.

6.3 Datapath block diagram

The TJA1104 data path consists of multiple connected building blocks: The CMOS IO pads[5] or SGMII SerDes[6], xMII rate-matching FIFO, the PTP, BIST and MACSec blocks and the 100BASE-T1 PHY.

Note: *After device reset, PTP and BIST and MACSec blocks are bypassed and xMII and CBT1 blocks are connected.*

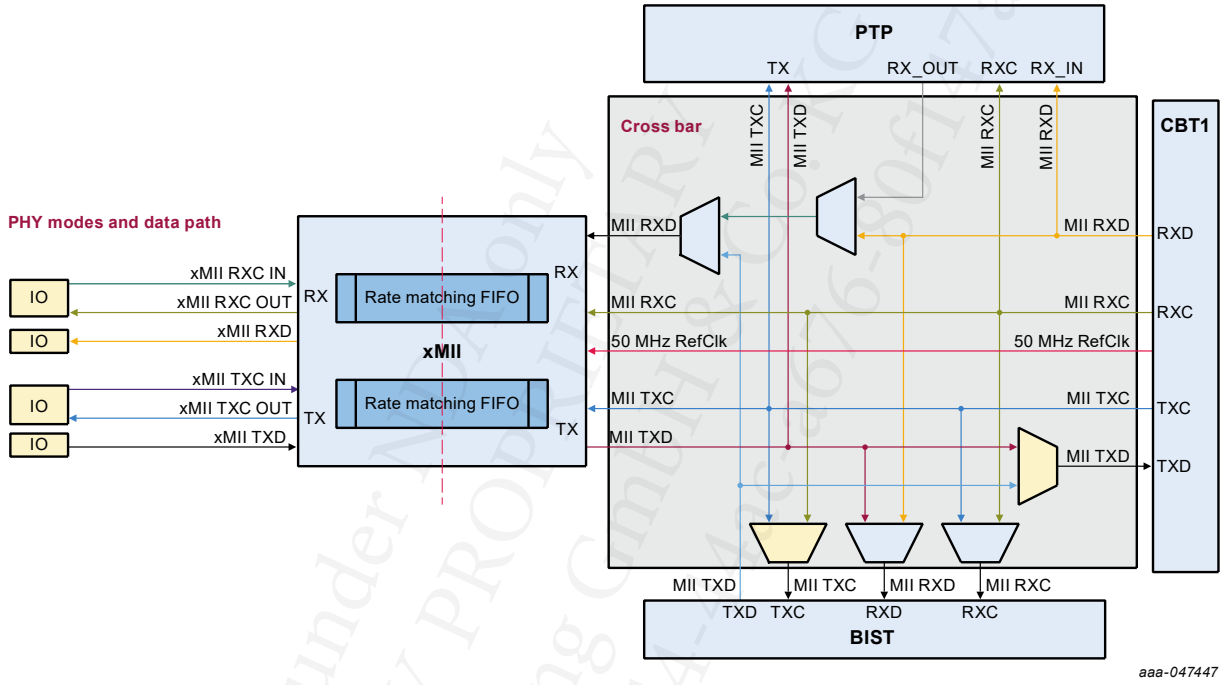


Figure 5. "Datapath block diagram (except MACsec, refer to 6.12 for details on MACsec)

5 Variant TJA1104A and TJA1104C
6 Variant TJA1104B and TJA1104D

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Table 4. xMII Datapath control

MODE	CMOS_OR_SERDES	xMII_MODE	ROLE	RMII_REFCLK	TXC dir.	RXC dir.	REFCLK_SRC	TX FIFO	RX FIFO
MII	0	100	0	0	out	out	XI pin	No	No
revMII	0	100	1	0	in	in	XI pin	Yes	Yes
RMII	0	101	0	0	in	-	XI pin	Yes	Yes
RMII clock mode	0	101	0	1	in	-	TXC	Yes	Yes
revRMII	0	101	1	0	out	-	XI pin	Yes	Yes
RGMII	0	111	0	0	in	out	XI pin	Yes	No
SGMII	1	001	0	0	-	-	XI pin	Yes	Yes

Table 5. BIST datapath control

BIST MODE	BIST_ENABLE	BIST_TX_TO_EPHY	BIST_TX_TO_XMII	BIST_RX_EPHY_OR_XMII	DRIVER_PORT	CAPTURE_PORT
Mode 1 (EPHY)	1	10	00	1	CBT1_TX	CBT1_RX
Mode 2 (xMII)	1	10	00	0	CBT1_TX	XMII_TX
Mode 3	1	00	10	0	XMII_RX	XMII_TX
Mode 4	1	00	10	1	XMII_RX	CBT1_RX
Disabled (default)	0	-	-	-	DISABLED	

The PTP block is added to the datapath by deasserting PORT_PTP_CONTROL[BYPASS].

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6.4 Startup self-test

The TJA1104 executes a self-test on startup. The test is split into two parts a device-level self-test and a port-level self-test. Both tests are part of the device and port startup sequence (see Section 6.1). The tests consists of several steps including voltage, clock, and latent fault checks.

Voltage supply inputs are tested for over-voltage (OV) and/or under-voltage (UV) conditions. The reference 25 MHz (Crystal or TXC pin in RMII-clkmode) are compared against the internal free-running oscillator.

When both the device and the port self-test checks are completed and pass, the DEVICE_STATUS[FUSA_STATUS] bit is set and the FUSA_PASS_IRQ is triggered automatically. In case a check has not passed successfully, the issue is reported via the related status register.

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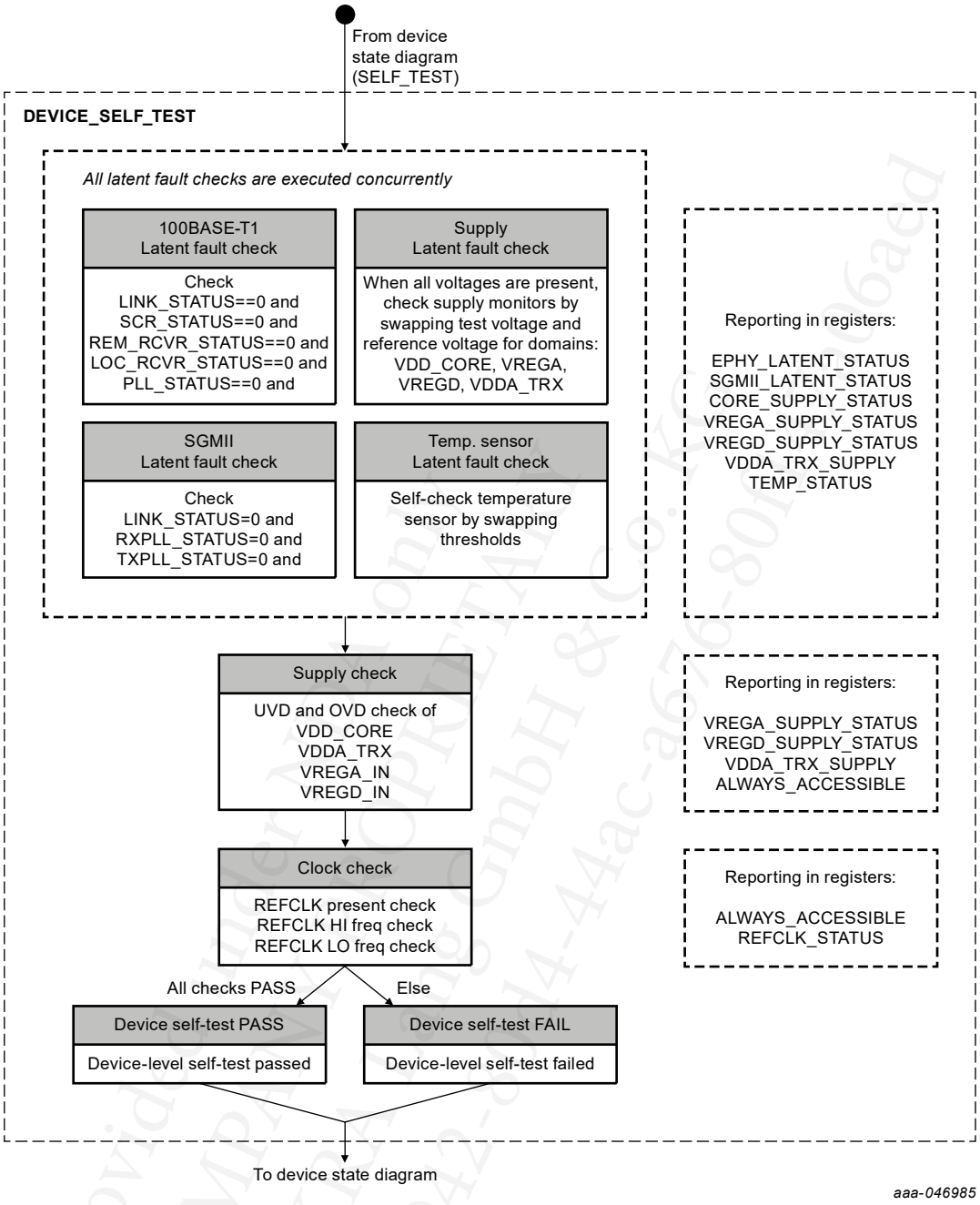


Figure 6. Device self-test sequence

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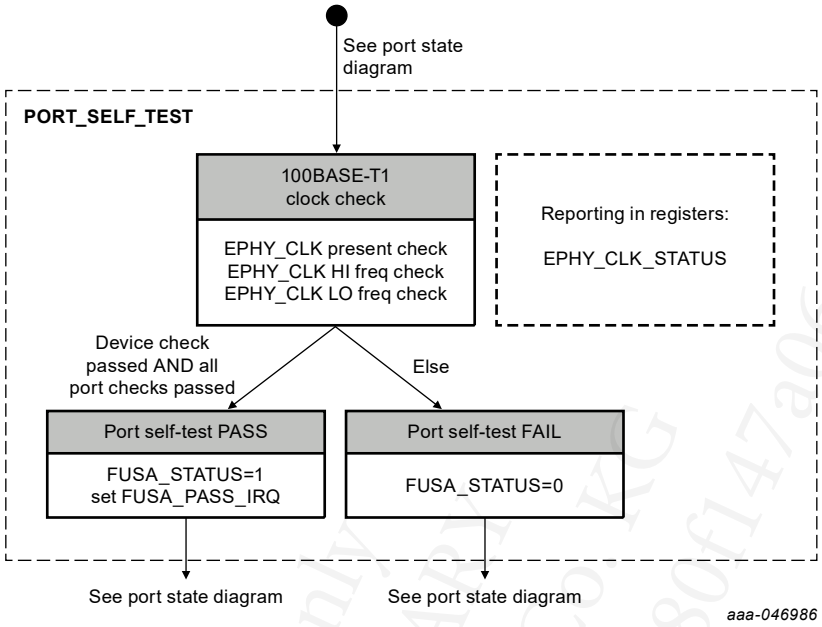


Figure 7. Port self-test sequence

6.5 xMII interfaces

The TJA1104 supports a number of xMII modes that can be selected via pin strapping and by MDIO registers.

Table 6. Interface support

Interface	TJA1104A and TJA1104C	TJA1104B and TJA1104D
MII	•	
RMII	•	
RGMII	•	
SGMII		•

For MII/RMII/RGMII most modes can be selected via pin strapping Section 6.7.1. All settings can be changed through register settings.

6.5.1 MII

The connections between the PHY and MAC are shown in more detail in Figure 8. Data is exchanged via 4-bit wide data nibbles on TXD[3:0] and RXD[3:0]. Transmit and receive data is synchronized with the transmit (TX_CLK) and receive (RX_CLK) clocks respectively. In MII operation, the clock signals are provided by the PHY and are typically derived from an external clock or crystal (master) or recovered from the MDI (slave). Both clocks have a nominal frequency of 25 MHz (± 100 ppm). Normal data transmission from MAC to PHY is initiated by asserting TX_EN, while data reception from PHY to MAC is indicated by asserting RX_DV.

Next to MII mode, the TJA1104 supports reverse MII (rev-MII). In rev-MII, the PHY's MII interface acts similar to a MAC. In this mode TX_CLK and RX_CLK are inputs to the PHY, while the roles of TX_EN, TX_ER, TXD, RX_DV, RXD an RX_ER remain identical. The rev-MII mode can be used to connect two PHY devices for media-converter or range-extender applications.

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Table 7. MII modes

MII mode	Description
MII	In this mode, the device behaves as per the IEEE 802.3 MII spec and TX_CLK and RX_CLK are outputs
rev-MII	In this mode, the device behaves similar to the MAC interface as described in IEEE 802.3 MII spec. TX_CLK and RX_CLK are inputs. As in MII, RXD, RX_DV and RX_ER remain outputs and TXD, TX_EN and TX_ER remain inputs to the TJA1104A. and TJA1104C.

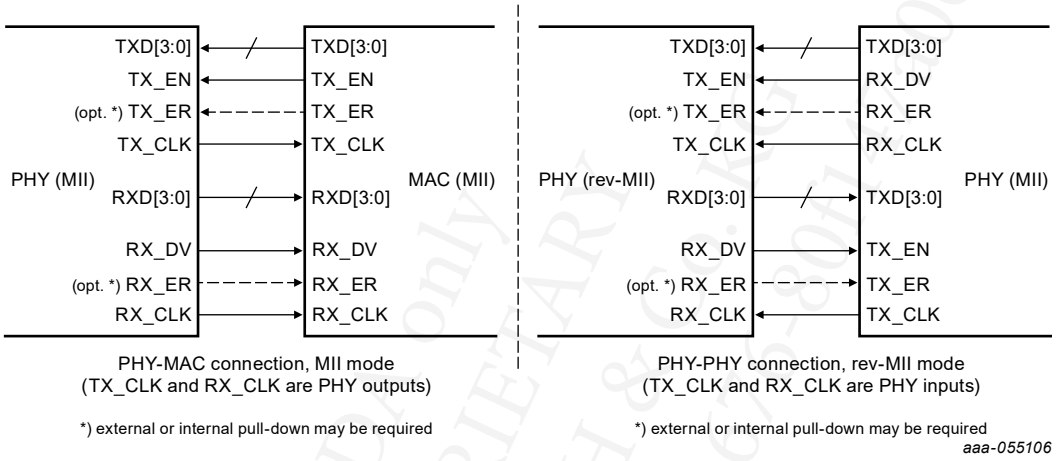


Figure 8. MII signaling

6.5.1.1 MII pinning

The following table shows the mapping of the MII interface pins to device pins.

Table 8. Pinning in MII-mode

MII Function	Pin	Symbol	Type[1]	Description
RX_ER	19	RX_ER / GPIO2 / TDO / CONFIG1	O	Receive error output[2]
RXD3	20	RXD3 / GPIO3 / CONFIG2	O	Receive data bit 3 of RXD[3:0] nibble
RXD2	21	RXD2 / GPIO4 / CONFIG3	O	Receive data bit 2 of RXD[3:0] nibble
RXD1	23	RXD1 / CONFIG4	O	Receive data bit 1 of RXD[3:0] nibble
RXD0	24	RXD0 / CONFIG5	O	Receive data bit 0 of RXD[3:0] nibble
RX_DV	25	RX_CTL / CONFIG6	O	Receive data valid output
RXC	26	RXC / GPIO8 / TX_TCLK	O[3] I[4]	Receive clock
TX_ER	28	TX_ER / GPIO9 / TCK	I, PD	Transmit error input. If unused, can be left open.[5]
TXC	29	TXC	O[3] I[4]	Transmit clock
TX_EN	30	TX_CTL	I, PD	Transmit data enable input
TXD0	31	TXD0	I, PD	Transmit data bit 0 of TXD[3:0] nibble

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Table 8. Pinning in MII-mode*...continued*

MII Function	Pin	Symbol	Type[1]	Description
TXD1	32	TXD1	I, PD	Transmit data bit 1 of TXD[3:0] nibble
TXD2	33	TXD2 / GPIO10	I, PD	Transmit data bit 2 of TXD[3:0] nibble
TXD3	34	TXD3 / GPIO11	I, PD	Transmit data bit 3 of TXD[3:0] nibble

[1] I: digital input; O: digital output; IO: digital input/output; PU: weak pull-up; PD: weak pull-down
[2] Optional, if not used pin GPIO2 is available; see RXER_GPIO_SWAP
[3] MII mode
[4] rev-MII mode
[5] Optional, if not used pin GPIO9 is available, see TXER_GPIO_SWAP.

6.5.2 RMII

RMII data is exchanged via 2-bit wide data bits on TXD[1:0] and RXD[1:0], as illustrated in Figure 9. To achieve 100 Mbit/s, the data is clocked at twice the frequency compared to MII.

Data and control signals are synchronous to a single clock signal, REF_CLK. This 50 MHz (± 50 ppm) clock is typically input to the PHY (RMII mode). Alternatively the PHY can output a 50 MHz REF_CLK (RMII-rev mode). This mode can be used in PHY back-to-back configurations or with a MAC that does not provide a 50 MHz REF_CLK.

The 100BASE-T1 is an active-idle, full-duplex transmission scheme and does not employ the CRS signal. The CRS_DV is a fully synchronous signal to REF_CLK and is identical to RX_DV. When, at the start of a frame TX_EN is asserted and RXD is found to be 00b, data is discarded until RXD is found do be 01b. This allows to connect the TX interface of this devices to a RX interface of another PHY.

The TJA1104 supports the following RMII modes:

Table 9. RMII modes

RMII mode	Description
RMII	In RMII mode, the REF_CLK is configured as an input and expects a 50 MHz clock.
RMII-rev	In RMII mode, the REF_CLK is configured as a 50 MHz output clock.
RMII-clkmode	This mode is as RMII, but the 50 MHz REF_CLK is used to clock the entire TJA1104. In this mode, an external XTAL is not required.

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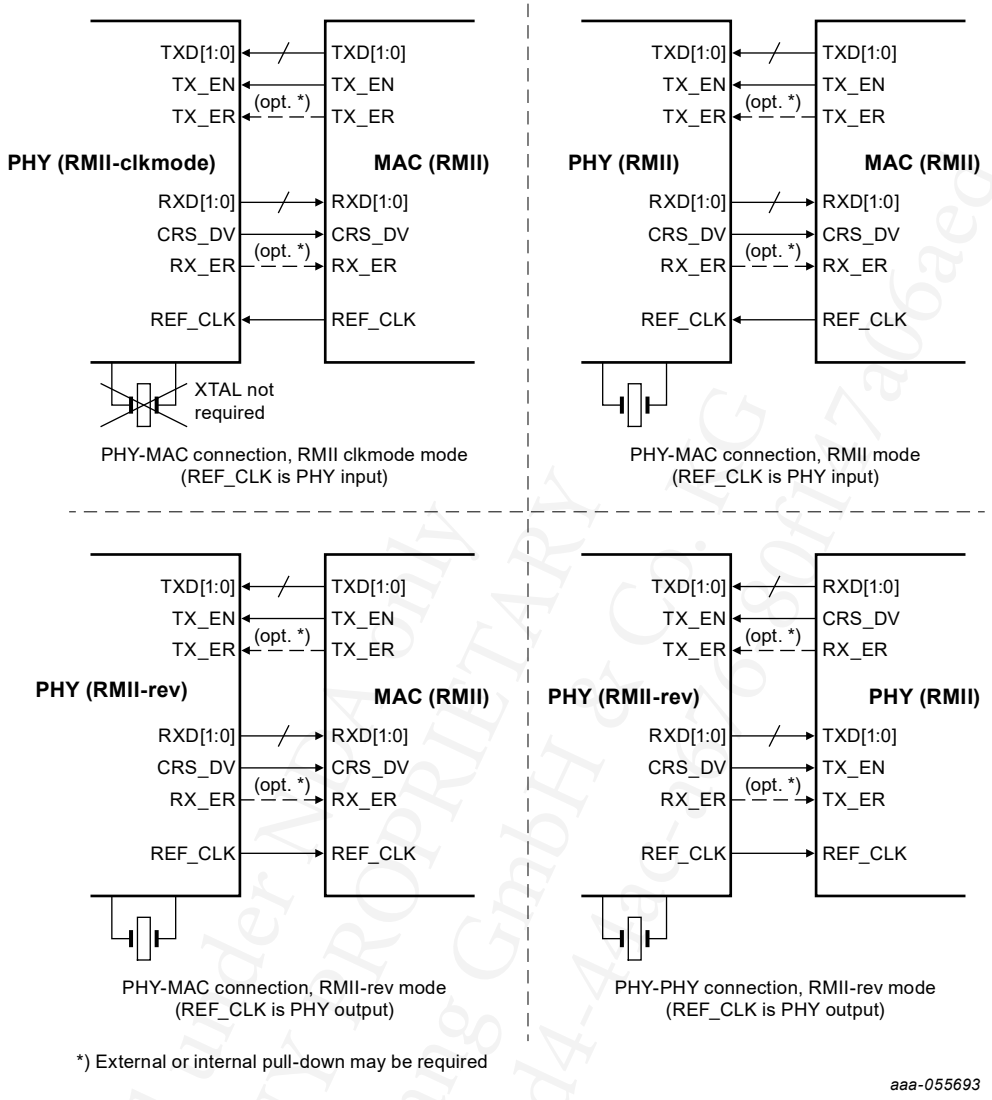


Figure 9. RMII signaling

Note: *REF_CLK can be also input to both MAC and PHY*

6.5.2.1 RMII pinning

The following table shows the mapping of the RMII interface pins to device pins. Unused GPIO can be left floating, consult Section 6.16.2 for details on how to use available GPIOs.

Table 10. Pinning in RMII-mode

Function	Pin	Symbol	Type[1]	Description
RX_ER	19	RX_ER / GPIO2 / TDO / CONFIG1	O	Receive error output[2]
—	20	RXD3 / GPIO3 / CONFIG2	IO	GPIO3
—	21	RXD2 / GPIO4 / CONFIG3	IO	GPIO4
RXD1	23	RXD1 / CONFIG4	O	Receive data bit 1 of RXD[3:0] nibble

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Table 10. Pinning in RMII-mode*...continued*

Function	Pin	Symbol	Type[1]	Description
RXD0	24	RXD0 / CONFIG5	O	Receive data bit 0 of RXD[3:0] nibble
CRS_DV	25	RX_CTL / CONFIG6	O	Carrier sense, receive data valid output
—	26	RXC / GPIO8 / TX_TCLK	IO	GPIO8
TX_ER	28	TX_ER / GPIO9 / TCK	I, PD	Transmit error input. If unused, can be left open.[3]
REF_CLK	29	TXC	O[4] I[5]	REF_CLK clock
TX_EN	30	TX_CTL	I, PD	Transmit data enable input
TXD0	31	TXD0	I, PD	Transmit data bit 0 of TXD[3:0] nibble
TXD1	32	TXD1	I, PD	Transmit data bit 1 of TXD[3:0] nibble
—	33	TXD2 / GPIO10	IO	GPIO10
—	34	TXD3 / GPIO11	IO	GPIO11

[1] I: digital input; O: digital output; IO: digital input/output; PU: weak pull-up; PD: weak pull-down
[2] Optional, if not used pin GPIO2 is available; see RXER_GPIO_SWAP
[3] Optional, if not used pin GPIO9 is available; see TXER_GPIO_SWAP
[4] RMII-rev
[5] RMII

6.5.3 RGMII

The RGMII PHY-MAC connection is shown in Figure 10. The RGMII protocol is intended as an alternative to the IEEE 802.3 GMII standard[7]. The objective is to reduce the number of pins between MAC and PHY in a cost-effective and technology independent way.

Control signals are multiplexed and transmitted data is synchronous to both clock edges (double data rate). The interface is intended for 1000 Mbit/s operation, but supports also 100 Mbit/s and 10 Mbit/s operation modes. This device only supports the 100 Mbit/s mode.

RGMII is a symmetrical interface and TXC is generated by the MAC and RXC is generated by the PHY. For 100 Mbit/s operation, the clock operates at 25 MHz (+/- 50 ppm) and data is duplicated on the falling edge of the appropriate clock.

RX_CTL and TX_CTL encode the RX_DV and RX_ER as well as TX_EN and TX_ER, respectively. RX_CTL/TX_CTL present the RX_DV/TX_EN on the rising edge and RXERR/TXERR on the falling edge. RXERR and TXERR are encoded to minimize the number of transitions during nominal data transmission.

TXERR <= TX_ER xor TX_EN

RXERR <= RX_ER xor RX_DV

This device implements RGMII v1.3 as well as RGMII 2.0 by supporting the RGMII integrated-delay option (RGMII-ID). Optional delay in the clock path can be added independently in the TXC and the RXC clock path.

Table 11. RGMII modes

RGMII mode[1]	Description
RGMII	In RGMII mode, the interface functions according to RGMII v1.3 (no internal delay is implemented)
RGMII-ID	In RGMII-ID mode the interface functions as per RGMII v2.0 (internal delay on RXC is enabled, no internal delay on TXC).

7 not supported

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Table 11. RGMII modes...*continued*

RGMII mode[1]	Description
RGMII-ID (TX/RX)	In RGMII-ID (TX/RX) mode the interface functions as per RGMII v2.0 (internal delay on both RXC and TXC is enabled).

[1] All possible TX/RX combinations for internal delay can be selected via register, but only a subset is available as pin strap options. See Section 6.7

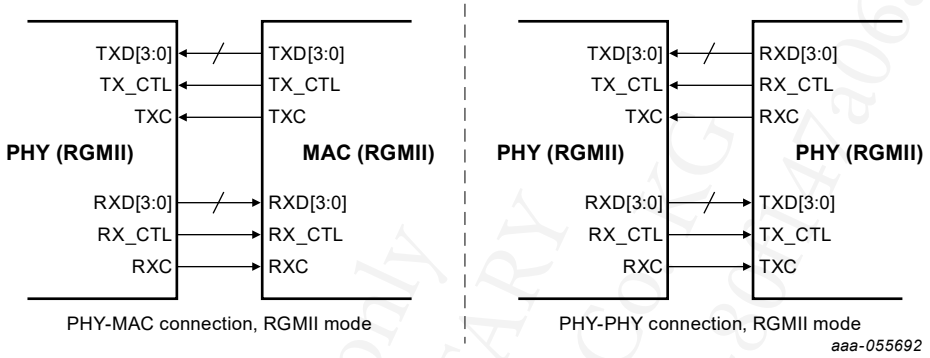


Figure 10. RGMII signaling

6.5.3.1 RGMII pinning

The following table shows the mapping of the RGMII interface pins to device pins. Unused GPIOs can be left floating, consult Section 6.16.2 for details on how to use available GPIOs.

Table 12. Pinning in RGMII-mode

Function	Pin	Symbol	Type[1]	Description
—	19	RX_ER / GPIO2 / TDO / CONFIG1	IO	GPIO2
RXD3	20	RXD3 / GPIO3 / CONFIG2	O	Receive data bit 3 of RXD[3:0] nibble
RXD2	21	RXD2 / GPIO4 / CONFIG3	O	Receive data bit 2 of RXD[3:0] nibble
RXD1	23	RXD1 / CONFIG4	O	Receive data bit 1 of RXD[3:0] nibble
RXD0	24	RXD0 / CONFIG5	O	Receive data bit 0 of RXD[3:0] nibble
RX_CTL	25	RX_CTL / CONFIG6	O	Receive data control output
RXC	26	RXC / GPIO8 / TX_TCLK	O	Receive clock
—	28	TX_ER / GPIO9 / TCK	IO	GPIO9
TXC	29	TXC	I	Transmit clock
TX_CTL	30	TX_CTL	I, PD	Transmit data control input
TXD0	31	TXD0	I, PD	Transmit data bit 0 of TXD[3:0] nibble
TXD1	32	TXD1	I, PD	Transmit data bit 1 of TXD[3:0] nibble
TXD2	33	TXD2 / GPIO10	I, PD	Transmit data bit 2 of TXD[3:0] nibble
TXD3	34	TXD3 / GPIO11	I, PD	Transmit data bit 3 of TXD[3:0] nibble

[1] I: digital input; O: digital output; IO: digital input/output; PU: weak pull-up; PD: weak pull-down

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6.5.3.2 RGMII clock delay

The RGMII integrated-delay is controlled by pin strapping and optionally register control. The most common modes can be selected directly from pin strapping, see Section 6.7.1.

The clock delay can be configured via registers RGMII_TXC_DELAY_CONFIG and RGMII_RXC_DELAY_CONFIG.

6.5.4 SGMII

6.5.4.1 Introduction

The Serial-GMII (SGMII) is a SerDes interface for PHY-MAC connection. The SGMII protocol is intended to be an alternative to the RGMII standard. It uses less interface signals and provides better EMC performance by using differential signaling.

The device provides the SGMII 4-wire interface[8] which implements clock data recovery so no additional clock pairs are required.

The transfer rate of SGMII is fixed at 1.25 GBaud. For 100 Mbit/s operation, the RX and TX interfaces "elongates" the frame by replicating each octet 10 times. The device automatically enables the SGMII transmitter (SON, SOP) on device startup in order to reduce time to communication.

6.5.4.2 Block diagram

A block diagram of the SGMII subsystem is shown in Figure 11.

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8 SGMII 8-wire interface is not supported.

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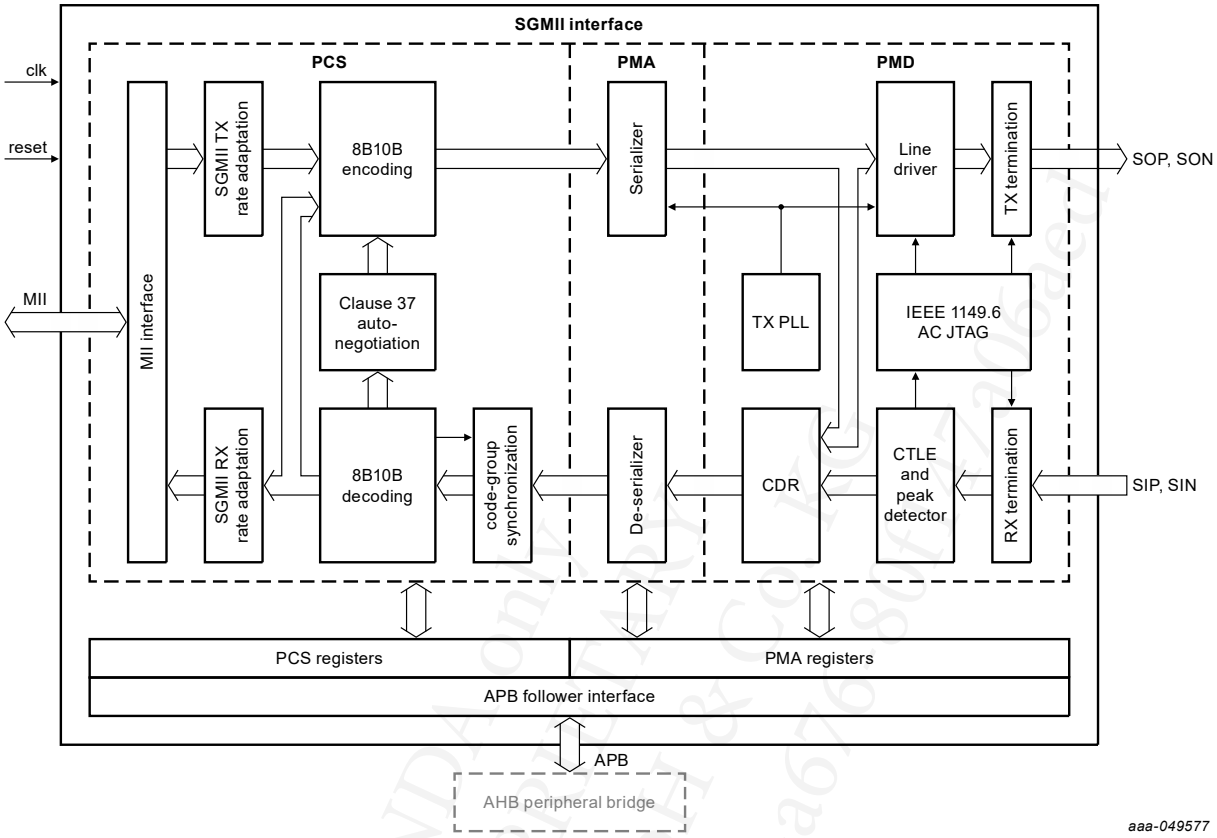


Figure 11. SGMII block diagram

6.5.4.3 Signaling

The SGMII block is AC-compliant to the SGMII 1.8 specification. The interface must always be AC coupled with capacitors of 100 nF.

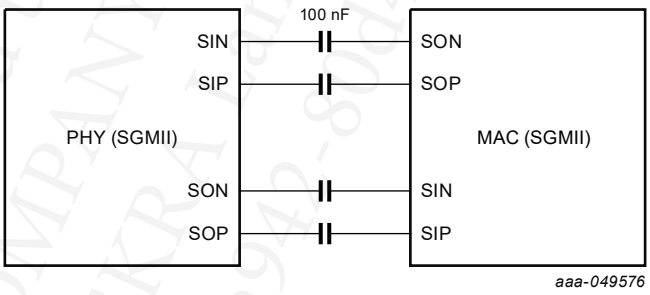


Figure 12. SGMII signalling

6.5.4.4 Polarity

The polarity of the differential pairs SI and SO can be flipped by the register bits SGMII_VR_MII_DIG_CTRL2[RX_POL_INV_0] and SGMII_VR_MII_DIG_CTRL2[TX_POL_INV_0], respectively.

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6.5.4.5 Auto-Negotiation

The SGMII interface implements (optional) Auto-Negotation. In this mode, PHY, and MAC handshake the supported interface capabilities to find the optimal operating conditions. The device advertise its speed, duplex mode (full-duplex or half-duplex).

In SGMII-PHY mode, the device communicates link speed as well as 100BASE-T1 link availability by passing tx_config_Reg from PHY to MAC. If configured in SGMII-MAC mode, the received autonegotiation data is ignored.

Auto-negotiation is automatically enabled and control information automatically exchanged from PHY to MAC via tx_config_Reg[15:0] as shown in Table 13. The PHY advertises the LINK_AVAILABILITY via tx_config_Reg[15]. Duplex mode and speed are fixed to full duplex and 100 Mbps.

Table 13. Auto-Negotiation control information passed between PHY and MAC

Bit Number	tx_config_Reg[15:0] sent from the PHY to the MAC	tx_config_Reg[15:0] sent from the MAC to the PHY
15	Link: 1 = link up, 0 = link down	0
14	Reserved for Auto-Negotiation acknowledge as specified in 802.3z	1
13	0	0
12	Duplex mode: 1 = full duplex, 0 = half duplex	0
11:10	Speed: Bit 11, 10: 1 1 = Reserved 1 0 = 1000 Mbps 0 1 = 100 Mbps 0 0 = 10 Mbps	0
9:1	0h	0h
0	1	1

6.5.4.6 Mode of operation

6.5.4.6.1 Normal operation

During normal operation, the transmit and receive data path of the SGMII interface are active. The transmit path accepts Ethernet data frames from the switch through the MII interfaces, applies rate adaptation for 100 Mbps Ethernet data and applies 1000BASE-X 8B10B encoding. After serialization the data is driven onto the physical medium by the line driver, which has an on-chip transmitter termination for a good impedance match to the medium. The TXPLL generates the half bit-rate clock necessary for serializing the data. The chip-internal MII interface clocks are derived from the clock generated by the TXPLL.

The receive path receives a serial data stream from the physical medium, to which is provides an impedance match with the on-chip receiver termination. The CTLE equalizes the received signals, to compensate for channel losses, before recovering the clock and data from the serial stream by the CDR. The recovered data is deserialized before aligning and decoding 8B10B code groups. Finally, rate adaption is applied for 100 Mbps Ethernet data before making the data frames available for on-chip MII interface.

The interface starts autonomously in normal mode. If the interface is in standby state, setting the SGMII_ADVANCED_CONTROL[START_OPERATION] bit transitions interface into normal mode.

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6.5.4.6.2 Standby

The SGMII interface can be disabled by setting the bit SGMII_ADVANCED_CONTROL[GOTO_STANDBY].

6.5.4.6.3 PCS loopback

The on-chip PCS RX-to-TX loopback is an operating mode where the received data is not applied by the PCS to the MII interface but is applied instead to the transmit path of the PCS. In this operating mode the SGMII interface effectively sends the serial data stream, received from the SGMII link partner, back out to the link partner through the transmitter data path.

6.5.4.6.4 PMA loopback

The on-chip PMA TX-to-RX loopback is an operating mode where the serial data stream of the transmit path is not provided to the line driver but is looped back instead to the CDR, which in turn does not receive the serial data from the SGMII link partner, through the CTLE, but recovers the data from the transmit path's serializer instead.

6.5.4.7 Boundary scan

In boundary scan mode the SGMII interface is used to perform AC-JTAG operations for testing the interconnect to the SGMII interface on other IC for manufacturing defects.

6.5.5 Alternate xMII pinning

The MII, RMII and RGMII standards do not dictate the physical pin arrangement of the xMII bus. Next to the default xMII pinning of the TJA1104A/TJA1104C, which are described in Section 5.1, the devices have the possibility for alternate mappings. These alternate mappings can be selected through registers TXD30_SWAP and RXCTL_SELECT. These two settings can be applied independently to allow more compact PCB layouts with less or no signal swaps.

If set, the TXD30_SWAP bit will flip the TXD bus from TXD[3:0] to TXD[0:3]. If set, the RXCTL_SELECT bit moves the RX_CTL function on the opposite side of the RXD[3:0] bus. The alternate pinning configurations are shown in Figure 13.

Any GPIO pin that is made available on RXD[2:3] and TX[2:3] moves respectively.

Note: *The alternate settings TXD30_SWAP and RXCTL_SELECT cannot be selected through pin strapping. Thus, on device startup the MII/RMII/RGMII pins are as per default configuration.*

Note: *The alternate pinning described in this section only applies to the TJA1104A/TJA1104C variants.*

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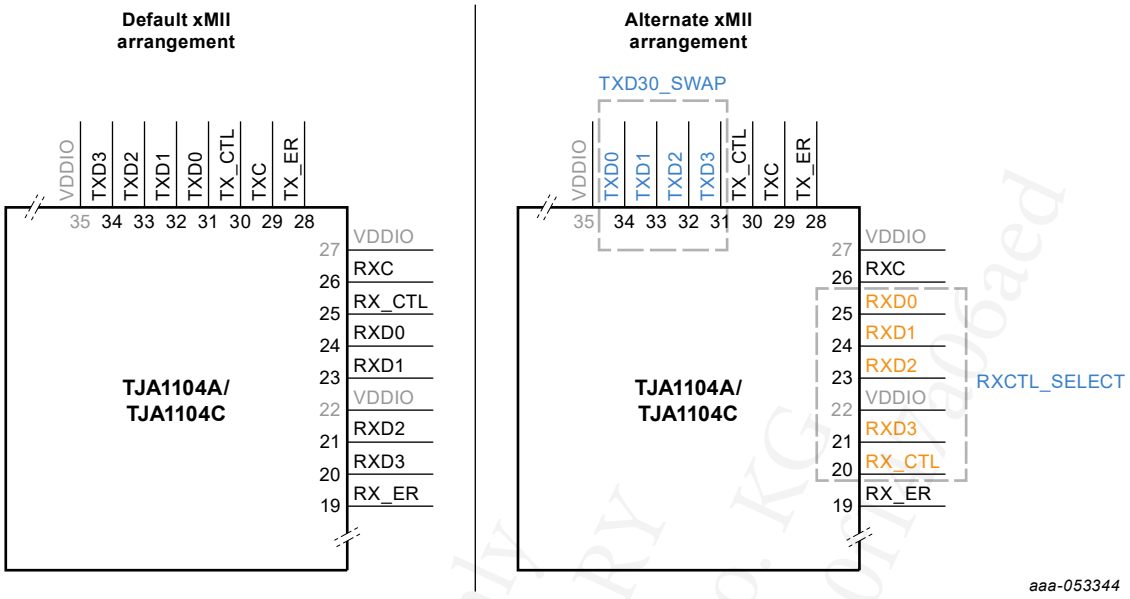


Figure 13. Default and alternate xMII pin arrangements

6.5.6 xMII elastic buffers

The xMII interface uses elasticity buffers to compensate between clock differences of the 100BASE-T1 link and the xMII clocks. These buffers are able to compensate rate differences of up to 100 ppm in all xMII modes. The clocking scheme and thus the need of buffering depends on the selected xMII mode.

Note: *When using a crystal that exceed the RMII and RGMII specification (50 ppm), the MAC must also be able to operate in such conditions.*

The latencies caused by these buffers are included in the overall xMII to MDI latency numbers reported in the dynamic characteristics. Table 14 shows in which mode buffers are added in the datapath and when they are bypassed.

Table 14. Elastic buffer scheme

xMII mode	TX Interface	RX Interface
RGMII	enabled	bypassed
RMII / revRMII / RMII-clkmode	enabled	enabled
MII	bypassed	bypassed
revMII	enabled	enabled

6.6 MDIO

The MDIO interface is a management interface to control the device through a two-wire shared bus interface. Individual PHYs are selected through a PHYAD identifier, configured through pin strapping. This device supports up to 27 addresses, see Section 6.7.1.

Note: *PHYAD zero is reserved and cannot be assigned. Optionally, the device can listen on PHYAD zero for broadcast read/writes.*

There are two protocol versions of MDIO, IEEE 802.3 Clause 22 and IEEE 802.3 Clause 45. The earlier Clause 22 standard allows to address 32 registers and up to 32 PHY devices. The more recent Clause 45

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standard allows to address up to 65536 registers per PHY per MMD. Both MDIO protocols are supported by the TJA1104, but only Clause 45 gives a full view on all registers, see Section 6.6.5 and Section 6.6.4.

Frame formats of both standards are provided in Section 6.6.2 and Section 6.6.1.

Note: *Do not interleave Clause 22 with Clause 45 accesses. Use Clause 22 or Clause 45 exclusively.*

Figure 14 shows a typical MDIO topology.

Note: *The TJA1104 does not have integrated pull-up or pull-down resistors in MDIO or MDC.*

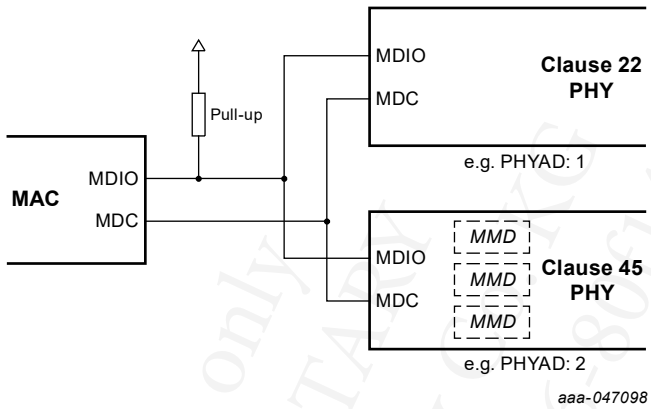
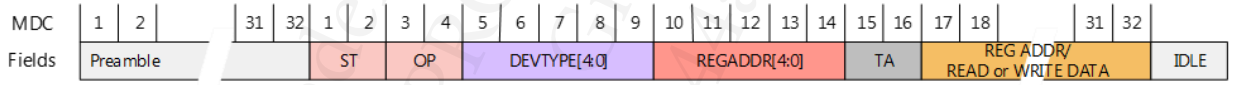


Figure 14. MDIO signaling

6.6.1 IEEE 802.3 Clause 45

Figure 15 shows the IEEE Clause 45 frame format and access types.



	Management Frame Fields							
	PRE	ST	OP	PRTAD	DEVAD	TA	ADDRESS/DATA	IDLE
Address	1...1	00	00	PPPPP	EEEEE	10	AAAAAAAAAAAAAAAA	Z
Write	1..1	00	01	PPPPP	EEEEE	10	DDDDDDDDDDDDDDDD	Z
Read	1..1	00	11	PPPPP	EEEEE	Z0	DDDDDDDDDDDDDDDD	Z
Post-read-increment-address	1..1	00	10	PPPPP	EEEEE	Z0	DDDDDDDDDDDDDDDD	Z

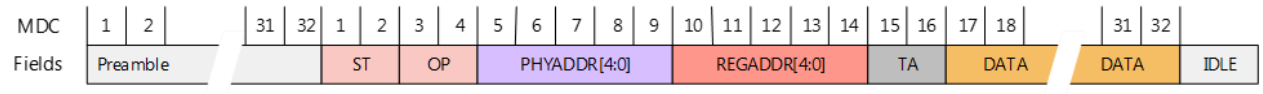
Figure 15. Clause 45 frame format and access types

6.6.2 IEEE 802.3 Clause 22

Figure 16 shows the IEEE Clause 22 frame format and access types.

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	Management Frame Fields							
	PRE	ST	OP	PHYAD	REGAD	TA	DATA	IDLE
Read	1...1	01	10	AAAAA	RRRRR	Z0	DDDDDDDDDDDDDDDD	Z
Write	1..1	01	01	AAAAA	RRRRR	10	DDDDDDDDDDDDDDDD	Z

Figure 16. Clause 22 frame format and access types

6.6.3 MDIO Broadcast

The device supports MDIO broadcast functionality which is disabled by default. If enabled, the TJA1104 responds on a configurable PHY address (by default PHYADR 0) to reads and writes.

This feature can be enabled by setting this bit BROADCAST_ENABLE in register MMD30.SMI_CONFIG.

6.6.4 Clause 45 MMD register organization

The device supports 4 MMDs:

- PMA (MMD 01)
- PCS (MMD 03)
- NXP Vendor Specific (MMD 30)
- NXP Vendor Specific (MMD 31)

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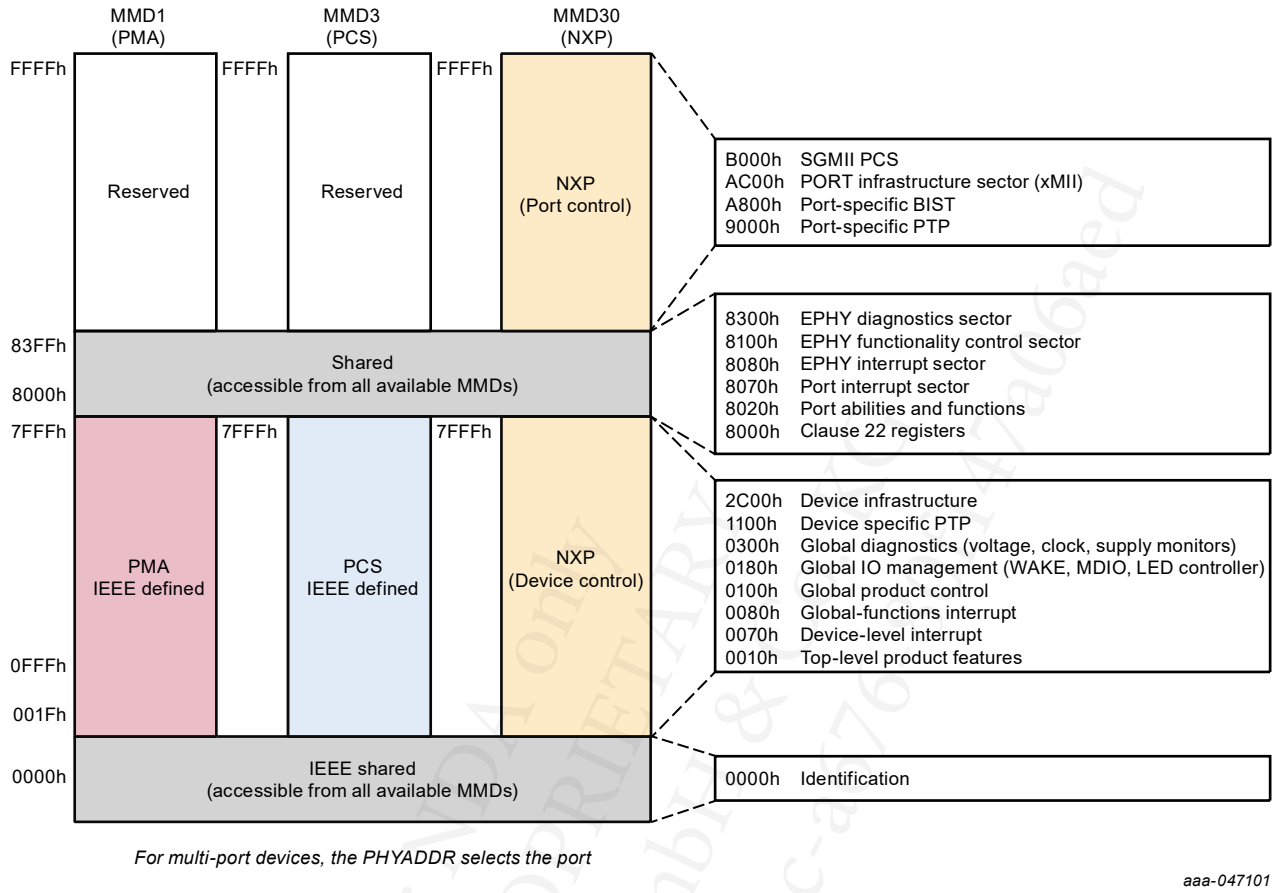


Figure 17. MMD Organization

The lower part of MMD 1 and MMD 3 following the IEEE 802.3 specification for PCS and PMA devices. The lower part of MMD 30 contains registers which are device or otherwise global control registers (e.g. LED, GPIO,...). The higher part of MMD 30 configures port related functions.

Note: *The register organization separates port from device control in order to support multi-port devices (not used in TJA1104).*

Common functionality is mapped into MMD1/3/30, see Shared MMD. In this sense, the Shared MMD is not a real MMD itself. The shared registers are mapped into the range between 0000h-001fh and 8000h-83ffh.

Standard Clause 22 registers are mapped in the vendor-specific register space of all MMDs from 8000h – 801Fh.

6.6.5 Clause 22 register organization

The device implements limited Clause 22 access as listed in Section 7.1 which are the only registers which are directly accessible through IEEE Clause 22 and IEEE Clause 45 type transactions. It must be noted that the C22 registers are not exclusive to IEEE Clause 22 accesses.

The CL22 registers are comprised of:

- PHY identification registers.
- FAVORITE_IRQS, which contains common IRQs. This register can be used to shortcut the IRQ tree.
- IEEE Clause 45 indirection registers, used if the MAC does not support Clause 45 MDIO.

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- ALWAYS_ACCESSIBLE register which is the only register accessible without presence of VDD_CORE. VDDIO and VDDAO must be present to access it.

The C22 register table lists the register offsets as mapped into Shared MMD Clause 45 space.

Notes:

- *The addresses as listed in the C22 register table cannot be used directly for a IEEE Clause 22 MDIO access. The C22 register table lists the register offsets as mapped into Shared MMD Clause 45 space. When accessing these registers through a physical Clause 22 transaction, 8000h must be subtracted from the offset as given in the register table. For example, PHY_ID_1 is located at 2h and ALWAYS_ACCESSIBLE at 1Fh.*
- *There is some limitation in switching between clause 22 and clause 45 access. If both, c22 and c45, are used, consult application notes for additional information.*

6.7 Pin strap

6.7.1 Pin strap configuration

The TJA1104 uses pin strap configuration to set critical configuration data. For this, the CONFIG pins of the device support a 3-level strapping which is sensed after device reset (power-on reset, RST_N pin or reset register). All configuration, except the PHYAD, can be changed at any time over MDIO.

During the sampling, it must be ensured that the CONFIG pins are not overruled by the microcontroller pin configuration. This can be achieved if

- The RX interface of the xMII microcontroller pins is high-ohmic during initialization phase (recommended),
- The RX interface of the xMII microcontroller pins is actively driving and are logically consistent with the specified by pin strapping or
- The RST_N pin is not released until the relevant microcontroller pins are configured as inputs (high-ohmic).

Figure 18 shows the application of the pull-up and pull-down configuration. Care must be taken if a CONFIG pin is used to drive loads that are pulling opposite to the configuration resistor. For LEDs, the polarity can be selected to match the direction of the configuration resistor. In case negative LED polarity is used (as shown in the pull-down case), the polarity of the LED driver must be adjusted via software.

The pin strap sequence has successfully latched the configuration latest 2 ms after RST_N is deasserted and power supplies are available.

Table 15. 3-Level Strap Resistor Values

Strap Setting	R_{pu}	R_{pd}
O	OPEN	OPEN
H	10 kΩ	OPEN
L	OPEN	10 kΩ

Note: *The sum of all resistances on a CONFIG pin must be greater than 145 kΩ in order to be reliably detected as O (OPEN).*

Note: *Consult the application note [7] for layout guidelines.*

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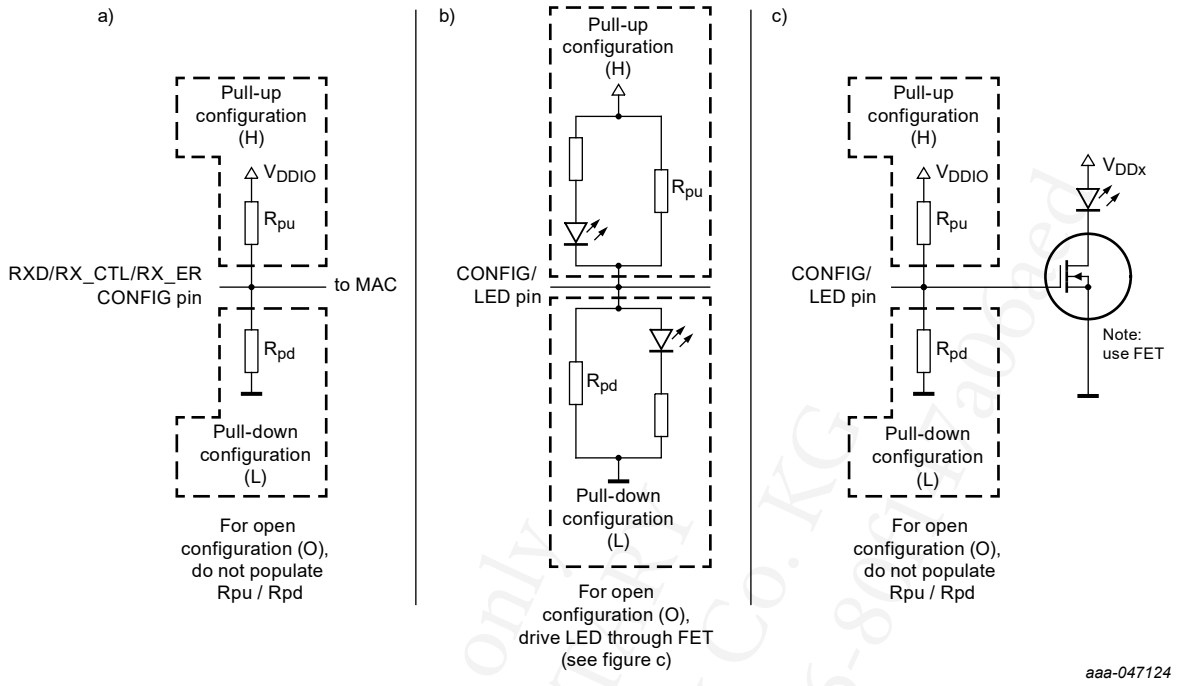


Figure 18. Pin strap configurations

6.7.2 Configuration options

The available CONFIG pins are grouped to configure MODE, PHYAD and miscellaneous settings.

Table 16. MODE group

xMII MODE (TJA1104A/ TJA1104C)	xMII MODE (TJA1104B/ TJA1104D)	Pin CONFIG3	Pin CONFIG4
RGMII-ID (RXC)	SGMII-PHY	O	O
RGMII	SGMII-MAC	O	H
RGMII-ID (both TXC/RXC)	*reserved*	O	L
RMII	*reserved*	H	O
RMII-clkmode	*reserved*	H	H
RMII-rev	*reserved*	H	L
MII	*reserved*	L	O
MII-rev	*reserved*	L	H
JTAG	JTAG	L	L

Table 17. PHYAD group

PHYAD	Pin CONFIG0	Pin CONFIG1	Pin CONFIG2
1	H	H	H
2	H	H	L
3	H	H	O

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Table 17. PHYAD group*...continued*

PHYAD	Pin CONFIG0	Pin CONFIG1	Pin CONFIG2
4	H	L	H
5	H	L	L
6	H	L	O
7	H	O	H
8	H	O	L
9	H	O	O
10	L	H	H
11	L	H	L
12	L	H	O
13	L	L	H
14	L	L	L
15	L	L	O
16	L	O	H
17	L	O	L
18	L	O	O
19	O	H	H
20	O	H	L
21	O	H	O
22	O	L	H
23	O	L	L
24	O	L	O
25	O	O	H
26	O	O	L
27	O	O	O

Table 18. Miscellaneous group

master/slave	AUTONOMOUS Mode	Polarity Correction	Pin CONFIG5	Pin CONFIG6
master	enabled	enabled	O	O
master	disabled	enabled	O	H
slave	enabled	enabled	O	L
slave	disabled	enabled	H	O
master	enabled	disabled	H	H
master	disabled	disabled	H	L
slave	enabled	disabled	L	O
slave	disabled	disabled	L	H
reserved	*reserved*	*reserved*	L	L

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6.8 Reconfiguration

Writing to the configuration registers is disabled by default to avoid unintended reconfiguration. To change the configuration, configuration register writing must be enabled. There are multiple, hierarchical CONFIG_ENABLE switches. Table 19 gives an overview which registers are protected in which way.

There is one device configuration enable bit (SUPER_CONFIG_ENABLE) which enables all reconfiguration. The configuration usually occurs after power-on as part of the software initialization routine. However, PHY reconfiguration is possible at any later time.

All other registers that are not listed in this table are not write reconfiguration protected.

Table 19. Reconfiguration protected registers

Level 1	Level 2	Level 3	Protected registers
MMD30.DEVICE_CONTROL[SUPER_CONFIG_ENABLE]			
	MMD30.DEVICE_CONTROL[GLOBAL_CONFIG]		
			MMD30.DEVICE_CONFIG
			MMD30.WISE_CONFIG
			MMD30.WISE_PARAMETERS
			MMD30.SMI_CONFIG
			MMD30.PORT_FUNC_ENABLES
	MMD30.GLOBAL_INFRA_CONTROL[CONFIG_ENABLE]		
			MMD30.GPIO0_FUNC_CONFIG — MMD30.GPIO11_FUNC_CONFIG
			GPIO_IO_CLK_CONFIG
	MMD30.PORT_CONTROL[CONFIG_ENABLE]		
		Shared.PHY_CONTROL[CONFIG_ENABLE]	
			MMD1.BASE_T1_PMA_CONTROL
			MMD1.E100BT1_PMA_TEST_CONTROL
			SharedMMD.PHY_CONFIG
			SharedMMD.PHY_PARAMETERS
			SharedMMD.WAKE_SLEEP_CONFIG
			SharedMMD.WAKE_SLEEP_PARAMETERS
			SharedMMD.PHY_COMPLIANCE_TEST
		MMD30.PORT_BIST_CONTROL[CONFIG_ENABLE]	
			MMD30.BIST_INTERCEPT_CONFIG
		MMD30.PORT_INFRA_CONTROL[CONFIG_ENABLE]	
			MMD30.INFRA_CONFIG
			MMD30.MII_BASIC_CONFIG
			MMD30.XMII_CLK_CONFIG
			MMD30.XMII_CLK_IO_CONFIG
			MMD30.XMII_DATA_CONFIG

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Table 19. Reconfiguration protected registers*...continued*

Level 1	Level 2	Level 3	Protected registers
			MMD30.XMII_DATA_IO_CONFIG
			MMD30.RGMII_RXC_DELAY_CONFIG
			MMD30.RGMII_TXC_DELAY_CONFIG
			MMD30.RX_PREAMBLE_COUNT
			MMD30.TX_PREAMBLE_COUNT
			MMD30.RX_IPG_LENGTH RX IPG
			MMD30.TX_IPG_LENGTH TX IPG
		MMD30.SGMII_ADVANCED_CONTROL[CONFIG_ENABLE]	
			MMD30.SGMII_ADVANCED_CONFIG

6.9 Polarity Detection/Correction

The 100BASE-T1 PHY in slave mode can automatically detect and flip the MDI polarity (polarity detection and correction). This feature can be enabled/disabled via pin strapping. Optionally, setting the POLARITY_CORRECT_DISABLE bit disables automatic polarity correction.

The MDI polarity can be swapped by setting the POL_SWAP bit. If set, the pins TRX_P and TRX_M are swapped and polarity detection and correction operate on the swapped pinning.

The status of the polarity inversion is reflected by the DETECTED_POLARITY bit. If polarity correction is disabled, the bit DETECTED_POLARITY flags a conflict between actual signal polarity and current MDI polarity. In that case, the pin POL_SWAP should be inverted.

A polarity conflict is also signaled via an IRQ in the PHY_FUNCTIONAL_IRQ_SOURCES group, consult Section 6.11.

6.10 Diagnosis

6.10.1 Link up / status

There are multiple flags that capture link up using different semantics.

The LINK_AVAILABILITY bit is asserted if all signals send_idle_or_data, rem_rcvr_status, scrambler_status, link_status, loc_rcvr_status are asserted. Optionally, glitches are filtered with an hysteresis. The hysteresis is configured with PHY_PARAM[LINK_AVAIL_HYSTERESIS_TIME]. For LINK_AVAIL_HYSTERESIS_TIME=0, this corresponds to the compliance 'link up condition'. This signal is then identical to communication ready (comm_ready) as defined in OPEN Alliance.

The SENDI_OR_DATA status bit indicates if the PHY Control state machine is in SEND_IDLE_OR_DATA state.

The LINK_MONITOR_STATUS bit indicates the link status from Link Monitor FSM.

The RECEIVE_LINK_STATUS flag shows if the PMA/PMD receive link is up. This flag is latched-low.

Note: *It is recommended to use the LINK_AVAILABILITY flag as a link state indication.*

6.10.2 SQI

The signal quality indicator (SQI) is part of the SIGNAL_QUALITY register. The field SQI determines the link quality between values 0 (worst, no-link) and 7 (best). The SQI value is only valid if the VALID bit in the

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SIGNAL_QUALITY register is set. Assuming gaussian noise, an SQI level of 3 and more indicates a BER of 10^{-10} and better.

In addition to the current SQI value, the lowest observed SQI value observed is stored in SQI_MIN_WATERMARK. The SQI_MIN_WATERMARK is cleared on read and set to the current SQI value. In order to avoid extensive SQI polling, an SQI threshold level can be defined. In case the threshold is exceeded an IRQ (SIGNAL_QUALITY_WARNING) is asserted. The threshold can be configured in register SQI_WARN_THRESHOLD. Setting this register to 0h disables the feature.

6.10.3 MSE

The mean square error is captured after channel equalization which has removed ISI, echo and attenuation effects. The MSE metric is capturing signal distortion caused by uncorrelated noise and interference. The MSE is captured after setting the MSE[ENABLE] bit.

The 8-bit MSE value as read from the register is scaled by 1024. To derive the mean square error in units of V^2 it must be divided by 1024. The resulting value is between 0.0 V^2 (best) and 0.25 V^2 (worst). The root mean square error can be derived as follows:

$$RMSE = \sqrt{\frac{MSE}{1024}}\ V_{rms} \quad (1)$$

6.10.4 Test Modes

The 100BASE-T1 PHY supports four test modes according to IEEE Std 802.3, Clause 96.

These test modes allow testing of the transmitter waveform, transmitter distortion, transmitter jitter, and transmitter droop.

The respective test mode is entered by setting MMD30.PHY_CONTROL[CONFIG_EN] enabling the test mode bit MMD30.PHY_CONFIG[TM_ENABLE] and writing the test mode number to in MMD1.B100T1_PMA_TEST_CONTROL[B100T1_TEST_MODE].

6.10.4.1 Test mode 1

Test mode 1 is for testing the transmitter droop. In Test mode 1, the PHY transmits '+1' symbols for 600 ns followed by '-1' symbols for a further 600 ns. This sequence is repeated continuously.

6.10.4.2 Test mode 2

Test mode 2 is for testing the transmitter timing jitter in either master/slave configuration.

In test mode 2, the PHY transmits the data symbol sequence {+1, -1} repeatedly. The transmission of the symbols is synchronized with the local external oscillator in master mode or with the recovered receiver clock in slave mode.

6.10.4.3 Test mode 4

Test mode 4 is for testing the transmitter distortion. In test mode 4, the PHY transmits the sequence of symbols generated by the scrambler polynomial $gs_1 = 1 + x^9 + x^{11}$.

The bit sequence x0n, x1n is derived from the scrambler according to the following equations:

x0n = Scrn[0]

x1n = Scrn[1] XOR Scrn[4]

This stream of 3-bit nibbles are mapped to a stream of ternary symbols according to the table below.

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Table 20. TestMode 4 ternary symbol mapping

X1n	X0n	PAM-3 Transmit Symbol
0	0	0
0	1	+1
1	0	0
1	1	–1

6.10.4.4 Test mode 5

Test mode 5 is for testing the transmit PSD mask. In test mode 5, the PHY transmits a random sequence of PAM-3 symbols. In this mode, PHY is forced to master Mode as per IEEE 802.3

6.10.4.5 Recovered clock (TCLK)

The recovered clock can be externalized on GPIO8. To enable TCLK, the following registers must be configured:

```
PHY_COMPLIANCE_TEST[TX_TCLK_EN] = 1
// Activate the test interface
COMPL_TEST_INTF[TEST_INTF_EN] = 1
COMPL_TEST_INTF[FUNC_SEL] = b0001
COMPL_TEST_INTF[COMPL_TCLK_EN] = 1
```

6.10.5 Loopback

Each PHY supports the following loopback modes:

- PMA local loopback
- PMA remote loopback
- PCS loopback
- xMII PHY loopback
- xMII MAC loopback

6.10.5.1 PMA local loopback

In this mode, the PMA is placed in local loopback mode. The PMA receive function utilizes the echo signals from the unterminated MDI. It decodes these signals to pass the data back to MII Receive Interface as shown in Figure 19.

The MAC can compare the packets sent through MII transmit function to the packets received from the MII receive function to validate PCS and PMA functions.

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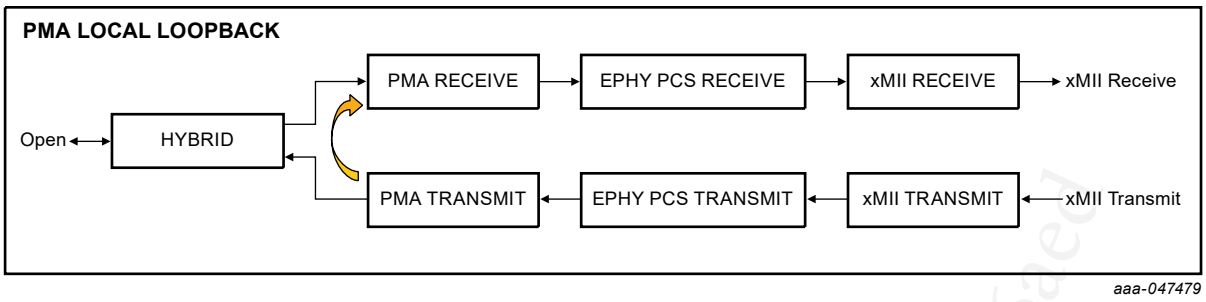


Figure 19. PMA local loopback

To enable PMA local loopback, the LOCAL_LOOPBACK bit in MMD1.PMA_CONTROL1 must be set. The PHY can be in any PHY_MODE. Data can be sent once the following flags in MMD30.PHY_STATUS are set:

- LINK_AVAILABLE
- SENDI_OR_DATA
- REM_RCVR_STATUS
- SCR_STATUS
- LINK_MONITOR_STATUS
- LOC_RCVR_STATUS

Note: *PCS loopback overrules PMA loopback.*

6.10.5.2 PMA remote loopback

In PMA Remote loopback mode, the data transmitted by the link partner on the MDI is mirrored back to the link partner. The received symbols are passed through the PMA receive and PCS receive function, forwarded to the PCS transmit function, which in turn sends it back to the link partner.

PMA Remote loopback allows the MAC to compare the packets sent to the MDI with the packets received back from the MDI and, therefore, to validate the functionality of the physical channel, including both 100BASE-T1 PHYs.

To enable PMA remote loopback the REMOTE_LOOPBACK bit in MMD1.PMA_CONTROL1 must be set.

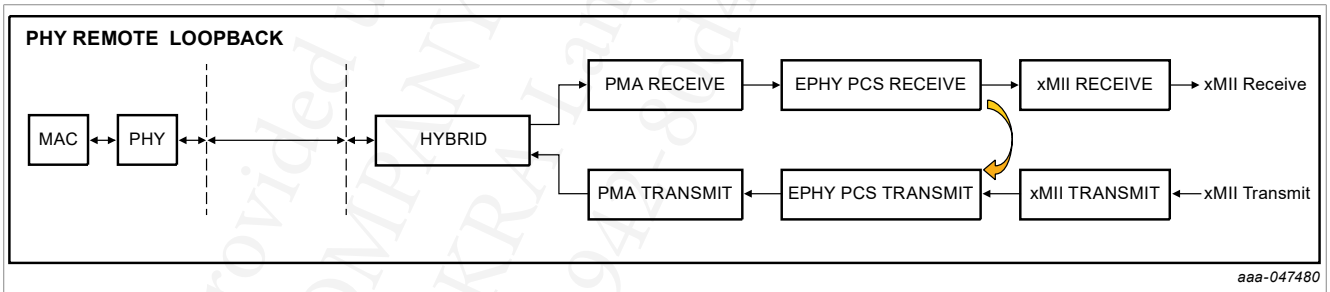


Figure 20. PMA remote loopback

6.10.5.3 PCS loopback

In Internal loopback, the PCS Receive function gets the ternary symbols An and Bn directly from the PCS Transmit function as shown in Figure 21. The xMII Receive data is transmitted through the PCS subsystem and mirrored to the xMII Transmit interface.

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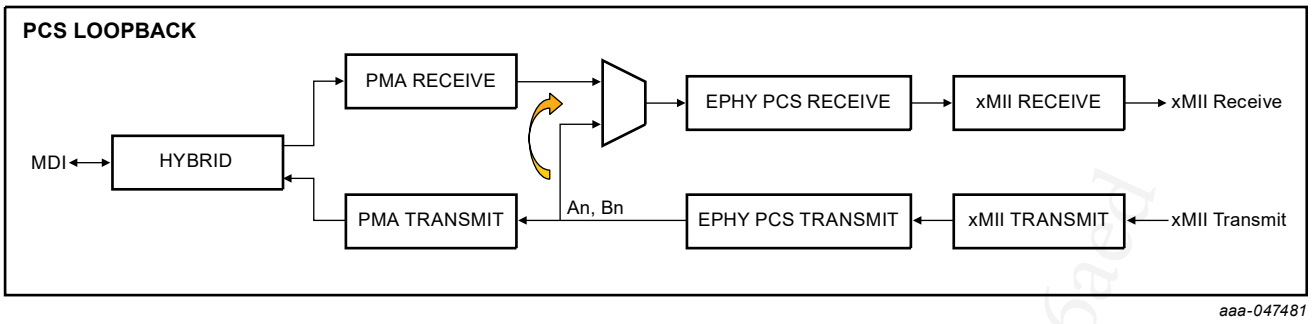


Figure 21. PCS Loopback

PCS loopback is enabled by setting the LOOPBACK bit in MMD3.PCS_CONTROL1. Data can be sent once the following flags in MMD30.PHY_STATUS are set:

- LINK_AVAILABLE
- SENDI_OR_DATA
- REM_RCVR_STATUS
- SCR_STATUS
- LINK_MONITOR_STATUS
- LOC_RCVR_STATUS

6.10.5.4 xMII PHY loopback

In xMII PHY loopback mode, the frames received by the link partner at the MDI are passed through the PMA receive and PCS receive functions and forwarded to the xMII transmit function, which in turn sends it back to the link partner from where it came.

xMII PHY loopback allows the MAC to compare the packets sent to the MDI with the packets received back from the MDI and, therefore, to validate the functionality of the physical channel, both 100BASE-T1 PHYs and the xMII function including rate matching FIFOs.

xMII PHY loopback is enabled by setting the XMII_CONTROL[PHY_LOOPBACK] bit.

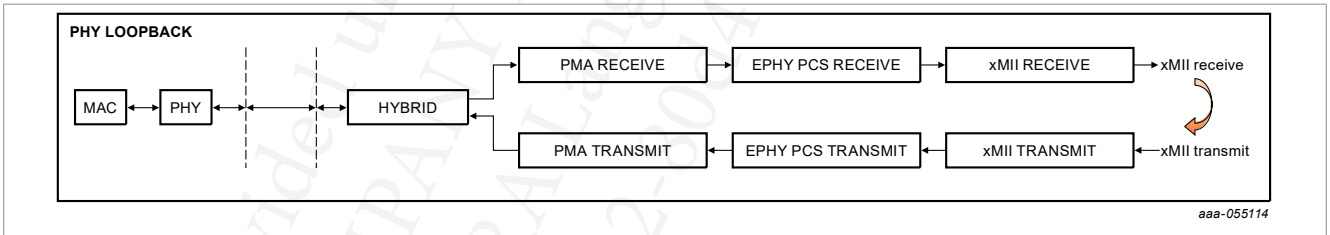


Figure 22. xMII PHY loopback

6.10.5.5 xMII MAC loopback

The xMII MAC loopback is similar to PCS loopback but directly loops back at the xMII interface.

xMII PHY loopback is enabled by setting the XMII_CONTROL[MAC_LOOPBACK] bit.

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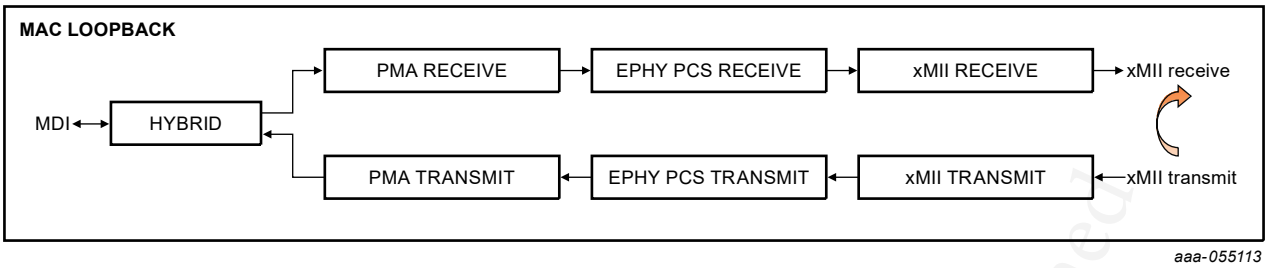


Figure 23. xMII MAC loopback

6.10.6 BIST Generator/Checker

The BIST (built-in self test) is an integrated Ethernet frame generator and checker intended for diagnosis. Both generator and checker can be enabled independently. BIST generator provides full configurability on Ethernet frame header with programmable inter packet gap (IPG).

BIST frame generator can generate mix of both good and bad frames. The BIST checker can also monitor the incoming Ethernet frames and provide statistics about number of good and bad frames received.

Together with loopback modes, TJA1104 supports a number of BIST configuration allowing self-testing of the full communication chain.

6.10.6.1 Introduction

The Built In Self-Test Generator and Checker (BIST) allows to test the data path without MAC connected to a PHY device.

Normally a MAC sends Ethernet frames to the PHY over an xMII interface. These frames are transmitted over the MDI and are eventually received by the remote PHY. The remote PHY transmits the frames through its xMII to the receiving MAC. For debugging purposes, the BIST module is able to replicates this behavior without a MAC being involved. The BIST module consists of a generator and a checker. The generator generates Ethernet frames and the checker receives these frames and checks for errors. The number of erred frames is captured in statistics counters.

Figure 24 shows the simplified block diagram.

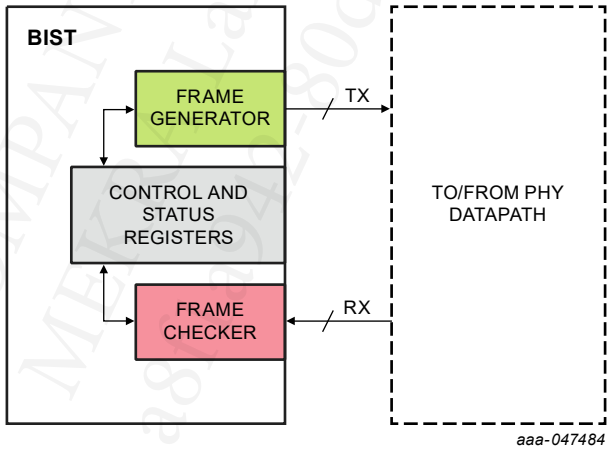


Figure 24. BIST block diagram

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6.10.6.2 Functional description

The BIST module is not active by default, and generator and checker need to be enabled (see registers BIST_CHECK_CTRL and BIST_GEN_CTRL).

The generator composes Ethernet frames with configurable length, payload content and other header fields. The generated frames are dispatched either xMII or MDI facing (see BIST_INTERCEPT_CONFIG[TX_INTERCEPT] and BIST_INTERCEPT_CONFIG[RX_INTERCEPT].

The BIST checker checks the incoming data based on MDI or xMII (RXD) direction (see CHECK_EPHY_OR_XMII). The checker validates the CRC and RX_ER and logs status and diagnostics information.

There are two modes of operation.

Production mode is intended for production test. In this mode, the total number of frames generated by the checker is configurable. The same number of frames is expected at the checker. The total number of frames is a configurable combination of good frames and error injected frames (erroneous FCS). The BIST checker gathers statistics until a configurable timeout has expired (BIST_WAIT_TIMER) or all expected frames were received. On completion of the test, the bit BIST_CHECK_DONE is set. The test verdict is then stored in register BIST_CHECK_FAIL.

Continuous mode is similar to stress test known from Ethernet test equipment. In this mode, the BIST generator generates continuous Ethernet frames and statistics are continuously collected. This mode can be used to generate Ethernet frames by the BIST module itself or it acts as a sink for frames generated by external test equipment.

6.10.6.3 Generator

The BIST Generator is capable of generating the IEEE 802.3 Ethernet frame format with varying frame length and different types of payload data. Once the BIST generator is set, it starts generating the Ethernet frame according to configuration.

Table 21. Custom BIST generator frame

Preamble 7 bytes	SFD 1 byte	DA 6 bytes	SA 6 bytes	EtherType 2 bytes	Frame Payload Data 46-1500 bytes	FCS 4 bytes	IPG 12 bytes

The number of preamble octets is configurable (PREAMBLE_LENGTH).

The IPG selection can be configured in one of the following modes:

- fixed value (IPG_LENGHT_TYPE=0). IPG_LENGTH controls the IPG for all frames.
- increment-by-one (IPG_LENGHT_TYPE=1) and roll-over to IPG_LENGTH when the current IP reaches 255.
- random IPG (IPG_LENGHT_TYPE=2). IPG_LENGTH acts as a minimum IPG as well as a LFSR seed.

The EtherType is configurable via the ETHER_TYPE register.

The generator supports various modes for the payload data length generation. It can by FIXED (PAYLOAD_DATA_TYPE=0), INCREMENT (PAYLOAD_DATA_TYPE=1) or RANDOM (PAYLOAD_DATA_TYPE=2).

In FIXED mode the number of octets is set by PAYLOAD_SIZE.

In INCREMENT mode, the data length starts with value PAYLOAD_SIZE[7:0]. The data length increments until a value of of PAYLOAD_SIZE[13:8]×256+255 is reached, then it rolls-over to PAYLOAD_SIZE[7:0].

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In RANDOM mode, the payload length is determined by a 14-bit LFSR $x^{14}+x^{13}+x^{12}+x^{2}+1$. PAYLOAD_SIZE is used as seed. PAYLOAD_SIZE[7:0] sets the minimum payload size. PAYLOAD[13:8] determined the maximum payload size.

If LFSR[13:8] ≤ PAYLOAD_SIZE[13:8], the full 14-bit LFSR[13:0] value determines the payload size.

If LFSR[13:8] > PAYLOAD_SIZE[13:8], the payload size is determined by LFSR[9:0].

If LFSR[13:0] < PAYLOAD_SIZE[7:0], the payload size is set to PAYLOAD_SIZE[7:0].

Frames in the standard range of 64 to 1535 octets are generated if PAYLOAD_SIZE[13:8] is set to 05h and PAYLOAD_SIZE[7:0] is set to 40h.

If PAYLOAD_SIZE[13:8]>05h, jumbo frames will be generated. Jumbo frames up to the max size of 16383 bytes will be generated if PAYLOAD_SIZE[13:8]=3Fh.

The payload data is configurable and the following types are supported:

- FIXED (PAYLOAD_DATA_TYPE = 0). The fixed data octet is controlled via FIXED_PAYLOAD_DATA.
- INCREMENT (PAYLOAD_DATA_TYPE = 1). The data starts with 00h, increments by one and rolls over at ffh.
- PRBS (PAYLOAD_DATA_TYPE = 2). Three different PRBS schemes can be selected via BIST_PRBS_SELECT (PRBS7, PRBS9, PRBS11, PRBS15). The random seed is configured via BIST_LFSR_SEED. It is configurable (LFSR_SEED_USAGE) if the PRBS sequence is reset to the seed for each frame or continues free-running.

The BIST generator calculates the FCS as specified in the Ethernet standard and appends it to the frame. All frames are transmitted with TX_ER=0.

The number of good frames and error frames is programmable. The total number of generated frames is the number of good frames plus number of error injected frames. The generator first produces all good frames, and then produces all bad frames.

In continuous mode, the generator only sends good frames and the good/bad frame plans are ignored. Setting the STOP bit stops the continues generation of frames. To restart frame generation set generation enable.

6.10.6.4 Checker

The BIST checker qualifies all received Ethernet frames and generate the status and statistical results. The checker is enabled by programming BIST_CHECKER_EN. Generator and checker work fully independent of each other. That means that the generator and checker functionality can be distributed across two physical devices.

The checker classified frames into good and bad frames. Good frames are frames with a correct FCS and bad frames with an incorrect FCS. The checker also counts all frames are are received with RX_ER set (R_RXER_FRAME_CNT).

The statistics (R_GOOD_FRAME_CNT and R_BAD_FRAME_CNT) can be reset by STATISTC_CNT_RST.

In production mode, status informations are available once the BIST test is done.

- Good frame error indicates that the number of good frames does not match the expected number according to the configured plan.
- Bad frame error indicates that the number of bad frames does not match the expected number according to the configured plan.
- RX_ER detected is set, if a frame is received with RX_ER indication asserted.
- RX_DV not detected is set if no data valid indicated is observed.

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6.11 Interrupts

The device is able to signal important events, for instance changes in link availability, via its IRQ_N pin. In order to efficiently deduce which of the available events has fired, an interrupt tree is implemented. The IRQ_N pin is driven low when at least one IRQ in the tree is active and enabled. Otherwise, the IRQ_N pin is high-Z.

Multiple devices can share the IRQ_N line using a wired-OR topology.

6.11.1 Interrupt tree

The interrupt tree concept is designed to support multi-port PHY devices and differentiates between port, global and device level interrupt sources. All port-level IRQs are present once, because this device has one port. Common functionality which is shared among multiple data ports is captures in the global section.

On device level the port and global level are joined together. The ALWAYS_ACCESSIBLE register serves a special purpose. This register is available for MDIO read and write access in cases when the VDD_CORE supply or the chip internal 25 MHz REFCLK is not present.

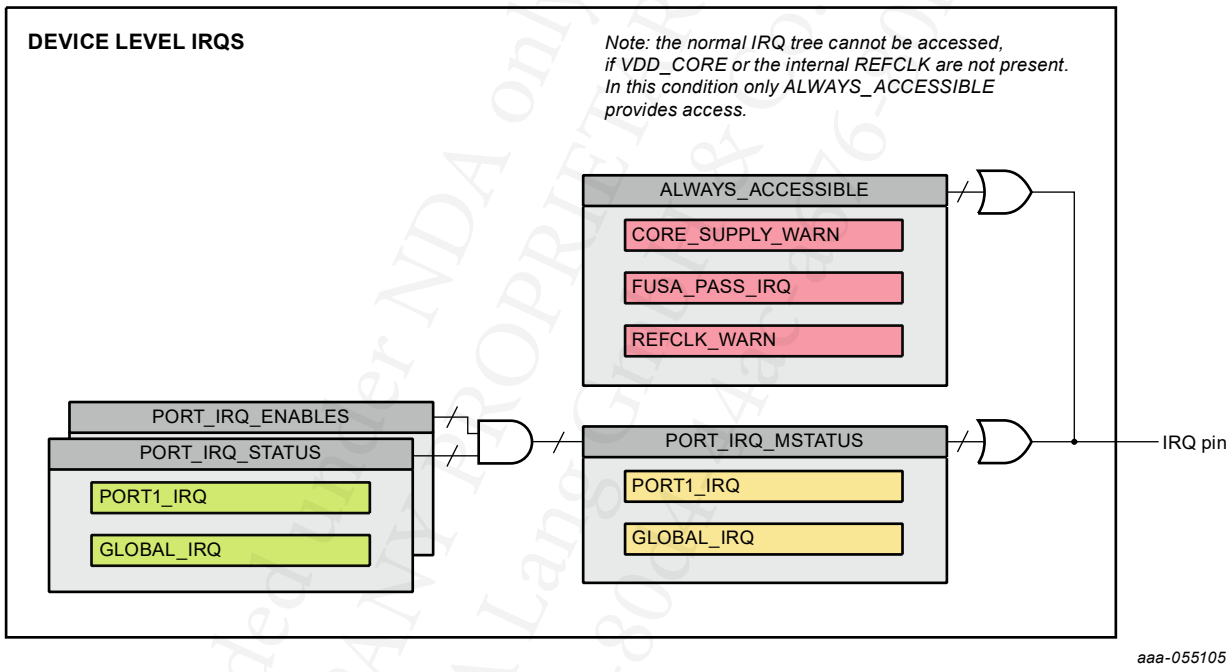


Figure 25. Device IRQ

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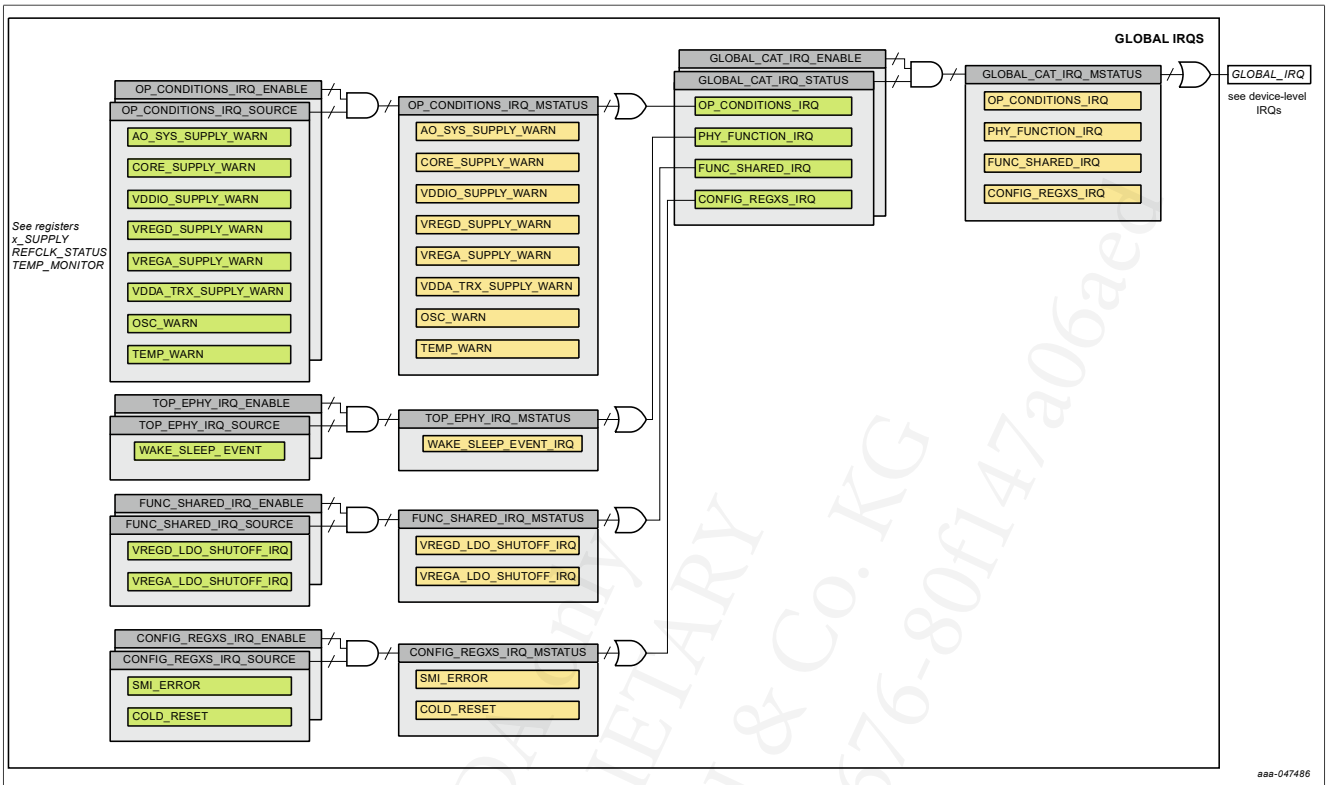


Figure 26. Global IRQs

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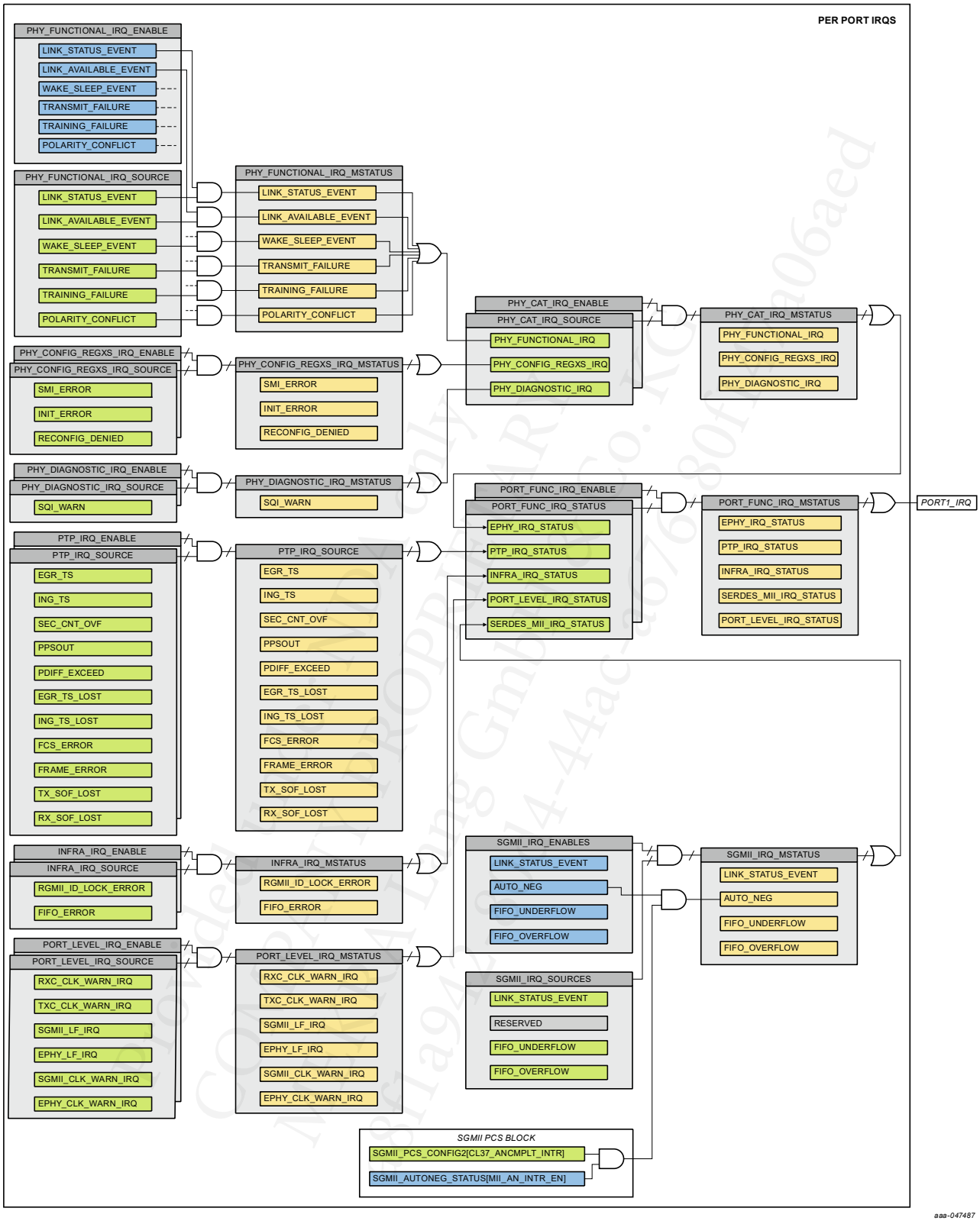


Figure 27. Port IRQs

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6.11.2 Non-maskable interrupts

The non-maskable FUSA_PASS_IRQ signals that all self-checks have passed as described in Section 6.4.

Several IRQs are non-maskable and act as an indication of severe malfunction. The interrupt enable bits that gate these IRQs are ignored.

Table 22. Critical non-maskable interrupts

Non-maskable interrupts	Description
FUSA_PASS_IRQ	The startup self-test has completed successfully
CORE_SUPPLY_WARN	Supply monitor for the core supply
VDDA_TRX_SUPPLY_WARN	Supply monitor for the VDDA_TRX supply
VREGA_SUPPLY_WARN	Supply monitor for the VREGA_IN supply
VREGD_SUPPLY_WARN	Supply monitor for the VREGD_IN supply
TEMP_WARN	Temperature monitor
REFCLK_WARN (not in RMII-clkmode)	XTAL frequency warning
TXC_CLK_WARN (only in RMII-clkmode)	TXC_CLK frequency warning
EPHY_LF	All interrupts inside the EPHY_LF group
EPHY_CLK_STATUS	All interrupts inside the EPHY_CLK_STATUS group
SGMII_LF	All interrupts inside the SGMII_LF group
SGMII_CLK_STATUS	All interrupts inside the EPHY_CLK_STATUS group

6.12 Timestamping

The PTP timestamping module is a generic module which supports both 100/1000 Mbit/s operation. In this device, the block is integrated for the 100 Mbit/s mode and internally connected to the on-chip MII bus as shown in Section 6.3.

To compensate the chip-specific pipeline delays, the two registers TX_PIPE_DLY and RX_PIPE_DLY must be configured as follows:

TX_PIPE_DLY = 77

RX_PIPE_DLY = 413

6.12.1 Introduction

IEEE 1588 v2 (PTP) is an industry standard that enables precise time synchronization of nodes in an Ethernet network. The synchronization is achieved by exchanging various protocol-defined PTP event messages. Based on these PTP event messages, the slave can correct offset and drift of its local clock to synchronize with the master clock.

The PTP module is sandwiched between the MII/GMII interface. The module provides a 1588 digital clock implementation, transmit and receive frame parsing, Start-of-Frame detection, transmit and receive timestamp units, event timestamp unit and a Pulse-Per-Second (PPS) generator.

6.12.1.1 Features

Following features are provided by the IEEE 1588 v2 PTP block:

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- Support for 100BASE-T1 and 1000BASE-T1 PHYs
- Supports hardware time stamping features of IEEE 1588 v2 PTP Clock
- Provides timestamping on ingress and egress PTP messages
- Hardware self-locking or software assisted rate correction to an externally provided reference clock
- Timestamping on external pin sync events
- Ingress and Egress timestamps stored in ring buffer for host processing
- Support for two-step operation
- Supports in-band timestamp delivery for ingress path
- Supports detection of PTP frame encoded in Ethernet both untagged and 802.1Q VLAN tagged (double tagged VLAN is not supported)
- Software configurable frame filter to match PTP event frames

6.12.2 Overview

The diagram below depicts the integration of the PTP module at system-level.

The timestamping module uses the PHY's RX_SOF and TX_SOF signals to accurately take a timestamp at the instant when a frame has ingressed/egressed the Ethernet PHY. The MACsec engine is between the PHY and the timestamping module and acts as a deep fifo of more than 64 octets. This implies that an RX_SOF/TX_SOF event cannot be easily correlated with a frame inside the timestamping module. To be able to correspond a frame with an SOF event internal sequence numbers are generated and passed through a sideband channel of the MACsec engine. This makes it possible to correlate SOF events with frames and keep track and frames that were discarded inside the MACsec subsystem.

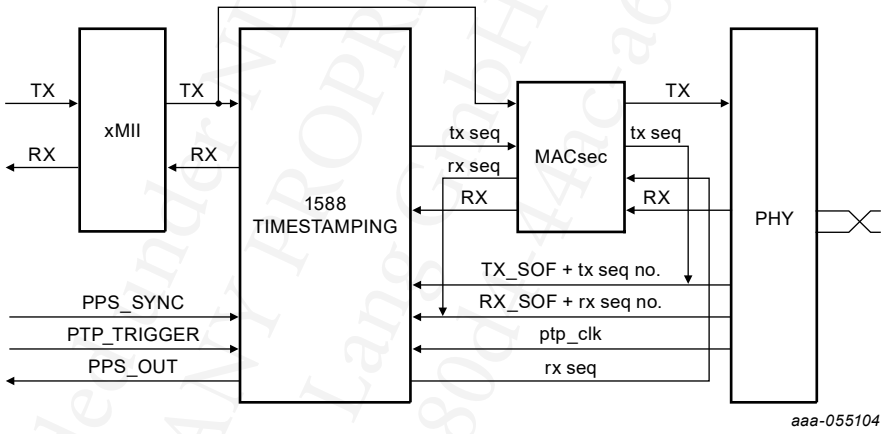


Figure 28. PTP PHY integration

The module consists of the following blocks:

- The actual clock, called local time counter (LTC)
- The transmit and receive frame parser and qualifier
- Transmit and receive timestamp FIFOs
- Other miscellaneous functions such as control and status registers, PPS output and syncing

The TX_SOF and RX_SOF indication comes from the PHY sublayer and usually exhibits a pipeline delay. This device-specific delay is constant and can be corrected automatically by configuring the registers TX_PIPE_DLY and RX_PIPE_DLY.

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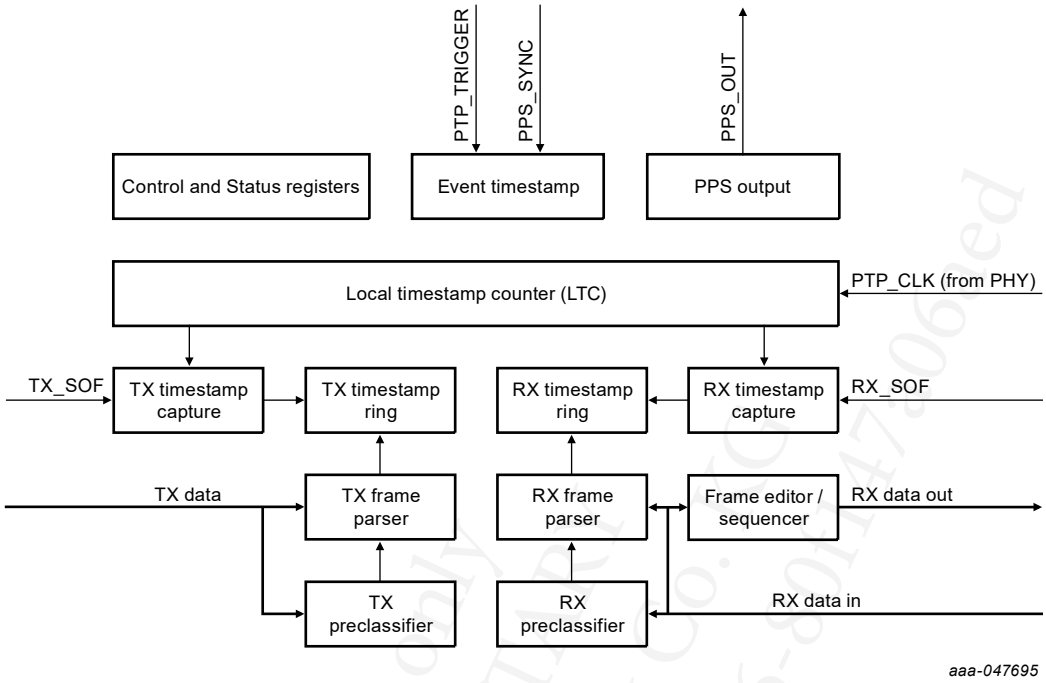


Figure 29. PTP block diagram

6.12.3 Clocking scheme

The IEEE 1588 PTP uses a timestamp clock that is derived from the Ethernet PHY block. The PTP_CLK frequency must be configured in the PTP_CLK_PERIOD register. For 100BASE-T1, the PTP_CLK frequency is 66.6 MHz, thus the PTP_CLK_PERIOD must be set to 15 ns.

6.12.4 Resets

By default, PTP module is inactive. All clocks are disabled and internal logic is under reset. The host can enable the PTP module by writing PTP_ENABLE=1.

6.12.5 Local time counter

The local time counter (LTC) module is the clock used for timestamping. It is local to the PHY and can be synchronized with other chips via a PPS_SYNC facility. The software can interact with the LTC to control absolute value, the rate as well as change the offset.

The ptp_clk is the digital clock that drives the LTC. Depending on the speed of the PHY, the ptp_clk is either 125 MHz (1000BASE-T1) or 66.66 MHz (100BASE-T1). A block diagram of the LTC structure is shown below:

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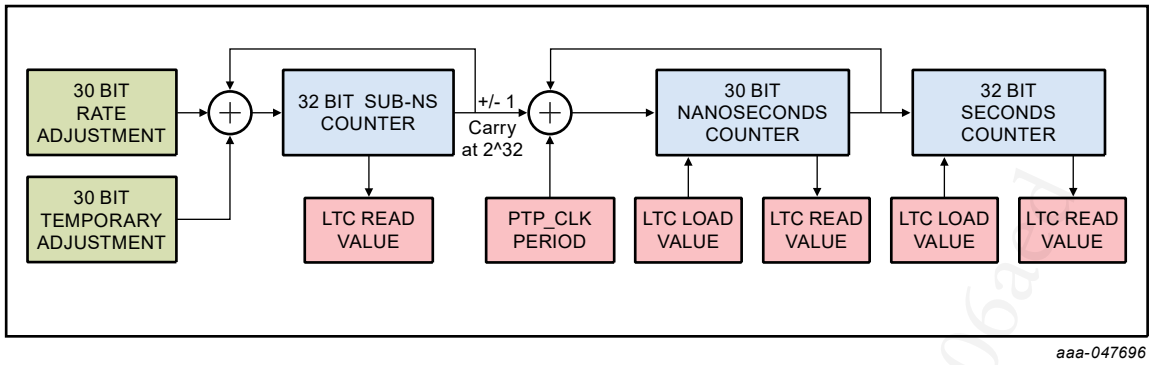


Figure 30. Block diagram of LTC

The local time counter consists of two counters, nanoseconds counter and seconds counter. The nanoseconds counter is 30 bits wide and seconds counter is 32 bits wide.

The ns counter increments by 8 (15) every ptp_clk cycle for 1000 Mbit/s (100 Mbit/s) operation. This translates to 8 ns (15 ns) per ptp_clk tick. Once the nanoseconds counter reaches the rollover value, the seconds counter increments by 1. The 32-bits seconds counter has a rollover time of 136 years.

A sub-ns counter is used for continuous and temporary rate-correction in fractions of a nanosecond per clock. Such ns fractions are accounted for in the 32-bit wide sub-ns counter, which has a granularity of 2^{-32}ns. When the sub-ns counter rolls over, an additional nanosecond is added or subtracted from the nanoseconds counter.

6.12.5.1 Direct read/write

A direct write will set the LTC immediately. The registers PTP_CLOCK_TIME_SEC and PTP_CLOCK_TIME_NSEC must be written to load internal seconds counter and nanoseconds counter. By setting the self-clearing load bit, these pre-programmed values are loaded into the LTC.

The seconds counter, nanoseconds counter, sub-nanoseconds counter can be read at any time by asserting the read bit.

6.12.5.1.1 Immediate LTC load

An immediate load causes the values as written to the nanoseconds and seconds register to be transferred to the LTC.

1. Program LTC_WR_NSEC_0, LTC_WR_NSEC_1, LTC_WR_SEC_0, LTC_WR_SEC_1
2. Load these values into internal counters by setting LD_LTC_NSEC and LD_LTC_SEC

6.12.5.1.2 PPS edge triggered load

The PPS edge loading transfers the time as loaded into the nanoseconds and seconds register into the LTC at the rising edge of the PPS_SYNC input.

1. Program LTC_WR_NSEC_0, LTC_WR_NSEC_1, LTC_WR_SEC_0, LTC_WR_SEC_1
2. Load these values into internal counters by setting LOAD_LTC_PPS
3. Internally these values are loaded into local counters on event of rising edge of external pps sync

When an external PPS Sync is chosen with a frequency of more than 1 Hz, loading the external PPS sync signal must be staged. The PPS must be paused before loading the LTC. Then the PPS can be started again. This is required in order to associate a particular edge with an absolute time value reliably. Pause the PPS signal, then follow step 1 and 2, and then restart the PPS Sync signal.

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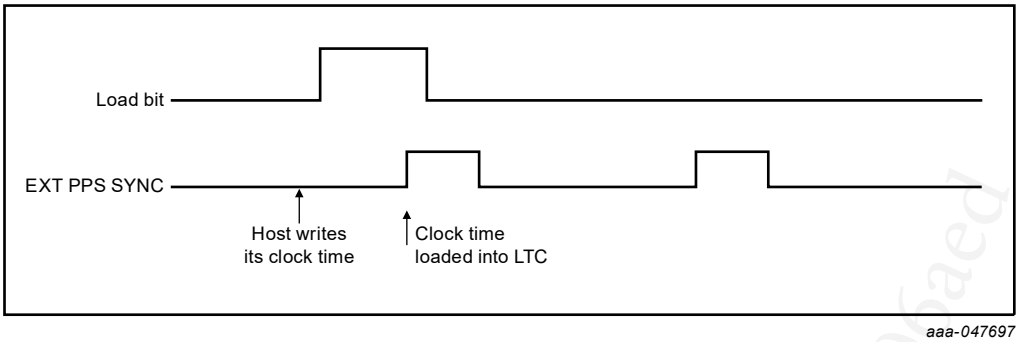


Figure 31. Loading LTC on rising edge of PPS

6.12.5.1.3 Immediate LTC read

1. Program READ_LTC to save the local clock time
2. Read LTC_RD_DATA_0 to LTC_RD_DATA_5
3. Same saved data can be read multiple times as new value is loaded only when READ_LTC is reprogrammed

6.12.5.1.4 PPS edge triggered read

The LTC value is latched at the rising or falling edge of the PPS input. It can then be read from the EXT_PPS_TS_DATAx registers. The timestamp remains latched until EXT_PPS_TS_DONE is set. Setting EXT_PPS_TS_DONE arms the hardware for latching the LTC at the next edge.

It is mandatory to read EXT_PPS_TS_DATA0 first and continue to EXT_PPS_TS_DATA5. It is not required to read the full set of six registers. The read sequence can be interrupted at any time by writing to EXT_PPS_TS_DONE in the EXT_PPS_TS_CTRL register. This frees the register for the next PPS edge event.

Note: *The timestamp is retained in the registers until the x_TS_DONE bit is set. Retaining the value allows reading the data multiple times.*

6.12.5.2 Rate correction

In order to match rate difference between PHY and an external clock source (for instance a microcontroller or switch), a digital rate correction can be applied. Every ptp_clk cycle, a rate correction value RATE_ADJ_SUBNS is added or subtracted to/from a hidden subns counter. On overflow/underflow of this subns counter, the carry is added/subtracted to/from the nanoseconds counter and integrated in the visible LTC. The register RATE_ADJ_SUBNS is 30 bit wide and in units of 2^{-32} ns. To control the direction (speed up or slow down) the tuning value can be added or subtracted. The direction is set with register RATE_ADJ_SUBNS_1[CLK_RATE_ADJ_DIR].

As an example, lets assume the clock rate of the LTC is too slow compared to the external PPS. This difference is 0.125 ns in the time interval of one ptp_clk cycle.

Then, PTP_CLOCK_RATE_SNS = rate * 2^{32} = 30'h2000 0000

6.12.5.3 Temporary rate correction

An LTC offset is corrected by temporarily speeding-up or slowing-down the LTC clock, therefore slowly integrating the offset error over multiple ptp_clk cycles. This avoids abrupt jumps and time discontinuities and guarantees a monotonic clock.

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A temporary correction is applied similarly as for rate-correction, except an additional time window is configured during which the rate change is applied. The register PTP_CLOCK_TEMP_ADJ_SNS holds the temporary rate-correction value and PTP_CLOCK_TEMP_ADJ_DUR is used to configure the number of ptp_clk clock cycles to be adjusted. The register PTP_CLOCK_TEMP_ADJ_DIR controls the direction of update (add/subtract). Setting the self-clearing bit PTP_CLOCK_LOAD_TEMP_ADJ triggers the temporary adjustment for one time.

The register PTP_CLOCK_TEMP_ADJ_SNS is 30 bit wide and in units of 2^{-32} ns.

The temporary rate correction acts additive, on-top of the regular rate-correction as described earlier.

For example, lets assume an offset error of 25.5 ns. With an integration rate of 0.125 ns per ptp_cycle, the temporary correction must be applied for 25.5/0.125 = 204 ptp_cycles. Thus the following registers are programmed:

PTP_CLOCK_TEMP_ADJ_SNS = rate * 2^{32} = 30'h2000 0000

PTP_CLOCK_TEMP_ADJ_DUR = 25.5/0.125 = 204

6.12.6 LTC synchronization

A typical system including MAC, IEEE 1588 protocol engine and a PHY is depicted in the below diagram. The diagram illustrates GMII, but any other interface such as MII, RMII, RGMII or SerDes are possible as well.

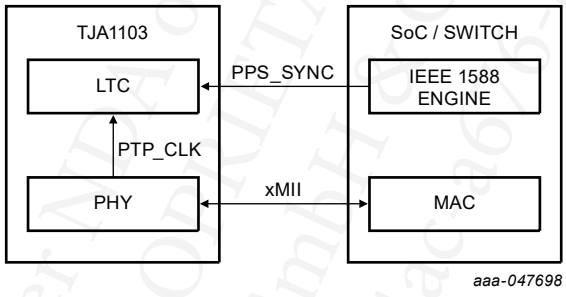


Figure 32. Example clocking scheme

Normally, the clock source used to drive the LTC is an external crystal. This provides a slightly different clock than used inside other timestamping engines. All timestamps taken in multi-port devices such as switches must be taken on the same timebase in order to be consistent.

A time difference between the LTC and any other timestamping clocks can have the following reasons:

1. The clock sources are different and have a relative frequency difference within the 100 ppm limit.
2. The timestamp clock in another device is itself a PTP slave clock, therefore constantly rate and offset corrected.

The PTP module can lock onto an external Pulse Per Second (PPS) signal. The PPS signal is then used to to correct the LTC automatically. The PPS does not need to be a pulse *per second*. The LTC block can track any low frequency signal, such as 1 Hz or 100 kHz. The rising edge or falling edge of external PPS sync is used to correct the timing drift. There are multiple ways to synchronize the LTC with the external reference PPS synchronization signal.

An external software tracking routine can be implemented to synchronize the LTC via a software driven control loop. Alternatively, the PTP block can track an external PPS synchronization pulse fully autonomously.

6.12.6.1 Software tracking

After initialization, an external host reads the clock time of the PHY at the instance of rising edges of consecutive pps_sync and finds out the rate difference between the PHY and IEEE 1588 protocol engine.

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Based on the calculations, the host can update LTC by using rate adjustment method to correct the rate difference and temporary adjustment to correct the offset. The following steps summarize the mechanism for a relatively slow PPS input:

1. The host pre-loads the expected LTC value (LTC_0) at the next PPS sync edge. Once the PPS sync input has fired, the LTC is running.
2. Capture LTC timestamp (LTC_1 at the following active edge of external PPS sync signal. The relative rate difference is calculated by dividing the expected PPS cycle time by the $LTC_1 – LTC_0$. The offset error can be determined through LTC_1.
3. Fix the rate-correction value as well as the offset error as described in Section 6.12.5.
4. Continuously repeat to fix rate-correction and offset based on LTC snapshots LTC_{n+1} and LTC_n.

6.12.6.2 Hardware tracking

The PTP block implements an optional hardware control loop to track an external PPT sync input. The rate difference and offset errors are detected automatically and corrected on-the-fly. Setting PTP_HW_LTC_LOCK_EN enables this feature.

The initialization works identical to the software tracking approach. The LTC value is pre-loaded and transferred into the working registers at the next active PPS edge. The PPS reference frequency must be programmed via registers HW_EXT_PPS.

Once LTC is loaded external PPS sync, the LTC is captured at the next PPY sync event. A phase comparator calculates the phase difference between observed LTC value and expected reference time. A direction signal is generated which tells if the LTC is lagging or leading. With this information, the rate difference and offset errors are computed and adjusted each PPS sync cycle.

If the hardware is not able to track the PPS input and the error exceeds a certain bound set by HW_LOCK_ER_0 and HW_LOCK_ER_1, an flag (PDIFF_EXCEED) is asserted and an IRQ is generated.

6.12.7 PPS output

It is possible to generate a pulse-per-second (PPS) signal based on the internal LTC and externalize it on a GPIO pin. This can be enabled by asserting PPS_OUT_EN. The hardware is able to generate a nominal a 1 Hz signal with a pulse at each one-second transition of time. If enabled, the PPS generator periodically keeps generating the PPS signal.

To change the polarity of the generated PPS, set the PTP_PPS_OUT_POL bit.

It is also possible to loop the PPS sync input back out on PPS_OUT for debugging purposes (see PPS_SYNC_RETURN). The PPS_OUT signal can be used to determine the LTC synchronization accuracy against an external reference clock.

6.12.8 Event timestamp

External events can be monitored and timestamped. A rising edge on the PTP_TRIGGER input latches the current LTC value into the registers EXT_TRG_TS_DATA.

Note: *The timestamp is retained in the registers until the x_TS_DONE bit is set. This allows to read the data multiple times.*

6.12.9 Egress timestamping unit

The egress timestamping is part of the transmit path. It is responsible for identifying a PTP message inside an Ethernet frame (classify), pull relevant information out of the PTP message (parse) and the actual timestamping.

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The captured timestamps are stored into a buffer (dataset capture and TS ring) for retrieval via MDIO. The diagram below depicts the data flow.

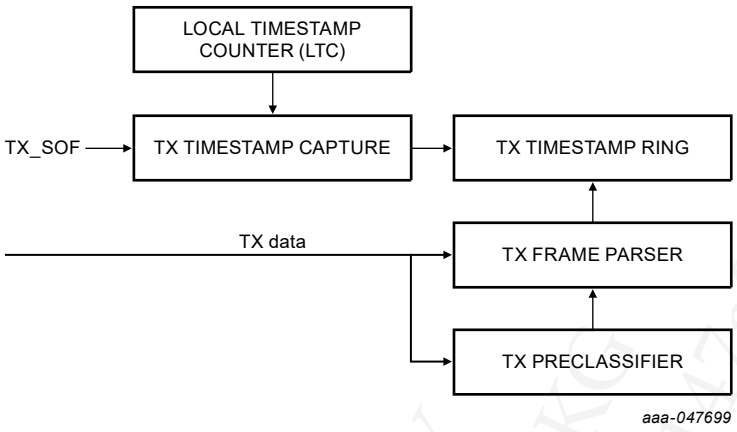


Figure 33. Egress timestamping unit

The TX_SOF is generated by the Ethernet PHY, not the PTP module. The TX_SOF is derived from PCS information and not the exact point of start of frame delimiter (SFD). The parser inside the PTP block detects the actual SFD point and compensates for the delay between TX_SOF and SFD.

6.12.9.1 Pre-classifier and TX parser

The TX parser interfaces with TX GMII/MII interface. It monitors the incoming transmit frames to detect relevant IEEE 1588 PTP event messages. It is able to recognize PTP event messages transported in layer 2 Ethernet frames[9]. The parser supports untagged and 802.1Q VLAN Tagging (single tagged) in Ethernet frames. PTP messages are qualified by the MAC address. Two PTP standard fixed MAC address (01:1B:19:00:00:00 and 01:80:C2:00:00:0E). Additionally, a single entry DA match rule can be used to support user-defined MAC addresses (USER_MAC_DA).

Once TX Enable is asserted on the internal MII/GMII bus the incoming frame is parsed to identify the PTP message. If a frame qualifies, all the necessary PTP message contents are extracted and stored for future use. It also sets a flag bit saying incoming detected message is PTP message.

The following frame filters can be individually enabled/disabled:

1. Destination MAC match (two fixed addresses, one configurable. See register USER_MAC_DA)
2. VLAN tagged (see register PTP_VLAN_TAG_ALLOW)
3. Message type: Sync, Delay_Req, Pdelay_Req, Pdelay_Resp event messages (see register PTP_EVENT_MSG_FILT)
4. Match a specific versionPTP (see register PTP_VERSION)
5. Match a specific domainNumber (see register PTP_DOMAIN_NUM)

It is possible to timestamp the following PTP event messages:

Table 23. PTP Event Messages

PTP Message Type	Value
Sync	0
Delay_Req	1
Pdelay_Req	2

9 IP encapsulated PTP messages are not recognized.

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Table 23. PTP Event Messages*...continued*

PTP Message Type	Value
PDelay_Resp	3

6.12.9.2 TX timestamp ring buffer

The TX timestamp (TS) Ring buffer stores the timestamping data in up to 4 available slots. Next to the timestamp additional metadata is stored to correlate the information with an Ethernet frame. This includes PTP message sequenceId, domain number, message type and valid flag to indicate the dataset is not yet read.

The buffer is organized as a ring buffer and the oldest valid entry is exposed to the registers EGR_RING_x for read. Once the timestamp has been read via the MDIO interface, the ring buffer pointer is advanced by writing the EGR_RING_DONE bit or any other of the data registers.

If the buffer is full, the oldest entry is flushed out to make room for a new entry. If enabled, an interrupt is generated on overflow.

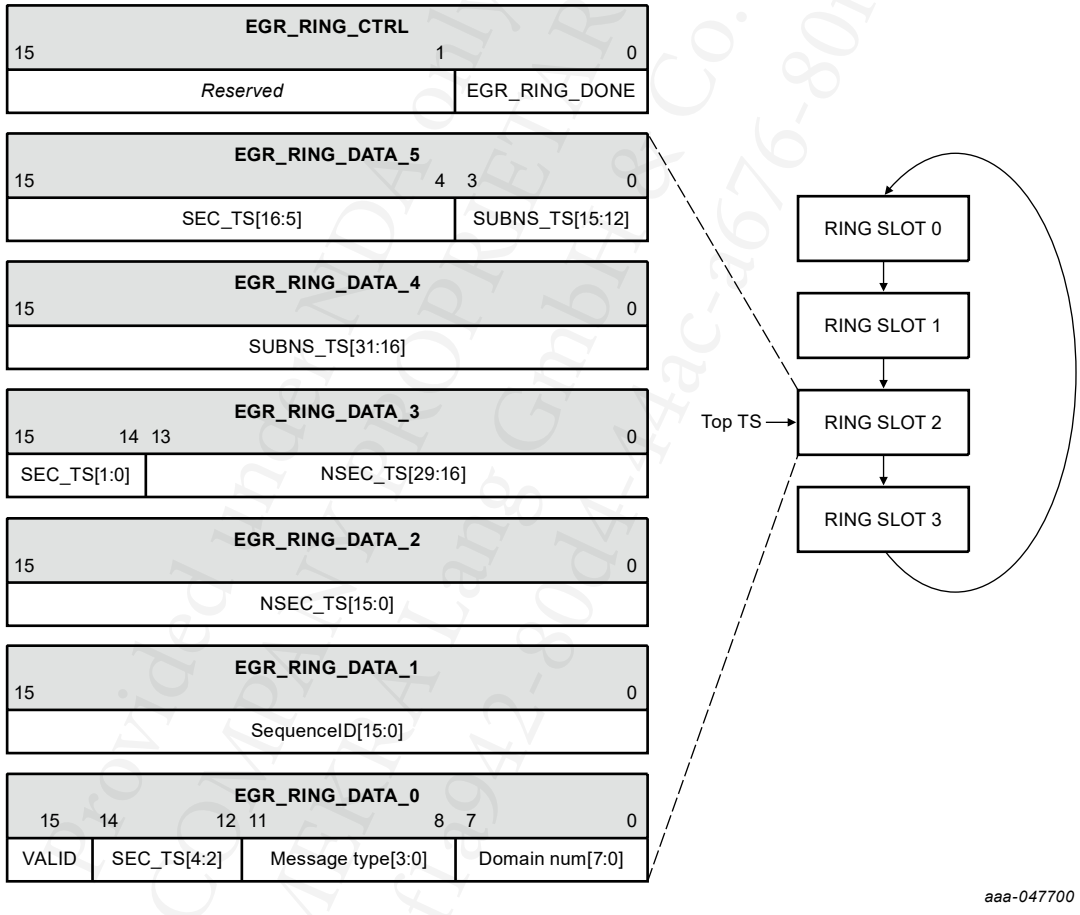


Figure 34. Egress TS ring buffer organization

6.12.10 Ingress timestamping unit

The ingress timestamping is part of the receive path. It is responsible for identifying a PTP message inside an Ethernet frame (classify), pull relevant information out of the PTP message (parse) and for timestamping the frame. The captured timestamps are stored into a buffer (dataset capture TS ring) for retrieval via MDIO.

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Alternatively, the timestamp can be embedded directly into the Ethernet frame (message editor). The diagram Figure 35 depicts the data flow.

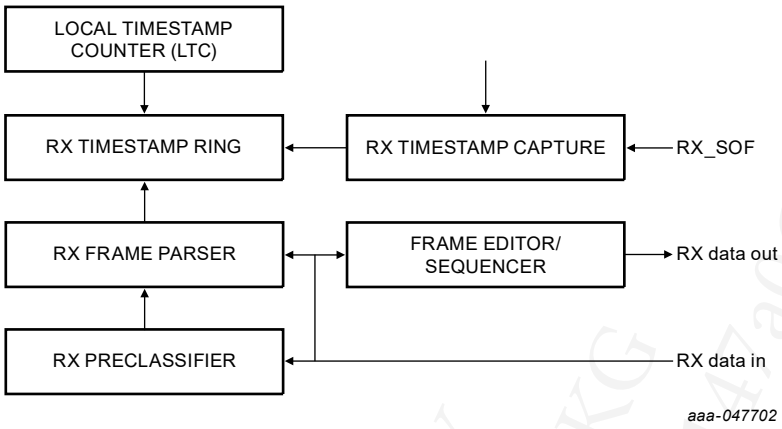


Figure 35. Ingress timestamping unit

The RX_SOF is generated by the Ethernet PHY, not the PTP module. The RX_SOF is derived from PCS information and not the exact point of start of frame delimiter (SFD). The parser inside the PTP block detects the actual SFD point and compensates for the delay between RX_SOF and SFD.

6.12.10.1 Pre-classifier and RX parser

The RX parser interfaces with RX GMII/MII interface. It monitors the incoming received frames to detect relevant IEEE 1588 PTP event messages. The parser and classifier logic is identical to Section 6.12.9.1.

6.12.10.2 RX timestamp ring buffer

The RX timestamp (TS) Ring buffer stores the timestamping data in up to 4 available slots. Next to the timestamp additional metadata is stored to correlate the information with an Ethernet frame. The organization of the ring structure and registers is identical as described in Section 6.12.9.2.

6.12.10.3 Frame message editor

Alternatively to delivering the RX timestamp via the ring buffer exposed via MDIO, it is possible to deliver the timestamp inside the PTP event frame itself. The message editor updates the PTP message header and the FCS in the Ethernet frame.

The engine is able to modify four reserved octets at PTP header offset 16 and overwrite these bits with 32 bits timestamp data. There are four supported timestamp formats that can be selected.

Seconds / Nanoseconds format 0

The inserted timestamp holds 2 bits of seconds counter and 30 bits of the nanoseconds counter: {SEC[1:0], NSEC[29:0]}

Nanoseconds format 1

The inserted timestamp 32 bit (integer) of ns precision. For this a 64 bit nanoseconds value is derived from base counters 32 bit seconds counter and 30 bit nanoseconds counter (seconds * 1000 + nanoseconds). Of the resulting 64 bit counter the lower 32 bit are inserted into the Ethernet frame.

Nanoseconds / subnanoseconds format 2

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The inserted timestamp 32 bit (fixed point) of ns and subns precision. For this a 64 bit nanoseconds value is derived from base counters 32 bit seconds counter and 30 bit nanoseconds counter (seconds * 1000 + nanoseconds). With this ns counter and the subns counter of the LTC a fixed point number is compiled. The resulting 32 bit value has (32-N) integer ns bits and N-bits fractional bits for subns position. The place of the decimal (N) can be configured via register RX_TS_INSRT_CTRL.ING_MSG_TS_PTR.

6.12.10.3.1 Checksum regeneration

Two parallel CRC-32 engines are used to check and regenerate a Frame Check Sequence (FCS) in case the frame is modified. Each frame is always checked for corrected by matching the residue with the magic number (32'hC704_DD7B). In parallel a new FCS is computed accounting for the modified 32 bits.

In case a frame qualifies as a PTP event frame and the original FCS is correct the timestamp is patched in and the FCS is regenerated. In case the original FCS was found to be incorrect, the frame will be left corrupted. This ensures that an FCS of an corrupted frame is not accidentally regenerated and the frame appears correct.

6.12.11 Interrupts

The PTP module is able to signal events via a PTP interrupt signal. This IRQ line is connected on device level to the device interrupt tree.

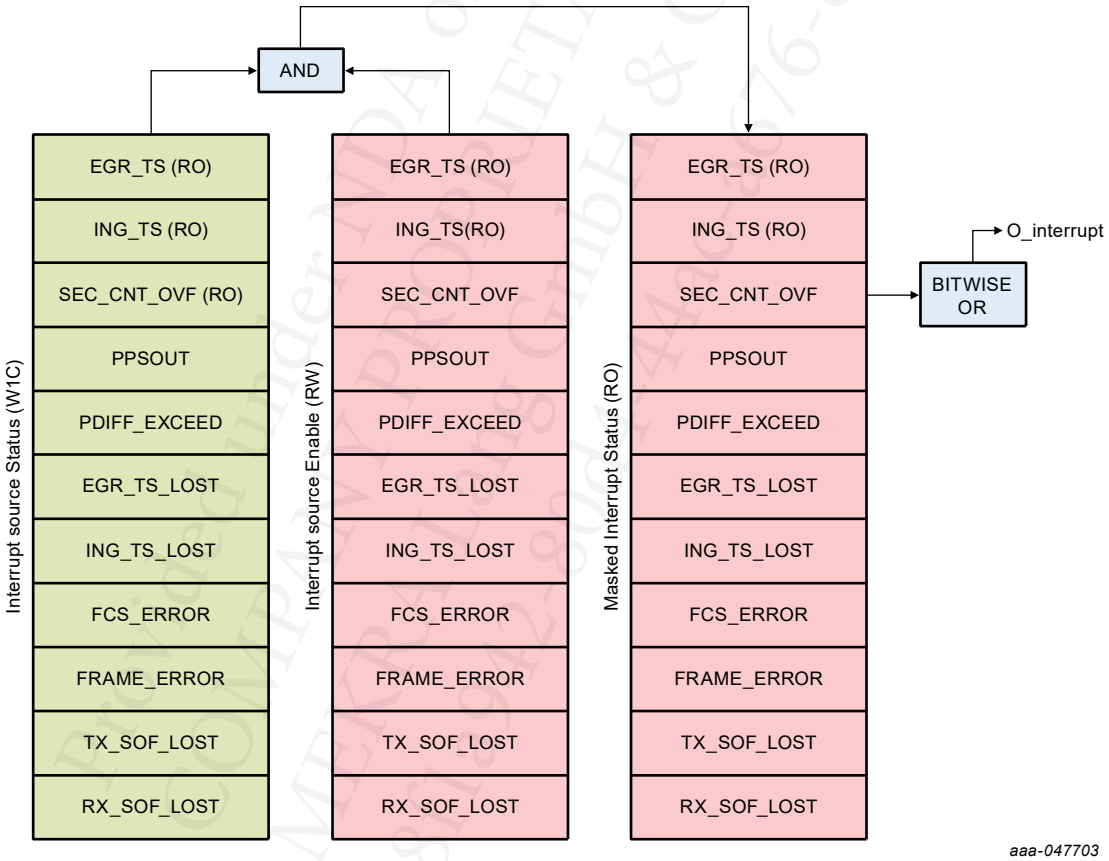


Figure 36. PTP interrupts

1. Egress timestamp event (Read only)
2. Ingress timestamp event (Read only)
3. PPS OUT event
4. Seconds counter saturates all 32bits (Read only)

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5. Hardware measured phase difference between external PPS sync and local LTC exceeds 250us (if PPS sync is 1 second period)
6. Egress TS lost due to overflow
7. Ingress TS lost due to overflow
8. Input FCS error detected (in-band signaling mode)
9. Ingress frame error (preamble detected but no SFD; Not complete PTP TS data is received)
10. TX SOF is not detected for a PTP event frame
11. RX SOF is not detected for a PTP event frame

6.12.12 Diagnostics

The PTP module captures the following statistics of the parsed frames:

- Total number of recognized received PTP event frames
- Number of received PTP event frames with FCS errors[10]

6.13 MACsec adapter (only TJA1104A and TJA1104B)

6.13.1 Introduction

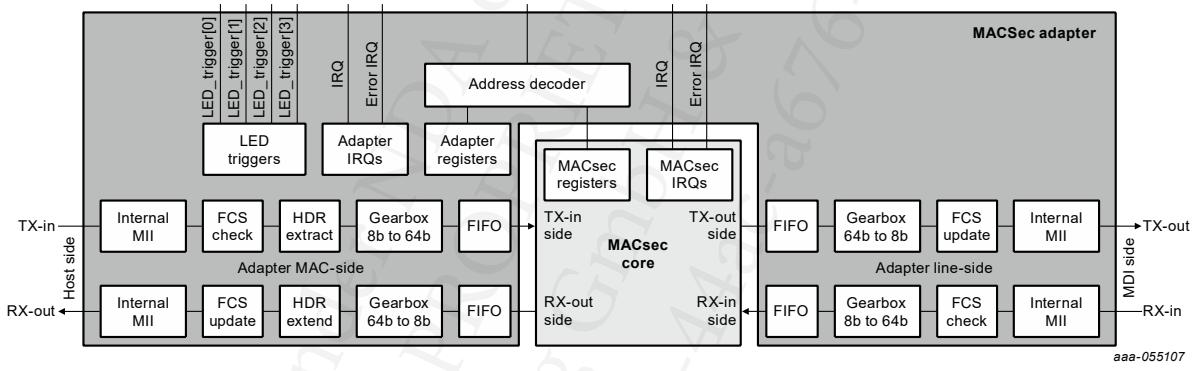


Figure 37. MACSec subsystem block diagram

The MACsec subsystem consists of two sub blocks. The MACsec core which is responsible for the 802.1AE data path and the MACsec adapter which connects the MACsec IP to the on-chip MII datapath.

The adapter implements rudimentary MAC functionality. This includes removal of the Ethernet preamble and SFD (start of frame delimiter) as well as FCS (frame check sequence) checking and recalculation. The MACsec IP is a synchronous IP and decoupled from the MII clock via asynchronous FIFOs.

The MACsec subsystem encompasses a cut-through architecture. This means that for each frame that ingresses the subsystem, after some initial delay, the frame starts to egress while ingress is not yet completed (except for certain error conditions and when policing is used).

The MACsec core processes each frame and, depending on direction appends or removes the 802.1 AE SecTag as well as ICV (integrity check vector). Optionally, the payload data of the Ethernet frame can be encrypted (confidentiality mode, red area below).

10 only available if the frame editor is enabled

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Figure 38. Basic Ethernet frame, without MACsec with VLAN tag

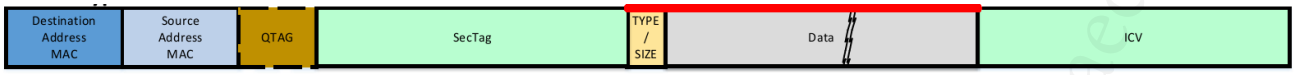


Figure 39. MACsec Ethernet frame with SecTag and VLAN-tag-in-clear

See UM12059 for detailed description of the core MACsec functionality.

Optionally, the HDR extraction block is able to extract additional control information from a proprietary header. Similarly, the HDR extend block is able to annotate received frames with additional meta data.

6.13.2 On-chip MII interface

The MACsec subsystem is connected to the rest of the system through an on-chip MII (Media-independent-Interface) connection. The MII interface implements 802.3 standard behavior. As part of the standard MII processing, the preamble is removed and the TX_ER/RX_ER indications are registered.

Any TX_ER/RX_ER event will be treated internally similar to an FCS error. Thus an TX_ER/RX_ER on one of the input interfaces will cause the frame to be corrupted and transmitted with a corrupted FCS on the respective output interface.

6.13.3 FCS checking and update

The TX_IN and RX_IN ingress stages of the adapter check the Frame check sequence (FCS) of each frame. FCS errors on TX/RX interfaces are counted through RX_STATS_FCS_ERRS and TX_STAT_FCS_ERRS respectively. Since the implementation is cut-through each frame is internally processed and start egressing before it is reliably known that the frame is correct. The egress stages TX_OUT and RX_OUT optionally update the FCS in case a frame was modified.

The FCS of a frame is modified in the following cases:

- A frame entered the device while TX_ERR/RX_ERR signals where asserted. In this case the FCS is corrupted on egress.
- The FCS check during RX/TX ingress did not pass. In this case the FCS is corrupted on egress.
- The frame passed the ingress FCS check and the frame was modified by the MACSec adapter or MACSec core. This can be addition or removal of SecTag/ICV or the addition/removal of a MACsec TX/RX control tag. In this case the FCS is regenerated to match the frame.

In case the FCS check on ingress has passed and a frame was not modified by the MACSec Adapter or MACSec core the FCS is retained as it was.

The FCS corruption on egress is done by calculating the correct FCS and then negating all bits.

6.13.4 Parity protection and ECC

The control and datapath of the MACsec adapter and MACsec core as well as register memories are protected with parity or ECC. Each pipeline stage in the adapter will check the parity and update the parity if needed.

An adapter internal data parity mismatch as well as a control parity mismatch is indicated via the PARITY_DATA_ERR and PARITY_CNTRL_ERR status flags.

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Parity errors inside the MACSec IP are flags via the of MACSEC_HARD_ERR and of MACSEC_SOFT_ERR status flags.

The MACsec subsystem is able to recover from data parity errors. It is is advised to reset the entire subsystem in case of control and hard errors.

6.13.5 Transmit flow control

On data transmission, the principle of the MACsec operation is to add new information (SecTag as well as ICV) to the Ethernet frame. The frame size grows by at most 32 octets. For example: A 64 octets frame *before* MACsec is extended to 76 bytes *after* MACsec[11]. This process presents a fundamental issue when MACsec is implemented at xMII level (i.e. in a PHY): When the MAC is sending data at a line-rate (e.g., 100 Mbps), the MACSec core cannot add the SecTag/ICV data, because it would exceed the line limit of 100 Mbps.

The following solutions available when using this device

1. **Cut-through pre-gapped:** The MAC must guarantee at least 12+length (SecTag + ICV) IPG octets. This is worst case 12+32 = 44 octets IPG between any two frames.
2. **Cut-through standard IPG with proprietary MACsec TX/RX control tag:** The software network stack can reserve bandwidth by the means of adding a proprietary header to reserve the required bit times. This header is remove by the TJA1104.
3. Accept frame-loss in case the medium is temporarily over-subscribed.

Without further configuration, the MAC is allowed to burst out frames at minimum IPG of 12 octets. The consequence of a large burst with minimum IPG is that frames are lost in the MACsec subsystem because of FIFO overruns. Such events are counted in the register ABORTED_FRMS. In case such drops occur, the IP_TRACKING register can be used identify critical IPG underrun scenarios.

Note: *The egress IPG of the adapter is actively shaped to be at least TX_OUT_IPG/RX_OUT_IPG octets and set to 12 by default. If the TX IPG enforcement strategy is used, the output IPG must match the input IPG minus MACsec overhead. If the Proprietary MACsec TX/RX control tag strategy is used, the output IPG must match the input IPG. Due to PPM clock differences, the input IPG can be smaller than 12 octets, thus it is recommended to set TX_OUT_IPG/RX_OUT_IPG to 10 to avoid a bottleneck.*

6.13.6 MACsec secure channel

The number of secure channels is fixed to four per direction. The MACsec core IP description [11] shows a higher number of channels, but is limited by the IP usage in the device.

6.13.7 MACsec TX frame classification

There are two distinct mechanism to classify frames on the TX-in interface for MACsec controlled or uncontrolled port as well as steer them to a secure channel.

UM12059, Section 3.3.2 has built-in lookup tables to match Layer-2 Ethernet fields with controlled/uncontrolled port and SCS. For more information, refer to UM12059, Section 3.3.1.

Second, controlled/uncontrolled and SCS can optionally be communicated in-band via a proprietary MACsec TX/RX control tag (MACsec TX/RX control tag). This tag is removed from the frame before sending it to the MACsec core IP and not visible on the medium.

The registers MACSEC_CFG_EXTSCS (see Section 6.14), TX_TAG_ENA, OVERRIDE_TLV_CP and OVERRIDE_TLV_SCI control where and how the frame classification occurs.

11 The actual growth depends on the configuration of the MACSec core.

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Table 24. MACsec controlled/uncontrolled port and secure channel selection overview

MACsec core Register	MACSec adapter Register			TX control tag present	Uncontrolled / controlled port / Secure Channel ID selection
MACSEC_CFG_EXTSCS	TX_TAG_ENA	OVERRIDE_TLV_CP	OVERRIDE_TLV_SCI		
0	0	—	—	—	Filtering based on MACsec core settings. MACsec TX/RX control tag parsing is disabled and frames are forward to the MACsec core as is.
0	1	—	—	—	Filtering based on MACsec core settings. If present, MACsec TX/RX control tag is stripped.
1	0	—	—	—	MACsec TX/RX control tag parsing is disabled and frames are forward to the MACsec core IP as is. Fixed CP / SCI settings as per reset values of TX_TAG_CNTRL1[TLV_CP] and TX_TAG_CNTRL1[TLV_SCI]
1	1	0	0	1	MACsec TX/RX control tag is stripped. CP / SCI extracted from MACsec TX/RX control tag
1	1	0	0	0	Fixed CP / SCI as per registers TX_TAG_CNTRL1[TLV_CP] and TX_TAG_CNTRL1[TLV_SCI]
1	1	1	0	1	MACsec TX/RX control tag is stripped. SCS as per control information expected via MACsec TX/RX control tag. Fixed CP as per register TX_TAG_CNTRL1[TLV_CP]
1	1	1	1	x	Fixed CP / SCS settings as per registers TX_TAG_CNTRL1[TLV_CP] and TX_TAG_CNTRL1[TLV_SCI]

6.13.8 MACsec TX/RX control tag

The MACsec TX/RX control tag is an optional proprietary header that serves two purposes:

- Reserve header space that can be later claimed by the MACsec core to add SecTag and ICV.
- Transfer controlled/uncontrolled port and SCS metadata to the MACsec core, thus bypassing its internal filters.

The MACsec TX/RX control tag is parsed by the HDR extract block as part of the TX datapath (see block diagram). This functionality is only enabled when both ADPTR_ENA and TX_TAG_ENA are set.

When TX_TAG_ENA is disabled, the header extraction block is in bypass and all frames including those with MACsec TX/RX control tags are forwarded to the MACsec core for further processing.

When TX_TAG_ENA is enabled, the control tag is parsed and the tag is stripped and the FCS updated. Then, the frame is forwarded to the MACsec IP.

The MACsec TX/RX control tag consists of an configurable Ethertype (see register TLV_ETHTYP) that is used to identify the header. The header is a TLV (tag length value) type and contains the header length as well as a sub-type specification as part of the header information.

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The supported sub-type in the TX direction is sub-type=1. This sub-type indicates that the header is a MACsec TX/RX control tag. Other sub-types are not supported and the entire frame is dropped if sub-type is found different than 1.

The length field of the header contains the number of octets excluding the TLV identification part (4 octets). Practically, the length field is set to 4+P, where P is the number of octets in the padding region.

The version field must be set to zero. A frame with a non-zero version field is dropped.

The SCS and CP fields are optional. See Table 24 for further details. The SCS field encodes the 4-bit logical secure channel identifier that is used to map the frame to a secure channel. The CP field encodes whether the frame is targeted the controlled port (CP=1). If set to zero (CP=0), the frame is send to the uncontrolled port.

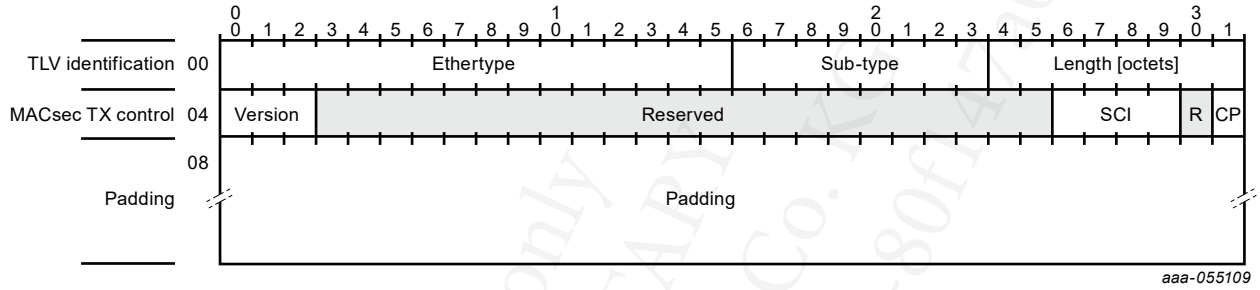


Figure 40. MACsec TX/RX control tag

Note: *If the MACsec TX/RX control tag is used, it is recommended to use it for all (controlled and uncontrolled port) frames to keep latencies constant.*

6.13.8.1 TX control tag padding

The TX control tag can be used to resolve the flow control problem. The purpose of the TX control tag is to make sure that the frame that leaves the MAC is the exact frame size that leaves the MACsec engine. If this is the case, all assumptions that are based on the frame size (such as AVB credit-based shaper, and TSN Qci, Qbv decisions) are still valid. Also no FIFO overflows or rate mismatches occur.

The padding length of the TX control tag must be selected so that the entire TX control tag length matches the added overhead by MACsec:

Length(TX control tag) = Length(Ethertype + SecTag + ICV)

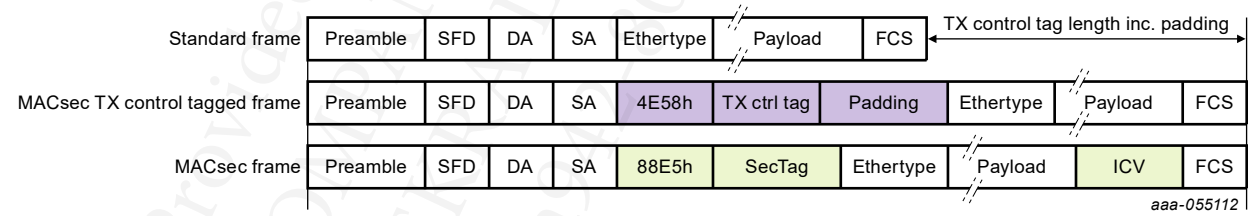


Figure 41. TX control length padding

The Length of the SecTag and ICV depend on the MACsec mode and can be between 16 and 32 octets.

6.13.8.2 TX TLV Policing

The device can be configured to enforce the use of the MACsec TX/RX control tag or reject it. In case the application makes use of this feature and it is expected that all Ethernet frames are MACsec TX/RX control tagged, the POLICE_FRMS register can optionally be set to DROP_STD. Any frame without a tag is then dropped.

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In case the TLV tag is not used, the POLICE_FRMS register can optionally be set to DROP_TLV. Any frames with the MACsec TX/RX control tag present are dropped.

All frames dropped due to this policing setting are reflected in the statistics counter TX_STATS_DISCARDED_FRMS.

6.13.8.3 TX Tag layering

The header extraction block is able to find the MACsec TX/RX control tag in case of a layered header structure. The following tags are recognized and can be skipped:

- VLAN tag (CTAG)
- QinQ tag (STAG)
- 802.1CB tag (RTAG)
- User-defined tag (MISC TAG) of up to 8 octets (see MISC_TAG_LENGTH)

The parser is searching for the first MACsec TX/RX control tag without checking the validity of the header structure itself. An arbitrary sequence of supported tags are accepted. For example, a frame with three VLAN tags is accepted. As soon as the first MACsec TX/RX control tag is found, it is stripped and all remaining octet are treated as payload.

By default, this feature is disabled and individual headers must be enabled (see CTAG_ENA, STAG_ENA, RTAG_ENA and MISC_TAG_ENA).

The header extractor always buffers the frame until the parser can definitely tell that the frame contains a MACsec TX/RX control tag or the maximum depth was reached. Thus, for applications which always place the MACsec TX/RX control tag as the first tag of the frame, keep the detection of additional headers disabled (this is the default setting). This allows the header parser to make decisions as early as possible. Frames that do not have a TX tag but one or multiple other tags are are not delayed because the header parser is waiting for a potential TLV tag after it has seen other supported tags.

The header extractor is able to find the MACsec TX/RX control tag within the first 64 octets (preamble and SFD are not counted).

Note: *If the layering changes from frame to frame, the latency of the MACsec adapter in TX direction is not constant anymore. This is because some frames are buffered longer than others. The latency is always predictable, if the header stack is known.*

6.13.8.4 Error conditions during header modification

If the MACsec TX/RX control tag processing fails, the frame is either fully dropped or invalidated.

Table 25.

Error condition	Manifestation
Unknown version	Frame dropped
Unknown sub-type	Frame dropped
TLV length field > actual frame length	Frame dropped
Frame stops unexpectedly during CTAG/STAG/CTAG/ MISC_TAG parsing	Frame dropped
MACsec TX/RX control tag too deep in the frame	Frame is processed as if no MACsec TX/RX control tag present
FCS error	Frame invalidated (illegal FCS)
MII TX_ERR/RX_ERR set	Frame invalidated (illegal FCS)

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Table 25. *...continued*

Data-parity error	Frame invalidated (illegal FCS)
Control-parity error	Frame either invalidated (illegal FCS) or dropped, depending when and where the error occurs

6.13.9 MACsec RX control tag

The MACsec RX control TAG is conceptually similar to the tag that is used for TX direction. The use of this header is optional and disabled by default. The feature can be enabled by setting RX_TAG_ENA to 1.

The purpose of the tag is to communicate MACsec metadata of the frame with the frame itself. The Ethertype of the MACsec RX control tag is as for the TX control tag set via register TLV_ETHTYP.

For this version of the module, the sub-type is set to 2, length always reports as 4 and version is 0.

The CP field reports the logical port. If CP is set to one, the frame came through the controlled port. If CP is set to zero, the frame came via the uncontrolled port.

The field SCI reports the secure channel index used to process the frame. The field VLD indicates if the SCI is valid. It is set to zero, when a frame came in via the uncontrolled port (CP=0) or a frame came in via the controlled port and the SCI is unknown.

When TAG_UCP_FRMS is set, uncontrolled port frames are tagged with this header. When TAG_CP_FRMS is set, controlled port frames are tagged with this header.

Note: *When TAG_UCP_FRMS is set, the IPG on the rx output side will shrink by 8 octets. Set RX_OUT_IPG to a value lower than 4 to avoid adapter overflow conditions. Consult the application note for details.*

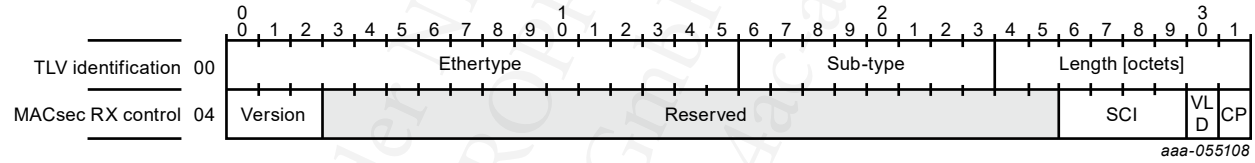


Figure 42. MACsec RX control tag

6.13.9.1 RX Tag layering

By default, the MACsec RX control tag is placed as the first tag, directly after the source MAC address in the Ethernet frame. For applications where the MAC offers hardware offloading and expects certain tags show first, the placement of the MACsec RX Control tag can be tuned.

The header extender block parses the frame on the fly and is able to skip recognized headers before inserting the RX tag. The following tags are recognized and can be skipped:

- VLAN tag (CTAG)
- QinQ tag (STAG)
- 802.1CB tag (RTAG)
- User-defined tag (MISC TAG) of up to 8 octets (see MISC_TAG_LENGTH)

The behavior is controlled via the two register bits SKIP_X and TAG_AFTER_X (where X is either CTAG, STAG, RTAG or MISC_TAG). If SKIP_X is set, header X is skipped and the RX TAG is placed later (but not immediately) and scanning continues. If a header is found that is not recognized, the RX tag is immediately placed.

If TAG_AFTER_X is set, the RX tag is placed immediately after header X. If both TAG_AFTER_X and SKIP_X are set, SKIP_X has precedence.

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6.13.10 Latency

The latency introduced by the MACsec subsystem consists of two components: First the MACsec adapter latencies and second the actual MACsec core latencies. The latency of the adapter is constant per-direction, if TX_TAG_ENA=0 and RX_TAG_ENA=0. When header parsing and extension is enabled, the delay can be data-dependent. The tag placement influences the latency. In these cases the latency is always predicable.

If MACsec TX/RX control tags are used for bandwidth reservation, it is recommended to use them for uncontrolled port and controlled port frames with the same padding settings. That keeps sure that the processing is identical for all frame types. If it is acceptable to have different (but predictable) latencies for UCP/CP this is not required.

The latency of the MACsec core is compensated for uncontrolled and controlled port frames and constant.

To summarize, the latency in TX and RX is constant per-direction and independent of:

- Payload size
- Tag structure (assuming TX TLV tag is placed first)
- MACSec action (encrypt, decrypt, verify, clear)

6.13.10.1 TX Latency

The MACSec core matches latency internally, when enabled. The delay is enabled by setting MACSEC_CFG[UFDCD]. By default the MACsec core delays uncontrolled port frames by a fixed number of cycles which is not fully matching the latency in this integration scenario. For best matching, the FIFO delay mode must be enabled MACSEC_CFG[UFDCM].

For TX controlled frame processing, the MACSec core will check if the input will create a MACSec short-length frame. For this, the core needs to accumulate up to 80 octets to make the decision if a frame is short-length or not. This will make latency matching impossible.

To disable short-length handling on transmission, MACSEC_CFG[SLD] must be set. This makes the internal buffering non-frame size dependent and allows constant latency. For this, the MAC on TX-in must guarantee that the MACsec payload data is more than 64 octets, thus not create incorrect short-length conditions.

The following table assumes UFDCM, UFDCD and SLD are set up as previously explained. TAG can be either S-TAG, C-TAG, R-TAG since they have the same length. See Section 6.13.8.2.

Table 26. TX latency

			MACsec subsystem latency	
TX_TAG_ENA	**TX control tag placement**	**Comment**	**MII-mode [MII cycles]**	**GMII-mode [GMII cycles]**
Disabled	—		176	108
Enabled	TLV always first tag	*Latency depends on number of padding octets*	176	108
Enabled	TAG, TLV		176	112
Enabled	TAG, TAG, TLV	*Order of all TAGs preceding TLV is irrelevant. MISC-TAG is 6 octets*	179	116
Enabled	TAG, TAG, MISC-AG, TLV		195	124
Enabled	TAG, TAG, TAG, MISC-TAG, TLV	*MACSec core only supports 3 tags in clear*	—	—

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6.13.10.2 RX Latency

For received frames, the MACsec core does not implement latency matching. Instead active latency compensation is implemented outside the MACsec core in the adapter logic. No additional settings are required.

Table 27. RX latency

			MACsec subsystem latency	
RX_TAG_ENA	**RX control tag placement**	**Comment**	**MACsec subsystem latency [MII cycles]**	**GMII-mode [GMII cycles]**
Disabled	—		179	122
Enabled	TLV always first tag		179	122
Enabled	TAG, TLV		179	122
Enabled	TAG, TAG, TLV	*Order of all TAGs preceding TLV is irrelevant*	179	122
Enabled	TAG, TAG, TAG, TLV		179	122
Enabled	TAG, TAG, TAG, MISC-TAG, TLV		179	122

6.13.11 Interrupts

The MACsec subsystem supports four interrupts. Two IRQs for the adapters and two IRQs for the actual MACsec IP. A detailed description of the MACsec core interrupts can be found in the MACsec core section. A detailed description of the adapter IRQs can be found in the register documentation. It is recommended to reset the MACsec adapter in case of MACSEC_HARD_ERR or PARITY_CNTRL_ERR interrupts.

Note: *In the TJA1104 integration, the MACsec Adapter IRQ (ADPTR_IRQ_INTSTAT) is connected to the device's IRQ hierarchy as PORT_FUNC_IRQ_STATUS[MAC_ADPTR_IRQ]. The ADPTR_ERR_IRQ is left unconnected because all functionality is also available through the normal IRQ. The MACsec core's event (MACSEC_EVR) and error IRQ (MACSEC_ERR) are ORed together and connected to PORT_FUNC_IRQ_STATUS[MACSEC_IRQ].*

6.13.12 Adapter Statistics

The adapter keeps track of various events through the means of counters.

- **TX control tag counter** TX_STATS_TLV_FRMS
- **FCS frame errors** TX_STATS_FCS_ERRS, RX_STATS_FCS_ERRS
- **Discarded frames due to policing rules** TX_STATS_DISCARDED_FRMS
- **Aborted frames due to error conditions** TX_STATS_ABORTED_FRMS, RX_STATS_ABORTED_FRMS
- **MII errors** TX_STATS_MII_ERRS , RX_STATS_MII_ERRS

6.13.13 LED interface

The adapter provides four configurable LED triggers to the LED controller. Each LED trigger can be configured individually (ADPTR_LED_TRIGn) to support the following events:

- TX_IN_EN: strobe when a frame is transmitted into the tx_in interface.
- TX_OUT_EN: strobe when a frame is transmitted out of the tx_out interface.
- RX_IN_DV: strobe when a frame is transmitted into the rx_in interface.
- RX_OUT_DV: strobe when a frame is transmitted out of the rx_out interface.
- TX_CP_EVENT: strobe when a frame is transmitted via a controlled port (tx)
- TX_UCP_EVENT: strobe when a frame is transmitted via a uncontrolled port (tx)
- RX_CP_EVENT: strobe when a frame is received via a controlled port (rx)

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- RX_UCP_EVENT: strobe when a frame is received via a uncontrolled port (rx)

The LED trigger signals are connected to the device's LED controller.

6.13.14 Reset

The adapter reset function accessible via the control register ADPTR_CNTRL[RESET]. Setting this bit resets the adapter, the reset is self-clearing.

The MACsec core is reset via the top-level reset triggered by disabling MACsec through PORT_FUNC[MACSEC_ENABLE]=0.

6.13.15 MDIO mapping

The MACsec adapter and core registers are 32-bit wide and mapped into a designated Clause 45 MDIO region in MMD31 between 9000h-9FFFh.

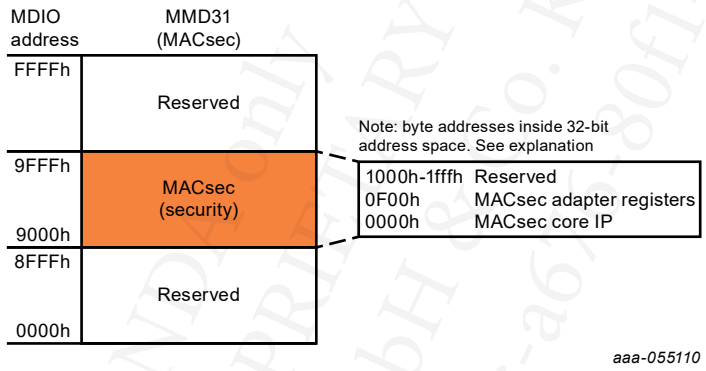


Figure 43. MACsec memory mapping

Since the MDIO access width is limited to 16-bit, each 32-bit register spans two neighboring 16-bit MDIO registers which need to be written/read consecutive from lower to upper 16-bit word. The read/write is only completed after both accesses occurred. The MDIO is resilient against incorrect accesses, but can return wrong data.

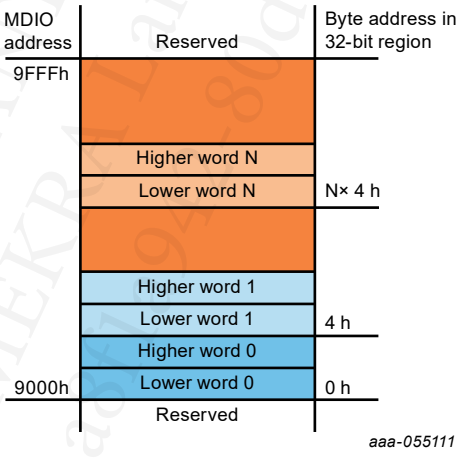


Figure 44. 32-bit register arrangement inside MMD31 space

To convert between the relative byte-addresses in the MACsec window and actual MDIO addresses the following equation can be used:

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mdio_address = relative_offset / 2 + 9000h

Example: Access the ADPTR_CNTRL register (relative address F00h). First read/write lower 16-bit from MDIO address 9780h (MMD31) and then the upper 16-bit from 9781h (MMD31).

Notes:

- *Lower and upper accesses must be atomic. Never interleave lower and upper accesses with other accesses.*
- *There is some limitation in the SMI access to the MACsec core , consult application notes for additional information.*

6.13.16 Register map

The following addresses are relative byte-addresses inside the MACsec address window.

Table 28.

Relative Byte-address inside 32-bit region	Description
0x000-0xEFF	MACsec core
0xF00-0xFFF	MACSec adapter registers

6.13.17 Configuration and test enable

In order to change the configuration of the MACsec adapter, the CONFIG_ENA bit must be set. This unlocks all configuration bits for change. The change is applied after CONFIG_ENA is deasserted.

Note: *After (re-)configuration is completed, CONFIG_ENA **must** be deasserted. Otherwise, settings are not committed.*

In this integration, TEST_ENA is not used, and test-modes are not available.

When TEST_ENA or CONFIG_ENA is asserted, ADPTR_STATUS[STATE] indicates TEST_CONFIG.

6.13.18 Initialization

After device reset or startup, the entire MACsec subsystem is externally bypassed and the MACsec adapter is by default in blocking mode (ADPTR_ENA=0). The following steps are required to enable the adapter.

The adapter is dropping frames until ADPTR_ENA is set to 1. To ensure that no frames accidentally pass through during a partial configuration, set ADPTR_ENA=1 as the last step.

1. Device specific initialization:
   To enable the MACsec function, PORT_FUNC_ENABLES[MACSEC_ENABLE] MMD30.8048h.5 must be set.
2. Set CONFIG_ENA=1 to allow adapter reconfiguration.
3. Optionally configure MACsec RX Control Tag and MACsec TX/RX control tag processing
   a. Set TX_TAG_ENA=1 for MACsec TX/RX control tag processing
   b. Set RX_TAG_ENA=1 for MACsec RX Control Tag processing
   c. Configure Ethertype via register TLV_ETHTYP
4. It is recommended to reduce RX_OUT_IPG and TX_OUT_IPG to 10 to achieve highest throughput.
   Note: *The TX and RX IPG as normally received by a PHY can already be below 12 due to IPG shrinkage caused by clock-domain-crossing inside the SoC / peer-PHY sides. Setting the RX_OUT_IPG/ TX_OUT_IPG to lower values adds some margin.*
   Note: *The settings RX_OUT_IPG=10 together with RX_TAG_ENA=1 can cause issues due to excessive IPG reduction since the RX control tag can increase the frame size.*
5. Enable IRQs, if used

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6. Configure MACsec core (see description of MACsec core)
7. Set ADPTR_ENA=1 and CONFIG_ENA=0 to commit the configuration

In this configuration, all frames are passed to the MACsec core for further processing.

6.13.19 Register Description

Table 29. Register overview: CNTRL_STATUS

Address	Name	Access	Reset	Description
0h	ADPTR_CNTRL	R/W	0h	MACSEC adapter control register
4h	ADPTR_STATUS	R/W	0h	MACSEC adapter status register
Ch	TX_TAG_CNTRL	R/W	0h	TX parser TAG control
10h	TX_TAG_CNTRL1	R/W	0h	TX parser TAG control0
14h	RX_TAG_CNTRL	R/W	0h	RX Extender TAG control
18h	ADPTR_PARAMS	R/W	0h	IPG related adapter parameters
1Ch	TAGS_PARAM	R/W	0h	Ethertype configurations
20h	TAGS_PARAM1	R/W	0h	Ethertype configurations
24h	TAGS_PARAM2	R/W	0h	Ethertype configurations
28h	IPG_TRACKING	R/W	0h	Tracking of IPG
40h	ADPTR_LED_TRIG0	R/W	0h	Configuration of MACSEC led output 0
44h	ADPTR_LED_TRIG1	R/W	0h	Configuration of MACSEC led output 1
48h	ADPTR_LED_TRIG2	R/W	0h	Configuration of MACSEC led output 2
4Ch	ADPTR_LED_TRIG3	R/W	0h	Configuration of MACSEC led output 3
50h	ADPTR_IRQ_STAT	R/W	0h	Adapter Raw Interrupt status
54h	ADPTR_IRQ_ENA	R/W	0h	Adapter interrupt enable
58h	ADPTR_IRQ_INTSTAT	R/W	0h	Adapter interrupt status
80h	TX_STATS_TLV_FRMS	R/W	0h	TX TLV frame counter
84h	TX_STATS_DISCARDED_FRMS	R/W	0h	TX discarded frames counter
88h	TX_STATS_FCS_ERRS	R/W	0h	TX FCS error counter
8Ch	TX_STATS_MII_ERRS	R/W	0h	TX MII error counter
90h	TX_STATS_ABORTED_FRMS	R/W	0h	TX Aborted frames counter
C0h	RX_STATS_FCS_ERRS	R/W	0h	RX FCS error counter
C4h	RX_STATS_MII_ERRS	R/W	0h	RX MII error counter
C8h	RX_STATS_ABORTED_FRMS	R/W	0h	RX Aborted frames counter
CCh	MACSEC_CFGMOD_SUMMARY	R/W	0h	Register modification status
F8h	MOD_STAT	R/W	0h	Module Configuration Status register
FCh	MOD_ID	R/W	0h	Module Identification register

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6.13.19.1 ADPTR_CNTRL register

Table 30. ADPTR_CNTRL register - MACSEC adapter control register (address 0h)

Bit	Symbol	Access	Value	Description
31:16	*reserved*	R	0h	Always write 0, ignore on read.
15	RESET	R/W	0h	Resets the adapter including all registers. Registers can be accessed again at least 10 clock cycles after reset.
14	CONFIG_ENA	R/W	0h	If set, the configuration registers can be changed. Changes in updated registers only take effect after CONFIG_ENA is set back to '0'.
13	TEST_ENA	R/W	0h	If set, test related registers can be changed. This bit is also gating the related test functionality. Thus, if this bit is deasserted, the related test-modes will be deactivated.
12	ADPTR_ENA	R/W	0h	If enabled, the MACSec adapter is operational and will forward frames. If disabled, the MII_RX_IN and MII_RX_OUT stages are disabled and the datapath towards the MACSec IP is gated. Frames which arrive on MII_RX_IN and MII_RX_OUT while the adapter is disabled are counted as aborted frames (see RX/TX_STATS_ABORTED_FRMS).
11:0	*reserved*	R	0h	Always write 0, ignore on read.

6.13.19.2 ADPTR_STATUS register

Table 31. ADPTR_STATUS register - MACSEC adapter status register (address 4h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
31:3	*reserved*	R	0h	Ignore on read.
2:0	STATE	R		Reports the executive state summary of this block.
			0*	Set if MACSEC IP is disabled or in powerdown
			1	MACSEC IP is enabled from top-level, but ADPTR_ENA=0
			2	Set if a test mode or configuration is enabled, here parity test
			3	Set if ENA=1 and TEST_ENA=0
			4	Parity error

6.13.19.3 TX_TAG_CNTRL register

Table 32. TX_TAG_CNTRL register - TX parser TAG control (address Ch)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
31	TX_TAG_ENA	R/W	0h	When disabled, the TX tag extraction / frame builder is disabled and operating in bypass. All frames are passing the TX header extraction unmodified
30:20	*reserved*	R	0h	Always write 0, ignore on read.
19	MISC_TAG_ENA	R/W	0h	If enabled, a configurable tag (see MISC_ETHTYP and MISC_TAG_LENGTH) is enabled. If this tag is detected at an

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Table 32. TX_TAG_CNTRL register - TX parser TAG control (address Ch) *...continued*

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
				arbitrary position, it will be skipped over and searching for the TLV is continued. If this bit is disabled, the parsing process is stopped once this header is seen.
18	RTAG_ENA	R/W	0h	If enabled FRER tag parsing is supported. If this tag is detected at an arbitrary position, it will be skipped over and searching for the TLV is continued. If this bit is disabled, the parsing process is stopped once this header is seen.
17	STAG_ENA	R/W	0h	If enabled, a QinQ tag parsing is enabled. If this tag is detected at an arbitrary position, it will be skipped over and searching for the TLV is continued. If this bit is disabled, the parsing process is stopped once this header is seen.
16	CTAG_ENA	R/W	0h	If enabled, a VLAN tag parsing is enabled. If this tag is detected at an arbitrary position, it will be skipped over and searching for the TLV is continued. If this bit is disabled, the parsing process is stopped once this header is seen.
15:8	TLV_LENGTH	R/W	0h	
7:4	*reserved*	R	0h	Always write 0, ignore on read.
3	POLICE_TLV_ LENGTH	R/W	0h	If set, the TLV parser asserts that the TLV length as seen in the tag has a length that matches TLV_LENGTH, if not the frame is dropped and counted as an ABORTED_FRMS.
2:0	POLICE_FRMS	R/W		If set, the TX parser is able to enforce certain frame types (TLV or STD). This setting controls if and which frames are dropped
			0*	No Policing
			3	Drops TLV frames. If set frames with a MACsec TX/RX control tag are dropped.
			6	If set, all frames without a TLV are dropped.

6.13.19.4 TX_TAG_CNTRL1 register

Table 33. TX_TAG_CNTRL1 register - TX parser TAG control0 (address 10h)

Bit	Symbol	Access	Value	Description
31:7	*reserved*	R	0h	Always write 0, ignore on read.
6:3	TLV_SCI	RW	0h	Secure channel setting as provided to the MACSec block when OVERRIDE_TLV_SCI is set. This value is also used to drive inputs to MACSec when no TLV is present
2	OVERRIDE_TLV_ SCI	RW	0h	If set the Secure-channel identifier (SCI) selection from TLV tag is not used but instead the value of register TLV_SCI is used.
1	TLV_CP	RW	0h	Uncontrolled/Controlled port setting as provided to the MACSec block when OVERRIDE_TLV_CP is set. This value is also used to drive inputs to MACSec when no TLV is present
0	OVERRIDE_TLV_ CP	RW	0h	If set, the controlled-port (CP) selection from TLV tag is not used but the value of register TLV_CP is used

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6.13.19.5 RX_TAG_CNTRL register

Table 34. RX_TAG_CNTRL register - RX Extender TAG control (address 14h)

Bit	Symbol	Access	Value	Description
31	RX_TAG_ENA	R/W	0h	When disabled, the RX header tag frame builder is disabled and all frames are forwarded unmodified.
30:12	*reserved*	R	0h	Always write 0, ignore on read.
11	SKIP_MISC_TAG	R/W	0h	MISC_TAG. For description see CTAG.
10	SKIP_RTAG	R/W	0h	RTAG. For description see CTAG.
9	SKIP_STAG	R/W	0h	STAG. For description see CTAG.
8	SKIP_CTAG	R/W	0h	If this bit is set, the CTAG will be detected and skipped over during RX header tag generation.
7:6	*reserved*	R	0h	Always write 0, ignore on read.
5	TAG_AFTER_ MISC_TAG	R/W	0h	MISC_TAG. For description see CTAG.
4	TAG_AFTER_RTAG	R/W	0h	RTAG. For description see CTAG.
3	TAG_AFTER_STAG	R/W	0h	STAG. For description see CTAG.
2	TAG_AFTER_CTAG	R/W	0h	If this bit is set and a CTAG is found, the RX header tag is added immediately after it, without scanning any further. If SKIP_CTAG is set, skipping has precedence. If multiple TAG_AFTER are set, the TLV tag is placed after the first tag found.
1	TAG_CP_FRMS	R/W	0h	If set, a RX header tag is added to controlled port frames. Attention: this shrinks the IPG!
0	TAG_UCP_FRMS	R/W	0h	If set, a RX header tag is added to uncontrolled port frames. Attention: this shrinks the IPG!

6.13.19.6 ADPTR_PARAMS register

Table 35. ADPTR_PARAMS register - IPG related adapter parameters (address 18h)

Bit	Symbol	Access	Value	Description
21:16	TX_OUT_IPG	R/W	Ch	Minimum RX IPG maintained on TX_OUT path
15:14	*reserved*	R	0h	Always write 0, ignore on read.
13:8	RX_OUT_IPG	R/W	Ch	Minimum RX IPG maintained on RX_OUT path
7:6	*reserved*	R	0h	Always write 0, ignore on read.
5:0	IPG_MON_THR	R/W	Ah	Set the minimum IPG threshold for which an IRQ is flagged. IF an IPG on RX/TX in path is seen with less than this value (in octets) the IRQ is set. Note, due to clock-domain-crossings the expected IPG is less than 12.

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6.13.19.7 TAGS_PARAM register

Table 36. TAGS_PARAM register - Ethertype configurations (address 1Ch)

Bit	Symbol	Access	Value	Description
31:16	STAG_ETHTYP	R/W	88A8h	Ethertype used to identify outer VLAN (STAG) tag
15:0	CTAG_ETHTYP	R/W	8100h	Ethertype used to identify VLAN (CTAG) tag

6.13.19.8 TAGS_PARAM1 register

Table 37. TAGS_PARAM1 register - Ethertype configurations (address 20h)

Bit	Symbol	Access	Value	Description
31:16	RTAG_ETHTYP	R/W	F1C1h	Ethertype used to identify RTAG tag
15:0	TLV_ETHTYP	R/W	4E58h	Ethertype used to identify RX/TX MACsec control TLV

6.13.19.9 TAGS_PARAM2 register

Table 38. TAGS_PARAM2 register - Ethertype configurations (address 24h)

Bit	Symbol	Access	Value	Description
18:16	MISC_TAG_ LENGTH	R/W	4h	Length used for MISC_ETHTYP
15:0	MISC_ETHTYP	R/W	0h	The MISC tag is catering for any header which is not natively supported. The MISC_ETHTYP is used to identify the tag and the MISC_TAG_LENGTH defines the length in octets not including the Ethertype itself. The supported maximum length is 6 octets. (so the header including the Ethertype is 64bits).

6.13.19.10 IPG_TRACKING register

Table 39. IPG_TRACKING register - Tracking of IPG (address 28h)

Bit	Symbol	Access	Value	Description
13:8	TX_IN_IPG_MIN	R/W	3Fh	Minimum number of IPG octets observed since last clear.
7:6	*reserved*	R	0h	Always write 0, ignore on read.
5:0	RX_IN_IPG_MIN	R/W	3Fh	Minimum number of IPG octets observed since last clear.

6.13.19.11 ADPTR_LED_TRIG0 register

Table 40. ADPTR_LED_TRIG0 register - Configuration of MACSEC led output 0 (address 40h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15	ENA	R/W	0h	Enable LED output
14:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	FUNCTION	R/W		Configures which internal signal to externalize as TRIGGER0
			0*	

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Table 40. ADPTR_LED_TRIG0 register - Configuration of MACSEC led output 0 (address 40h) *...continued*

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
			1	If selected, the internal TX_EN signal at the TX_IN interface is selected
			2	If selected, the internal TX_EN signal at the TX_OUT interface is selected
			3	If selected, the internal RX_DV signal at the RX_IN interface is selected
			4	If selected, the internal RX_DV signal at the RX_OUT interface is selected
			5	If selected, the trigger signal is strobed every time a frame is transmitted on the controlled port
			6	If selected, the trigger signal is strobed every time a frame is transmitted on the uncontrolled port
			7	If selected, the trigger signal is strobed every time a frame is received on the controlled port
			8	If selected, the trigger signal is strobed every time a frame is received on the uncontrolled port

6.13.19.12 ADPTR_LED_TRIG1 register

Table 41. ADPTR_LED_TRIG1 register - Configuration of MACSEC led output 1 (address 44h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15	ENA	R/W	0h	Enable LED output
14:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	FUNCTION	R/W		Configures which internal signal to externalize as TRIGGER0
			0*	
			1	If selected, the internal TX_EN signal at the TX_IN interface is selected
			2	If selected, the internal TX_EN signal at the TX_OUT interface is selected
			3	If selected, the internal RX_DV signal at the RX_IN interface is selected
			4	If selected, the internal RX_DV signal at the RX_OUT interface is selected
			5	If selected, the trigger signal is strobed every time a frame is transmitted on the controlled port
			6	If selected, the trigger signal is strobed every time a frame is transmitted on the uncontrolled port
			7	If selected, the trigger signal is strobed every time a frame is received on the controlled port
			8	If selected, the trigger signal is strobed every time a frame is received on the uncontrolled port

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6.13.19.13 ADPTR_LED_TRIG2 register

Table 42. ADPTR_LED_TRIG2 register - Configuration of MACSEC led output 2 (address 48h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15	ENA	R/W	0h	Enable LED output
14:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	FUNCTION	R/W		Configures which internal signal to externalize as TRIGGER0
			0*	
			1	If selected, the internal TX_EN signal at the TX_IN interface is selected
			2	If selected, the internal TX_EN signal at the TX_OUT interface is selected
			3	If selected, the internal RX_DV signal at the RX_IN interface is selected
			4	If selected, the internal RX_DV signal at the RX_OUT interface is selected
			5	If selected, the trigger signal is strobed every time a frame is transmitted on the controlled port
			6	If selected, the trigger signal is strobed every time a frame is transmitted on the uncontrolled port
			7	If selected, the trigger signal is strobed every time a frame is received on the controlled port
			8	If selected, the trigger signal is strobed every time a frame is received on the uncontrolled port

6.13.19.14 ADPTR_LED_TRIG3 register

Table 43. ADPTR_LED_TRIG3 register - Configuration of MACSEC led output 3 (address 4Ch)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15	ENA	R/W	0h	Enable LED output
14:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	FUNCTION	R/W		Configures which internal signal to externalize as TRIGGER0
			0*	
			1	If selected, the internal TX_EN signal at the TX_IN interface is selected
			2	If selected, the internal TX_EN signal at the TX_OUT interface is selected
			3	If selected, the internal RX_DV signal at the RX_IN interface is selected
			4	If selected, the internal RX_DV signal at the RX_OUT interface is selected
			5	If selected, the trigger signal is strobed every time a frame is transmitted on the controlled port

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Table 43. ADPTR_LED_TRIG3 register - Configuration of MACSEC led output 3 (address 4Ch)*...continued*

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
			6	If selected, the trigger signal is strobed every time a frame is transmitted on the uncontrolled port
			7	If selected, the trigger signal is strobed every time a frame is received on the controlled port
			8	If selected, the trigger signal is strobed every time a frame is received on the uncontrolled port

6.13.19.15 ADPTR_IRQ_STAT register

Table 44. ADPTR_IRQ_STAT register - Adapter Raw Interrupt status (address 50h)

Bit	Symbol	Access	Value	Description
10	MACSEC_SOFT_ERR	R/W	0h	If set, the MACsec core is in a recoverable error condition which could have caused frame loss Write one to clear.
9	MACSEC_HARD_ERR	R/W	0h	If set, the MACsec core is in an unrecoverable error condition and requires reset. Write one to clear.
8	MACSEC_CFGMOD_EVT	R/W	1h	if set, MACSEC core context register changed due software action or an error caused by internal bit-flips. Write one to clear.
7	DISCARDED_FRM	R/W	0h	If set, a frame was discarded because of the POLICE feature. See TX_STATS_DISCARDED_FRMS register. Write one to clear.
6	ABORTED_FRM	R/W	0h	If set, a frame was aborted. See register TX_STATS_ABORTED_FRMS for the conditions. Write one to clear.
5	PARITY_CNTRL_ERR	R/W	0h	If set, a control parity error in the adapter occurred Write one to clear.
4	PARITY_DATA_ERR	R/W	0h	If set, a data parity error in the adapter occurred Write one to clear.
3	TX_IPG_THR_EXCEEDED	R/W	0h	If set, the IPG on the TX_IN interface fell below the threshold Write one to clear.
2	RX_IPG_THR_EXCEEDED	R/W	0h	If set, the IPG on the RX_IN interface fell below the threshold Write one to clear.
1	TX_IN_ERR	R/W	0h	If set, an MII error on the TX_IN interface was observed Write one to clear.
0	RX_IN_ERR	R/W	0h	If set, an MII error on the RX_IN interface was observed Write one to clear.

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6.13.19.16 ADPTR_IRQ_ENA register

Table 45. ADPTR_IRQ_ENA register - Adapter interrupt enable (address 54h)

Bit	Symbol	Access	Value	Description
10	MACSEC_SOFT_ERR	R/W	0h	If set, the respective IRQ is enabled
9	MACSEC_HARD_ERR	R/W	0h	If set, the respective IRQ is enabled
8	MACSEC_CFGMOD_EVT	R/W	0h	If set, the respective IRQ is enabled
7	DISCARDED_FRM	R/W	0h	If set, the respective IRQ is enabled
6	ABORTED_FRM	R/W	0h	If set, the respective IRQ is enabled
5	PARITY_CNTRL_ERR	R/W	0h	If set, the respective IRQ is enabled
4	PARITY_DATA_ERR	R/W	0h	If set, the respective IRQ is enabled
3	TX_IPG_THR_EXCEEDED	R/W	0h	If set, the respective IRQ is enabled
2	RX_IPG_THR_EXCEEDED	R/W	0h	If set, the respective IRQ is enabled
1	TX_IN_ERR	R/W	0h	If set, the respective IRQ is enabled
0	RX_IN_ERR	R/W	0h	If set, the respective IRQ is enabled

6.13.19.17 ADPTR_IRQ_INTSTAT register

Table 46. ADPTR_IRQ_INTSTAT register - Adapter interrupt status (address 58h)

Bit	Symbol	Access	Value	Description
10	MACSEC_SOFT_ERR	R	0h	Masked status of the respective interrupt
9	MACSEC_HARD_ERR	R	0h	Masked status of the respective interrupt
8	MACSEC_CFGMOD_EVT	R	0h	Masked status of the respective interrupt
7	DISCARDED_FRM	R	0h	Masked status of the respective interrupt
6	ABORTED_FRM	R	0h	Masked status of the respective interrupt
5	PARITY_CNTRL_ERR	R	0h	Masked status of the respective interrupt
4	PARITY_DATA_ERR	R	0h	Masked status of the respective interrupt
3	TX_IPG_THR_EXCEEDED	R	0h	Masked status of the respective interrupt
2	RX_IPG_THR_EXCEEDED	R	0h	Masked status of the respective interrupt
1	TX_IN_ERR	R	0h	Masked status of the respective interrupt
0	RX_IN_ERR	R	0h	Masked status of the respective interrupt

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6.13.19.18 TX_STATS_TLV_FRMS register

Table 47. TX_STATS_TLV_FRMS register - TX TLV frame counter (address 80h)

Bit	Symbol	Access	Value	Description
31:0	TLV_FRMS	R/W	0h	Counts the number of TLV frames. • This counter is overflowing. • Writing 0x0 clears the counter

6.13.19.19 TX_STATS_DISCARDED_FRMS register

Table 48. TX_STATS_DISCARDED_FRMS register - TX discarded frames counter (address 84h)

Bit	Symbol	Access	Value	Description
31:0	DISCARDED_FRM	R/W	0h	Counts the number of discarded frames because of the POLICE feature. • This counter is overflowing. • Writing 0x0 clears the counter

6.13.19.20 TX_STATS_FCS_ERRS register

Table 49. TX_STATS_FCS_ERRS register - TX FCS error counter (address 88h)

Bit	Symbol	Access	Value	Description
31:0	FCS_ERRS	R/W	0h	Counts the number of frames where the FCS-check on ingress failed. • This counter is overflowing. • Writing 0x0 clears the counter

6.13.19.21 TX_STATS_MII_ERRS register

Table 50. TX_STATS_MII_ERRS register - TX MII error counter (address 8Ch)

Bit	Symbol	Access	Value	Description
31:0	MII_ERRS	R/W	0h	Counts the number of frames where the TX_ER is set. • This counter is overflowing. • Writing 0x0 clears the counter

6.13.19.22 TX_STATS_ABORTED_FRMS register

Table 51. TX_STATS_ABORTED_FRMS register - TX Aborted frames counter (address 90h)

Bit	Symbol	Access	Value	Description
31:0	ABORTED_FRMS	R/W	0h	Counts the number of frames that are aborted inflight because of an error condition. • This counter is overflowing. • Writing 0x0 clears the counterThere are the following reasons for aborts. 1. Input fifo overflow 2. Output fifo underflow 3. Parity error in data or control 4. Frames that are dropped on TX_IN because ENA = 0

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6.13.19.23 RX_STATS_FCS_ERRS register

Table 52. RX_STATS_FCS_ERRS register - RX FCS error counter (address C0h)

Bit	Symbol	Access	Value	Description
31:0	FCS_ERRS	R/W	0h	Counts the number of frames where the FCS-check on ingress failed. • This counter is overflowing. • Writing 0x0 clears the counter.

6.13.19.24 RX_STATS_MII_ERRS register

Table 53. RX_STATS_MII_ERRS register - RX MII error counter (address C4h)

Bit	Symbol	Access	Value	Description
31:0	MII_ERRS	R/W	0h	Counts the number of frames where the RX_ER is set. • This counter is overflowing. • Writing 0x0 clears the counter.

6.13.19.25 RX_STATS_ABORTED_FRMS register

Table 54. RX_STATS_ABORTED_FRMS register - RX Aborted frames counter (address C8h)

Bit	Symbol	Access	Value	Description
31:0	ABORTED_FRMS	R/W	0h	Counts the number of frames that are aborted inflight because of an error condition. • This counter is overflowing. • Writing 0x0 clears the counter The following are reasons for aborts. • Input fifo overflow • Output fifo underflow • Parity error in data or control • Frames that are dropped on RX_IN because ENA = 0

6.13.19.26 MACSEC_CFGMOD_SUMMARY register

Table 55. MACSEC_CFGMOD_SUMMARY register - Register modification status (address CCh)

Bit	Symbol	Access	Value	Description
18	TX_ADPTR	R/W	1h	If set, MACSEC TX Adapter settings have changed since this flag was cleared. Write one to clear.
17	RX_ADPTR	R/W	1h	If set, the MACSEC RX Adapter settings have changed since this flag was cleared. Write one to clear.
16	GLOBAL	R/W	0h	If set, the MACSEC global context and uncontrolled port filtering has changed since this flag was cleared. Write one to clear.
15:12	*reserved*	R	0h	Always write 0, ignore on read.
11	TX_SC4	R/W	0h	If set, the MACSEC TX secure channel 4 context has changed since this flag was cleared

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Table 55. MACSEC_CFGMOD_SUMMARY register - Register modification status (address CCh)*...continued*

Bit	Symbol	Access	Value	Description
				Write one to clear.
10	TX_SC3	R/W	0h	If set, the MACSEC TX secure channel 3 context has changed since this flag was cleared Write one to clear.
9	TX_SC2	R/W	0h	If set, the MACSEC TX secure channel 2 context has changed since this flag was cleared Write one to clear.
8	TX_SC1	R/W	0h	If set, the MACSEC TX secure channel 1 context has changed since this flag was cleared Write one to clear.
7:4	*reserved*	R	0h	Always write 0, ignore on read.
3	RX_SC4	R/W	0h	If set, the MACSEC RX secure channel 4 context has changed since this flag was cleared Write one to clear.
2	RX_SC3	R/W	0h	If set, the MACSEC RX secure channel 3 context has changed since this flag was cleared Write one to clear.
1	RX_SC2	R/W	0h	If set, the MACSEC RX secure channel 2 context has changed since this flag was cleared Write one to clear.
0	RX_SC1	R/W	0h	If set, the MACSEC RX secure channel 1 context has changed since this flag was cleared. Note: For all of the following flags in this register, a change can occur due to a deliberate software (re)-configuration or because of an internal error caused by bit-flips. Write one to clear.

6.13.19.27 MOD_STAT register

Table 56. MOD_STAT register - Module Configuration Status register (address F8h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
31:24	N_W	R	0h	Implemented number of write-only registers
23:16	N_R	R	5h	Implemented number of read-only registers
15:8	N_RW	R	12h	Implemented number of read/write registers
7:0	TYPE	R		Indicates type of configuration register that is implemented
			7*	Configuration Register module

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6.13.19.28 MOD_ID register

Table 57. MOD_ID register - Module Identification register (address FCh)

Bit	Symbol	Access	Value	Description
31:16	ID	R	A09Bh	Module Identifier: Unique 16-bit code, assigned through NXP internal registration
15:12	MAJOR_REV	R	1h	Major Revision: major revision of module implementation, starting at 0. There may not be software compatibility between major revisions.
11:8	MINOR_REV	R	0h	Minor Revision: minor revision of module implementation, starting at 0. Minor revision of module implementation, starting at 0. Software compatibility is expected between minor revisions.
7:0	APERTURE	R	0h	Aperture Size: Encoded as (aperture size/4K) - 1, so 0 means 4 kByte

6.14 MACsec core

The MACsec memory map and register definition of the MACsec core are described in UM12059. The application notes give additional information on the limitation on MACsec in specific use cases.

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6.15 TC10

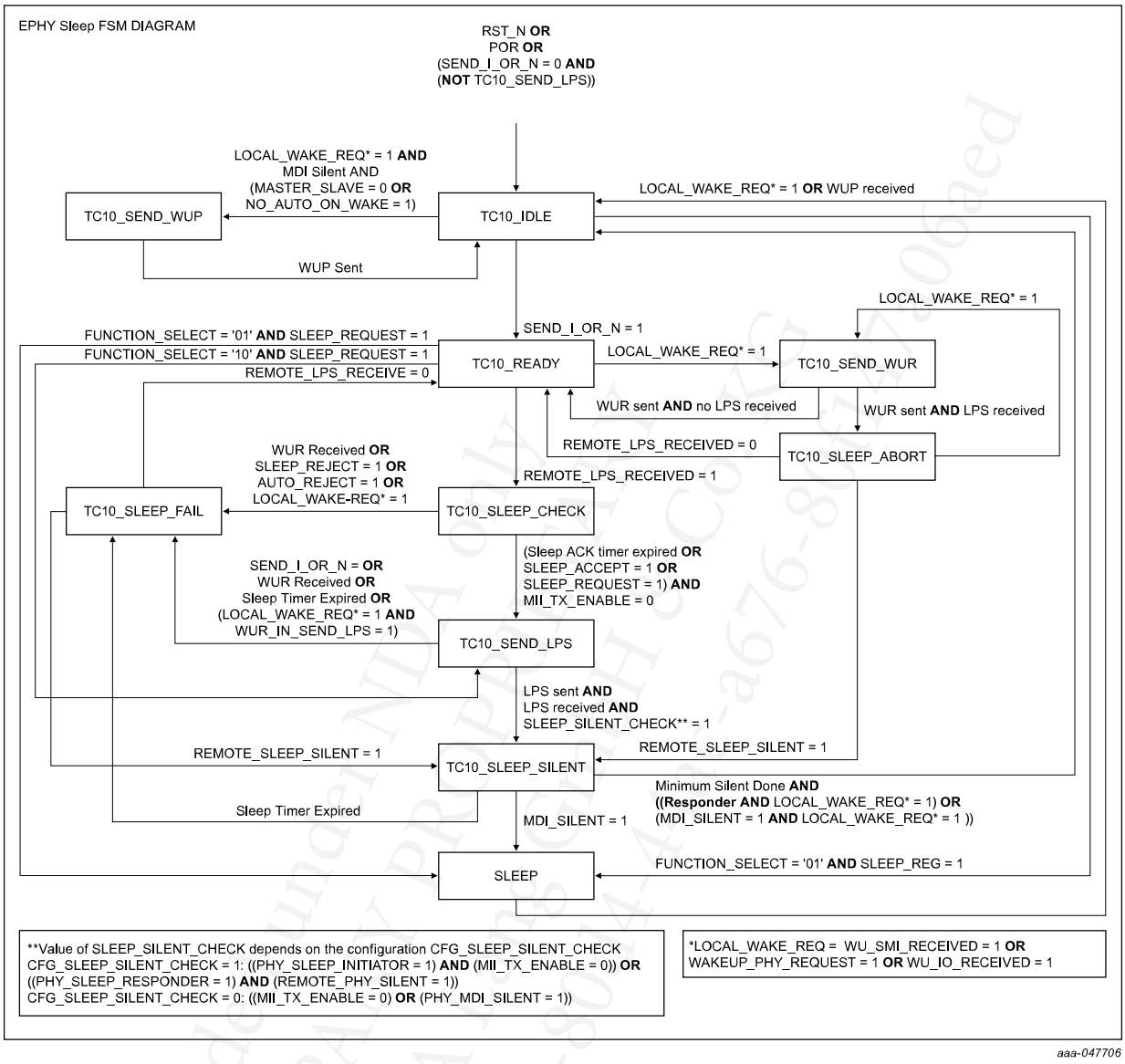


Figure 45. TC10 Ethernet PHY state transitions

6.16 Miscellaneous

6.16.1 JTAG

The TJA1104 supports IEEE1149.1 and IEEE1149.6 compliant JTAG interface for boundary scan. A boundary scan description file (BSDL) is available on request.

JTAG mode is entered via pin-strapping the MODE selection configuration accordingly. In this mode the JTAG TAP interface is accessible on pins TMS, TDI, TDO and TCK as shown in Table 58. Other pins remain in functional mode until a IEEE1149 instruction is executed.

JTAG mode can only be entered by pin strap configuration and left via a device reset though the RST_N pin.

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Table 58. JTAG interface

JTAG Function	Symbol	Pin	Type[1]	Description
TMS	TMS / GPIO0 / CONFIG0	17	IO, PU	JTAG Test mode select (VDDIO domain)
TDI	TDI / GPIO1	18	I, PU	JTAG Test data in (VDDIO domain)
TDO	RX_ER / GPIO2 / TDO / CONFIG1	19	O	JTAG Test data out (VDDIO domain)
TCK	TX_ER / GPIO9 / TCK	28	I	JTAG Test clock (VDDIO domain)

[1] I: digital input; O: digital output; IO: digital input/output; OD: digital open-drain output; AO: Analog output; AI: Analog Input; AIO: Analog input/output; PU: weak pull-up; P: power supply. G: ground

6.16.2 General-purpose I/O

The device has 12 GPIOs multiplexed on functional device pins. Not all pins are available in all modes. The following table gives an overview of which GPIOs are available in which mode of operation.

Table 59. Available GPIOs across xMII modes.

GPIO	MII	RMII	RGMII	SGMII
GPIO0	Available	Available	Available	Available
GPIO1	Available	Available	Available	Available
GPIO2	Available[1]	Available[1]	Available	Available
GPIO3	—	•	—	Available
GPIO4	—	•	—	Available
GPIO5	—	—	—	Available
GPIO6	—	—	—	Available
GPIO7	—	—	—	Available
GPIO8	—	—	—	Available
GPIO9	Available[2]	Available[2]	Available	Available
GPIO10	—	Available	—	—
GPIO11	—	Available	—	—

[1] if RX_ER is not used
[2] if TX_ER is not used

Only GPIOs which are free in the configured xMII mode can be used. The configuration settings for GPIOs that are not available are ignored.

Each GPIO can be freely assigned to a function. The available functions are LED, PTP (timestamping) and miscellaneous. The assignment is done through the FUNC_SELECT field of the associated GPIOx_FUNC_CONFIG register.

Some functions, for example the LED function, have multiple internal signals to which a GPIO can be connected, which is configured via SIGNAL_SELECT.

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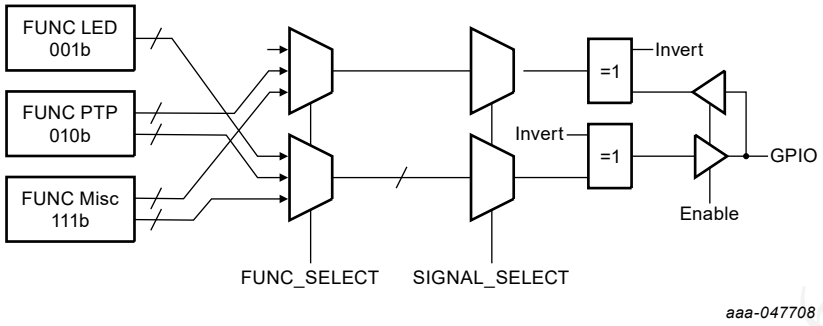


Figure 46. GPIO controller function/signal hierarchy

The table below lists the FUNC_SELECT to function mapping as well as the SIGNAL_SELECT to signal mapping for each function.

Table 60. GPIO function and signal selection (see register GPIOx_FUNC_CONFIG)

Function	Signal	FUNC_ SELECT	SIGNAL_ SELECT	Dir.[1]	Description
LED	LED0_A	001b	00000b	O	LED controller 0, channel A
	LED0_B	001b	00001b	O	LED controller 0, channel B
	LED1_A	001b	00010b	O	LED controller 1, channel A
	LED1_B	001b	00011b	O	LED controller 1, channel B
	LED2_A	001b	00100b	O	LED controller 2, channel A
	LED2_B	001b	00101b	O	LED controller 2, channel B
	LED3_A	001b	00110b	O	LED controller 3, channel A
	LED3_B	001b	00111b	O	LED controller 3, channel B
	reserved	001b	*others*	Z	—
PTP	PTP_SYNC	010b	00000b	I	PTP synchronization input (PPS in)
	PTP_TRIGGER	010b	00001b	I	external PTP timestamp trigger signal
	TX_SOF	010b	10000b	I	External transmit start of frame (SOF) timestamp trigger
	RX_SOF	010b	10001b	I	External receive start of frame (SOF) timestamp trigger
	PPS_OUT	010b	10010b	O	Recovered PPS output
	reserved	010b	*others*		
Misc	REGISTER_OUT	111b	10000b	O	GPIO_OUTPUTS register drives pin
	REF_CLK_OUT	111b	10001b	O	25 MHz REF_CLK_OUT output is enabled. Only available on GPIO9
	TX_CLK_OUT	111b	10010b	O	Test clock output. Only available on GPIO9
	reserved	111b	*others*	Z	—

[1] O: output, I: input, Z: high-Z

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The GPIOx_FUNC_CONFIG.ENABLE acts as a qualifier to and from the GPIO pin. If the selected function is an output function and ENABLE=0, then the GPIO will drive zero. If the selected function is an input function and ENABLE=1, then the GPIO provides a low signal to the selected function.

If GPIOx_FUNC_CONFIG.FUNC_SELECT is set to to RESERVED, the GPIO respective GPIO is configured to input and the pin is tristated.

If multiple GPIOs are assigned to the same function/signal, the inputs are ORed.

Note: *In order to disable RX_ER and TX_ER MII/RMII function and free these pins for GPIO function, XMII_DATA_PIN_CONFIG[RXER_GPIO_SWAP] and XMII_DATA_PIN_CONFIG[TXER_GPIO_SWAP] must be set. After reset, the MII/RMII function defaults to TX_ER and RX_ER function enabled.*

6.16.3 IO driver configuration

MDIO, xMII, and GPIO CMOS drivers are software configurable. The slew rate (SLEW_RATE) and drive strength (DRIVE_STRENGTH) can be adjusted. The following table shows an overview of which registers configure which pins:

Table 61. IO driver configuration

Register	Pins / Function
GPIO_IO_CONFIG	All other GPIO function
GPIO_IO_CLK_CONFIG	GPIO configured for REF_CLK_OUT, TX_CLK_OUT modes
SMI_DATA_IO_CFG	MDIO pin
XMII_CLK_IO_CONFIG	MII: TX_CLK, RX_CLK revRMII: REF_CLK RGMII: RXC
XMII_DATA_IO_CONFIG	MII/RMII/RGMII RXD and TXD pins

The device automatically selects the drive characteristics for the selected xMII/VDDIO combination. Using default settings, the device is able to drive 25 pF capacitive load. If the MAC device is physically close to the PHY and trace length is low (< 15 pF capacitive load), an optimized driver configuration can be used. This is configured by asserting XMII_DATA_IO_CONFIG[ALTERNATE_CONFIG]. This bit changes the configuration for all data and control xMII pins.

To override the default configuration, MANUAL_CONFIG in the respective register must be asserted. The configuration can be changed by setting SLEW_RATE to a value between 0 (slowest) and 7 (fastest) and DRIVE_STRENGTH to a value between 0 (weakest) and 7 (strongest).

The pins INT_N, INH, WAKE_IN_OUT cannot be configured.

Note: *Changing the slew rate and/or drive strength impacts the xMII/MDIO timing. Custom settings do not guarantee compliance to MDIO/MII/RMII/RGMII standards.*

7 Registers

Ignore reserved register fields on read. For reserved fields in write-access registers, always write 0. For singlebit fields, reset values are indicated in the Value column. For multibit fields, reset values are indicated, where applicable, by an asterisk (*).

For reference the notation MMD.REGISTER[BIT] is used.

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7.1 CL22

Consult Section 6.6.5 for detailed explanation of the C22 register mapping and how to access these registers using either Clause 22 or Clause 45 accesses.

7.1.1 Register Description

Table 62. Register overview: C22

Address	Name	Access	Reset	Description
8002h	PHY_ID_1	R/W	1Bh	PHY Identification 1 register
8003h	PHY_ID_2	R/W	B013h	PHY Identification 2 register
800Dh	CL45_ACCESS_CONTROL	R/W	0h	CL45 access control register
800Eh	CL45_ADDRESS_DATA	R/W	0h	CL45 address data register
8014h	FAVORITE_IRQS	R/W	0h	Favorite IRQs register
801Eh	CL45_ADDRESS	R/W	0h	CL45 address register
801Fh	ALWAYS_ACCESSIBLE	R/W	0h	Always-accessible global functions register. This register is accessible also in sleep mode, assuming the VDDIO is powered.

7.1.1.1 PHY_ID_1 register

Table 63. PHY_ID_1 register - PHY Identification 1 register (address 8002h)

Bit	Symbol	Access	Value	Description
15:0	OUI_BITS_3_TO_18	R	1Bh	OUI bits [3:18] OUI = 00:60:37

7.1.1.2 PHY_ID_2 register

Table 64. PHY_ID_2 register - PHY Identification 2 register (address 8003h)

Bit	Symbol	Access	Value	Description
15:10	OUI_BITS_19_TO_24	R	2Ch	OUI bits [19:24] OUI = 00:60:37
9:4	MODEL	R	1h	6-bit PHY model number
3:0	REVISION	R	3h	4-bit PHY revision number

7.1.1.3 CL45_ACCESS_CONTROL register

Table 65. CL45_ACCESS_CONTROL register - CL45 access control register (address 800Dh)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15:14	OP	R/W	0h	MDIO operation 00* = address 01 = data, no post increment 10 = data, post increment on reads and writes 11 = data, post increment on writes only

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Table 65. CL45_ACCESS_CONTROL register - CL45 access control register (address 800Dh)*...continued*

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
13:5	*reserved*	R	0h	Always write 0, ignore on read.
4:0	MMD	R/W	0h	MMD selection 1Fh = NXP-specific (A/B) 1Eh = NXP-specific 1h = PMA/PMD 3h = PCS others = Reserved

7.1.1.4 CL45_ADDRESS_DATA register

Table 66. CL45_ADDRESS_DATA register - CL45 address data register (address 800Eh)

Bit	Symbol	Access	Value	Description
15:0	ADDRESS_DATA	R/W	0h	Address or data of the CL45 transaction, depending on OP selection.

7.1.1.5 FAVORITE_IRQS register

Table 67. FAVORITE_IRQS register - Favorite IRQs register (address 8014h)

Bit	Symbol	Access	Value	Description
15:9	*reserved*	R	0h	Always write 0, ignore on read.
8	SGMII_LINK_EVENT	R/W	0h	If set, change in SGMII Link Status
7	*reserved*	R	0h	Always write 0, ignore on read.
6	INGRESS_TS_IRQ	R	0h	If set, time-stamp(s) available of received PTP frame(s) in INGRESS buffer.
5	EGRESS_TS_IRQ	R	0h	If set, time-stamp(s) available of transmitted PTP frame(s) in EGRESS buffer.
4	WAKE_SLEEP_EVENT	R/W	0h	If set, change of PHY Wake-Sleep status bit(s)
3	LINK_AVAILABLE_EVENT	R/W	0h	If set, change of LINK_AVAILABILITY
2	*reserved*	R	0h	Always write 0, ignore on read.
1	PORT_IRQ	R	0h	If set, Interrupts pending for this PORT
0	GLOBAL_IRQS	R	0h	If set, Interrupts pending in Global interrupt tree

7.1.1.6 CL45_ADDRESS register

Table 68. CL45_ADDRESS register - CL45 address register (address 801Eh)

Bit	Symbol	Access	Value	Description
15:0	ADDRESS	R/W	0h	Current address pointer of the MMD selected by CL45_ACCESS_CONTROL[MMD]

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7.1.1.7 ALWAYS_ACCESSIBLE register

Table 69. ALWAYS_ACCESSIBLE register - Always-accessible global functions register. This register is accessible also in sleep mode, assuming the VDDIO is powered. (address 801Fh)

Bit	Symbol	Access	Value	Description
15	WU_SMI	R/W	0h	If set, Wake-up the device/sub-system by SMI. VDDIO must be present
14:6	*reserved*	R	0h	Always write 0, ignore on read.
5	REFCLK_WARN	R	0h	If this bit is set, a warning is present in OP_CONDITIONS_IRQ_MSTATUS[OSC_WARN] or in PORT_LEVEL_IRQ_MSTATUS[TXC_CLK_WARN] in case of RMII-clkmode. To clear this bit, the respective source IRQs must be cleared.
4	FUSA_PASS_IRQ	R/W	0h	If set, FUSA check has successfully passed.
3	CORE_SUPPLY_WARN	R/W	0h	This bit is asserted upon detection of a fault condition by the core supply monitor. Status register MMD30.CORE_SUPPLY_STATUS located in MMD30 contains details about the fault type. Note that reading register MMD30.CORE_SUPPLY_STATUS requires CORE_SUPPLY_STATUS=1. The IRQ is edge triggered. If the trigger condition persist the IRQ will not be immediately refired when cleared.
2	CORE_SUPPLY_STATUS	R	0h	This bit shows the actual status of the core supply voltage. If the bit is asserted, the voltage is within the tolerated voltage range.
1	LIMITED_ACCESS	R	0h	Indicates whether there is limited device access because the device is sleeping. 0* All registers are available for read and write 1, If CORE_SUPPLY_STATUS=0, registers (except the ALWAYS_ACCESSIBLE) are inaccessible and read zero. If CORE_SUPPLY_STATUS=1 and device is not in sleep with INH deasserted, all registers are readable, the ALWAYS_ACCESSIBLE register is writable.
0	INHIBIT_STATUS	R	0h	This bit indicates the overall inhibit status, that determines if this device drives the INH pin high.

7.2 MMD1

7.2.1 Register Description

Table 70. Register overview: MMD1

Address	Name	Access	Reset	Description
0h	PMA_CONTROL1	R/W	2000h	PMA control 1 register
1h	PMA_STATUS1	R/W	2h	PMA status 1 register
4h	PMA_SPEED_ABILITY	R/W	20h	PMA speed ability register
7h	PMA_CONTROL2	R/W	3Dh	PMA control 2 register
8h	PMA_STATUS2	R/W	9301h	PMA status 2 register
9h	PMD_TRANSMIT_DISABLE	R/W	0h	PMD transmit disable register
Ah	PMD_RCV_DETECT	R/W	0h	PMD receive signal detect register

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Table 70. Register overview: MMD1*...continued*

Address	Name	Access	Reset	Description
Bh	PMA_EXTENDED_ ABILITIES	R/W	800h	PMA extended abilities
12h	BASE_T1_PMA_XTD_ ABILITY	R/W	1h	BASE-T1 PMA extended abilities register
834h	BASE_T1_PMA_CONTROL	R/W	8000h	BASE-T1 PMA control register
836h	E100BT1_PMA_TEST_ CONTROL	R/W	0h	100BASE-T1 PMA test control register

7.2.1.1 PMA_CONTROL1 register

Table 71. PMA_CONTROL1 register - PMA control 1 register (address 0h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15	RESET	R/W	0h	Reset This bit is self-clearing. 0* = Normal operation 1 = On write a PHY reset is asserted. On read this indicates that a reset is still in progress
14	*reserved*	R	0h	Always write 0, ignore on read.
13	SPEED_SELECT_ LSB	R	1h	see SPEED_SELECT_MSB
12	*reserved*	R	0h	Always write 0, ignore on read.
11	LOW_POWER	R/W	0h	If set, the PMA enters low power mode ***Note:*** *PMA LOW_POWER (MMD1) and PCS LOW_ POWER (MMD3) controls have mirrored behaviour. If one is written the other follows.*
10:7	*reserved*	R	0h	Always write 0, ignore on read.
6	SPEED_SELECT_ MSB	R	0h	Speed selection [SPEED_SELECT_MSB, SPEED_SELECT_LSB] are encoded as follows: 11 = Reserved (not supported) 10 = 1000 Mb/s (not supported) 01 = 100 Mb/s 00 = 10 Mb/s (not supported)
5:2	*reserved*	R	0h	Always write 0, ignore on read.
1	REMOTE_ LOOPBACK	R/W	0h	Enable PMA remote loopback mode.
0	LOCAL_ LOOPBACK	R/W	0h	Enable PMA loopback mode.

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7.2.1.2 PMA_STATUS1 register

Table 72. PMA_STATUS1 register - PMA status 1 register (address 1h)

Bit	Symbol	Access	Value	Description
15:10	*reserved*	R	0h	Ignore on read.
9	PIASA	R	0h	PMA ingress AUI stop ability
8	PEASA	R	0h	PMA egress AUI stop ability
7	FAULT	R	0h	If set, fault condition detected
6:3	*reserved*	R	0h	Ignore on read.
2	RECEIVE_LINK_STATUS	R	0h	If set, PMA/PMD receive link up. This bit is latched low. This bit is latched-low.
1	LOW_POWER_ABILITY	R	1h	If set, PCS supports low-power mode
0	*reserved*	R	0h	Ignore on read.

7.2.1.3 PMA_SPEED_ABILITY register

Table 73. PMA_SPEED_ABILITY register - PMA speed ability register (address 4h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	If set, PMA/PMD is capable of operating at 400 Gbit/s Ignore on read.
14	*reserved*	R	0h	If set, PMA/PMD is capable of operating at 5 Gbit/s Ignore on read.
13	*reserved*	R	0h	If set, PMA/PMD is capable of operating at 2.5 Gbit/s Ignore on read.
12	*reserved*	R	0h	If set, PMA/PMD is capable of operating at 200 Gbit/s Ignore on read.
11	*reserved*	R	0h	If set, PMA/PMD is capable of operating at 25 Gbit/s Ignore on read.
10	*reserved*	R	0h	If set, PMA/PMD is capable of operating as 10GPASS-XR Ignore on read.
9	*reserved*	R	0h	If set, PMA/PMD is capable of operating at 100 Gbit/s Ignore on read.
8	*reserved*	R	0h	If set, PMA/PMD is capable of operating at 40 Gbit/s Ignore on read.
7	*reserved*	R	0h	If set, PMA/PMD is capable of operating at 10 Gbit/s downstream and 1 Gbit/s upstream Ignore on read.
6	*reserved*	R	0h	If set, PMA/PMD is capable of operating at 10 Mbit/s Ignore on read.
5	ABILITY_100M	R	1h	If set, PMA/PMD is capable of operating at 100 Mbit/s
4	*reserved*	R	0h	If set, PMA/PMD is capable of operating at 1000 Mbit/s Ignore on read.
3	*reserved*	R	0h	Ignore on read.

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Table 73. PMA_SPEED_ABILITY register - PMA speed ability register (address 4h)*...continued*

Bit	Symbol	Access	Value	Description
2	*reserved*	R	0h	If set, PMA/PMD is capable of operating as 10PASS-TS Ignore on read.
1	*reserved*	R	0h	If set, PMA/PMD is capable of operating as 2BASE-TL Ignore on read.
0	*reserved*	R	0h	If set, PMA/PMD is capable of operating at 10 Gbit/s Ignore on read.

7.2.1.4 PMA_CONTROL2 register

Table 74. PMA_CONTROL2 register - PMA control 2 register (address 7h)

Bit	Symbol	Access	Value	Description
15:7	*reserved*	R	0h	Ignore on read.
6:0	TYPE_SELECT	R	3Dh	Type selection others = Reserved, ignore on read 3Dh = BASE-T1 PMA/PMD

7.2.1.5 PMA_STATUS2 register

Table 75. PMA_STATUS2 register - PMA status 2 register (address 8h)

Bit	Symbol	Access	Value	Description
15:14	DEVICE_PRESENT	R	2h	Device present 10 = Device responding at this address x1 = No device responding at this address 0x = No device responding at this address
13	TRANSMIT_FAULT_ABILITY	R	0h	If set, PMA/PMD has the ability to detect a fault condition on the transmit path
12	RECEIVE_FAULT_ABILITY	R	1h	If set, PMA/PMD has the ability to detect a fault condition on the receive path
11	*reserved*	R	0h	Ignore on read.
10	RECEIVE_FAULT	R	0h	If set, fault condition on receive path This flag is cleared on read. This bit is latched-high.
9	EXTENDED_ABILITIES	R	1h	If set, PMA/PMD has extended abilities listed in PMA_EXTENDED_ABILITIES
8	PMD_TRANSMIT_DISABLE	R	1h	If set, PMD has the ability to disable the transmit path
7:1	*reserved*	R	0h	Ignore on read.
0	PMA_LOCAL_LOOPBACK_ABILITY	R	1h	If set, PMA has the ability to perform a local loopback function

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7.2.1.6 PMD_TRANSMIT_DISABLE register

Table 76. PMD_TRANSMIT_DISABLE register - PMD transmit disable register (address 9h)

Bit	Symbol	Access	Value	Description
15:1	*reserved*	R	0h	Always write 0, ignore on read.
0	TX_DISABLE	R/W	0h	Transmit disable This setting is test protected.

7.2.1.7 PMD_RCV_DETECT register

Table 77. PMD_RCV_DETECT register - PMD receive signal detect register (address Ah)

Bit	Symbol	Access	Value	Description
15:1	*reserved*	R	0h	Ignore on read.
0	RCV_SIGNAL_ DETECT	R	0h	If set, signal detected on receive

7.2.1.8 PMA_EXTENDED_ABILITIES register

Table 78. PMA_EXTENDED_ABILITIES register - PMA extended abilities (address Bh)

Bit	Symbol	Access	Value	Description
15	XTD_ABILITIES_ BASE_H	R	0h	If set, PMA/PMD has BASE-H extended abilities
14	XTD_ABILITIES_2 G5_5G	R	0h	If set, PMA/PMD has 2.5G/5G extended abilities
13	XTD_ABILITIES_ 200G_400G	R	0h	If set, PMA/PMD has 200G/400G extended abilities
12	XTD_ABILITIES_25 G	R	0h	f set, PMA/PMD has 25G extended abilities
11	XTD_ABILITIES_ BASE_T1	R	1h	If set, PMA/PMD has BASE-T1 extended abilities
10	XTD_ABILITIES_40 G_100G	R	0h	If set, PMA/PMD has 40G/100G extended abilities
9	ABILITY_P2MP	R	0h	If set, PMA/PMD has P2MP abilities
8	ABILITY_10BASE_T	R	0h	If set, PMA/PMD is able to perform 10BASE-T
7	ABILITY_100BASE_ TX	R	0h	If set, PMA/PMD is able to perform 100BASE-TX
6	ABILITY_1000 BASE_KX	R	0h	If set, PMA/PMD is able to perform 1000BASE-KX
5	ABILITY_1000 BASE_T	R	0h	If set, PMA/PMD is able to perform 1000BASE-T
4	ABILITY_10 GBASE_KR	R	0h	If set, PMA/PMD is able to perform 10GBASE-KR
3	ABILITY_10 GBASE_KX4	R	0h	If set, PMA/PMD is able to perform 10GBASE-KX4
2	ABILITY_10 GBASE_T	R	0h	If set, PMA/PMD is able to perform 10GBASE-T

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Table 78. PMA_EXTENDED_ABILITIES register - PMA extended abilities (address Bh)*...continued*

Bit	Symbol	Access	Value	Description
1	ABILITY_10 GBASE_LRM	R	0h	If set, PMA/PMD is able to perform 10GBASE-LRM
0	ABILITY_10 GBASE_CX4	R	0h	If set, PMA/PMD is able to perform 10GBASE-CX4

7.2.1.9 BASE_T1_PMA_XTD_ABILITY register

Table 79. BASE_T1_PMA_XTD_ABILITY register - BASE-T1 PMA extended abilities register (address 12h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Ignore on read.
1	ABILITY_1000 BASE_T1	R	0h	If set, PMA/PMD is able to perform 1000BASE-T1
0	ABILITY_100BASE_ T1	R	1h	If set, PMA/PMD is able to perform 100BASE-T1

7.2.1.10 BASE_T1_PMA_CONTROL register

Table 80. BASE_T1_PMA_CONTROL register - BASE-T1 PMA control register (address 834h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15	MANUAL_ MASTER_SLAVE_ CONFIG	R	1h	Returns a one to indicate that master or slave configuration is set manually. This value is always 1 This setting is configuration protected.
14	MASTER_SLAVE	R/W	0h	Master slave selection This setting is preserved during deep sleep. This setting is configuration protected. 0 = SLAVE 1 = MASTER
13:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	MODE	R	0h	only 100BASE-T1 is supported This setting is configuration protected. 0000* = 100BASE-T1 0001 = 1000BASE-T1 others = Reserved

7.2.1.11 E100BT1_PMA_TEST_CONTROL register

Table 81. E100BT1_PMA_TEST_CONTROL register - 100BASE-T1 PMA test control register (address 836h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15:13	PMA_TEST_MODE	R/W	0h	Selects the PMA test mode. Ignore PHY_STATUS if a test mode is enabled. This setting is configuration protected. 000* = Normal operation 001 = Test mode 1

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Table 81. E100BT1_PMA_TEST_CONTROL register - 100BASE-T1 PMA test control register (address 836h)*...continued*

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
				010 = Test mode 2 100 = Test mode 4 101 = Test mode 5 111 = Drive SEND_Z others = Reserved
12:0	*reserved*	R	0h	Always write 0, ignore on read.

7.3 MMD3

7.3.1 Register Description

Table 82. Register overview: MMD3

Address	Name	Access	Reset	Description
0h	PCS_CONTROL1	R/W	2000h	PCS control 1 register
1h	PCS_STATUS1	R/W	2h	PCS status 1 register
4h	PCS_SPEED_ABILITY	R/W	0h	PCS speed ability register
8h	PCS_STATUS2	R/W	8000h	PCS status 2 register

7.3.1.1 PCS_CONTROL1 register

Table 83. PCS_CONTROL1 register - PCS control 1 register (address 0h)

Bit	Symbol	Access	Value	Description
15	RESET	R/W	0h	Reset This bit is self-clearing. 0* = Normal operation 1 = On write a PHY reset is asserted. On read this indicates that a reset is still in progress ***Note:*** *In slave mode with polarity detection and correction enabled, use PMA reset instead.*
14	LOOPBACK	R/W	0h	If set, Enable loopback mode Note, PCS loopback will overrule all PMA loopback modes
13	SPEED_SELECT_LSB	R	1h	see SPEED_SELECT_MSB
12	*reserved*	R	0h	Always write 0, ignore on read.
11	LOW_POWER	R/W	0h	If set, the PCS enters low power mode ***Note:*** *PMA LOW_POWER (MMD1) and PCS LOW_POWER (MMD3) controls have mirrored behaviour. If one is written the other follows.*
10	*reserved*	R	0h	Always write 0, ignore on read.
9:7	*reserved*	R	0h	Always write 0, ignore on read.
6	SPEED_SELECT_MSB	R	0h	Speed selection [SPEED_SELECT_MSB, SPEED_SELECT_LSB] are encoded as follows:

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Table 83. PCS_CONTROL1 register - PCS control 1 register (address 0h)*...continued*

Bit	Symbol	Access	Value	Description
				11 = bits 5:2 select speed 10 = 1000 Mb/s 01 = 100 Mb/s 00 = 10 Mb/s
5:2	*reserved*	R	0h	Extended speed selection Always write 0, ignore on read.
1:0	*reserved*	R	0h	Always write 0, ignore on read.

7.3.1.2 PCS_STATUS1 register

Table 84. PCS_STATUS1 register - PCS status 1 register (address 1h)

Bit	Symbol	Access	Value	Description
15:8	*reserved*	R	0h	Ignore on read.
7	FAULT_CONDITION	R	0h	If set, fault condition detected
6:3	*reserved*	R	0h	Ignore on read.
2	RCV_LINK_ STATUS_LL	R	0h	If set, PCS receive link up. Latched low This bit is latched-low.
1	LOW_POWER_ ABILITY	R	1h	If set, PCS supports low-power mode
0	*reserved*	R	0h	Ignore on read.

7.3.1.3 PCS_SPEED_ABILITY register

Table 85. PCS_SPEED_ABILITY register - PCS speed ability register (address 4h)

Bit	Symbol	Access	Value	Description
15:10	*reserved*	R	0h	Ignore on read.
9	ABILITY_400G	R	0h	If set, PCS is capable of operating at 400 Gbit/s
8	ABILITY_200G	R	0h	If set, PCS is capable of operating at 200 Gbit/s
7	ABILITY_5G	R	0h	If set, PCS is capable of operating at 5 Gbit/s
6	ABILITY_2500M	R	0h	If set, PCS is capable of operating at 2.5 Gbit/s
5	*reserved*	R	0h	Ignore on read.
4	ABILITY_25G	R	0h	If set, PCS is capable of operating at 25 Gbit/s
3	ABILITY_100G	R	0h	If set, PCS is capable of operating at 100 Gbit/s
2	ABILITY_40G	R	0h	If set, PCS is capable of operating at 40 Gbit/s
1	ABILITY_10P_2B	R	0h	If set, PCS is capable of operating as the 10P/2B PCS
0	ABILITY_10G	R	0h	If set, PCS is capable of operating at 10 Gbit/s

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7.3.1.4 PCS_STATUS2 register

Table 86. PCS_STATUS2 register - PCS status 2 register (address 8h)

Bit	Symbol	Access	Value	Description
15:14	DEVICE_PRESENT	R	2h	Device present 10 = Device responding at this address x1 = No device responding at this address 0x = No device responding at this address
13:12	*reserved*	R	0h	Ignore on read.
11	TRANSMIT_FAULT	R	0h	If set, PCS transmit fault conditions detected. Latched high • TX_ER detected HIGH while TX_EN was set HIGH This flag is cleared on read. This bit is latched-high.
10	RECEIVE_FAULT	R	0h	If set, PCS receive fault conditions detected. Latched high • RX_ER was high during RX_DV HIGH period • SSD/ESD error detection • Jab_detect asserted • Scrambler lock drop This flag is cleared on read. This bit is latched-high.
9:0	*reserved*	R	0h	Ignore on read.

7.4 MMD30

7.4.1 Register Description

Table 87. Register overview: MMD30

Address	Name	Access	Reset	Description
4h	DEVICE_IDENTIFIER3	R/W	3401h	Device identifier 3 register
8h	MMD30_STATUS	R/W	8000h	MMD30 status register
40h	DEVICE_CONTROL	R/W	0h	Device control register
42h	DEVICE_STATUS	R/W	0h	Device status register
44h	DEVICE_STATUS_ LATCHED	R/W	0h	Device status latched register
46h	DEVICE_ABILITIES	R/W	A/B: 0868h C/D: 0808h	Device abilities register
47h	DEVICE_PROPERTIES	R/W	1h	Device properties register
48h	DEVICE_CONFIG	R/W	8000h	Device config register
49h	DEVICE_CONFIG_ EXTENDED	R/W	0h	Device configuration extended register
70h	PORT_IRQ_STATUS	R/W	0h	PHY IRQ status register
72h	PORT_IRQ_ENABLES	R/W	3h	PORT IRQ enables register
74h	PORT_IRQ_MSTATUS	R/W	0h	Port masked IRQ status register

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Table 87. Register overview: MMD30*...continued*

Address	Name	Access	Reset	Description
80h	GLOBAL_CAT_IRQ_ STATUS	R/W	0h	Global category IRQ status register
81h	GLOBAL_CAT_IRQ_ ENABLES	R/W	134h	Global category IRQ enable register
82h	GLOBAL_CAT_IRQ_ MSTATUS	R/W	0h	GLobal category IRQ masked status
90h	OP_CONDITIONS_IRQ_ SOURCES	R/W	0h	Operating conditions IRQ sources
91h	OP_CONDITIONS_IRQ_ ENABLES	R/W	0h	Operating conditions IRQ enables
92h	OP_CONDITIONS_IRQ_ MSTATUS	R/W	0h	Operating conditions IRQ masked status
A0h	TOP_EPHY_IRQ_SOURCES	R/W	0h	Top Ethernet PHY IRQ sources
A1h	TOP_EPHY_IRQ_ENABLES	R/W	0h	Top Ethernet PHY IRQ enables
A2h	TOP_EPHY_IRQ_MSTATUS	R/W	0h	Top Ethernet PHY IRQ masked status
B0h	FUNC_SHARED_IRQ_ SOURCES	R/W	0h	Global shared function IRQ sources
B1h	FUNC_SHARED_IRQ_ ENABLES	R/W	0h	Global shared function IRQ enables
B2h	FUNC_SHARED_IRQ_ MSTATUS	R/W	0h	Global shared function IRQ masked status
C0h	CONFIG_REGXS_IRQ_ SOURCES	R/W	0h	Configuration register access IRQ sources
C1h	CONFIG_REGXS_IRQ_ ENABLES	R/W	0h	Configuration register access IRQ enable
C2h	CONFIG_REGXS_IRQ_ MSTATUS	R/W	0h	Configuration register access IRQ masked status
100h	BROADCAST	R/W	0h	Broadcast control
101h	PORT1_CONFIG	R/W	C001h	Port 1 configuration
180h	WISE_CONTROL	R/W	1h	Wake, inhibit and sleep control
181h	WISE_STATUS	R/W	0h	Wake, inhibit and sleep status
182h	WISE_CONFIG	R/W	3000h	Wake, inhibit and sleep configuration
184h	WISE_PARAMETERS	R/W	1h	Wake, inhibit and sleep parameters
189h	SMI_STATUS	R/W	0h	SMI status
18Ah	SMI_CONFIG	R/W	8000h	SMI config
18Ch	SMI_DATA_IO_CONFIG	R/W	0h	SMI Data IO configuration
190h	GLOBAL_LED_TRIGGER0	R/W	0h	LED trigger 0
191h	GLOBAL_LED_TRIGGER1	R/W	0h	LED trigger 1
192h	GLOBAL_LED_TRIGGER2	R/W	0h	LED trigger 2
193h	GLOBAL_LED_TRIGGER3	R/W	0h	LED trigger 3

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Table 87. Register overview: MMD30*...continued*

Address	Name	Access	Reset	Description
1A0h	LED0_CONFIG	R/W	C00Ch	LED Controller 0 Configuration
1A1h	LED0_TRIG_SOURCE	R/W	21h	LED controller 0 trigger source selection
1A2h	LED0_TRIG01_CONFIG	R/W	E0E0h	LED controller 0 Trigger 0 and 1 configuration
1A3h	LED0_TRIG23_CONFIG	R/W	00E0h	LED controller 0 Trigger 2 and 3 configuration
1A4h	LED1_CONFIG	R/W	C00Ch	LED Controller 1 Configuration
1A5h	LED1_TRIG_SOURCE	R/W	21h	LED controller 1 trigger source selection
1A6h	LED1_TRIG01_CONFIG	R/W	E0E0h	LED controller 1 Trigger 0 and 1 configuration
1A7h	LED1_TRIG23_CONFIG	R/W	00E0h	LED controller 1 Trigger 2 and 3 configuration
1A8h	LED2_CONFIG	R/W	C00Ch	LED Controller 2 Configuration
1A9h	LED2_TRIG_SOURCE	R/W	21h	LED controller 2 trigger source selection
1AAh	LED2_TRIG01_CONFIG	R/W	E0E0h	LED controller 2 Trigger 0 and 1 configuration
1ABh	LED2_TRIG23_CONFIG	R/W	00E0h	LED controller 2 Trigger 2 and 3 configuration
1ACh	LED3_CONFIG	R/W	C00Ch	LED Controller 3 Configuration
1ADh	LED3_TRIG_SOURCE	R/W	21h	LED controller 3 trigger source selection
1AEh	LED3_TRIG01_CONFIG	R/W	E0E0h	LED controller 3 Trigger 0 and 1 configuration
1AFh	LED3_TRIG23_CONFIG	R/W	00E0h	LED controller 3 Trigger 2 and 3 configuration
311h	AO_SYSTEM_SUPPLY_STATUS	R/W	0h	Always on system supply
313h	CORE_SUPPLY_STATUS	R/W	0h	Core supply status
315h	VDDIO_SUPPLY_STATUS	R/W	0h	VDDIO supply status
318h	VREGD_SUPPLY_STATUS	R/W	0h	VREGD supply status
319h	VREGA_SUPPLY_STATUS	R/W	0h	VREGA supply status
31Ah	VDDA_TRX_SUPPLY	R/W	0h	VDDA_TRX supply status
31Eh	OSC_STATUS	R/W	0h	XO status
31Fh	TEMP_STATUS	R/W	A1h	Temperature status
3F4h	COMPL_TEST_INTF	R/W	100h	Compliance test interface
1100h	PTP_IDENTIFIER	R/W	10h	PTP IP identifier register
1101h	PTP_CONTROL	R/W	0h	General control
1102h	PTP_CONFIG	R/W	0h	General control
1103h	PTP_ABILITY	R/W	6h	PTP IP Ability registers
1104h	PTP_CLK_PERIOD	R/W	8h	Clock period of ptp_clk

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Table 87. Register overview: MMD30*...continued*

Address	Name	Access	Reset	Description
1105h	LTC_LOAD_CTRL	R/W	0h	LTC load and read control
1106h	LTC_WR_NSEC_0	R/W	0h	Loading nanoseconds counter value
1107h	LTC_WR_NSEC_1	R/W	0h	Loading nanoseconds counter value
1108h	LTC_WR_SEC_0	R/W	0h	Loading seconds counter value
1109h	LTC_WR_SEC_1	R/W	0h	Loading seconds counter value
110Ah	LTC_RD_DATA_0	R/W	0h	Local LTC nanoseconds part. The value read is the saved value of local counter when READ_LTC is set.
110Bh	LTC_RD_DATA_1	R/W	0h	Local LTC nanoseconds part. The value read is the saved value of local counter when READ_LTC is set.
110Ch	LTC_RD_DATA_2	R/W	0h	Local LTC seconds part. The value read is the saved value of local counter when READ_LTC is set.
110Dh	LTC_RD_DATA_3	R/W	0h	Local LTC seconds part. The value read is the saved value of local counter when READ_LTC is set.
110Eh	LTC_RD_DATA_4	R/W	0h	Local LTC sub-nanoseconds part. The value read is the saved value of local counter when READ_LTC is set.
110Fh	RATE_ADJ_SUBNS_0	R/W	0h	SW Rate adjust value
1110h	RATE_ADJ_SUBNS_1	R/W	0h	SW Rate adjust value
1111h	TEMP_ADJ_SUBNS_0	R/W	0h	SW temporary rate adjust value
1112h	TEMP_ADJ_SUBNS_1	R/W	0h	SW temporary rate adjust value
1113h	TEMP_ADJ_DUR_0	R/W	0h	Temporary adjustment window
1114h	TEMP_ADJ_DUR_1	R/W	0h	Temporary adjustment window
1115h	HW_LTC_LOCK_CTRL	R/W	1h	HW LTC self locking.control
1116h	HW_EXT_PPS_NS_0	R/W	CA00h	Period of external PPS sync in nanosecond. Used for HW LTC self locking.
1117h	HW_EXT_PPS_NS_1	R/W	3B9Ah	Period of external PPS sync in nanosecond. Used for HW LTC self locking.
1118h	HW_LOCK_ER_LMT_0	R/W	D090h	Error limitation by HW LTC self locking engine
1119h	HW_LOCK_ER_LMT_1	R/W	3h	Error limitation by HW LTC self locking engine
111Ah	LTC_LOOP_CONTROL	R/W	303h	This register defines control parameters to adapt behavior of the LTC loop if HW_LTC_ LOCK_EN=1
111Bh	EXT_PPS_TS_DATA0	R/W	0h	Timestamping data on the event of rising edge of external pps signal. Reading this register copied into snapshot register for multiple reads of EXT_PPS_TS_ DATA0/1/2/3/4/5 register

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Table 87. Register overview: MMD30*...continued*

Address	Name	Access	Reset	Description
111Ch	EXT_PPS_TS_DATA1	R/W	0h	Timestamping data on the event of rising edge of external pps signal.
111Dh	EXT_PPS_TS_DATA2	R/W	0h	Timestamping data on the event of rising edge of external pps signal.
111Eh	EXT_PPS_TS_DATA3	R/W	0h	Timestamping data on the event of rising edge of external pps signal.
111Fh	EXT_PPS_TS_DATA4	R/W	0h	Timestamping data on the event of rising edge of external pps signal.
1120h	EXT_PPS_TS_CTRL	R/W	0h	External PPS timestamp reading done
1121h	EXT_TRG_TS_DATA0	R/W	0h	Timestamping data on the event of rising edge of external trigger event signal. Reading this register copied into snapshot register for multiple reads of EXT_TRG_TS_DATA0/1/2/3/4/5 register
1122h	EXT_TRG_TS_DATA1	R/W	0h	Timestamping data of external trigger event signal.
1123h	EXT_TRG_TS_DATA2	R/W	0h	Timestamping data of external trigger event signal.
1124h	EXT_TRG_TS_DATA3	R/W	0h	Timestamping data of external trigger event signal.
1125h	EXT_TRG_TS_DATA4	R/W	0h	Timestamping data of external trigger event signal.
1126h	EXT_TRG_TS_CTRL	R/W	0h	External trigger event timestamp reading done
1130h	PTP_IRQ_SOURCES	R/W	0h	Interrupt status signals
1131h	PTP_IRQ_ENABLES	R/W	0h	Interrupt enable control
1132h	PTP_IRQ_MSTATUS	R/W	0h	Interrupt status signals
1140h	PKT_FILT_CTRL	R/W	0h	Ingress or egress packet filtering
1141h	DA_FILT_CTRL	R/W	3h	Ingress or egress packet DA filtering.
1142h	USER_MAC_DA_0	R/W	0h	[15:0] of User defined MAC address
1143h	USER_MAC_DA_1	R/W	0h	[31:16] of User defined MAC address
1144h	USER_MAC_DA_2	R/W	0h	[47:32] of User defined MAC address
1145h	USER_DA_MASK_0	R/W	0h	User MAC DA Mask control
1146h	USER_DA_MASK_1	R/W	0h	User MAC DA Mask control
1147h	USER_DA_MASK_2	R/W	0h	User MAC DA Mask control
1148h	EVENT_MSG_FILT	R/W	Fh	PTP event message consideration
1149h	TX_PIPE_DLY_NS	R/W	0h	Egress data path latency from timestamping point of SOF signal to mdi (nanoseconds part).
114Ah	TX_PIPEDLY_SUBNS	R/W	0h	Egress data path latency from timestamping point of SOF signal to mdi (sub-nanoseconds part)

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Table 87. Register overview: MMD30*...continued*

Address	Name	Access	Reset	Description
114Bh	RX_PIPE_DLY_NS	R/W	0h	Ingress data path latency of SOF signal from MDI line to timestamping point (nanoseconds part).
114Ch	RX_PIPEDLY_SUBNS	R/W	0h	Ingress data path latency of SOF signal from MDI line to timestamping point (sub-nanoseconds part)
114Dh	RX_TS_INSRT_CTRL	R/W	11h	Inband timestamp insertion controls at ingress path
114Eh	EGR_RING_DATA_0	R/W	0h	Timestamping data of egress ptp event message. Reading this register fetches and latches buffered timestamp data set and puts into EGR_RING_DATA_0/1/2/3/4/5
114Fh	EGR_RING_DATA_1	R/W	0h	Timestamping data of egress ptp event message.
1150h	EGR_RING_DATA_2	R/W	0h	Timestamping data of egress ptp event message.
1151h	EGR_RING_DATA_3	R/W	0h	Timestamping data of egress ptp event message.
1152h	EGR_RING_DATA_4	R/W	0h	Timestamping data of egress ptp event message.
1153h	EGR_RING_DATA_5	R/W	0h	Timestamping data of egress ptp event message.
1154h	EGR_RING_CTRL	R/W	0h	Egress timestamp termination control
1155h	ING_RING_DATA_0	R/W	0h	Timestamping data of egress ptp event message. Reading this register fetches and latches buffered timestamp data set and puts into ING_RING_DATA_0/1/2/3/4/5
1156h	ING_RING_DATA_1	R/W	0h	Timestamping data of Ingress ptp event message.
1157h	ING_RING_DATA_2	R/W	0h	Timestamping data of Ingress ptp event message.
1158h	ING_RING_DATA_3	R/W	0h	Timestamping data of Ingress ptp event message.
1159h	ING_RING_DATA_4	R/W	0h	Timestamping data of Ingress ptp event message.
115Ah	ING_RING_DATA_5	R/W	0h	Timestamping data of Ingress ptp event message.
115Bh	ING_RING_CTRL	R/W	0h	Ingress timestamp termination control
115Fh	PTP_PKT_RCV_CNT	R/W	0h	Number of received ptp event packet
1160h	PTP_FCS_ERR_CNT	R/W	0h	Number of received ptp event packet has fcs error.
2C00h	GLOBAL_INFRA_CONTROL	R/W	0h	Global infrastructure control
2C20h	GPIO_OUTPUTS	R/W	0h	GPIO output
2C21h	GPIO_STATUS	R/W	0h	GPIO status

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Table 87. Register overview: MMD30*...continued*

Address	Name	Access	Reset	Description
2C22h	GPIO_IO_CONFIG	R/W	0h	GPIO IO configuration
2C23h	GPIO_IO_CLK_CONFIG	R/W	0h	GPIO IO CLK configuration
2C40h	GPIO0_FUNC_CONFIG	R/W	8020h	GPIO0 function configuration (GPIOx function registers are identical to this one)
2C41h	GPIO1_FUNC_CONFIG	R/W	8000h	see GPIO0_FUNC_CONFIG
2C42h	GPIO2_FUNC_CONFIG	R/W	8000h	see GPIO0_FUNC_CONFIG
2C43h	GPIO3_FUNC_CONFIG	R/W	8000h	see GPIO0_FUNC_CONFIG
2C44h	GPIO4_FUNC_CONFIG	R/W	8000h	see GPIO0_FUNC_CONFIG
2C45h	GPIO5_FUNC_CONFIG	R/W	8000h	see GPIO0_FUNC_CONFIG
2C46h	GPIO6_FUNC_CONFIG	R/W	8000h	see GPIO0_FUNC_CONFIG
2C47h	GPIO7_FUNC_CONFIG	R/W	8000h	see GPIO0_FUNC_CONFIG
2C48h	GPIO8_FUNC_CONFIG	R/W	8000h	see GPIO0_FUNC_CONFIG
2C49h	GPIO9_FUNC_CONFIG	R/W	8000h	see GPIO0_FUNC_CONFIG
2C4Ah	GPIO10_FUNC_CONFIG	R/W	8000h	see GPIO0_FUNC_CONFIG
2C4Bh	GPIO11_FUNC_CONFIG	R/W	8000h	see GPIO0_FUNC_CONFIG
9000h	PORT_PTP_CONTROL	R/W	800h	Port PTP control
9200h	PTP_TS_CAPABILITY	R/W	3h	PTP TS capabilities
9201h	PTP_TS_TX_MAX_DELAY_LSB	R/W	0h	Maximum PTP pcs transmit path data delay LSBs
9202h	PTP_TS_TX_MAX_DELAY_MSB	R/W	0h	Maximum PTP pcs transmit path data delay MSBs
9203h	PTP_TS_TX_MIN_DELAY_LSB	R/W	0h	Minimum PTP pcs transmit path data delay MSBs
9204h	PTP_TS_TX_MIN_DELAY_MSB	R/W	0h	Minimum PTP pcs transmit path data delay MSBs
9205h	PTP_TS_RX_MAX_DELAY_LSB	R/W	0h	Maximum PTP pcs receive path data delay LSBs
9206h	PTP_TS_RX_MAX_DELAY_MSB	R/W	0h	Maximum PTP pcs receive path data delay MSBs
9207h	PTP_TS_RX_MIN_DELAY_LSB	R/W	0h	Minimum PTP pcs receive path data delay MSBs
9208h	PTP_TS_RX_MIN_DELAY_MSB	R/W	0h	Minimum PTP pcs receive path data delay MSBs
A800h	PORT_BIST_CONTROL	R/W	0h	Port BIST control
A807h	BIST_INTERCEPT_CONFIG	R/W	0h	BIST intercept config
A880h	BIST_GEN_CTRL	R/W	0h	BIST generator control register
A881h	BIST_GEN_STATUS	R/W	0h	BIST generator status
A887h	PREAMBLE_IPG_SIZE	R/W	700Ch	Preamble and IPG lengths
A888h	BIST_DA_0	R/W	0h	lower 16 bits of DA value[15:0]

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Table 87. Register overview: MMD30*...continued*

Address	Name	Access	Reset	Description
A889h	BIST_DA_1	R/W	0h	middle 16 bits of DA value[31:16]
A88Ah	BIST_DA_2	R/W	0h	higher 16 bits of DA value[47:32]
A88Bh	BIST_SA_0	R/W	0h	lower 16 bits of SA value[15:0]
A88Ch	BIST_SA_1	R/W	0h	middle 16 bits of SA value[31:16]
A88Dh	BIST_SA_2	R/W	0h	higher 16 bits of SA value[47:32]
A88Fh	BIST_PTP_CONFIG	R/W	0h	PTP frame generation configuration
A890h	BIST_ETHER_TYPE	R/W	0h	Ethernet type (T/L)
A891h	PAYLOAD_CONFIG	R/W	20A5h	Payload data controls
A892h	PAYLOAD_SIZE	R/W	2Eh	Length of frame payload data in bytes
A893h	PRBS_DATA_CONFIG	R/W	0h	PRBS data configuration
A894h	BIST_LFSR_SEED	R/W	7FFFh	LFSR Seed value
A8A2h	GOOD_FRAMES_PLAN	R/W	64h	Number of good frames to be generated by BIST generator
A8A4h	G_GOOD_FRAME_CNT	R/W	0h	Generated good frames in continuous mode.
A8A6h	BAD_FRAMES_PLAN	R/W	0h	Number of bad frames to be generated by BIST generator
A8C0h	BIST_CHECK_CTRL	R/W	0h	BIST checker control register
A8C1h	BIST_PROD_STATUS	R/W	0h	BIST status used when BIST is in production mode and NOT valid for continuous mode
A8C4h	BIST_WAIT_TIMER	R/W	FFh	Timer timeout value for checking all status in production mode. This timeout is useful when expected number of generated frames is never received.
A8D0h	R_GOOD_FRAME_CNT	R/W	0h	Received good frame count
A8D2h	R_BAD_FRAME_CNT	R/W	0h	Received bad frame count
A8D4h	R_RXER_FRAME_CNT	R/W	0h	RX_ER detected when rx_dv is set.
AC00h	PORT_INFRA_CONTROL	R/W	0h	Port infrastructure control
AC04h	INFRA_ABILITY	R/W	8000h	Infrastsructure ability
AC06h	INFRA_CONFIG	R/W	0h	Infrastructure configuration
AC08h	INFRA_IRQ_SOURCES	R/W	0h	Infrastructure IRQ sources
AC0Ah	INFRA_IRQ_ENABLES	R/W	0h	Infrastructure IRQ enables
AC0Ch	INFRA_IRQ_MSTATUS	R/W	0h	Infrastructure IRQ masked status
AFC0h	XMII_CONTROL	R/W	0h	XMII control
AFC2h	XMII_STATUS	R/W	0h	XMII status
AFC4h	XMII_ABILITIES	R/W	A/C: CF00h B/D: 1h	XMII abilities
AFC5h	SERDES_MII_ABILITIES	R/W	A/C: 0h B/D: 2h	SERDES MII abilities
AFC6h	MII_BASIC_CONFIG	R/W	0h	MII basic configuration

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Table 87. Register overview: MMD30*...continued*

Address	Name	Access	Reset	Description
AFC8h	XMII_CLK_CONFIG	R/W	0h	XMII clock configuration
AFC9h	XMII_CLK_IO_CONFIG	R/W	0h	XMII CLK IO configuration
AFCAh	XMII_DATA_CONFIG	R/W	0h	XMII data configuration
AFCBh	XMII_DATA_IO_CONFIG	R/W	0h	XMII Data IO configuration
AFCCh	RGMII_TXC_DELAY_ CONFIG	R/W	12h	RGMII TXC delay configuration
AFCDh	RGMII_RXC_DELAY_ CONFIG	R/W	12h	RGMII RXC delay configuration
AFCEh	RX_PREAMBLE_COUNT	R/W	0h	RX preamble counter
AFCFh	TX_PREAMBLE_COUNT	R/W	0h	TX preamble counter
AFD0h	RX_IPG_LENGTH	R/W	0h	RX IPG length capture
AFD1h	TX_IPG_LENGTH	R/W	0h	TX IPG length capture
B000h	SGMII_BASIC_CONTROL	R/W	3000h	SGMII PCS register: SR MII MMD Control Register.
B001h	SGMII_BASIC_STATUS	R/W	129h	SGMII PCS register: SR MII MMD Status Register.
B004h	SGMII_AUTONEG_ ADVERTISE	R/W	20h	SGMII PCS register: SR MII MMD AN Advertisement Register.
B005h	SGMII_AUTONEG_LP_ ABILITY	R/W	0h	SGMII PCS register: SR MII MMD AN Link Partner Base Ability Register.
B006h	SGMII_AUTONEG_ EXPANSION	R/W	0h	SGMII PCS register: SR MII MMD AN Expansion Register.
B00Fh	SGMII_EXTENDED_ STATUS	R/W	C000h	SGMII PCS register: SR MII MMD Extended Status Register.
B010h	SGMII_ADVANCED_ CONTROL	R/W	0h	Advanced control
B011h	SGMII_ADVANCED_ STATUS	R/W	0h	Advanced status
B013h	SGMII_ADVANCED_ CONFIG	R/W	0h	Advanced configuration
B014h	SGMII_IRQ_SOURCES	R/W	0h	SGMII interrupt source
B015h	SGMII_IRQ_ENABLES	R/W	4h	SGMII interrupt enables
B016h	SGMII_IRQ_MSTATUS	R/W	0h	SGMII interrupt mask status. This is AND of IRQ_SOURCE and IRQ_ENABLE.
B040h	SGMII_LED_TRIGGER0	R/W	0h	SGMII LED trigger 0
B041h	SGMII_LED_TRIGGER1	R/W	0h	see SGMII_LED_TRIGGER0
B042h	SGMII_LED_TRIGGER2	R/W	0h	see SGMII_LED_TRIGGER0
B043h	SGMII_LED_TRIGGER3	R/W	0h	see SGMII_LED_TRIGGER0
B070	SGMII_PCS_CONTROL	R/W	0h	see SGMII_PCS_CONTROL
B073	SGMII_PCS_CONFIG1	R/W	0h	see SGMII_PCS_CONFIG1

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Table 87. Register overview: MMD30*...continued*

Address	Name	Access	Reset	Description
B074	SGMII_PCS_CONFIG2	R/W	0h	see SGMII_PCS_CONFIG2
B075	SGMII_AUTONEG_STATUS	R/W	0h	see SGMII_AUTONEG_STATUS
B0A1	SGMII_LINK_DROP_ COUNTER	R/W	0h	SGMII link drop counter register

7.4.1.1 DEVICE_IDENTIFIER3 register

Table 88. DEVICE_IDENTIFIER3 register - Device identifier 3 register (address 4h)

Bit	Symbol	Access	Value	Description
15:12	SILICON_VERSION	R	3h	Silicon version
11:8	SAMPLE_VERSION	R	4h	R - Released version
7:0	NVM_VERSION	R	1h	NVM version

7.4.1.2 MMD30_STATUS register

Table 89. MMD30_STATUS register - MMD30 status register (address 8h)

Bit	Symbol	Access	Value	Description
15:14	DEVICE_PRESENT	R	2h	Device present 10 = Device responding at this address x1 = No device responding at this address 0x = No device responding at this address
13:0	*reserved*	R	0h	Ignore on read.

7.4.1.3 DEVICE_CONTROL register

Table 90. DEVICE_CONTROL register - Device control register (address 40h)

Bit	Symbol	Access	Value	Description
15	DEVICE_RESET	R/W	0h	Hard device reset. The entire device is reset all settings are lost and the device initializes and executes pin strapping. The behavior is similar to asserting the reset pin. This bit is self-clearing. 0* = Normal operation 1 = On write a PHY reset is asserted. On read this indicates that a reset is still in progress
14	GLOBAL_CONFIG_ ENABLE	R/W	0h	If set, the device global configuration registers are unlocked for reconfiguration.
13	SUPER_CONFIG_ ENABLE	R/W	0h	If set, all configuration registers and bits can be changed for any PHY and the Global sub-system. This bit overrules all respective port CONFIG_ENABLEs and GLOBAL_CONFIG_ENABLE.
12:2	*reserved*	R	0h	Always write 0, ignore on read.
1	GOTO_STANDBY	R/W	0h	If set, the device is directed to go to STANDBY. This bit is self clearing. On read it returns 1, if the device is still in transition. This bit is self-clearing.

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Table 90. DEVICE_CONTROL register - Device control register (address 40h)*...continued*

Bit	Symbol	Access	Value	Description
0	START_ OPERATION	R/W	0h	If set, exit standby state and start device operation. This bit is self clearing. On read it returns 1, if the device is still in transition. This bit is self-clearing.

7.4.1.4 DEVICE_STATUS register

Table 91. DEVICE_STATUS register - Device status register (address 42h)

Bit	Symbol	Access	Value	Description
15	FUSA_STATUS	R	0h	If set, the FUSA startup test has passed successfully
14:0	*reserved*	R	0h	Ignore on read.

7.4.1.5 DEVICE_STATUS_LATCHED register

Table 92. DEVICE_STATUS_LATCHED register - Device status latched register (address 44h)

Bit	Symbol	Access	Value	Description
15	PIN_RESET	R/W	0h	If set, the device was reset via the RST_N pin. Write one to clear. This bit is latched-high.
14:0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.6 DEVICE_ABILITIES register

Table 93. DEVICE_ABILITIES register - (address 46h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Ignore on read.
11	BIST_ABILITY	R	1h	If set, BIST function is available
10:7	*reserved*	R	0h	Ignore on read.
6	MAC_ADAPTR_ ABILITY	R	A/B: 1h C/D: 0h	If set, MAC adapter function is available. Note, reset value depends on variant.
5	MACSEC_ABILITY	R	A/B: 1h C/D: 0h	If set, MACsec function is available. Note, reset value depends on variant.
4	*reserved*	R	0h	Ignore on read.
3	PTP_ABILITY	R	1h	If set, PTP function is available
2:0	*reserved*	R	0h	Ignore on read.

7.4.1.7 DEVICE_PROPERTIES register

Table 94. DEVICE_PROPERTIES register - Device properties register (address 47h)

Bit	Symbol	Access	Value	Description
15:5	*reserved*	R	0h	Ignore on read.
4:0	NUMBER_OF_ PORTS	R	1h	Binary value represents number of ports of this device.

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7.4.1.8 DEVICE_CONFIG register

Table 95. DEVICE_CONFIG register - Device config register (address 48h)

Bit	Symbol	Access	Value	Description
15	DEVICE_ENABLE	R/W	1h	If deasserted, the device is disabled. Re-asserting this bit triggers a soft reset of all subsystem control logic. The pin strap process is not restarted, previously latched settings are reapplied.
14:2	*reserved*	R	0h	Always write 0, ignore on read.
1	SUPER_TEST_ ENABLE	R/W	0h	By default, entering test modes is restricted and guarded by respective TEST_ENABLE bits. This bit enables all test modes of the entire device and all ports. Note, the TEST_ENABLEs through the hierarchy (super, port, port sub-functions) are ORed. This setting is configuration protected.
0	TEST_ENABLE	R/W	0h	By default, entering test modes is restricted and guarded by respective TEST_ENABLE bits. This bit enables test modes on device global level only, port test modes remain restricted. This setting is configuration protected.

7.4.1.9 DEVICE_CONFIG_EXTENDED register

Table 96. DEVICE_CONFIG_EXTENDED register - Device configuration extended register (address 49h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14	OP_IRQ_ RECOVERY	R/W	0h	By default operating condition interrupts are only asserted on occurrence of a fault condition. Optionally the recovery from fault conditions can be flagged by the same interrupt bits if this bit is set to one.
13:0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.10 PORT_IRQ_STATUS register

Table 97. PORT_IRQ_STATUS register - PHY IRQ status register (address 70h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Ignore on read.
1	PORT1_IRQ	R	0h	Status bit indicating a non-masked interrupt from PortNr 1.
0	GLOBAL_IRQ	R	0h	Status bit indicating a non-masked interrupt from global functions.

7.4.1.11 PORT_IRQ_ENABLES register

Table 98. PORT_IRQ_ENABLES register - PORT IRQ enables register (address 72h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Always write 0, ignore on read.
1	PORT1_IRQ	R/W	1h	Enable for interrupt from PortNr 1.

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Table 98. PORT_IRQ_ENABLES register - PORT IRQ enables register (address 72h)*...continued*

Bit	Symbol	Access	Value	Description
				This setting is preserved during deep sleep.
0	GLOBAL_IRQ	R/W	1h	Enable for interrupt from GLOBAL functions This setting is preserved during deep sleep.

7.4.1.12 PORT_IRQ_MSTATUS register

Table 99. PORT_IRQ_MSTATUS register - Port masked IRQ status register (address 74h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Ignore on read.
1	PORT1_IRQ	R	0h	Masked-status of PortNr 1 interrupt
0	GLOBAL_IRQ	R	0h	Masked-status of GLOBAL interrupt

7.4.1.13 GLOBAL_CAT_IRQ_STATUS register

Table 100. GLOBAL_CAT_IRQ_STATUS register - Global category IRQ status register (address 80h)

Bit	Symbol	Access	Value	Description
15:9	*reserved*	R	0h	Ignore on read.
8	CONFIG_REGXS_IRQ	R	0h	If set, interrupt(s) are pending in the global 'Config & Register Access' interrupts source register
7:6	*reserved*	R	0h	Ignore on read.
5	FUNC_SHARED_IRQ	R	0h	If set, interrupt(s) are pending in the global shared functional Interrupt status
4	PHY_FUNCTION_IRQ	R	0h	If set, interrupt(s) are pending in the global 'Function' interrupt masked-status register
3	*reserved*	R	0h	Ignore on read.
2	OP_CONDITIONS_IRQ	R	0h	If set, interrupt(s) are pending in the global 'Operating condition interrupts' source register
1:0	*reserved*	R	0h	Ignore on read.

7.4.1.14 GLOBAL_CAT_IRQ_ENABLES register

Table 101. GLOBAL_CAT_IRQ_ENABLES register - Global category IRQ enable register (address 81h)

Bit	Symbol	Access	Value	Description
15:9	*reserved*	R	0h	Always write 0, ignore on read.
8	CONFIG_REGXS_IRQ	R/W	1h	If set, the respective IRQ is enabled This setting is preserved during deep sleep.
7:6	*reserved*	R	0h	Always write 0, ignore on read.
5	FUNC_SHARED_IRQ	R/W	1h	If set, the respective IRQ is enabled This setting is preserved during deep sleep.
4	PHY_FUNCTION_IRQ	R/W	1h	If set, the respective IRQ is enabled This setting is preserved during deep sleep.
3	*reserved*	R	0h	Always write 0, ignore on read.

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Table 101. GLOBAL_CAT_IRQ_ENABLES register - Global category IRQ enable register (address 81h)*...continued*

Bit	Symbol	Access	Value	Description
2	OP_CONDITIONS_IRQ	R/W	1h	If set, the respective IRQ is enabled This setting is preserved during deep sleep.
1:0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.15 GLOBAL_CAT_IRQ_MSTATUS register

Table 102. GLOBAL_CAT_IRQ_MSTATUS register - GLobal category IRQ masked status (address 82h)

Bit	Symbol	Access	Value	Description
15:9	*reserved*	R	0h	Ignore on read.
8	CONFIG_REGXS_IRQ	R	0h	Masked-status for the respective IRQ
7:6	*reserved*	R	0h	Ignore on read.
5	FUNC_SHARED_IRQ	R	0h	Masked-status for the respective IRQ
4	PHY_FUNCTION_IRQ	R	0h	Masked-status for the respective IRQ
3	*reserved*	R	0h	Ignore on read.
2	OP_CONDITIONS_IRQ	R	0h	Masked-status for the respective IRQ
1:0	*reserved*	R	0h	Ignore on read.

7.4.1.16 OP_CONDITIONS_IRQ_SOURCES register

Table 103. OP_CONDITIONS_IRQ_SOURCES register - Operating conditions IRQ sources (address 90h)

Bit	Symbol	Access	Value	Description
15	TEMP_WARN	R/W	0h	If set, temperature has exceeded the limit. Write one to clear. This bit is latched-high.
14	OSC_WARN	R/W	0h	If set, a OSC clock warning is pending Write one to clear. This bit is latched-high.
13:11	*reserved*	R	0h	Always write 0, ignore on read.
10	VDDA_TRX_SUPPLY_WARN	R/W	0h	If set, VDDA_TRX supply voltage is not reliable Write one to clear. This bit is latched-high.
9	VREGA_SUPPLY_WARN	R/W	0h	If set, VREGA supply voltage is not reliable Write one to clear. This bit is latched-high.
8	VREGD_SUPPLY_WARN	R/W	0h	If set, VREGD supply voltage is not reliable Write one to clear. This bit is latched-high.
7:6	*reserved*	R	0h	Always write 0, ignore on read.

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Table 103. OP_CONDITIONS_IRQ_SOURCES register - Operating conditions IRQ sources (address 90h)*...continued*

Bit	Symbol	Access	Value	Description
5	VDDIO_SUPPLY_WARN	R/W	0h	If set, VDDIO supply voltage is not reliable Write one to clear. This bit is latched-high.
4	*reserved*	R	0h	Always write 0, ignore on read.
3	CORE_SUPPLY_WARN	R/W	0h	If set, core supply has crossed limits Write one to clear. This bit is latched-high.
2	*reserved*	R	0h	Always write 0, ignore on read.
1	AO_SYS_SUPPLY_WARN	R/W	0h	If set, always-on system supply has crossed limits. This bit is set at whenever a main always-on supply power cycle occurs. Write one to clear. This bit is latched-high.
0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.17 OP_CONDITIONS_IRQ_ENABLES register

Table 104. OP_CONDITIONS_IRQ_ENABLES register - Operating conditions IRQ enables (address 91h)

Bit	Symbol	Access	Value	Description
15	TEMP_WARN	R/W	0h	If set, enable temperature warning interrupt This setting is preserved during deep sleep.
14	OSC_WARN	R/W	0h	If set, enable for OSC clock warning interrupt This setting is preserved during deep sleep.
13:11	*reserved*	R	0h	Always write 0, ignore on read.
10	VDDA_TRX_SUPPLY_WARN	R/W	0h	If set, enable for VDDA_TRX supply warning interrupt This setting is preserved during deep sleep.
9	VREGA_SUPPLY_WARN	R/W	0h	If set, enable for VREGA supply warning interrupt This setting is preserved during deep sleep.
8	VREGD_SUPPLY_WARN	R/W	0h	If set, enable for VREGD supply warning interrupt This setting is preserved during deep sleep.
7:6	*reserved*	R	0h	Always write 0, ignore on read.
5	VDDIO_SUPPLY_WARN	R/W	0h	If set, enable for VDDIO supply interrupt This setting is preserved during deep sleep.
4	*reserved*	R	0h	Always write 0, ignore on read.
3	CORE_SUPPLY_WARN	R/W	0h	If set, enable Core supply interrupt This setting is preserved during deep sleep.
2	*reserved*	R	0h	Always write 0, ignore on read.
1	AO_SYSTEM_SUPPLY_WARN	R/W	0h	If set, enable Always-on system supply interrupt This setting is preserved during deep sleep.
0	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.

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7.4.1.18 OP_CONDITIONS_IRQ_MSTATUS register

Table 105. OP_CONDITIONS_IRQ_MSTATUS register - Operating conditions IRQ masked status (address 92h)

Bit	Symbol	Access	Value	Description
15	TEMP_WARN	R	0h	Masked-status of High-Temp Warning IRQ
14	OSC_WARN	R	0h	Masked-status of OSC clock Monitor warning IRQ
13:11	*reserved*	R	0h	Ignore on read.
10	VDDA_TRX_ SUPPLY_WARN	R	0h	Masked-status of VDDA_TRX supply warning IRQ
9	VREGA_SUPPLY_ WARN	R	0h	Masked-status of VREGA supply warning IRQ
8	VREGD_SUPPLY_ WARN	R	0h	Masked-status of VREGD supply warning IRQ
7:6	*reserved*	R	0h	Ignore on read.
5	VDDIO_SUPPLY_ WARN	R	0h	Masked-status of VDDIO supply warning IRQ
4	*reserved*	R	0h	Ignore on read.
3	CORE_SUPPLY_ WARN	R	0h	Masked-status of digital core supply warning interrupt
2	*reserved*	R	0h	Ignore on read.
1	AO_SYSTEM_ SUPPLY_WARN	R	0h	Masked-status of Always-on system supply warning interrupt
0	*reserved*	R	0h	Ignore on read.

7.4.1.19 TOP_EPHY_IRQ_SOURCES register

Table 106. TOP_EPHY_IRQ_SOURCES register - Top Ethernet PHY IRQ sources (address A0h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Always write 0, ignore on read.
1	WAKE_SLEEP_ EVENT	R/W	0h	If set, an event is detected by WISE_STATUS register. This interrupt does not get asserted by clearing latched-status bits. Write one to clear. This bit is latched-high.
0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.20 TOP_EPHY_IRQ_ENABLES register

Table 107. TOP_EPHY_IRQ_ENABLES register - Top Ethernet PHY IRQ enables (address A1h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Always write 0, ignore on read.
1	WAKE_SLEEP_ EVENT	R/W	0h	If set, the wake sleep event IRQ is enabled This setting is preserved during deep sleep.
0	*reserved*	R	0h	Always write 0, ignore on read.

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7.4.1.21 TOP_EPHY_IRQ_MSTATUS register

Table 108. TOP_EPHY_IRQ_MSTATUS register - Top Ethernet PHY IRQ masked status (address A2h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Ignore on read.
1	WAKE_SLEEP_EVENT_IRQ	R	0h	Masked-status of wake sleep event IRQ
0	*reserved*	R	0h	Ignore on read.

7.4.1.22 FUNC_SHARED_IRQ_SOURCES register

Table 109. FUNC_SHARED_IRQ_SOURCES register - Global shared function IRQ sources (address B0h)

Bit	Symbol	Access	Value	Description
15:3	*reserved*	R	0h	Always write 0, ignore on read.
2	VREGA_LDO_SHUTOFF_IRQ	R/W	0h	If set, VREGA LDO shutoff condition detected due to short circuit on LDO output Write one to clear.
1	VREGD_LDO_SHUTOFF_IRQ	R/W	0h	If set, VREGD LDO shutoff condition detected due to short circuit on LDO output Write one to clear.
0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.23 FUNC_SHARED_IRQ_ENABLES register

Table 110. FUNC_SHARED_IRQ_ENABLES register - Global shared function IRQ enables (address B1h)

Bit	Symbol	Access	Value	Description
15:3	*reserved*	R	0h	Always write 0, ignore on read.
2	VREGA_LDO_SHUTOFF_IRQ	R/W	0h	If set, the VREGA_LDO_SHUTOFF_IRQ is enabled This setting is preserved during deep sleep.
1	VREGD_LDO_SHUTOFF_IRQ	R/W	0h	If set, the VREGD_LDO_SHUTOFF_IRQ is enabled This setting is preserved during deep sleep.
0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.24 FUNC_SHARED_IRQ_MSTATUS register

Table 111. FUNC_SHARED_IRQ_MSTATUS register - Global shared function IRQ masked status (address B2h)

Bit	Symbol	Access	Value	Description
15:3	*reserved*	R	0h	Always write 0, ignore on read.
2	VREGA_LDO_SHUTOFF_IRQ	R/W	0h	Masked status of VREGA_LDO_SHUTOFF_IRQ
1	VREGD_LDO_SHUTOFF_IRQ	R/W	0h	Masked status of VREGD_LDO_SHUTOFF_IRQ
0	*reserved*	R	0h	Always write 0, ignore on read.

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7.4.1.25 CONFIG_REGXS_IRQ_SOURCES register

Table 112. CONFIG_REGXS_IRQ_SOURCES register - Configuration register access IRQ sources (address C0h)

Bit	Symbol	Access	Value	Description
15	COLD_RESET	R/W	0h	If set, all configuration settings reset to defaults due to RST_N Pin. Write one to clear. This bit is latched-high.
14:2	*reserved*	R	0h	Always write 0, ignore on read.
1	SMI_ERROR	R/W	0h	If set, invalid SMI transaction. This bit flags syntax errors in SMI commands and invalid SMI transactions to global register addresses, like writing to read-only registers and reading or writing non-existent registers. Write one to clear. This bit is latched-high.
0	*reserved*	R	0h	If set, Additional bits in part b Always write 0, ignore on read.

7.4.1.26 CONFIG_REGXS_IRQ_ENABLES register

Table 113. CONFIG_REGXS_IRQ_ENABLES register - Configuration register access IRQ enable (address C1h)

Bit	Symbol	Access	Value	Description
15	COLD_RESET	R/W	0h	Enables the COLD_RESET interrupt This setting is preserved during deep sleep.
14:2	*reserved*	R	0h	Always write 0, ignore on read.
1	SMI_ERROR	R/W	0h	Enables the SMI_ERROR interrupt This setting is preserved during deep sleep.
0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.27 CONFIG_REGXS_IRQ_MSTATUS register

Table 114. CONFIG_REGXS_IRQ_MSTATUS register - Configuration register access IRQ masked status (address C2h)

Bit	Symbol	Access	Value	Description
15	COLD_RESET	R	0h	Masked-status of COLD_RESET interrupt
14:2	*reserved*	R	0h	Ignore on read.
1	SMI_ERROR	R	0h	Masked-status of SMI_ERROR interrupt
0	*reserved*	R	0h	Ignore on read.

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7.4.1.28 BROADCAST register

Table 115. BROADCAST register - Broadcast control (address 100h)

Bit	Symbol	Access	Value	Description
15:5	*reserved*	R	0h	Always write 0, ignore on read.
4:0	ADDRESS	R/W	0h	5-bit SMI PHY address used for broadcasting. See SMI_ CONFIG register. This setting is preserved during deep sleep.

7.4.1.29 PORT1_CONFIG register

Table 116. PORT1_CONFIG register - Port 1 configuration (address 101h)

Bit	Symbol	Access	Value	Description
15	PRESENT	R	1h	If set, port is present
14	ENABLE	R/W	1h	If set, port is enabled This setting is preserved during deep sleep.
13:5	*reserved*	R	0h	Always write 0, ignore on read.
4:0	ADDRESS	R	1h	5-bit SMI PHY address assigned to this port This setting is preserved during deep sleep.

7.4.1.30 WISE_CONTROL register

Table 117. WISE_CONTROL register - Wake, inhibit and sleep control (address 180h)

Bit	Symbol	Access	Value	Description
15:1	*reserved*	R	0h	Always write 0, ignore on read.
0	INH_OVERRIDE	R/W	1h	If asserted, drive inhibit signal high at any time as long as the device is powered. When set to zero, the inhibit pin is HIGH-Z when this does does not requests power.

7.4.1.31 WISE_STATUS register

Table 118. WISE_STATUS register - Wake, inhibit and sleep status (address 181h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14	WU_IO_RECEIVED	R/W	0h	If set, wake-up request received on wakeup IO pin Write one to clear. This bit is latched-high.
13	WU_SMI_ RECEIVED	R/W	0h	If set, a wake-up is received via the WU_SMI bit Write one to clear. This bit is latched-high.
12:1	*reserved*	R	0h	Always write 0, ignore on read.
0	DEV_SLEEP_ STATUS	R	0h	If set, device in SLEEP

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7.4.1.32 WISE_CONFIG register

Table 119. WISE_CONFIG register - Wake, inhibit and sleep configuration (address 182h)

Bit	Symbol	Access	Value	Description
15	WAKE_SLEEP_ ENABLE	R/W	0h	If set, Wake-Sleep functionality enabled This setting is preserved during deep sleep. This setting is configuration protected.
14	*reserved*	R	0h	Always write 0, ignore on read.
13:12	WU_IO	R/W	3h	Wakeup IO configuration This setting is preserved during deep sleep. This setting is configuration protected. 00 = Wake-Up IO is disabled 01 = Wake-Up IO is input only, output disabled 10 = Wake-Up IO is output only, input disabled 11 = Wake-Up IO is input and output
11	*reserved*	R	0h	Always write 0, ignore on read.
10	FWD_WU_SMI_TO_ IO	R/W	0h	If set, forward SMI wakeup request (WU_SMI) to wakeup IO pin This setting is preserved during deep sleep. This setting is configuration protected.
9:0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.33 WISE_PARAMETERS register

Table 120. WISE_PARAMETERS register - Wake, inhibit and sleep parameters (address 184h)

Bit	Symbol	Access	Value	Description
15:5	*reserved*	R	0h	Always write 0, ignore on read.
4	WU_IO_DETECT_ LIMIT	R/W	0h	Wakeup IO detection pulse length This setting is preserved during deep sleep. This setting is configuration protected. 0* = The detection window for wake-up pulses is 10-40 µs. Wakeup pulses shorter than 10 µs are ignored and pulses longer than 40 µs are detected. 1 = If this bit is set to one, the detection window for wake-up pulses is 10 to 20 ms. This higher detection window is meant to support non-TC10 wake-up mechanisms, such as a dedicated wake-up wire.
3:0	WU_IO_PULSE_ LENGTH	R/W	1h	Length of a generated wake-up pulse on the wake-up IO pin in steps of 50 µs*2^n. If set to 0h, the pulse length is 50 µs. If set to Fh, the pulse length is 1.6 s. This setting is configuration protected. This setting is preserved during deep sleep.

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7.4.1.34 SMI_STATUS register

Table 121. SMI_STATUS register - SMI status (address 189h)

Bit	Symbol	Access	Value	Description
15	CL45_DETECT	R/W	0h	If set, Clause 45 transaction detected. Latched high Write one to clear. This bit is latched-high.
14	CL22_INDIRECT_DETECT	R/W	0h	If set, Clause 22 registers 0Dh or 0Eh accessed. Latched high Write one to clear. This bit is latched-high.
13	CL22_DIRECT_DETECT	R/W	0h	If set, Regular Clause 22 transaction detected (excluding access to registers 0Dh and 0Eh). Latched high Write one to clear. This bit is latched-high.
12	SMI_TIMEOUT	R/W	0h	If set, an SMI transaction did not complete due to timeout. Latched high Write one to clear. This bit is latched-high.
11:0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.35 SMI_CONFIG register

Table 122. SMI_CONFIG register - SMI config (address 18Ah)

Bit	Symbol	Access	Value	Description
15	ACCESS_ENABLE	R	1h	Enable register access via SMI. Note that this is read-only, if accessed from SMI side. This register is read-write if accessed internally via APB (only available in switch devices). ***Note:*** *Broadcasting can be enabled even if this field is set to zero. If set, Enable register access via SMI for all individual Phy's* This setting is configuration protected.
14	SIDE_EFFECTS_DISABLE	R/W	0h	Side-effects are effects on a register value due to a read action. This include Clear-on-Read for Latch-High, Latch-Low bit and Multi-Word reads. If set, Disable side-effects (for debugging purposes) This setting is configuration protected.
13	IDLE_PRECHARGE	R/W	0h	If set, Drive MDIO high before return to idle (high-Z) This setting is configuration protected. This setting is preserved during deep sleep.
12	SYNC_ON_16_ONES	R/W	0h	If set, the device does not expect a preamble for SMI synchronization This setting is configuration protected. This setting is preserved during deep sleep.
11:2	*reserved*	R	0h	Always write 0, ignore on read.

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Table 122. SMI_CONFIG register - SMI config (address 18Ah)*...continued*

Bit	Symbol	Access	Value	Description
1	BROADCAST_ READ_FUNC	R/W	0h	This bit configures the behavior of a read/write SMI transaction to the BROADCAST address. This setting is preserved during deep sleep. This setting is configuration protected. 0* = Broadcast write to all present PHYs Broadcast read of full address space Response on reads to PHY address space is a bit-wise OR function of that register for all present PHYs. 1 = Broadcast write to all present PHYs Broadcast read of shared address space only all-zero response for read of remaining address space
0	BROADCAST_ ENABLE	R/W	0h	When BROADCAST_ENABLE is set, SMI broadcast is enabled. If enabled, SMI access over the broadcasting mechanism is independent of the value of ACCESS_ENABLE. The broadcast address is configured with BROADCAST[ADDRESS]. This setting is configuration protected. This setting is preserved during deep sleep.

7.4.1.36 SMI_DATA_IO_CONFIG register

Table 123. SMI_DATA_IO_CONFIG register - SMI Data IO configuration (address 18Ch)

Bit	Symbol	Access	Value	Description
15	MANUAL_CONFIG	R/W	0h	If set, manual config of MDIO cells; otherwise settings are automatically determined. Caution: a manual override may violate the SMI timing in some or all operating conditions. The timing values in the dynamic characteristics section of the data sheet are valid under default settings only.
14:11	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
10:8	DRIVE_STRENGTH	R/W	0h	Select drive strength of MDIO output IO cells 0 = weakest 7 = strongest
7:3	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
2:0	SLEW_RATE	R/W	0h	Select slew rate of DATA MDIO output pins. 0 = weakest 7 = strongest

7.4.1.37 GLOBAL_LED_TRIGGER0 register

Table 124. GLOBAL_LED_TRIGGER0 register - LED trigger 0 (address 190h)

Bit	Symbol	Access	Value	Description
15	ENABLE	R/W	0h	Enable the LED trigger output
14:4	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.

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Table 124. GLOBAL_LED_TRIGGER0 register - LED trigger 0 (address 190h)*...continued*

Bit	Symbol	Access	Value	Description
3:0	FUNCTION	R/W	0h	Selects the function muxed on the device global LED trigger 0 0 = Safety status 1 = Supplies valid (*) 2 = Reserved 3 = XTAL oscillator running / clock available 4 to 15 = Reserved

7.4.1.38 GLOBAL_LED_TRIGGER1 register

Table 125. GLOBAL_LED_TRIGGER1 register - LED trigger 1 (address 191h)

Bit	Symbol	Access	Value	Description
15	ENABLE	R/W	0h	Enable the LED trigger output
14:4	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
3:0	FUNCTION	R/W	0h	Selects the function muxed on the device global LED trigger 1

7.4.1.39 GLOBAL_LED_TRIGGER2 register

Table 126. GLOBAL_LED_TRIGGER2 register - LED trigger 2 (address 192h)

Bit	Symbol	Access	Value	Description
15	ENABLE	R/W	0h	Enable the LED trigger output
14:4	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
3:0	FUNCTION	R/W	0h	Selects the function muxed on the device global LED trigger 2

7.4.1.40 GLOBAL_LED_TRIGGER3 register

Table 127. GLOBAL_LED_TRIGGER3 register - LED trigger 3 (address 193h)

Bit	Symbol	Access	Value	Description
15	ENABLE	R/W	0h	Enable the LED trigger output
14:4	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
3:0	FUNCTION	R/W	0h	Selects the function muxed on the device global LED trigger 3

7.4.1.41 LED0_CONFIG register

Table 128. LED0_CONFIG register - LED Controller 0 Configuration (address 1A0h)

Bit	Symbol	Access	Value	Description
15	ENABLE	R/W	1h	If set, LED controller is enabled.
14	INVERT	R/W	1h	If set, LED0A becomes high to drive the LED.
13	SPLIT_UP	R/W	0h	Split up

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Table 128. LED0_CONFIG register - LED Controller 0 Configuration (address 1A0h)*...continued*

Bit	Symbol	Access	Value	Description
				0 = Control one LED (blink/PWM/multi-color); One output A or two correlated outputs A/B. 1 = Control two LEDs (blink only) with two synchronous, but uncorrelated outputs A/B.
12:4	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
3:0	BLINKING_PERIOD	R/W	Ch	Sets the blinking period. 0000 = Reserved 0001 = 250 µs 0010 = 500 µs 0011 = 1 ms 0100 = 2 ms 0101 = 4 ms 0110 = 8 ms 0111 = 16 ms 1000 = 32 ms 1001 = 64 ms 1010 = 128 ms 1011 = 256 ms (~0.25 s) 1100 = 512 ms (~0.5 s) 1101 = 1024 ms (~1 s) 1110 = 2048 ms (~2 s) 1111 = latched (reset by disable)

7.4.1.42 LED0_TRIG_SOURCE register

Table 129. LED0_TRIG_SOURCE register - LED controller 0 trigger source selection (address 1A1h)

Bit	Symbol	Access	Value	Description
15:13	FUNC_SELECT_B	R/W	0h	If FUNC_SELECT_B≠0, the selected PORT subfunction is used from the PortNr as configured by the PORTNR_(OR_ TOPFUNC)_B field. If FUNC_SELECT_B=000=GLOBAL, the (PORTNR_OR)_TOPFUNC_B field is used to select the desired GLOBAL subfunction. 0* = Global LED trigger 1 = Port EPHY LED trigger 2h = Port PTP LED trigger 3h = Port MACsec LED Trigger 7h = Port xMII/SGMII LED Trigger others = Reserved
12:8	PORTNR_OR_ TOPFUNC_B	R/W	0h	If FUNC_SELECT_A≠0: Select PORT number 1-32 (00000=32) If FUNC_SELECT_A=0: Select GLOBAL sub-function, as follows 00000 = Global shared resources Otherwise = Reserved
7:5	FUNC_SELECT_A	R/W	1h	If FUNC_SELECT_A≠0, the selected PORT sub-function is used from the PortNr as configured by the PORTNR_(OR_ TOPFUNC)_A field. If FUNC_SELECT_A=000=GLOBAL,

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Table 129. LED0_TRIG_SOURCE register - LED controller 0 trigger source selection (address 1A1h) *...continued*

Bit	Symbol	Access	Value	Description
				the (PORTNR_OR)_TOPFUNC_A field is used to select the desired GLOBAL sub-function. 0* = Global LED trigger 1 = Port EPHY LED trigger 2h = Port PTP LED trigger 3h = Port MACsec LED Trigger 7h = Port xMII/SGMII LED Trigger others = Reserved
4:0	PORTNR_OR_ TOPFUNC_A	R/W	1h	If FUNC_SELECT_A≠0: Select PORT number 1-32 (00000=32) If FUNC_SELECT_A=0: Select GLOBAL sub-function, as follows 00000 = Global shared resources Otherwise = Reserved

7.4.1.43 LED0_TRIG01_CONFIG register

Table 130. LED0_TRIG01_CONFIG register - LED controller 0 Trigger 0 and 1 configuration (address 1A2h)

Bit	Symbol	Access	Value	Description
15	TRIG1_ENABLE	R/W	1h	If set, this trigger is enabled
14	TRIG1_POLARITY	R/W	1h	If this bit is deasserted, the polarity of the input signal is inverted after the event detector.
13	TRIG1_HILO_ PRIORITY	R/W	1h	If this bit is set, the event detector is sensitive on high-level events and short high-level pulses are stretched. Otherwise, the detector is sensitive on low-level events and low level pulses are stretched.
12	TRIG1_SELECT	R/W	0h	Selects if FUNC_SELECT_A or B is selected 0* = A input is used for the trigger 1 = B input is used for the trigger
11:8	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
7	TRIG0_ENABLE	R/W	1h	If set, this trigger is enabled
6	TRIG0_POLARITY	R/W	1h	If this bit is deasserted, the polarity of the input signal is inverted after the event detector.
5	TRIG0_HILO_ PRIORITY	R/W	1h	If this bit is set, the event detector is sensitive on high-level events and short high-level pulses are stretched. Otherwise, the detector is sensitive on low-level events and low level pulses are stretched.
4	TRIG0_SELECT	R/W	0h	Selects if FUNC_SELECT_A or B is selected 0* = A input is used for the trigger 1 = B input is used for the trigger
3	TRIG0_RAW	R/W	0h	If set the raw trigger signal is used and pulse stretching is disabled.
2:0	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.

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7.4.1.44 LED0_TRIG23_CONFIG register

Table 131. LED0_TRIG23_CONFIG register - LED controller 0 Trigger 2 and 3 configuration (address 1A3h)

Bit	Symbol	Access	Value	Description
15	TRIG3_ENABLE	R/W	0	If set, this trigger is enabled
14	TRIG3_POLARITY	R/W	0h	If this bit is deasserted, the polarity of the input signal is inverted after the event detector.
13	TRIG3_HILO_ PRIORITY	R/W	0h	If this bit is set, the event detector is sensitive on high-level events and short high-level pulses are stretched. Otherwise, the detector is sensitive on low-level events and low level pulses are stretched.
12	TRIG3_SELECT	R/W	0h	Selects if FUNC_SELECT_A or B is selected 0* = A input is used for the trigger 1 = B input is used for the trigger
11:10	TRIG3_FUNCTION	R/W	0h	This field has no effect when SPLIT_UP is set and TRIG2/3 determine output-B with similar behavior as TRIG0/1 for output-A. If SPLIT_UP = 0: 00 = TRIG3 is AND'ed with TRIG0 event. if TRIG2_MODULATION = 1: 01 = Blink color modulation by TRIG2/3 1x = Reserved If TRIG2_MODULATION = 0: 01 = ColorB replaces OFF for TRIG3 event 10 = Direct color modulation by TRIG3 11 = Reserved
9:8	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
7	TRIG2_ENABLE	R/W	1h	If set, this trigger is enabled
6	TRIG2_POLARITY	R/W	1h	If this bit is deasserted, the polarity of the input signal is inverted after the event detector.
5	TRIG2_HILO_ PRIORITY	R/W	1h	If this bit is set, the event detector is sensitive on high-level events and short high-level pulses are stretched. Otherwise, the detector is sensitive on low-level events and low level pulses are stretched.
4	TRIG2_SELECT	R/W	0h	Selects if FUNC_SELECT_A or B is selected 0* = A input is used for the trigger 1 = B input is used for the trigger
3	TRIG2_RAW	R/W	0h	If set the raw trigger signal is used and pulse stretching is disabled.
2	TRIG2_ MODULATION	R/W	0h	If set, enables multi-color blink on ColorA and ColorB. This bit has no effect when SPLIT_UP=1 or TRIG2_RAW=1. If SPLIT_UP=1, TRIG2/3 determine output-B with similar behavior as TRIG0/1 for output-A.
1:0	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.

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7.4.1.45 LED1_CONFIG register

Table 132. LED1_CONFIG register - LED Controller 1 Configuration (see LED controller 0) (address 1A4h)

Bit	Symbol	Access	Value	Description
15	ENABLE	R/W	1h	see LED0_CONFIG[ENABLE]
14	INVERT	R/W	1h	see LED0_CONFIG[INVERT]
13	SPLIT_UP	R/W	0h	see LED0_CONFIG[SPLIT_UP]
12:4	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
3:0	BLINKING_PERIOD	R/W	Ch	see LED0_CONFIG[BLINKING_PERIOD]

7.4.1.46 LED1_TRIG_SOURCE register

Table 133. LED1_TRIG_SOURCE register - LED controller 1 trigger source selection (see LED controller 0) (address 1A5h)

Bit	Symbol	Access	Value	Description
15:13	FUNC_SELECT_B	R/W	0h	see LED0_TRIG_SOURCE[FUNC_SELECT_B]
12:8	PORTNR_OR_ TOPFUNC_B	R/W	0h	see LED0_TRIG_SOURCE[PORTNR_OR_TOPFUNC_B]
7:5	FUNC_SELECT_A	R/W	1h	see LED0_TRIG_SOURCE[FUNC_SELECT_A]
4:0	PORTNR_OR_ TOPFUNC_A	R/W	1h	see LED0_TRIG_SOURCE[PORTNR_OR_TOPFUNC_A]

7.4.1.47 LED1_TRIG01_CONFIG register

Table 134. LED1_TRIG01_CONFIG register - LED controller 1 Trigger 0 and 1 configuration (see LED controller 0) (address 1A6h)

Bit	Symbol	Access	Value	Description
15	TRIG1_ENABLE	R/W	1h	see LED0_TRIG01_CONFIG[TRIG1_ENABLE]
14	TRIG1_POLARITY	R/W	1h	see LED0_TRIG01_CONFIG[TRIG1_POLARITY]
13	TRIG1_HILO_ PRIORITY	R/W	1h	see LED0_TRIG01_CONFIG[TRIG1_PRIORITY]
12	TRIG1_SELECT	R/W	0h	see LED0_TRIG01_CONFIG[TRIG1_SELECT]
11:8	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
7	TRIG0_ENABLE	R/W	1h	see LED0_TRIG01_CONFIG[TRIG0_ENABLE]
6	TRIG0_POLARITY	R/W	1h	see LED0_TRIG01_CONFIG[TRIG0_POLARITY]
5	TRIG0_HILO_ PRIORITY	R/W	1h	see LED0_TRIG01_CONFIG[TRIG0_HILO_PRIORITY]
4	TRIG0_SELECT	R/W	0h	see LED0_TRIG01_CONFIG[TRIG0_SELECT]
3	TRIG0_RAW	R/W	0h	see LED0_TRIG01_CONFIG[TRIG0_RAW]
2:0	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.

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7.4.1.48 LED1_TRIG23_CONFIG register

Table 135. LED1_TRIG23_CONFIG register - LED controller 1 Trigger 2 and 3 configuration (see LED controller 0) (address 1A7h)

Bit	Symbol	Access	Value	Description
15	TRIG3_ENABLE	R/W	0h	see LED0_TRIG23_CONFIG[TRIG3_ENABLE]
14	TRIG3_POLARITY	R/W	0h	see LED0_TRIG23_CONFIG[TRIG3_POLARITY]
13	TRIG3_HILO_ PRIORITY	R/W	0h	see LED0_TRIG23_CONFIG[TRIG3_HILO_PRIORITY]
12	TRIG3_SELECT	R/W	0h	see LED0_TRIG23_CONFIG[TRIG3_SELECT]
11:10	TRIG3_FUNCTION	R/W	0h	see LED0_TRIG23_CONFIG[TRIG3_FUNCTION]
9:8	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
7	TRIG2_ENABLE	R/W	1h	see LED0_TRIG23_CONFIGTRIG2_ENABLE]
6	TRIG2_POLARITY	R/W	1h	see LED0_TRIG23_CONFIG[TRIG2_POLARITY]
5	TRIG2_HILO_ PRIORITY	R/W	1h	see LED0_TRIG23_CONFIG[TRIG2_HILO_PRIORITY]
4	TRIG2_SELECT	R/W	0h	see LED0_TRIG23_CONFIG[TRIG2_SELECT]
3	TRIG2_RAW	R/W	0h	see LED0_TRIG23_CONFIG[TRIG2_RAW]
2	TRIG2_ MODULATION	R/W	0h	see LED0_TRIG23_CONFIG[TRIG2_MODULATION]
1:0	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.

7.4.1.49 LED2_CONFIG register

Table 136. LED2_CONFIG register - LED Controller 2 Configuration (see LED controller 0) (address 1A8h)

Bit	Symbol	Access	Value	Description
15	ENABLE	R/W	1h	see LED0_CONFIG[ENABLE]
14	INVERT	R/W	1h	see LED0_CONFIG[INVERT]
13	SPLIT_UP	R/W	0h	see LED0_CONFIG[SPLIT_UP]
12:4	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
3:0	BLINKING_PERIOD	R/W	Ch	see LED0_CONFIG[BLINKING_PERIOD]

7.4.1.50 LED2_TRIG_SOURCE register

Table 137. LED2_TRIG_SOURCE register - LED controller 2 trigger source selection (see LED controller 0) (address 1A9h)

Bit	Symbol	Access	Value	Description
15:13	FUNC_SELECT_B	R/W	0h	see LED0_TRIG_SOURCE[FUNC_SELECT_B]
12:8	PORTNR_OR_ TOPFUNC_B	R/W	0h	see LED0_TRIG_SOURCE[PORTNR_OR_TOPFUNC_B]
7:5	FUNC_SELECT_A	R/W	1h	see LED0_TRIG_SOURCE[FUNC_SELECT_A]
4:0	PORTNR_OR_ TOPFUNC_A	R/W	1h	see LED0_TRIG_SOURCE[PORTNR_OR_TOPFUNC_A]

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7.4.1.51 LED2_TRIG01_CONFIG register

Table 138. LED2_TRIG01_CONFIG register - LED controller 2 Trigger 0 and 1 configuration (see LED controller 0) (address 1AAh)

Bit	Symbol	Access	Value	Description
15	TRIG1_ENABLE	R/W	1h	see LED0_TRIG01_CONFIG[TRIG1_ENABLE]
14	TRIG1_POLARITY	R/W	1h	see LED0_TRIG01_CONFIG[TRIG1_POLARITY]
13	TRIG1_HILO_ PRIORITY	R/W	1h	see LED0_TRIG01_CONFIG[TRIG1_PRIORITY]
12	TRIG1_SELECT	R/W	0h	see LED0_TRIG01_CONFIG[TRIG1_SELECT]
11:8	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
7	TRIG0_ENABLE	R/W	1h	see LED0_TRIG01_CONFIG[TRIG0_ENABLE]
6	TRIG0_POLARITY	R/W	1h	see LED0_TRIG01_CONFIG[TRIG0_POLARITY]
5	TRIG0_HILO_ PRIORITY	R/W	1h	see LED0_TRIG01_CONFIG[TRIG0_HILO_PRIORITY]
4	TRIG0_SELECT	R/W	0h	see LED0_TRIG01_CONFIG[TRIG0_SELECT]
3	TRIG0_RAW	R/W	0h	see LED0_TRIG01_CONFIG[TRIG0_RAW]
2:0	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.

7.4.1.52 LED2_TRIG23_CONFIG register

Table 139. LED2_TRIG23_CONFIG register - LED controller 2 Trigger 2 and 3 configuration (see LED controller 0) (address 1ABh)

Bit	Symbol	Access	Value	Description
15	TRIG3_ENABLE	R/W	0h	see LED0_TRIG23_CONFIG[TRIG3_ENABLE]
14	TRIG3_POLARITY	R/W	0h	see LED0_TRIG23_CONFIG[TRIG3_POLARITY]
13	TRIG3_HILO_ PRIORITY	R/W	0h	see LED0_TRIG23_CONFIG[TRIG3_HILO_PRIORITY]
12	TRIG3_SELECT	R/W	0h	see LED0_TRIG23_CONFIG[TRIG3_SELECT]
11:10	TRIG3_FUNCTION	R/W	0h	see LED0_TRIG23_CONFIG[TRIG3_FUNCTION]
9:8	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
7	TRIG2_ENABLE	R/W	1h	see LED0_TRIG23_CONFIGTRIG2_ENABLE]
6	TRIG2_POLARITY	R/W	1h	see LED0_TRIG23_CONFIG[TRIG2_POLARITY]
5	TRIG2_HILO_ PRIORITY	R/W	1h	see LED0_TRIG23_CONFIG[TRIG2_HILO_PRIORITY]
4	TRIG2_SELECT	R/W	0h	see LED0_TRIG23_CONFIG[TRIG2_SELECT]
3	TRIG2_RAW	R/W	0h	see LED0_TRIG23_CONFIG[TRIG2_RAW]
2	TRIG2_ MODULATION	R/W	0h	see LED0_TRIG23_CONFIG[TRIG2_MODULATION]
1:0	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.

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7.4.1.53 LED3_CONFIG register

Table 140. LED3_CONFIG register - LED Controller 3 Configuration (see LED controller 0) (address 1ACh)

Bit	Symbol	Access	Value	Description
15	ENABLE	R/W	1h	see LED0_CONFIG[ENABLE]
14	INVERT	R/W	1h	see LED0_CONFIG[INVERT]
13	SPLIT_UP	R/W	0h	see LED0_CONFIG[SPLIT_UP]
12:4	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
3:0	BLINKING_PERIOD	R/W	Ch	see LED0_CONFIG[BLINKING_PERIOD]

7.4.1.54 LED3_TRIG_SOURCE register

Table 141. LED3_TRIG_SOURCE register - LED controller 3 trigger source selection (see LED controller 0) (address 1ADh)

Bit	Symbol	Access	Value	Description
15:13	FUNC_SELECT_B	R/W	0h	see LED0_TRIG_SOURCE[FUNC_SELECT_B]
12:8	PORTNR_OR_ TOPFUNC_B	R/W	0h	see LED0_TRIG_SOURCE[PORTNR_OR_TOPFUNC_B]
7:5	FUNC_SELECT_A	R/W	1h	see LED0_TRIG_SOURCE[FUNC_SELECT_A]
4:0	PORTNR_OR_ TOPFUNC_A	R/W	1h	see LED0_TRIG_SOURCE[PORTNR_OR_TOPFUNC_A]

7.4.1.55 LED3_TRIG01_CONFIG register

Table 142. LED3_TRIG01_CONFIG register - LED controller 3 Trigger 0 and 1 configuration (see LED controller 0) (address 1AEh)

Bit	Symbol	Access	Value	Description
15	TRIG1_ENABLE	R/W	1h	see LED0_TRIG01_CONFIG[TRIG1_ENABLE]
14	TRIG1_POLARITY	R/W	1h	see LED0_TRIG01_CONFIG[TRIG1_POLARITY]
13	TRIG1_HILO_ PRIORITY	R/W	1h	see LED0_TRIG01_CONFIG[TRIG1_PRIORITY]
12	TRIG1_SELECT	R/W	0h	see LED0_TRIG01_CONFIG[TRIG1_SELECT]
11:8	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
7	TRIG0_ENABLE	R/W	1h	see LED0_TRIG01_CONFIG[TRIG0_ENABLE]
6	TRIG0_POLARITY	R/W	1h	see LED0_TRIG01_CONFIG[TRIG0_POLARITY]
5	TRIG0_HILO_ PRIORITY	R/W	1h	see LED0_TRIG01_CONFIG[TRIG0_HILO_PRIORITY]
4	TRIG0_SELECT	R/W	0h	see LED0_TRIG01_CONFIG[TRIG0_SELECT]
3	TRIG0_RAW	R/W	0h	see LED0_TRIG01_CONFIG[TRIG0_RAW]
2:0	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.

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7.4.1.56 LED3_TRIG23_CONFIG register

Table 143. LED3_TRIG23_CONFIG register - LED controller 3 Trigger 2 and 3 configuration (see LED controller 0) (address 1AFh)

Bit	Symbol	Access	Value	Description
15	TRIG3_ENABLE	R/W	0h	see LED0_TRIG23_CONFIG[TRIG3_ENABLE]
14	TRIG3_POLARITY	R/W	0h	see LED0_TRIG23_CONFIG[TRIG3_POLARITY]
13	TRIG3_HILO_ PRIORITY	R/W	0h	see LED0_TRIG23_CONFIG[TRIG3_HILO_PRIORITY]
12	TRIG3_SELECT	R/W	0h	see LED0_TRIG23_CONFIG[TRIG3_SELECT]
11:10	TRIG3_FUNCTION	R/W	0h	see LED0_TRIG23_CONFIG[TRIG3_FUNCTION]
9:8	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
7	TRIG2_ENABLE	R/W	1h	see LED0_TRIG23_CONFIGTRIG2_ENABLE]
6	TRIG2_POLARITY	R/W	1h	see LED0_TRIG23_CONFIG[TRIG2_POLARITY]
5	TRIG2_HILO_ PRIORITY	R/W	1h	see LED0_TRIG23_CONFIG[TRIG2_HILO_PRIORITY]
4	TRIG2_SELECT	R/W	0h	see LED0_TRIG23_CONFIG[TRIG2_SELECT]
3	TRIG2_RAW	R/W	0h	see LED0_TRIG23_CONFIG[TRIG2_RAW]
2	TRIG2_ MODULATION	R/W	0h	see LED0_TRIG23_CONFIG[TRIG2_MODULATION]
1:0	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.

7.4.1.57 AO_SYSTEM_SUPPLY_STATUS register

Table 144. AO_SYSTEM_SUPPLY_STATUS register - Always on system supply (address 311h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14	UV_EVENT	R/W	0h	If set, an under-voltage event occurred since last clear Write one to clear. This bit is latched-high.
13:0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.58 CORE_SUPPLY_STATUS register

Table 145. CORE_SUPPLY_STATUS register - Core supply status (address 313h)

Bit	Symbol	Access	Value	Description
15	OV_EVENT	R/W	0h	This bit is asserted if a voltage monitor for this supply detects an over-voltage. This bit is asserted if a voltage monitor for this supply detects an over-voltage. Write one to clear. This bit is latched-high.
14	UV_EVENT	R/W	0h	This bit is asserted if a voltage monitor for this supply detects an under-voltage. Write one to clear.

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Table 145. CORE_SUPPLY_STATUS register - Core supply status (address 313h)*...continued*

Bit	Symbol	Access	Value	Description
				This bit is latched-high.
13	OV_STATUS	R	0h	This bit shows the actual over-voltage status as measured by the voltage monitor of this supply. If set, there is an over-voltage condition present. If there is no voltage monitor implemented for this supply, this bit reads zero.
12	UV_STATUS	R	0h	This bit shows the actual under-voltage status as measured by the voltage monitor of this supply. If set, there is an under-voltage condition present. If there is no voltage monitor implemented for this supply, this bit reads zero.
11	OV_LATENT_ FAULT	R/W	0h	This bit gets asserted if no over-voltage is detected by an over-voltage latent-fault test. This bit can only get asserted during the FuSa test procedure at device power-up or wake-up from sleep. A detected latent-fault creates an interrupt trigger pulse Write one to clear. This bit is latched-high.
10	UV_LATENT_FAULT	R/W	0h	This bit gets asserted if no under-voltage is detected by an under-voltage latent fault test. This bit can only get asserted during the FuSa test procedure at device power-up or wake-up from sleep. A detected latent-fault creates an interrupt trigger pulse Write one to clear. This bit is latched-high.
9:0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.59 VDDIO_SUPPLY_STATUS register

Table 146. VDDIO_SUPPLY_STATUS register - VDDIO supply status (address 315h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14	UV_EVENT	R/W	0h	If set, an under-voltage occurred. Write one to clear. This bit is latched-high.
13:2	*reserved*	R	0h	Always write 0, ignore on read.
1:0	VDDIO_LEVEL	R	0h	Detected IO supply level 00* = Reserved 01 = 1.8 V 10 = 2.5 V 11 = 3.3 V

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7.4.1.60 VREGD_SUPPLY_STATUS register

Table 147. VREGD_SUPPLY_STATUS register - VREGD supply status (address 318h)

Bit	Symbol	Access	Value	Description
15	OV_EVENT	R/W	0h	This bit is asserted if a voltage monitor for this supply detects an under-voltage. This bit is asserted if a voltage monitor for this supply detects an over-voltage. Write one to clear. This bit is latched-high.
14	*reserved*	R	0h	Always write 0, ignore on read.
13	OV_STATUS	R	0h	This bit shows the actual over-voltage status as measured by the voltage monitor of this supply. If set, there is an over-voltage condition present. If there is no voltage monitor implemented for this supply, this bit reads zero.
12	*reserved*	R	0h	Always write 0, ignore on read.
11	OV_LATENT_FAULT	R/W	0h	This bit gets asserted if no over-voltage is detected by an over-voltage latent-fault test. This bit can only get asserted during the FuSa test procedure at device power-up or wake-up from sleep. A detected latent-fault creates an interrupt trigger pulse Write one to clear. This bit is latched-high.
10:0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.61 VREGA_SUPPLY_STATUS register

Table 148. VREGA_SUPPLY_STATUS register - VREGA supply status (address 319h)

Bit	Symbol	Access	Value	Description
15	OV_EVENT	R/W	0h	This bit is asserted if a voltage monitor for this supply detects an under-voltage. This bit is asserted if a voltage monitor for this supply detects an over-voltage. Write one to clear. This bit is latched-high.
14	*reserved*	R	0h	Always write 0, ignore on read.
13	OV_STATUS	R	0h	This bit shows the actual over-voltage status as measured by the voltage monitor of this supply. If set, there is an over-voltage condition present. If there is no voltage monitor implemented for this supply, this bit reads zero.
12	*reserved*	R	0h	Always write 0, ignore on read.
11	OV_LATENT_FAULT	R/W	0h	This bit gets asserted if no over-voltage is detected by an over-voltage latent-fault test. This bit can only get asserted during the FuSa test procedure at device power-up or wake-up from sleep. A detected latent-fault creates an interrupt trigger pulse Write one to clear. This bit is latched-high.
10:0	*reserved*	R	0h	Always write 0, ignore on read.

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7.4.1.62 VDDA_TRX_SUPPLY register

Table 149. VDDA_TRX_SUPPLY register - VDDA_TRX supply status (address 31Ah)

Bit	Symbol	Access	Value	Description
15	OV_EVENT	R/W	0h	This bit is asserted if a voltage monitor for this supply detects an under-voltage. This bit is asserted if a voltage monitor for this supply detects an over-voltage. Write one to clear. This bit is latched-high.
14	UV_EVENT	R/W	0h	This bit is asserted if a voltage monitor for this supply detects an over-voltage. Write one to clear. This bit is latched-high.
13	OV_STATUS	R	0h	This bit shows the actual over-voltage status as measured by the voltage monitor of this supply. If set, there is an over-voltage condition present. If there is no voltage monitor implemented for this supply, this bit reads zero.
12	UV_STATUS	R	0h	This bit shows the actual under-voltage status as measured by the voltage monitor of this supply. If set, there is an under-voltage condition present. If there is no voltage monitor implemented for this supply, this bit reads zero.
11	OV_LATENT_FAULT	R/W	0h	This bit gets asserted if no over-voltage is detected by an over-voltage latent-fault test. This bit can only get asserted during the FuSa test procedure at device power-up or wake-up from sleep. A detected latent-fault creates an interrupt trigger pulse Write one to clear. This bit is latched-high.
10	UV_LATENT_FAULT	R/W	0h	This bit gets asserted if no under-voltage is detected by an under-voltage latent fault test. This bit can only get asserted during the FuSa test procedure at device power-up or wake-up from sleep. A detected latent-fault creates an interrupt trigger pulse Write one to clear. This bit is latched-high.
9:0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.63 OSC_STATUS register

Table 150. OSC_STATUS register - XO status (address 31Eh)

Bit	Symbol	Access	Value	Description
15	HIGH_FQ_EVENT	R/W	0h	If set, ratio between the oscillator and FRO frequencies was too high. Write one to clear. This bit is latched-high.
14	LOW_FQ_EVENT	R/W	0h	If set, ratio between the oscillator and FRO frequencies was too low. Write one to clear. This bit is latched-high.

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Table 150. OSC_STATUS register - XO status (address 31Eh)*...continued*

Bit	Symbol	Access	Value	Description
13	HIGH_FQ_NOW	R	0h	If set, current ratio between oscillator and FRO frequencies is too high
12	LOW_FQ_NOW	R	0h	If set, current ratio between oscillator and FRO frequencies is too low
11	CLK_PRESENT	R	0h	Set if oscillator clock is detected to be present
10:0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.64 TEMP_STATUS register

Table 151. TEMP_STATUS register - Temperature status (address 31Fh)

Bit	Symbol	Access	Value	Description
15	HIGH_TEMP_EVENT	R/W	0h	If set, a temperature beyond TEMP_LIMIT has occurred since last clear. Write one to clear. This bit is latched-high.
14	*reserved*	R	0h	Always write 0, ignore on read.
13	HIGH_TEMP_NOW	R	0h	If set, the die temperature is above TEMP_LIMIT now
12	*reserved*	R	0h	Always write 0, ignore on read.
11	HIGH_TEMP_LATENT_FAULT	R/W	0h	If set, the temperature sensor latent-fault test failed Write one to clear. This bit is latched-high.
10:8	*reserved*	R	0h	Always write 0, ignore on read.
7:0	TEMP_LIMIT	R/W	A1h	Sets the current temperature warning limit. The temperature needs to be converted as follows: TEMP_LIMIT=7.22*x^2+38.8*x+65.3 with x = T/100 where T is the temperature in degree Celsius. See application note AN13662

7.4.1.65 COMPL_TEST_INTF register

Table 152. COMPL_TEST_INTF register - Compliance test interface (address 3F4h)

Bit	Symbol	Access	Value	Description
15	TEST_INTF_EN	R/W	0h	If set, enables device test interface
14	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
11:8	FUNC_SEL	R/W	1h	Test function select 0001* = Selects Ethernet PHY for test others = Reserved
7:6	*reserved*	R	0h	Always write 0, ignore on read.
5	COMPL_TCLK_EN	R/W	0h	If set, TCLK out is enabled on the test interface
4:0	TEST_PORT_NR	R/W	1h	Selected port number for the test. This must be set to 1.

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7.4.1.66 PTP_IDENTIFIER register

Table 153. PTP_IDENTIFIER register - PTP IP identifier register (address 1100h)

Bit	Symbol	Access	Value	Description
15:8	*reserved*	R	0h	Ignore on read.
7:4	PTP_VERSION	R	1h	PTP IP version.
3:0	PTP_REVISION	R	0h	PTP module revision

7.4.1.67 PTP_CONTROL register

Table 154. PTP_CONTROL register - General control (address 1101h)

Bit	Symbol	Access	Value	Description
15:1	*reserved*	R	0h	Always write 0, ignore on read.
0	SOFT_RESET	R/W	0h	If asserted, the PTP block is reset. This is self-clearing bit.

7.4.1.68 PTP_CONFIG register

Table 155. PTP_CONFIG register - General control (address 1102h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15:5	*reserved*	R	0h	Always write 0, ignore on read.
4	PPS_SYNC_RETURN	R/W		Return external PPS_Sync on PPS_OUT
			0*	Synchronized (ptp_clk) external pps_sync is not routed back on PPS_OUT
			1	Synchronized (ptp_clk) external pps_sync is routed back on PPS_OUT
3	PPS_OUT_EN	R/W		Enable the PPS_OUT signal generation
			0*	PPS_OUT is not generated
			1	PPS_OUT is generated
2	PPS_OUT_POL	R/W		Inverts the polarity of generated output 1 PPS signal.
			0*	No inversion
			1	Invert a 1 PPS out
1	EXT_TRG_EDGE_SEL	R/W		Inversion control of external trigger event
			0*	Rising edge of external trigger event is timestamped
			1	Falling edge of external trigger event is timestamped
0	EXT_PPS_EDGE_SEL	R/W		When switching this bit, inverted PPS Sync may be disturbed for first pulse after bit is switched. However switching this bit will not have any disturbance on PPS sync timestamping. This bit is not dynamically changing bit.
			0*	Rising edge of external PPS sync is timestamped and used in all places
			1	Falling edge of external PPS sync is timestamped and inverted PPS sync is used in all places

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7.4.1.69 PTP_ABILITY register

Table 156. PTP_ABILITY register - PTP IP Ability registers (address 1103h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15:3	*reserved*	R	0h	Ignore on read.
2	INGRESS_IN_BAND	R		Indicates PTP IP supports insertion of timestamping information in "reserved" field.
			0	Does not support in-band deliver
			1*	Supports in-band deliver
1	PTP_TWO_STEP	R		Indicates PTP IP supports two-step operation.
			0	Does not support two-step operation
			1*	Supports TWO-step operation
0	PTP_ONE_STEP	R		Indicates PTP IP supports one-step operation.
			0*	Does not support one-step operation
			1	Supports one-step operation

7.4.1.70 PTP_CLK_PERIOD register

Table 157. PTP_CLK_PERIOD register - Clock period of ptp_clk (address 1104h)

Bit	Symbol	Access	Value	Description
15:8	*reserved*	R	0h	Always write 0, ignore on read.
7:0	PTP_CLK_PERIOD	R/W	8h	Clock period of ptp_clk. Local LTC increment its counters by this value for every ptp clock tick. In 100BT1, it is 15 ns (66.66 MHz). In 1000BT1, it is 8 ns (125 MHz)

7.4.1.71 LTC_LOAD_CTRL register

Table 158. LTC_LOAD_CTRL register - LTC load and read control (address 1105h)

Bit	Symbol	Access	Value	Description
15:3	*reserved*	R	0h	Always write 0, ignore on read.
2	READ_LTC	R/W	0h	When set, captures the current internal LTC clock time in LTC_RD_NSEC_0/1, LTC_RD_SEC_0/1 & LTC_RD_SUB_NS_0/1 register. This bit is cleared on capture.
1	LOAD_LTC_PPS	R/W	0h	When this bit is set and on next rising edge of external pps_sync signal, LTC_WR_SEC_0/1 & LTC_WR_NSEC_0/1 is loaded into the LTC. This bit is cleared after the loading at PPS edge(with additional 4-5 APB clocks (CDC latency)) This bit should be programmed only after LTC_WR_SEC_0/1 & LTC_WR_NSEC_0/1 is written.

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Table 158. LTC_LOAD_CTRL register - LTC load and read control (address 1105h)*...continued*

Bit	Symbol	Access	Value	Description
0	LOAD_LTC	R/W	0h	Self-clearing when LTC_WR_SEC_0/1 & LTC_WR_NSEC_0/1 is loaded into the LTC. This should be set only after LTC_WR_SEC_0/1 & LTC_WR_NSEC_0/1 is written.

7.4.1.72 LTC_WR_NSEC_0 register

Table 159. LTC_WR_NSEC_0 register - Loading nanoseconds counter value (address 1106h)

Bit	Symbol	Access	Value	Description
15:0	LTC_WR_NSEC_0	R/W	0h	[15:0] of LTC_WR_NSEC. This is nanoseconds part of clock time loaded into the internal LTC.

7.4.1.73 LTC_WR_NSEC_1 register

Table 160. LTC_WR_NSEC_1 register - Loading nanoseconds counter value (address 1107h)

Bit	Symbol	Access	Value	Description
15:14	*reserved*	R	0h	Always write 0, ignore on read.
13:0	LTC_WR_NSEC_1	R/W	0h	[29:16] of LTC_WR_NSEC. This is nanoseconds part of clock time loaded into the internal LTC.

7.4.1.74 LTC_WR_SEC_0 register

Table 161. LTC_WR_SEC_0 register - Loading seconds counter value (address 1108h)

Bit	Symbol	Access	Value	Description
15:0	LTC_WR_SEC_0	R/W	0h	[15:0] of LTC_WR_SEC. This is seconds part of clock time loaded into the internal LTC.

7.4.1.75 LTC_WR_SEC_1 register

Table 162. LTC_WR_SEC_1 register - Loading seconds counter value (address 1109h)

Bit	Symbol	Access	Value	Description
15:0	LTC_WR_SEC_1	R/W	0h	[31:16] of LTC_WR_SEC. This is seconds part of clock time loaded into the internal LTC.

7.4.1.76 LTC_RD_DATA_0 register

Table 163. LTC_RD_DATA_0 register - Local LTC nanoseconds part. The value read is the saved value of local counter when READ_LTC is set. (address 110Ah)

Bit	Symbol	Access	Value	Description
15:0	LTC_RD_NSEC_0	R	0h	[15:0] of LTC_RD_NSEC. This reads the value of internal nanoseconds counter.

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7.4.1.77 LTC_RD_DATA_1 register

Table 164. LTC_RD_DATA_1 register - Local LTC nanoseconds part. The value read is the saved value of local counter when READ_LTC is set. (address 110Bh)

Bit	Symbol	Access	Value	Description
15:14	*reserved*	R	0h	Ignore on read.
13:0	LTC_RD_NSEC_1	R	0h	[29:16] of LTC_RD_NSEC. This reads the value of internal nanoseconds counter.

7.4.1.78 LTC_RD_DATA_2 register

Table 165. LTC_RD_DATA_2 register - Local LTC seconds part. The value read is the saved value of local counter when READ_LTC is set. (address 110Ch)

Bit	Symbol	Access	Value	Description
15:0	LTC_RD_SEC_0	R	0h	[15:0] of LTC_RD_SEC. This reads the value of internal seconds counter.

7.4.1.79 LTC_RD_DATA_3 register

Table 166. LTC_RD_DATA_3 register - Local LTC seconds part. The value read is the saved value of local counter when READ_LTC is set. (address 110Dh)

Bit	Symbol	Access	Value	Description
15:0	LTC_RD_SEC_1	R	0h	[31:16] of LTC_RD_SEC. This reads the value of internal seconds counter.

7.4.1.80 LTC_RD_DATA_4 register

Table 167. LTC_RD_DATA_4 register - Local LTC sub-nanoseconds part. The value read is the saved value of local counter when READ_LTC is set. (address 110Eh)

Bit	Symbol	Access	Value	Description
15:0	LTC_RD_SUBNS	R	0h	[31:16] of sub-ns counter. This reads the value of internal sub-nanoseconds counter.

7.4.1.81 RATE_ADJ_SUBNS_0 register

Table 168. RATE_ADJ_SUBNS_0 register - SW Rate adjust value (address 110Fh)

Bit	Symbol	Access	Value	Description
15:0	RATE_ADJ_ SUBNS_0	R/W	0h	[15:0] of RATE_ADJ_SUBNS. For every clock ticks, this value is added to the sub-nanoseconds counter. When the sub-nanoseconds counter rolls over past zero, then value 1 will be added or subtracted to the nanoseconds counter based on direction bit.

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7.4.1.82 RATE_ADJ_SUBNS_1 register

Table 169. RATE_ADJ_SUBNS_1 register - SW Rate adjust value (address 1110h)

Bit	Symbol	Access	Value	Description
15	CLK_RATE_ADJ_LD	R/W	0h	Loads new value of rate correction by SW. For every clock ticks, value in RATE_ADJ_SUBNS is applied for rate correction. This bit is cleared after 4-5 APB clocks (CDC latency). This bit should be programmed after all required configuration registers has been programmed including SW adjustment is selected.
14	CLK_RATE_ADJ_DIR	R/W	0h	Clock rate adjust direction 0* = For every clock ticks, value in RATE_ADJ_SUBNS is added to the sub-nanoseconds counter. Value 1 is subtracted from the nanoseconds counter when sub-nanoseconds counter rolls over past zero. 1 = For every clock ticks, value in RATE_ADJ_SUBNS is added to the sub-nanoseconds counter. Value 1 is added to the nanoseconds counter when sub-nanoseconds counter rolls over past zero.
13:0	RATE_ADJ_SUBNS_1	R/W	0h	[29:16] of RATE_ADJ_SUBNS. For every clock ticks, this value is added to the sub-nanoseconds counter. When the sub-nanoseconds counter rolls over past zero, then value 1 will be added or subtracted to the nanoseconds counter based on direction bit.

7.4.1.83 TEMP_ADJ_SUBNS_0 register

Table 170. TEMP_ADJ_SUBNS_0 register - SW temporary rate adjust value (address 1111h)

Bit	Symbol	Access	Value	Description
15:0	TEMP_ADJ_SUBNS_0	R/W	0h	[15:0] of TEMP_ADJ_SUBNS. For a programmable clock duration, this value is added to the sub-nanoseconds counter. When the sub-nanoseconds counter rolls over past zero, then value 1 will be added or subtracted to the nanoseconds counter based on direction bit.

7.4.1.84 TEMP_ADJ_SUBNS_1 register

Table 171. TEMP_ADJ_SUBNS_1 register - SW temporary rate adjust value (address 1112h)

Bit	Symbol	Access	Value	Description
15	CLK_LD_TEMP_ADJ	R/W	0h	Setting a bit triggers the temporary adjustment for a programmed duration (TEMP_ADJ_DUR). This bit is cleared after the programmed duration is elapsed (with additional 4-5 APB clocks (CDC latency)). This bit should be programmed after all required configuration registers has been programmed including SW adjustment is selected.
14	CLK_TEMP_ADJ_DIR	R/W	0h	Clock temporary adjustment direction 0* = Value 1 is subtracted from the nanoseconds counter when sub-nanoseconds counter rolls over past zero.

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Table 171. TEMP_ADJ_SUBNS_1 register - SW temporary rate adjust value (address 1112h)*...continued*

Bit	Symbol	Access	Value	Description
				1 = Value 1 is added to the nanoseconds counter when subnanoseconds counter rolls over past zero.
13:0	TEMP_ADJ_SUBNS_1	R/W	0h	[29:16] of TEMP_ADJ_SUBNS. For a programmable clock duration, this value is added to the sub-nanoseconds counter. When the sub-nanoseconds counter rolls over past zero, then value 1 will be added or subtracted to the nanoseconds counter based on direction bit.

7.4.1.85 TEMP_ADJ_DUR_0 register

Table 172. TEMP_ADJ_DUR_0 register - Temporary adjustment window (address 1113h)

Bit	Symbol	Access	Value	Description
15:0	TEMP_ADJ_DUR_0	R/W	0h	[15:0] of TEMP_ADJ_DUR. Number of clock ticks during which temporary adjustment is applied.

7.4.1.86 TEMP_ADJ_DUR_1 register

Table 173. TEMP_ADJ_DUR_1 register - Temporary adjustment window (address 1114h)

Bit	Symbol	Access	Value	Description
15:0	TEMP_ADJ_DUR_1	R/W	0h	[31:16] of TEMP_ADJ_DUR. Number of clock ticks during which temporary adjustment is applied.

7.4.1.87 HW_LTC_LOCK_CTRL register

Table 174. HW_LTC_LOCK_CTRL register - HW LTC self locking.control (address 1115h)

Bit	Symbol	Access	Value	Description
15:1	*reserved*	R	0h	Always write 0, ignore on read.
0	HW_LTC_LOCK_EN	R/W	1h	Rate correction and offset correction is handled by HW. This takes the precedence over SW adjustment.

7.4.1.88 HW_EXT_PPS_NS_0 register

Table 175. HW_EXT_PPS_NS_0 register - Period of external PPS sync in nanosecond. Used for HW LTC self locking. (address 1116h)

Bit	Symbol	Access	Value	Description
15:0	HW_EXT_PPS_NS_0	R/W	CA00h	[15:0] of period of external PPS sync in terms of nanosecond.

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7.4.1.89 HW_EXT_PPS_NS_1 register

Table 176. HW_EXT_PPS_NS_1 register - Period of external PPS sync in nanosecond. Used for HW LTC self locking. (address 1117h)

Bit	Symbol	Access	Value	Description
15:14	*reserved*	R	0h	Always write 0, ignore on read.
13:0	HW_EXT_PPS_NS_1	R/W	3B9Ah	[29:16] of period of external PPS sync in terms of nanosecond.

7.4.1.90 HW_LOCK_ER_LMT_0 register

Table 177. HW_LOCK_ER_LMT_0 register - Error limitation by HW LTC self locking engine (address 1118h)

Bit	Symbol	Access	Value	Description
15:0	HW_LOCK_ER_LMT_0	R/W	D090h	[15:0] of HW_LOCK_ER_LMT. This register limits the number of nanoseconds beyond which HW can not self-lock with external PPS sync. Interrupt is asserted if the error correction limit is exceeded.

7.4.1.91 HW_LOCK_ER_LMT_1 register

Table 178. HW_LOCK_ER_LMT_1 register - Error limitation by HW LTC self locking engine (address 1119h)

Bit	Symbol	Access	Value	Description
15:8	*reserved*	R	0h	Always write 0, ignore on read.
7:0	HW_LOCK_ER_LMT_1	R/W	3h	[23:16] of HW_LOCK_ER_LMT. This register limits the number of nanoseconds beyond which HW can not self-lock with external PPS sync. Interrupt is asserted if the error correction limit is exceeded.

7.4.1.92 LTC_LOOP_CONTROL register

Table 179. LTC_LOOP_CONTROL register - This register defines control parameters to adapt behavior of the LTC loop if HW_LTC_LOCK_EN=1 (address 111Ah)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Always write 0, ignore on read.
11:8	PROP_GAIN	R/W	3h	Proportional path gain as 2^-PROP_GAIN
7:5	*reserved*	R	0h	Always write 0, ignore on read.
4:0	INTEGRAL_GAIN	R/W	3h	Integrating path gain as 2^-INTEGRAL_GAIN

7.4.1.93 EXT_PPS_TS_DATA0 register

Table 180. EXT_PPS_TS_DATA0 register - Timestamping data on the event of rising edge of external pps signal. Reading this register copied into snapshot register for multiple reads of EXT_PPS_TS_DATA0/1/2/3/4/5 register (address 111Bh)

Bit	Symbol	Access	Value	Description
15:0	EXT_PPS_TS_DATA0	R	0h	[15:0] of EXT_PPS_TS_NSEC. Holds nanoseconds part of clock time at the event of rising edge of external pps sync.

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7.4.1.94 EXT_PPS_TS_DATA1 register

Table 181. EXT_PPS_TS_DATA1 register - Timestamping data on the event of rising edge of external pps signal. (address 111Ch)

Bit	Symbol	Access	Value	Description
15:14	*reserved*	R	0h	Ignore on read.
13:0	EXT_PPS_TS_DATA1	R	0h	[29:16] of EXT_PPS_TS_NSEC. Holds nanoseconds part of clock time at the event of rising edge of external pps sync.

7.4.1.95 EXT_PPS_TS_DATA2 register

Table 182. EXT_PPS_TS_DATA2 register - Timestamping data on the event of rising edge of external pps signal. (address 111Dh)

Bit	Symbol	Access	Value	Description
15:0	EXT_PPS_TS_DATA2	R	0h	[15:0] of PTP_EXT_PPS_TS_SEC. Holds seconds part of clock time at the event of rising edge of external pps sync.

7.4.1.96 EXT_PPS_TS_DATA3 register

Table 183. EXT_PPS_TS_DATA3 register - Timestamping data on the event of rising edge of external pps signal. (address 111Eh)

Bit	Symbol	Access	Value	Description
15:0	EXT_PPS_TS_DATA3	R	0h	[31:16] of PTP_EXT_PPS_TS_SEC. Holds seconds part of clock time at the event of rising edge of external pps sync.

7.4.1.97 EXT_PPS_TS_DATA4 register

Table 184. EXT_PPS_TS_DATA4 register - Timestamping data on the event of rising edge of external pps signal. (address 111Fh)

Bit	Symbol	Access	Value	Description
15:0	EXT_PPS_TS_DATA4	R	0h	[31:16] of sub-ns counter. Holds sub-nanoseconds part of clock time at the event of rising edge of external pps sync.

7.4.1.98 EXT_PPS_TS_CTRL register

Table 185. EXT_PPS_TS_CTRL register - External PPS timestamp reading done (address 1120h)

Bit	Symbol	Access	Value	Description
15:1	*reserved*	R	0h	Always write 0, ignore on read.
0	EXT_PPS_TS_DONE	R/W	0h	Once EXT_PPS_TS_DATA is read and there is no need to further re-read the same data set, then this bit should be set to '1'. This is self-clearing bit.

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7.4.1.99 EXT_TRG_TS_DATA0 register

Table 186. EXT_TRG_TS_DATA0 register - Timestamping data on the event of rising edge of external trigger event signal. Reading this register copied into snapshot register for multiple reads of EXT_TRG_TS_DATA0/1/2/3/4/5 register (address 1121h)

Bit	Symbol	Access	Value	Description
15:0	EXT_TRG_TS_ DATA0	R	0h	[15:0] of EXT_TRG_TS_NSEC Holds nanoseconds part of clock time at the event of rising edge of external trigger event signal.

7.4.1.100 EXT_TRG_TS_DATA1 register

Table 187. EXT_TRG_TS_DATA1 register - Timestamping data of external trigger event signal. (address 1122h)

Bit	Symbol	Access	Value	Description
15:14	*reserved*	R	0h	Ignore on read.
13:0	EXT_TRG_TS_ DATA1	R	0h	[29:16] of EXT_TRG_TS_NSEC Holds nanoseconds part of clock time at the event of rising edge of external trigger event signal.

7.4.1.101 EXT_TRG_TS_DATA2 register

Table 188. EXT_TRG_TS_DATA2 register - Timestamping data of external trigger event signal. (address 1123h)

Bit	Symbol	Access	Value	Description
15:0	EXT_TRG_TS_ DATA2	R	0h	[15:0] of EXT_TRG_TS_SEC. Holds seconds part of clock time at the event of rising edge of external trigger event signal.

7.4.1.102 EXT_TRG_TS_DATA3 register

Table 189. EXT_TRG_TS_DATA3 register - Timestamping data of external trigger event signal. (address 1124h)

Bit	Symbol	Access	Value	Description
15:0	EXT_TRG_TS_ DATA3	R	0h	[31:16] of EXT_TRG_TS_SEC. Holds seconds part of clock time at the event of rising edge of external trigger event signal.

7.4.1.103 EXT_TRG_TS_DATA4 register

Table 190. EXT_TRG_TS_DATA4 register - Timestamping data of external trigger event signal. (address 1125h)

Bit	Symbol	Access	Value	Description
15:0	EXT_TRG_TS_ DATA4	R	0h	[31:16] of sub-ns counter. Holds sub-nanoseconds part of clock time at the event of rising edge of external trigger event signal.

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7.4.1.104 EXT_TRG_TS_CTRL register

Table 191. EXT_TRG_TS_CTRL register - External trigger event timestamp reading done (address 1126h)

Bit	Symbol	Access	Value	Description
15:1	*reserved*	R	0h	Always write 0, ignore on read.
0	EXT_TRG_TS_DONE	R/W	0h	Once EXT_TRG_TS_DATA is read and no need to further read the same data set, then this bit should be set to '1'. This is self-clearing bit.

7.4.1.105 PTP_IRQ_SOURCES register

Table 192. PTP_IRQ_SOURCES register - Interrupt status signals (address 1130h)

Bit	Symbol	Access	Value	Description
15:11	*reserved*	R	0h	Always write 0, ignore on read.
10	RX_SOF_LOST	R/W	0h	RX SOF is not detected when ptp event packet is received. Interrupt status is cleared when it is written to 1 (self-clear bit).
9	TX_SOF_LOST	R/W	0h	TX SOF is not detected when ptp event packet is received. Interrupt status is cleared when it is written to 1 (self-clear bit).
8	FRAME_ERROR	R/W	0h	Detected errors in ingress path: 1) Detected pre-amble, but no detection of an SFD. 2) Prematurely ended PTP frame before all required fields for the time-stamp have been received. Interrupt status is cleared when it is written to 1 (self-clear bit).
7	FCS_ERROR	R/W	0h	Incorrect FCS value observed on a received PTP frame. Note that this is only available if ingress in-band time-stamping is enabled. Interrupt status is cleared when it is written to 1 (self-clear bit).
6	ING_TS_LOST	R/W	0h	New ingress time-stamp has overwritten the oldest unread time-stamp in the buffer. Interrupt status is cleared when it is written to 1 (self-clear bit).
5	EGR_TS_LOST	R/W	0h	New egress time-stamp has overwritten the oldest unread time-stamp in the buffer. Interrupt status is cleared when it is written to 1 (self-clear bit).
4	PDIFF_EXCEED	R/W	0h	Interrupt status when hardware measured phase diff between external PPS sync and local LTC is exceeded the limit. (250us is limit if 1 PPS period is 1 second time). Interrupt status is cleared when it is written to 1 (self-clear bit).
3	PPSOUT	R/W	0h	Interrupt status when PPS out is generated. Interrupt status is cleared when it is written to 1 (self-clear bit)
2	SEC_CNT_OVF	R	0h	Interrupt status when 32 bit seconds counter is saturated. Interrupt status is cleared when LTC counters are re-initialized or soft reset is applied.
1	ING_TS	R	0h	This bit is signaling the presence of time-stamps in the ingress buffer. It has been intentionally made non-latching as otherwise a new stamp could be taken between emptying the buffer and resetting this interrupt bit, in which case the new time-stamp wouldn't be flagged. This bit indicating the actual

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Table 192. PTP_IRQ_SOURCES register - Interrupt status signals (address 1130h)*...continued*

Bit	Symbol	Access	Value	Description
				presence of time-stamps in the buffer doesn't require a reset as it will automatically disappear when the buffer becomes empty, and it will be set again when a new time-stamp is taken.
0	EGR_TS	R	0h	This bit is signaling the presence of time-stamps in the egress buffer. It has been intentionally made non-latching as otherwise a new stamp could be taken between emptying the buffer and resetting this interrupt bit, in which case the new time-stamp wouldn't be flagged. This bit indicating the actual presence of time-stamps in the buffer doesn't require a reset as it will automatically disappear when the buffer becomes empty, and it will be set again when a new time-stamp is taken.

7.4.1.106 PTP_IRQ_ENABLES register

Table 193. PTP_IRQ_ENABLES register - Interrupt enable control (address 1131h)

Bit	Symbol	Access	Value	Description
15:11	*reserved*	R	0h	Always write 0, ignore on read.
10	RX_SOF_LOST	R/W	0h	Enable for respective interrupt
9	TX_SOF_LOST	R/W	0h	Enable for respective interrupt
8	FRAME_ERROR	R/W	0h	Enable for respective interrupt
7	FCS_ERROR	R/W	0h	Enable for respective interrupt
6	ING_TS_LOST	R/W	0h	Enable for respective interrupt
5	EGR_TS_LOST	R/W	0h	Enable for respective interrupt
4	PDIFF_EXCEED	R/W	0h	Enable for respective interrupt
3	PPSOUT	R/W	0h	Enable for respective interrupt
2	SEC_CNT_OVF	R/W	0h	Enable for respective interrupt
1	ING_TS	R/W	0h	Enable for respective interrupt
0	EGR_TS	R/W	0h	Enable for respective interrupt

7.4.1.107 PTP_IRQ_MSTATUS register

Table 194. PTP_IRQ_MSTATUS register - Interrupt status signals (address 1132h)

Bit	Symbol	Access	Value	Description
15:11	*reserved*	R	0h	Ignore on read.
10	RX_SOF_LOST	R	0h	Masked status of the respective interrupt
9	TX_SOF_LOST	R	0h	Masked status of the respective interrupt
8	FRAME_ERROR	R	0h	Masked status of the respective interrupt
7	FCS_ERROR	R	0h	Masked status of the respective interrupt
6	ING_TS_LOST	R	0h	Masked status of the respective interrupt
5	EGR_TS_LOST	R	0h	Masked status of the respective interrupt

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Table 194. PTP_IRQ_MSTATUS register - Interrupt status signals (address 1132h)*...continued*

Bit	Symbol	Access	Value	Description
4	PDIFF_EXCEED	R	0h	Masked status of the respective interrupt
3	PPSOUT	R	0h	Masked status of the respective interrupt
2	SEC_CNT_OVF	R	0h	Masked status of the respective interrupt
1	ING_TS	R	0h	Masked status of the respective interrupt
0	EGR_TS	R	0h	Masked status of the respective interrupt

7.4.1.108 PKT_FILT_CTRL register

Table 195. PKT_FILT_CTRL register - Ingress or egress packet filtering (address 1140h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14	VLAN_TAG_ALLOW	R/W	0h	Controls whether VLAN tagged PTP frame is are considered for timestamp. If set, VLAN tagged frames are allowed.
13:6	DOMAIN_NUM	R/W	0h	Domain number to be compared with received versionPTP field to qualify PTP frame.
5	DOMAIN_FILT_EN	R/W	0h	Controls whether domainNumber to be checked against DOMAIN_NUM.
4:1	VERSION	R/W	0h	This value is compared against received versionPTP field to qualify PTP frame.
0	VERSION_FILT_EN	R/W	0h	Controls whether received versionPTP to be checked against programmed version to qualify PTP frame. If set, the version PTP field is checked against VERSION.

7.4.1.109 DA_FILT_CTRL register

Table 196. DA_FILT_CTRL register - Ingress or egress packet DA filtering. (address 1141h)

Bit	Symbol	Access	Value	Description
15:3	*reserved*	R	0h	Always write 0, ignore on read.
2	DA_FILT_EN_USR	R/W	0h	Controls whether user DA can be used for honoring PTP message. 0* = If any DA filter bits are set, then the received DA should match either one of the specified DA of the respective DA filter bit is set. If none of the DA filter bits are set, then all PTP messages are honored. 1 = Match the received DA with programmed user DA. If matched, then honor the PTP message.
1	DA_FILT_EN_FIX2	R/W	1h	Controls whether fixed DA (01:1B:19:00:00:00) can be used for honoring PTP message. 0 = If any DA filter bits are set, then the received DA should match either one of the specified DA of the respective DA filter bit is set. If none of the DA filter bits are set, then all PTP messages are honored. 1* = Match the received DA with (01:1B:19:00:00:00). If matched, then honor the PTP message.

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Table 196. DA_FILT_CTRL register - Ingress or egress packet DA filtering. (address 1141h)*...continued*

Bit	Symbol	Access	Value	Description
0	DA_FILT_EN_FIX1	R/W	1h	Controls whether fixed DA (01:80:C2:00:00:0E) can be used for honoring PTP message. 0 = If any DA filter bits are set, then the received DA should match either one of the specified DA of the respective DA filter bit is set. If none of the DA filter bits are set, then all PTP messages are honored. 1* = Match the received DA with (01:80:C2:00:00:0E). If matched, then honor the PTP message.

7.4.1.110 USER_MAC_DA_0 register

Table 197. USER_MAC_DA_0 register - [15:0] of User defined MAC address (address 1142h)

Bit	Symbol	Access	Value	Description
15:0	USER_MAC_DA_0	R/W	0h	[15:0] of User defined MAC address

7.4.1.111 USER_MAC_DA_1 register

Table 198. USER_MAC_DA_1 register - [31:16] of User defined MAC address (address 1143h)

Bit	Symbol	Access	Value	Description
15:0	USER_MAC_DA_1	R/W	0h	[31:16] of User defined MAC address

7.4.1.112 USER_MAC_DA_2 register

Table 199. USER_MAC_DA_2 register - [47:32] of User defined MAC address (address 1144h)

Bit	Symbol	Access	Value	Description
15:0	USER_MAC_DA_2	R/W	0h	[47:32] of User defined MAC address

7.4.1.113 USER_DA_MASK_0 register

Table 200. USER_DA_MASK_0 register - User MAC DA Mask control (address 1145h)

Bit	Symbol	Access	Value	Description
15:0	USER_DA_MASK_0	R/W	0h	Controls [15:0] of User MAC DA Mask bits 0* = DA bits are compared for matching. 1 = DA bits are ignored from being matched. Hence DA range comparison is possible.

7.4.1.114 USER_DA_MASK_1 register

Table 201. USER_DA_MASK_1 register - User MAC DA Mask control (address 1146h)

Bit	Symbol	Access	Value	Description
15:0	USR_DA_MASK_1	R/W	0h	Controls [31:16] of User MAC DA Mask bits 0* = DA bits are compared for matching. 1 = DA bits are ignored from being matched. Hence, DA range comparison is possible

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7.4.1.115 USER_DA_MASK_2 register

Table 202. USER_DA_MASK_2 register - User MAC DA Mask control (address 1147h)

Bit	Symbol	Access	Value	Description
15:0	USR_DA_MASK_2	R/W	0h	Controls [47:32] of User MAC DA Mask bits 0* = DA bits are compared for matching. 1 = DA bits are ignored from being matched. Hence, DA range comparison is possible

7.4.1.116 EVENT_MSG_FILT register

Table 203. EVENT_MSG_FILT register - PTP event message consideration (address 1148h)

Bit	Symbol	Access	Value	Description
15:4	*reserved*	R	0h	Always write 0, ignore on read.
3	PDLY_RES_FILT	R/W	1h	Controls whether Pdelay_Resp event should be considered for timestamping or not.
2	PDLY_REQ_FILT	R/W	1h	Controls whether Pdelay_Request event should be considered for timestamping or not.
1	DLY_REQ_FILT	R/W	1h	Controls whether Delay_Request event should be considered for timestamping or not.
0	SYNC_FILT	R/W	1h	Controls whether sync event should be considered for timestamping or not.

7.4.1.117 TX_PIPE_DLY_NS register

Table 204. TX_PIPE_DLY_NS register - Egress data path latency from timestamping point of SOF signal to mdi (nanoseconds part). (address 1149h)

Bit	Symbol	Access	Value	Description
15:0	TX_PIPE_DLY_NS	R/W	0h	Egress data path latency from timestamping point of SOF signal to mdi. This is nanoseconds part.

7.4.1.118 TX_PIPEDLY_SUBNS register

Table 205. TX_PIPEDLY_SUBNS register - Egress data path latency from timestamping point of SOF signal to mdi (sub-nanoseconds part) (address 114Ah)

Bit	Symbol	Access	Value	Description
15:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	TX_PIPEDLY_SUBNS	R/W	0h	Egress data path latency from timestamping point of SOF signal to mdi. This is sub-nanoseconds part

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7.4.1.119 RX_PIPE_DLY_NS register

Table 206. RX_PIPE_DLY_NS register - Ingress data path latency of SOF signal from MDI line to timestamping point (nanoseconds part). (address 114Bh)

Bit	Symbol	Access	Value	Description
15:0	RX_PIPE_DLY_NS	R/W	0h	Ingress data path latency of SOF signal from MDI line to timestamping point. This is nanoseconds part.

7.4.1.120 RX_PIPEDLY_SUBNS register

Table 207. RX_PIPEDLY_SUBNS register - Ingress data path latency of SOF signal from MDI line to timestamping point (sub-nanoseconds part) (address 114Ch)

Bit	Symbol	Access	Value	Description
15:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	RX_PIPEDLY_SUBNS	R/W	0h	Ingress data path latency of SOF signal from MDI line to timestamping point. This is sub-nanoseconds part.

7.4.1.121 RX_TS_INSRT_CTRL register

Table 208. RX_TS_INSRT_CTRL register - Inband timestamp insertion controls at ingress path (address 114Dh)

Bit	Symbol	Access	Value	Description
15:9	*reserved*	R	0h	Always write 0, ignore on read.
8	ING_TS_INS_BYPS	R/W	0h	Ingress timestamp insertion bypass 0* = Ingress timestamp is inserted into the “reserved” field of ptp header. 1 = Ingress timestamp is not inserted into the “reserved” field of ptp header
7	ENDIANNESS	R/W	0h	Sequence of insertion of calculated timestamp into the “reserved” field 0* = Little endian: MSByte first, LSByte last 1 = Big endian: LSByte first, MSByte last
6:2	ING_MSG_TS_PTR	R/W	4h	A register holds the binary pointer position of the timestamp. Pointer register defines number of bits of subnanoseconds is inserted along with nanosecond. Max is 20.
1:0	ING_TS_INS_FRMT	R/W	1h	Defines the format of timestamp to be inserted in reserved 4 octets in the PTP message header. 0h = 64-bit nanoseconds counter format1: Inserted timestamp holds lower 32-bit of NANOSECONDS. 64-bit NANOSECONDS is derived from base counters 32-bit seconds counter and 30-bit nanoseconds counter. Base 32bit seconds counter is converted to nanoseconds format and this value is added to base 30bit nanoseconds counter. The end result is in 64-bit NANOSECONDS value. [4.29s rollover] For ex: seconds counter = 32’h0005; nanoseconds counter = 30’h12 3456 Seconds counter value in nanoseconds = 5 * 30’h3B9ACA00 = 62’h1 2A05 F200 Hence, 64-bit NAN

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Table 208. RX_TS_INSRT_CTRL register - Inband timestamp insertion controls at ingress path (address 114Dh)*...continued*

Bit	Symbol	Access	Value	Description
				OSECONDS = 62'h1 2A05 F200 + 30'h123456 = 64'h1 2A18 2656 INSERTED TIMESTAMP = 32'h2A18 2656 1h* = 64-bit nanoseconds counter format2: Inserted timestamp holds lower 32-N bits of NANOSECONDS and MSB Nbits of sub-ns counter value. 64-bit NANOSECONDS is derived from base counters 32-bit seconds counter and 30-bit nanoseconds counter. Base 32bit seconds counter is converted to nanoseconds format and this value is added to base 30bit nanoseconds counter. The end result is in 64-bit NANOSECONDS value. A dedicated register holds the binary pointer position of the timestamp (N). For example if the register value is 0, then inserted timestamp would be lower 32bits of NANOSECONDS; if N=4 then inserted timestamp would be 28 bits of NANOSECONDS and MSB 4-bits of sub-ns counter. [0.268s rollover]. For ex: 64-bit NAN OSECONDS = 64'h1 2A18 2656; 32-bit sub-ns counters = 32'h8000; N = 4 INSERTED TIMESTAMP = 32'hA18 2656 8 2h = 32-bit seconds counter and 30-bit nanoseconds counter format: Inserted timestamp holds lower 2 bits of seconds counter field and 30 bits nanoseconds counter field. (Timestamp = {SEC[1:0], NSEC[29:0]}) 3h = Not used

7.4.1.122 EGR_RING_DATA_0 register

Table 209. EGR_RING_DATA_0 register - Timestamping data of egress ptp event message. Reading this register fetches and latches buffered timestamp data set and puts into EGR_RING_DATA_0/1/2/3/4/5 (address 114Eh)

Bit	Symbol	Access	Value	Description
15:0	EGR_RING_DATA_0	R	0h	[7:0] - Domain number corresponding to the reading TS ring buffer. [11:8] - Message type corresponding to the reading TS ring buffer. [14:12] - [4:2] of seconds counter value of a corresponding reading TS ring buffer. [15] - Indicates valid timestamp information is present in reading TS ring buffer.

7.4.1.123 EGR_RING_DATA_1 register

Table 210. EGR_RING_DATA_1 register - Timestamping data of egress ptp event message. (address 114Fh)

Bit	Symbol	Access	Value	Description
15:0	EGR_RING_DATA_1	R	0h	SequenceId corresponding to the reading TS ring buffer.

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7.4.1.124 EGR_RING_DATA_2 register

Table 211. EGR_RING_DATA_2 register - Timestamping data of egress ptp event message. (address 1150h)

Bit	Symbol	Access	Value	Description
15:0	EGR_RING_DATA_2	R	0h	[15:0] of a nanoseconds counter value of a corresponding reading TS ring buffer.

7.4.1.125 EGR_RING_DATA_3 register

Table 212. EGR_RING_DATA_3 register - Timestamping data of egress ptp event message. (address 1151h)

Bit	Symbol	Access	Value	Description
15:0	EGR_RING_DATA_3	R	0h	[13:0] - [29:16] of a nanoseconds counter value of a corresponding reading TS ring buffer. [15:14] - [1:0] of a seconds counter value of a corresponding reading TS ring buffer.

7.4.1.126 EGR_RING_DATA_4 register

Table 213. EGR_RING_DATA_4 register - Timestamping data of egress ptp event message. (address 1152h)

Bit	Symbol	Access	Value	Description
15:0	EGR_RING_DATA_4	R	0h	[31:16] of a sub-nanoseconds counter value of a corresponding reading TS ring buffer.

7.4.1.127 EGR_RING_DATA_5 register

Table 214. EGR_RING_DATA_5 register - Timestamping data of egress ptp event message. (address 1153h)

Bit	Symbol	Access	Value	Description
15:0	EGR_RING_DATA_5	R	0h	[3:0] - [15:12] of a sub-nanoseconds counter value of a corresponding reading TS ring buffer. [15:4] - [16:5] of a seconds counter value of a corresponding reading TS ring buffer.

7.4.1.128 EGR_RING_CTRL register

Table 215. EGR_RING_CTRL register - Egress timestamp termination control (address 1154h)

Bit	Symbol	Access	Value	Description
15:1	*reserved*	R	0h	Always write 0, ignore on read.
0	EGR_RING_DONE	R/W	0h	Once EGR_RING_DATA is read and no need to further read the same data set, then this bit should be set to '1'. This is self-clearing bit.

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7.4.1.129 ING_RING_DATA_0 register

Table 216. ING_RING_DATA_0 register - Timestamping data of egress ptp event message. Reading this register fetches and latches buffered timestamp data set and puts into ING_RING_DATA_0/1/2/3/4/5 (address 1155h)

Bit	Symbol	Access	Value	Description
15:0	ING_RING_DATA_0	R	0h	[7:0] - Domain number corresponding to the reading TS ring buffer. [11:8] - Message type corresponding to the reading TS ring buffer. [14:12] - [4:2] of seconds counter value of a corresponding reading TS ring buffer. [15] - Indicates valid timestamp information is present in reading TS ring buffer.

7.4.1.130 ING_RING_DATA_1 register

Table 217. ING_RING_DATA_1 register - Timestamping data of Ingress ptp event message. (address 1156h)

Bit	Symbol	Access	Value	Description
15:0	ING_RING_DATA_1	R	0h	SequenceId corresponding to the reading TS ring buffer.

7.4.1.131 ING_RING_DATA_2 register

Table 218. ING_RING_DATA_2 register - Timestamping data of Ingress ptp event message. (address 1157h)

Bit	Symbol	Access	Value	Description
15:0	ING_RING_DATA_2	R	0h	[15:0] of a nanoseconds counter value of a corresponding reading TS ring buffer.

7.4.1.132 ING_RING_DATA_3 register

Table 219. ING_RING_DATA_3 register - Timestamping data of Ingress ptp event message. (address 1158h)

Bit	Symbol	Access	Value	Description
15:0	ING_RING_DATA_3	R	0h	[13:0] - [29:16] of a nanoseconds counter value of a corresponding reading TS ring buffer. [15:14] - [1:0] of a seconds counter value of a corresponding reading TS ring buffer.

7.4.1.133 ING_RING_DATA_4 register

Table 220. ING_RING_DATA_4 register - Timestamping data of Ingress ptp event message. (address 1159h)

Bit	Symbol	Access	Value	Description
15:0	ING_RING_DATA_4	R	0h	[31:16] of a sub-nanoseconds counter value of a corresponding reading TS ring buffer.

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7.4.1.134 ING_RING_DATA_5 register

Table 221. ING_RING_DATA_5 register - Timestamping data of Ingress ptp event message. (address 115Ah)

Bit	Symbol	Access	Value	Description
15:0	ING_RING_DATA_5	R	0h	[3:0] - [15:12] of a sub-nanoseconds counter value of a corresponding reading TS ring buffer. [15:4] - [16:5] of a seconds counter value of a corresponding reading TS ring buffer.

7.4.1.135 ING_RING_CTRL register

Table 222. ING_RING_CTRL register - Ingress timestamp termination control (address 115Bh)

Bit	Symbol	Access	Value	Description
15:1	*reserved*	R	0h	Always write 0, ignore on read.
0	ING_RING_DONE	R/W	0h	Once ING_RING_DATA is read and no need to further read the same data set, then this bit should be set to '1'. This is self-clearing bit.

7.4.1.136 PTP_PKT_RCV_CNT register

Table 223. PTP_PKT_RCV_CNT register - Number of received ptp event packet (address 115Fh)

Bit	Symbol	Access	Value	Description
15:8	*reserved*	R	0h	Ignore on read.
7:0	PTP_PKT_RCV_CNT	R	0h	Number of received ptp event packet

7.4.1.137 PTP_FCS_ERR_CNT register

Table 224. PTP_FCS_ERR_CNT register - Number of received ptp event packet has fcs error. (address 1160h)

Bit	Symbol	Access	Value	Description
15:8	*reserved*	R	0h	Ignore on read.
7:0	PTP_FCS_ERR_CNT	R	0h	Number of received ptp event packet has fcs error. Note that this is only available if ingress in-band time-stamping is enabled.

7.4.1.138 GLOBAL_INFRA_CONTROL register

Table 225. GLOBAL_INFRA_CONTROL register - Global infrastructure control (address 2C00h)

Bit	Symbol	Access	Value	Description
15	RESET	R/W	0h	This resets GPIO controller and LED controller This bit is self-clearing.
14	CONFIG_ENABLE	R/W	0h	When this bit is zero (default), the GLOBAL INFRA configuration registers are write-protected. Note that if bit 30.0040h.13 is set to one, this bit overruled and all GLOBAL configuration registers can be changed.

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Table 225. GLOBAL_INFRA_CONTROL register - Global infrastructure control (address 2C00h)*...continued*

Bit	Symbol	Access	Value	Description
13:0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.139 GPIO_OUTPUTS register

Table 226. GPIO_OUTPUTS register - GPIO output (address 2C20h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Always write 0, ignore on read.
11:0	GPIO_OUTS	R/W	0h	GPIO output register. Bit driven to if FUNC_SELECT=MISC=111 and SIGNAL_SELECT=REGISTER_OUT=10000 for the corresponding GPIO. If the GPIO is configured to any other function, the corresponding bit in this register is ignored. This register determines the logical value driven on one of the GPIOS in GPIO_OUTS[11:0]. A GPIO is driven from this register if FUNC_SELECT is set to MISC and SIGNAL_SELECT is set to REGISTER_OUT

7.4.1.140 GPIO_STATUS register

Table 227. GPIO_STATUS register - GPIO status (address 2C21h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Ignore on read.
0	GPIO_STATUS	R	0h	This register reflects, the current logic level as read from the GPIO pins. Bit GPIO_STATUS[n] corresponds to GPIOn

7.4.1.141 GPIO_IO_CONFIG register

Table 228. GPIO_IO_CONFIG register - GPIO IO configuration (address 2C22h)

Bit	Symbol	Access	Value	Description
15	MANUAL_CONFIG	R/W	0h	If set, SLEW_RATE and DRIVE_STRENGTH are used to set the IO configuration. Otherwise default values are used.
14:11	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
10:8	DRIVE_STRENGTH	R/W	0h	Select drive strength 0 = weakest 7 = strongest
7:3	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
2:0	SLEW_RATE	R/W	0h	Select slew rate of IOs configured for GPIO mode output. 0 = weakest 7 = strongest

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7.4.1.142 GPIO_IO_CLK_CONFIG register

Table 229. GPIO_IO_CLK_CONFIG register - GPIO IO CLK configuration (address 2C23h)

Bit	Symbol	Access	Value	Description
15	MANUAL_CONFIG	R/W	0h	If set, SLEW_RATE and DRIVE_STRENGTH are used to set the IO configuration. Otherwise default values are used.
14:11	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
10:8	DRIVE_STRENGTH	R/W	0h	Select drive strength 0 = weakest 7 = strongest
7:3	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
2:0	SLEW_RATE	R/W	0h	Select slew rate of IOs configured for GPIO CLK mode. 0 = weakest 7 = strongest

7.4.1.143 GPIO0_FUNC_CONFIG register

Table 230. GPIO0_FUNC_CONFIG register - GPIO0 function configuration (GPIOx function registers are identical to this one) (address 2C40h)

Bit	Symbol	Access	Value	Description
15	ENABLE	R/W	1h	If set, GPIO is enabled
14	INVERT	R/W	0h	If set, the input and output is invertered
13	PIN_SNIF	R/W	0h	If set, pin status readable from register GPIO_STATUS.
12:8	*reserved*	R	0h	Always write 0, ignore on read.

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Table 230. GPIO0_FUNC_CONFIG register - GPIO0 function configuration (GPIOx function registers are identical to this one) (address 2C40h)*...continued*

Bit	Symbol	Access	Value	Description
7:5	FUNC_SELECT	R/W	1h	Function select 000 According to default PIN assignment 001* LED (output) 010 PTP 111 MISC others = Reserved
4:0	SIGNAL_SELECT	R/W	0h	Signal select If FUNC_SELECT=LED: xx000 = LED0_A xx001 = LED0_B xx010 = LED1_A xx011 = LED1_B xx100 = LED2_A xx101 = LED2_B xx110 = LED3_A xx111 = LED3_B If FUNC_SELECT=PTP: 0xx00 = PTP_SYNC 0xx01 = PTP_TRIGGER 0xx1x = Reserved 1xx00 = TX_SOF 1xx01 = RX_SOF 1xx10 = PPS_OUT 1xx11 = Reserved If FUNC_SELECT=MISC: 0xx00 = SNIF/REGISTER_IN 1xx00 = REGISTER_OUT 1xx01 = REF_CLK_OUT (only GPIO9) 1xx10 = TX_CLK_OUT (test) others = Reserved

7.4.1.144 PORT_PTP_CONTROL register

Table 231. PORT_PTP_CONTROL register - Port PTP control (address 9000h)

Bit	Symbol	Access	Value	Description
15	RESET	R/W	0h	If set the PTP block is reset This bit is self-clearing.
14:12	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
11	BYPASS	R/W	1h	If set, PTP of this port is in bypass
10:0	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.

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7.4.1.145 PTP_TS_CAPABILITY register

Table 232. PTP_TS_CAPABILITY register - PTP TS capabilities (address 9200h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Ignore on read.
1	TX_DELAY_PATH_CAP	R	1h	If set, PHY provides information on transmit path data delay in registers 3.9201 through 3.9204
0	RX_DELAY_PATH_CAP	R	1h	If set, PHY provides information on receive path data delay in registers 3.9205 through 3.9208

7.4.1.146 PTP_TS_TX_MAX_DELAY_LSB register

Table 233. PTP_TS_TX_MAX_DELAY_LSB register - Maximum PTP pcs transmit path data delay LSBs (address 9201h)

Bit	Symbol	Access	Value	Description
15:0	TX_MAX_DELAY_LSB	R	0h	Maximum PCS transmit path data delay in ns, lower

7.4.1.147 PTP_TS_TX_MAX_DELAY_MSB register

Table 234. PTP_TS_TX_MAX_DELAY_MSB register - Maximum PTP pcs transmit path data delay MSBs (address 9202h)

Bit	Symbol	Access	Value	Description
15:0	TX_MAX_DELAY_MSB	R	0h	Maximum PCS transmit path data delay in ns, upper

7.4.1.148 PTP_TS_TX_MIN_DELAY_LSB register

Table 235. PTP_TS_TX_MIN_DELAY_LSB register - Minimum PTP pcs transmit path data delay MSBs (address 9203h)

Bit	Symbol	Access	Value	Description
15:0	TX_MIN_DELAY_LSB	R	0h	Minimum PCS transmit path data delay in ns, lower

7.4.1.149 PTP_TS_TX_MIN_DELAY_MSB register

Table 236. PTP_TS_TX_MIN_DELAY_MSB register - Minimum PTP pcs transmit path data delay MSBs (address 9204h)

Bit	Symbol	Access	Value	Description
15:0	TX_MIN_DELAY_MSB	R	0h	Minimum PCS transmit path data delay in ns, upper

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7.4.1.150 PTP_TS_RX_MAX_DELAY_LSB register

Table 237. PTP_TS_RX_MAX_DELAY_LSB register - Maximum PTP pcs receive path data delay LSBs (address 9205h)

Bit	Symbol	Access	Value	Description
15:0	RX_MAX_DELAY_LSB	R	0h	Maximum PCS receive path data delay in ns, lower

7.4.1.151 PTP_TS_RX_MAX_DELAY_MSB register

Table 238. PTP_TS_RX_MAX_DELAY_MSB register - Maximum PTP pcs receive path data delay MSBs (address 9206h)

Bit	Symbol	Access	Value	Description
15:0	RX_MAX_DELAY_MSB	R	0h	Maximum PCS receive path data delay in ns, upper

7.4.1.152 PTP_TS_RX_MIN_DELAY_LSB register

Table 239. PTP_TS_RX_MIN_DELAY_LSB register - Minimum PTP pcs receive path data delay MSBs (address 9207h)

Bit	Symbol	Access	Value	Description
15:0	RX_MIN_DELAY_LSB	R	0h	Minimum PCS receive path data delay in ns, lower

7.4.1.153 PTP_TS_RX_MIN_DELAY_MSB register

Table 240. PTP_TS_RX_MIN_DELAY_MSB register - Minimum PTP pcs receive path data delay MSBs (address 9208h)

Bit	Symbol	Access	Value	Description
15:0	RX_MIN_DELAY_MSB	R	0h	Minimum PCS receive path data delay in ns, Upper

7.4.1.154 PORT_BIST_CONTROL register

Table 241. PORT_BIST_CONTROL register - Port BIST control (address A800h)

Bit	Symbol	Access	Value	Description
15	RESET	R/W	0h	Reset This bit is self-clearing.
14	CONFIG_ENABLE	R/W	0h	If set BIST configuration can be changed. Otherwise the BIST configuration is write protected.
13:0	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.

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7.4.1.155 BIST_INTERCEPT_CONFIG register

Table 242. BIST_INTERCEPT_CONFIG register - BIST intercept config (address A807h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
11	CHECK_EPHY_OR_XMII	R/W	0h	If set, EPHY received data to PORT BIST RXD This setting is configuration protected.
10:6	*reserved*	R	0h	Always write 0, ignore on read.
5:4	TX_INTERCEPT	R/W	0h	TX intercept 00 = Normal operation(default) 10 = Intercept CBT1 TXD-path with PORT BIST TXD others = Reserved This setting is configuration protected.
3:2	*reserved*	R	0h	Always write 0, ignore on read.
1:0	RX_INTERCEPT	R/W	0h	RX intercept 00 = Normal operation(default) 10 = Intercept XMII RXD-path with PORT BIST TXD others = Reserved This setting is configuration protected.

7.4.1.156 BIST_GEN_CTRL register

Table 243. BIST_GEN_CTRL register - BIST generator control register (address A880h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14	BIST_GEN_EN	R/W	0h	If set, BIST generator is enabled
13	BIST_GEN_MODE	R/W	0h	Controls whether BIST is operating in continuous mode or production mode. 0* = Production mode. BIST generator generates number of programmed good frame and number of programmed bad frames. 1 = BIST generator continuously generate good frames only. Hence, GOOD_FRAMES_PLAN and BAD_FRAMES_PLAN are ignored.
12	STOP	R/W	0h	STOP control. Stop generating new frames after the currently generated one has completed. This is self-clearing bit.
11:0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.157 BIST_GEN_STATUS register

Table 244. BIST_GEN_STATUS register - BIST generator status (address A881h)

Bit	Symbol	Access	Value	Description
15	BIST_GEN_DONE	R	0h	This status flag is set when programmed number of good frame and bad frame are completely generated and sent. This bit will not get asserted when continuous mode is enabled.

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Table 244. BIST_GEN_STATUS register - BIST generator status (address A881h)*...continued*

Bit	Symbol	Access	Value	Description
14:0	*reserved*	R	0h	Ignore on read.

7.4.1.158 PREAMBLE_IPG_SIZE register

Table 245. PREAMBLE_IPG_SIZE register - Preamble and IPG lengths (address A887h)

Bit	Symbol	Access	Value	Description
15:12	PREAMBLE_ LENGTH	R/W	7h	Length of preamble in bytes.
11:10	*reserved*	R	0h	Always write 0, ignore on read.
9:8	IPG_LENGTH_ TYPE	R/W	0h	Selection of type of IPG length 0h* = IPG length is fixed to IPG_LENGTH 1h = IPG length starts with IPG_LENGTH bytes, incrementing by one every next IPG until 255 is reached. The IPG length rolls-over to IPG_LENGTH bytes after reaching an IPG length of 255 bytes, and starts incrementing from there again. 2h = IPG length is determined by 8bit LFSR. If LFSR value is less than IPG_LENGTH, then IPG length would be IPG_ LENGTH. If LFSR value is greater than or equal to IPG_ LENGTH, then IPG length would be LFSR value. IPG_ LENGTH also acts as seed for 8 bit LFSR.
7:0	IPG_LENGTH	R/W	Ch	Defines minimum length of IPG in bytes and maximum length is decided by 8 bit LFSR. If LFSR value is less than IPG_ LENGTH register then IPG length would be IPG_LENGTH. If LFSR value is greater than or equal to IPG_LENGTH register then IPG length would be LFSR value. IPG_LENGTH also acts as seed for LFSR.

7.4.1.159 BIST_DA_0 register

Table 246. BIST_DA_0 register - lower 16 bits of DA value[15:0] (address A888h)

Bit	Symbol	Access	Value	Description
15:0	BIST_DA_0	R/W	0h	[15:0] of DA.

7.4.1.160 BIST_DA_1 register

Table 247. BIST_DA_1 register - middle 16 bits of DA value[31:16] (address A889h)

Bit	Symbol	Access	Value	Description
15:0	BIST_DA_1	R/W	0h	[31:16] of DA.

7.4.1.161 BIST_DA_2 register

Table 248. BIST_DA_2 register - higher 16 bits of DA value[47:32] (address A88Ah)

Bit	Symbol	Access	Value	Description
15:0	BIST_DA_2	R/W	0h	[47:32] of DA.

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7.4.1.162 BIST_SA_0 register

Table 249. BIST_SA_0 register - lower 16 bits of SA value[15:0] (address A88Bh)

Bit	Symbol	Access	Value	Description
15:0	BIST_SA_0	R/W	0h	[15:0] of SA.

7.4.1.163 BIST_SA_1 register

Table 250. BIST_SA_1 register - middle 16 bits of SA value[31:16] (address A88Ch)

Bit	Symbol	Access	Value	Description
15:0	BIST_SA_1	R/W	0h	[31:16] of SA

7.4.1.164 BIST_SA_2 register

Table 251. BIST_SA_2 register - higher 16 bits of SA value[47:32] (address A88Dh)

Bit	Symbol	Access	Value	Description
15:0	BIST_SA_2	R/W	0h	[47:32] of SA.

7.4.1.165 BIST_PTP_CONFIG register

Table 252. BIST_PTP_CONFIG register - PTP frame generation configuration (address A88Fh)

Bit	Symbol	Access	Value	Description
15:8	PTP_MSG_ INTERVAL	R/W	0h	Configures the interval of ptp event message to be transmitted. For every programmed N frames, one PTP event frame is generated. (N = PTP_MSG_INTERVAL). For ex: for every 10 frames, 9 normal frames are sent and 1 ptp frame is sent. 16'h88F7 is sent as ether type during ptp frame.
7:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	PTP_MSG_TYPE	R/W	0h	PTP message type as per IEEE 1588 PTP header. This configuration enables flexibility of selecting ptp message type. When all condition is met to send PTP frame over Ethernet frame, PTP_MSG_TYPE replaces the first nibble of selected data(constant data/incremental pattern/prbs data) after ether type is sent. PTP frame generation enables basic functional checks of PTP IP and to exercise PTP IP esp. for HTOL testing.

7.4.1.166 BIST_ETHER_TYPE register

Table 253. BIST_ETHER_TYPE register - Ethernet type (T/L) (address A890h)

Bit	Symbol	Access	Value	Description
15:0	ETHER_TYPE	R/W	0h	Ethernet type (T/L).

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7.4.1.167 PAYLOAD_CONFIG register

Table 254. PAYLOAD_CONFIG register - Payload data controls (address A891h)

Bit	Symbol	Access	Value	Description
15:14	*reserved*	R	0h	Always write 0, ignore on read.
13:12	PAYLOAD_DATA_TYPE	R/W	2h	Payload data type 0h = Programmable fixed data is sent 1h = Ramp data is sent(increment by 1) 2h* = PRBS output is sent (see BIST_PRBS_SELECT) 3h = Not used
11:10	*reserved*	R	0h	Always write 0, ignore on read.
9:8	PAYLOAD_SIZE_TYPE	R/W	0h	Indicates the increment in payload size. 0h = No increment in payload size (Always PAYLOAD_SIZE) 1h = Increment always from PAYLOAD_SIZE[7:0] until {PAYLOAD_SIZE[13:8], 8'hFF} and rolls-over to PAYLOAD_SIZE[7:0] again. The limit is between PAYLOAD_SIZE[7:0] & {PAYLOAD_SIZE[13:8], 8'hFF} – where PAYLOAD_SIZE is programmable 2h = For PAYLOAD_SIZE_TYPE= '10'=RANDOM, the payload data length is determined by a 14-bit LFSR (x^14+x^13+x^12+x^2+1). PAYLOAD_SIZE is used as seed. PAYLOAD_SIZE[7:0] sets the minimum payload size. PAYLOAD[13:8] determined the maximum payload size. If LFSR[13:8} =< PAYLOAD_SIZE[13:8], the LFSR[13:0] value determines the payload size. If LFSR[13:8}> PAYLOAD_SIZE[13:8], the payload size is determined by LFSR[9:0]. For PAYLOAD_SIZE[13:8] =05h, the max payload size is 1535 bytes (5FFh). The number of payload data bytes equals PAYLOAD_SIZE[7:0] if LFSR[13:0]<={00h & PAYLOAD_SIZE[7:0]}. This generates frames with a pseudorandom payload size ranging from PAYLOAD_SIZE[7:0] to {PAYLOAD_SIZE[13:8] & FFh} bytes. If PAYLOAD_SIZE[13:8]>05h, jumbo frames will be generated. Jumbo frames up to the max size of 16383 bytes will be generated if PAYLOAD_SIZE[13:8]=3Fh. PAYLOAD_SIZE is used as seed for 14-bit LFSR which is used to derive random payload length. 3h = Not used
7:0	FIXED_PAYLOAD_DATA	R/W	A5h	Byte value that will repeatedly send as frame payload.

7.4.1.168 PAYLOAD_SIZE register

Table 255. PAYLOAD_SIZE register - Length of frame payload data in bytes (address A892h)

Bit	Symbol	Access	Value	Description
15:14	*reserved*	R	0h	Always write 0, ignore on read.
13:0	PAYLOAD_SIZE	R/W	2Eh	Maximum of 1500d=5DCh according to Ethernet standard and 9216d=2400h for jumbo-frames. This 14-bit field allows to send up to 3FFFh=16383 payload bytes in a frame. The minimum number of payload data bytes for Ethernet is 46d=2Eh. If a value smaller than 2Eh is programmed the generator

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Table 255. PAYLOAD_SIZE register - Length of frame payload data in bytes (address A892h)*...continued*

Bit	Symbol	Access	Value	Description
				will work, but runt frames will be generated. The usage of this field depends on PAYLOAD_SIZE_TYPE: PAYLOAD_SIZE_TYPE=FIXED: this field sets a fixed payload size. PAYLOAD_SIZE_TYPE=INCREMENT or RANDOM: bits[7:0] determine the minimum payload size, and bits[13:8] determine the maximum payload size.

7.4.1.169 PRBS_DATA_CONFIG register

Table 256. PRBS_DATA_CONFIG register - PRBS data configuration (address A893h)

Bit	Symbol	Access	Value	Description
15:13	BIST_PRBS_SELECT	R/W	0h	Different type of PRBS is selected 0h = PRBS7 1h = PRBS9 2h = PRBS11 3h = PRBS15 4h = Not used all other conditions
12	*reserved*	R	0h	Always write 0, ignore on read.
11	LFSR_SEED_USAGE	R/W	0h	This bit selects the seed value of the LFSR. 0* = The seed of the LFSR returns to the programmed value for every packet. Seed value is taken from BIST_LFSR_SEED register 1 = First generated ethernet frame uses the programmed seed, and the seed keeps progressing for next frames.
10:0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.170 BIST_LFSR_SEED register

Table 257. BIST_LFSR_SEED register - LFSR Seed value (address A894h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14:0	BIST_LFSR_SEED	R/W	7FFFh	Initial value to be loaded into the LSFR for PRBS data.

7.4.1.171 GOOD_FRAMES_PLAN register

Table 258. GOOD_FRAMES_PLAN register - Number of good frames to be generated by BIST generator (address A8A2h)

Bit	Symbol	Access	Value	Description
15:0	GOOD_FRAMES_PLAN	R/W	64h	Number of good frames to be generated by BIST generator.

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7.4.1.172 G_GOOD_FRAME_CNT register

Table 259. G_GOOD_FRAME_CNT register - Generated good frames in continuous mode. (address A8A4h)

Bit	Symbol	Access	Value	Description
15:0	G_GOOD_FRAME_CNT	R	0h	Number of generated 'good frames' with a correct FCS value. This register stores the number of generated good frames when the packet generator is stopped by the STOP bit. This register wraps on overflow. In production mode, this register counts all frames (good and bad).

7.4.1.173 BAD_FRAMES_PLAN register

Table 260. BAD_FRAMES_PLAN register - Number of bad frames to be generated by BIST generator (address A8A6h)

Bit	Symbol	Access	Value	Description
15:0	BAD_FRAMES_PLAN	R/W	0h	Number of bad frames to be generated by BIST generator

7.4.1.174 BIST_CHECK_CTRL register

Table 261. BIST_CHECK_CTRL register - BIST checker control register (address A8C0h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14	BIST_CHECKER_EN	R/W	0h	BIST checker enable 0* = BIST checker is disabled 1 = BIST checker is enabled
13	STATISTC_CNT_RST	R/W	0h	If this bit is set, then resets all status – BIST_STATUS, GOOD_FRAME_CNT, BAD_FRAME_CNT. This bit resets good frame counter, bad frame counter and rx_er detected frame counter. After reset, all status registers hold fresh values. This bit is self-clearing bit. This bit is used only when BIST is in continuous mode.
12:1	*reserved*	R	0h	Always write 0, ignore on read.
0	BIST_CHECK_MODE	R/W	0h	Controls whether BIST is operating in continuous mode or production mode. 0* = Production mode. BIST checker checks for BIST done and BIST fail status. 1 = Continuous mode. BIST checker keeps checking the good frames, bad frames and rx_er when rx_dv is set. Results are available in statistical registers.

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7.4.1.175 BIST_PROD_STATUS register

Table 262. BIST_PROD_STATUS register - BIST status used when BIST is in production mode and NOT valid for continuous mode (address A8C1h)

Bit	Symbol	Access	Value	Description
15	BIST_CHECK_DONE	R	0h	This bit indicates BIST status is ready for reading. This bit is set when expected good number of frames and bad number of frames is received or timed out.
14:5	*reserved*	R	0h	Ignore on read.
4	BIST_CHECK_FAIL	R	0h	Indicates at least one of the following error status is set: 1. Good frame error: If the expected number of good frames are not received, then bit flag is set. 2. Bad frame error: If the expected number of bad frames are not received, then bit flag is set (Error is injected, but not received) 3. rx_dv is not detected: If RX_DV is not received by the BIST checker, then this status is set. 4. RX_ER detected: This status is set when rx_er is set with rx_dv is high If BIST_CHECK_DONE is set and BIST_CHECK_FAIL is logic zero, then it is said to be PASSED. If BIST_CHECK_DONE is set and BIST_CHECK_FAIL is logic high, then it is said to be FAIL.
3	RX_ER_DETECT_ER	R	0h	This status is set when rx_er is set with rx_dv is high
2	RXDV_DETECT_ER	R	0h	If RX_DV is not received by the BIST checker, then this status is set.
1	BAD_FRAME_ERROR	R	0h	If the expected number of bad frames are not received, then bit flag is set.
0	GOOD_FRAME_ERROR	R	0h	If the expected number of good frames are not received, then bit flag is set

7.4.1.176 BIST_WAIT_TIMER register

Table 263. BIST_WAIT_TIMER register - Timer timeout value for checking all status in production mode. This timeout is useful when expected number of generated frames is never received. (address A8C4h)

Bit	Symbol	Access	Value	Description
15:0	BIST_WAIT_TIMER	R/W	FFh	Timer timeout value for checking all status in production mode. This count begins when BIST checker is enabled. Hence this register should be programed before BIST checker is enabled. Ideally this value should be- time taken to send all frames by the BIST generator plus latency from BIST generator to BIST checker plus additional time to program BIST generator and checker enable bit. This is time out register and is useful when expected good number of frames and bad number of frames is not received. In this condition, time out asserts BIST_CHECK_DONE status.

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Table 263. BIST_WAIT_TIMER register - Timer timeout value for checking all status in production mode. This timeout is useful when expected number of generated frames is never received. (address A8C4h)

Bit	Symbol	Access	Value	Description
				This count is in terms of microseconds. ***Note:*** *This is 16 bit value hence can support up to 65 ms only.*

7.4.1.177 R_GOOD_FRAME_CNT register

Table 264. R_GOOD_FRAME_CNT register - Received good frame count (address A8D0h)

Bit	Symbol	Access	Value	Description
15:0	R_GOOD_FRAME_CNT	R	0h	Counts the number received frames for which CRC magic number is matched. This can be used in both production mode and continuous mode. This counter can be reset by programmable bit to count fresh good frame (in continuous mode).

7.4.1.178 R_BAD_FRAME_CNT register

Table 265. R_BAD_FRAME_CNT register - Received bad frame count (address A8D2h)

Bit	Symbol	Access	Value	Description
15:0	R_BAD_FRAME_CNT	R	0h	Counts the number received frames for which CRC magic number is NOT matched. This can be used in both production mode and continuous mode. This counter can be reset by programmable bit to count fresh bad frame (in continuous mode)

7.4.1.179 R_RXER_FRAME_CNT register

Table 266. R_RXER_FRAME_CNT register - RX_ER detected when rx_dv is set. (address A8D4h)

Bit	Symbol	Access	Value	Description
15:0	R_RXER_FRAME_CNT	R	0h	Counts the number received frames for which rx_er is detected when rx_dv is high. This can be used in both production mode and continuous mode. This counter can be reset by programmable bit to count fresh bad frame (in continuous mode).

7.4.1.180 PORT_INFRA_CONTROL register

Table 267. PORT_INFRA_CONTROL register - Port infrastructure control (address AC00h)

Bit	Symbol	Access	Value	Description
15	RESET	R/W	0h	Resets CMOS xMII. Registers which are set by pin strap values, are set to what was latched during the initial pin strap sequence. This bit is self-clearing.

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Table 267. PORT_INFRA_CONTROL register - Port infrastructure control (address AC00h)*...continued*

Bit	Symbol	Access	Value	Description
14	CONFIG_ENABLE	R/W	0h	If set, the infrastructure configuration can be changed. Otherwise writes are ignored.
13:0	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.

7.4.1.181 INFRA_ABILITY register

Table 268. INFRA_ABILITY register - Infrastsructure ability (address AC04h)

Bit	Symbol	Access	Value	Description
15	XMII_ABILITY	R	1h	InfraStruture xMII Ability
14:1	*reserved*	R	0h	Ignore on read.
0	*reserved*	R	0h	Ignore on read.

7.4.1.182 INFRA_CONFIG register

Table 269. INFRA_CONFIG register - Infrastructure configuration (address AC06h)

Bit	Symbol	Access	Value	Description
15:1	*reserved*	R	0h	Always write 0, ignore on read.
0	TEST_ENABLE	R/W	0h	If enabled, Infrasstructure related test modes can be enabled This setting is configuration protected.

7.4.1.183 INFRA_IRQ_SOURCES register

Table 270. INFRA_IRQ_SOURCES register - Infrastructure IRQ sources (address AC08h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14	FIFO_ERROR	R/W	0h	If set, elastic FIFO underflow or overflow. Write one to clear. This bit is latched-high.
13	RGMII_ID_LOCK_ERROR	R/W	0h	If set, Lock-error of internal delay(s) of RGMII. Write one to clear. This bit is latched-high.
12:0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.184 INFRA_IRQ_ENABLES register

Table 271. INFRA_IRQ_ENABLES register - Infrastructure IRQ enables (address AC0Ah)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14	FIFO_ERROR	R/W	0h	If set, the respective IRQ is enabled This setting is preserved during deep sleep.

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Table 271. INFRA_IRQ_ENABLES register - Infrastructure IRQ enables (address AC0Ah)*...continued*

Bit	Symbol	Access	Value	Description
13	RGMII_ID_LOCK_ERROR	R/W	0h	If set, the respective IRQ is enabled This setting is preserved during deep sleep.
12:0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.185 INFRA_IRQ_MSTATUS register

Table 272. INFRA_IRQ_MSTATUS register - Infrastructure IRQ masked status (address AC0Ch)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Ignore on read.
14	FIFO_ERROR	R	0h	Masked status of the respective interrupt
13	RGMII_ID_LOCK_ERROR	R	0h	Masked status of the respective interrupt
12:0	*reserved*	R	0h	Ignore on read.

7.4.1.186 XMII_CONTROL register

Table 273. XMII_CONTROL register - XMII control (address AFC0h)

Bit	Symbol	Access	Value	Description
15	RESET	R/W	0h	Reset xMII block. Reloads all OTP configuration and reads pinstrap settings This bit is self-clearing.
14:3	*reserved*	R	0h	Always write 0, ignore on read.
2	LOGICAL_ISOLATE	R/W	0h	When set, xMII data and control signals become inactive. Output clocks (if any) keep running. TX data/control pins on the xMII interface are ignored. RX data/control pins are tied to low.
1	MAC_LOOPBACK	R/W	0h	If set, data received on external xMII port is looped on internal (G)MII, and re-transmitted on external xMII port. This setting is test protected.
0	PHY_LOOPBACK	R/W	0h	If set, data received on internal (G)MII is looped on xMII, and re-transmitted on internal (G)MII. This setting is test protected.

7.4.1.187 XMII_STATUS register

Table 274. XMII_STATUS register - XMII status (address AFC2h)

Bit	Symbol	Access	Value	Description
15	TXD_FIFO_OVERFLOW	R/W	0h	If set, Overflow of transmit path FIFO Write one to clear. This bit is latched-high.
14	TXD_FIFO_UNDERFLOW	R/W	0h	If set, Underflow of transmit path FIFO Write one to clear. This bit is latched-high.

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Table 274. XMII_STATUS register - XMII status (address AFC2h)*...continued*

Bit	Symbol	Access	Value	Description
13	RXD_FIFO_ OVERFLOW	R/W	0h	If set, Overflow of receive path FIFO Write one to clear. This bit is latched-high.
12	RXD_FIFO_ UNDERFLOW	R/W	0h	If set, Underflow of receive path FIFO Write one to clear. This bit is latched-high.
11	RGMII_TXID_ LOCK_ERROR	R/W	0h	If set, Lock-error of integrated delay in TXC Write one to clear. This bit is latched-high.
10	RGMII_RXID_ LOCK_ERROR	R/W	0h	If set, Lock-error of integrated delay in RXC Write one to clear. This bit is latched-high.
9:0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.188 XMII_ABILITIES register

Table 275. XMII_ABILITIES register - XMII abilities (address AFC4h)

Bit	Symbol	Access	Value	Description
15	RGMII_ID_ABILITY	R	A/C: 1h B/D: 0h	If set, Ability to add Internal Delay in RXC/TXC clock signals for RGMII
14	RGMII_ABILITY	R	A/C: 1h B/D: 0h	If set, Ability to support RGMII for this PORT
13:12	*reserved*	R	0h	Ignore on read.
11	RMII_master_ ABILITY	R	A/C: 1h B/D: 0h	If set, Ability to support RMII with clock output (MAC-master) for this PORT
10	RMII_slave_ABILITY	R	A/C: 1h B/D: 0h	If set, Ability to support RMII with clock input (PHY or MAC-slave) for this PORT
9	MII_master_ABILITY	R	A/C: 1h B/D: 0h	If set, Ability to support MII with clock outputs (clock-master, PHY-mode) for this PORT
8	MII_slave_ABILITY	R	A/C: 1h B/D: 0h	If set, Ability to support MII with clock input (clock-slave, MAC-mode) for this PORT
7:1	*reserved*	R	0h	Ignore on read.
0	SGMII_ABILITY	R	A/C: 0h B/D: 1h	If set, Ability to support SERDES-MII on this PORT. Specific SERDES abilities in AFC5h.

7.4.1.189 SERDES_MII_ABILITIES register

Table 276. SERDES_MII_ABILITIES register - SERDES MII abilities (address AFC5h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Ignore on read.
1	SGMII_ABILITY	R	A/C : 0h B/D : 1h	If set, SGMII is available on this PORT

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Table 276. SERDES_MII_ABILITIES register - SERDES MII abilities (address AFC5h)*...continued*

Bit	Symbol	Access	Value	Description
0	*reserved*	R	0h	Ignore on read.

7.4.1.190 MII_BASIC_CONFIG register

Table 277. MII_BASIC_CONFIG register - MII basic configuration (address AFC6h)

Bit	Symbol	Access	Value	Description
15:10	*reserved*	R	0h	Always write 0, ignore on read.
9	*reserved*	R/W	0h	Always write 0, ignore on read.
8	JUMBO_EN	R/W	0h	If set, enable support for jumbo packets (16K) This setting is configuration protected.
7:5	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
4	ROLE	R/W	0h	For A/C • If set, the device will operate in "reversed" role mode. • If XMII_MODE is set to MII, and ROLE is set, the device operates in revMII mode (TXCLK and RXCLK are input). • If XMII_MODE is set to RMII, and ROLE is set, the device operates in revRMII mode (REF_CLK is output). For B/D • If ROLE is set, SGMII operate in MAC mode. This setting is configuration protected. This setting is preserved during deep sleep.
3	CMOS_OR_SERDES	R	A/C = 0h B/D = 1h	CMOS or SERDES This setting is configuration protected. 0 = CMOS 1 = SERDES
2:0	XMII_MODE	R/W	0h	The values of the fields CMOS_OR_SERDES, XMII_MODE determine the XMII mode of operation. All possible mode encodings are listed in Table 4. Do not write undefined values. This setting is configuration protected. 0000 = MII 0101 = RMII 0111 = RGMII 1000 = SGMII This setting is preserved during deep sleep.

7.4.1.191 XMII_CLK_CONFIG register

Table 278. XMII_CLK_CONFIG register - XMII clock configuration (address AFC8h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
14	CLOCK_DISABLE	R/W	0h	If set, xMII clock output(s) disabled This setting is configuration protected.
13:12	*reserved*	R	0h	Always write 0, ignore on read.

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Table 278. XMII_CLK_CONFIG register - XMII clock configuration (address AFC8h)*...continued*

Bit	Symbol	Access	Value	Description
11	RMII_CLK_REF	R/W	0h	If RMII and ROLE=1, the output clock signal is synchronous with the device REF_CLK input (XTAL or externally provided reference clock) If set, If RMII and and ROLE=1, the input clock signal is also used as clock reference for the transceiver. If RMII in MAC-mode, the output clock signal is synchronous with transceiver communication (PHY-locked) This setting is configuration protected. This setting is preserved during deep sleep.
10:0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.192 XMII_CLK_IO_CONFIG register

Table 279. XMII_CLK_IO_CONFIG register - XMII CLK IO configuration (address AFC9h)

Bit	Symbol	Access	Value	Description
15	MANUAL_CONFIG	R/W	0h	If set, SLEW_RATE and DRIVE_STRENGTH are used to set the xMII clock output IO configuration. Otherwise default values are used. Caution: a manual override may violate the xMII timing in some or all operating conditions. The timing values in the dynamic characteristics section of the data sheet are valid under default or ALTERNATE=1 settings, only. This setting is configuration protected.
14:11	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
10:8	DRIVE_STRENGTH	R/W	0h	Select drive strength selection This setting is configuration protected.
7:3	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
2:0	SLEW_RATE	R/W	0h	Select slew rate selection 0 = weakest 7 = strongest This setting is configuration protected.

7.4.1.193 XMII_DATA_CONFIG register

Table 280. XMII_DATA_CONFIG register - XMII data configuration (address AFCAh)

Bit	Symbol	Access	Value	Description
15	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
14	TXD30_SWAP	R/W	0h	TXD swap. If set, the TXD nibble is reversed to TXD[0:3] This setting is configuration protected. This setting is preserved during deep sleep.
13:11	*reserved*	R	0h	Always write 0, ignore on read.
10	TXER_GPIO_SWAP	R/W	0h	Controls TX_ER pin function This setting is configuration protected. 0* = classic TX_ER functionality 1 = TX_ER is available as GPIO9

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Table 280. XMII_DATA_CONFIG register - XMII data configuration (address AFCAh)*...continued*

Bit	Symbol	Access	Value	Description
9:8	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
7	RXD_DISABLE	R/W	0h	If set, RXD and XMII RXCTL,RXER outputs drive a low value except RXC This setting is configuration protected.
6	*reserved*	R	0h	Always write 0, ignore on read.
5:4	RXCTL_SELECT	R/W	0h	Alternate RXCTL selection This setting is configuration protected. 00* = Default RXCTL pin position. RXCTL next to RXC 11 = Alternate RXCTL position. RXCTL at other side of RXD[3:0] others = Reserved This setting is preserved during deep sleep.
3	*reserved*	R	0h	Always write 0, ignore on read.
2	RXER_GPIO_SWAP	R/W	0h	Controls RX_ER pin function This setting is configuration protected. 0* = classic RX_ER functionality 1 = RX_ER is available as GPIO2
1:0	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.

7.4.1.194 XMII_DATA_IO_CONFIG register

Table 281. XMII_DATA_IO_CONFIG register - XMII Data IO configuration (address AFCBh)

Bit	Symbol	Access	Value	Description
15	MANUAL_CONFIG	R/W	0h	If set, SLEW_RATE and DRIVE_STRENGTH are used to set the xMII data output IO configuration. Otherwise default values are used. Caution: a manual override may violate the xMII timing in some or all operating conditions. The timing values in the dynamic characteristics section of the data sheet are valid under default or ALTERNATE=1 settings, only. This setting is configuration protected.
14	ALTERNATE_CONFIG	R/W	0h	If set, reduced emission selection for XMII data and clock pads (up to 15 pF load) This setting is configuration protected.
13:11	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
10:8	DRIVE_STRENGTH	R/W	0h	Select drive strength of DATA output IO cells used for the configured CMOS xMII mode This setting is configuration protected.
7:3	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
2:0	SLEW_RATE	R/W	0h	Select slew rate of IOs configured for CMOS xMII mode. 0 = weakest 7 = strongest This setting is configuration protected.

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7.4.1.195 RGMII_TXC_DELAY_CONFIG register

Table 282. RGMII_TXC_DELAY_CONFIG register - RGMII TXC delay configuration (address AFCCh)

Bit	Symbol	Access	Value	Description
15	ENABLE	R/W	0h	If set, internal TXC delay enabled This setting is configuration protected. This setting is preserved during deep sleep.
14:5	*reserved*	R	0h	Always write 0, ignore on read.
4:0	PHASE_SHIFT	R/W	12h	The Phase shift or Internal delay is as per below equation Phase-shift = 73.8 + 0.9*PHASE_SHIFT[4:0] degrees This setting is configuration protected. This setting is preserved during deep sleep.

7.4.1.196 RGMII_RXC_DELAY_CONFIG register

Table 283. RGMII_RXC_DELAY_CONFIG register - RGMII RXC delay configuration (address AFCDh)

Bit	Symbol	Access	Value	Description
15	ENABLE	R/W	0h	If set, internal RXC delay enabled This setting is configuration protected. This setting is preserved during deep sleep.
14:5	*reserved*	R	0h	Always write 0, ignore on read.
4:0	PHASE_SHIFT	R/W	12h	The Phase shift or Internal delay is as per below equation Phase-shift = 73.8 + 0.9*PHASE_SHIFT[4:0] degrees This setting is configuration protected. This setting is preserved during deep sleep.

7.4.1.197 RX_PREAMBLE_COUNT register

Table 284. RX_PREAMBLE_COUNT register - RX preamble counter (address AFCEh)

Bit	Symbol	Access	Value	Description
15	ENABLE	R/W	0h	Enable for capturing RX Preamble Count This setting is configuration protected.
14:6	*reserved*	R	0h	Always write 0, ignore on read.
5:0	COUNT	R	0h	Captures Lowest Preamble count on receive side of the PHY's PCS interface. If read, the counter gets cleared, and again starts looking for lowest preamble count. The count is in terms of number of nibbles for 100Base-T1 This flag is cleared on read. This setting is configuration protected.

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7.4.1.198 TX_PREAMBLE_COUNT register

Table 285. TX_PREAMBLE_COUNT register - TX preamble counter (address AFCFh)

Bit	Symbol	Access	Value	Description
15	ENABLE	R/W	0h	Enable for capturing TX Preamble Count This setting is configuration protected.
14:6	*reserved*	R	0h	Always write 0, ignore on read.
5:0	COUNT	R	0h	Captures Lowest Preamble count on transmit side of the PHY's PCS interface. If read, the counter gets cleared, and again starts looking for lowest preamble count. The count is in terms of number of nibbles for 100Base-T1 This flag is cleared on read.

7.4.1.199 RX_IPG_LENGTH register

Table 286. RX_IPG_LENGTH register - RX IPG length capture (address AFD0h)

Bit	Symbol	Access	Value	Description
15	ENABLE	R/W	0h	Enable for capturing RX IPG Length This setting is configuration protected.
14:9	*reserved*	R	0h	Always write 0, ignore on read.
8:0	LENGTH	R	0h	Captures Lowest IPG Length on receive side of the PHY's PCS interface. If read, the IPG length gets cleared, and again starts looking for lowest IPG length. The length is reported in multiple of 4bits. This flag is cleared on read.

7.4.1.200 TX_IPG_LENGTH register

Table 287. TX_IPG_LENGTH register - TX IPG length capture (address AFD1h)

Bit	Symbol	Access	Value	Description
15	ENABLE	R/W	0h	Enable for capturing TX IPG Length This setting is configuration protected.
14:9	*reserved*	R	0h	Always write 0, ignore on read.
8:0	LENGTH	R	0h	Captures Lowest IPG Length on transmit side of the PHY's PCS interface. If read, the IPG length gets cleared, and again starts looking for lowest IPG Length. The length is in terms of The length is reported in multiple of 4bits. This flag is cleared on read.

7.4.1.201 SGMII_BASIC_CONTROL register

Table 288. SGMII_BASIC_CONTROL register - SGMII PCS register (address B000h)

Bit	Symbol	Access	Value	Description
15	RST	RW	0h	Soft Reset.

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Table 288. SGMII_BASIC_CONTROL register - SGMII PCS register (address B000h)*...continued*

Bit	Symbol	Access	Value	Description
				When the host sets this bit, this triggers the software reset process in which all internal blocks are reset, except the APB interface block. The registers are reset to their default values. When this bit is set, it also resets the PMA & PMD. This bit is self-cleared after 32 clock cycles of the APB bus clock. Type: Self Clearing.
14	LBE	RW	0h	Not used
13	SS13	RW	1h	Speed Selection (LSB). This bit, along with the SS6 bit of this register, indicates the speed. • SS6 SS13 = Speed • 1 0 = 1000 Mbps • 0 1 = 100 Mbps • 0 0 = 10 Mbps
12	AN_ENABLE	RW	1h	Enable Auto-Negotiation. When set to 1, this bit enables the Clause 37 auto-negotiation process.
11	LPM	RW	0h	Power-Down Mode. This bit controls the powerdown mode of the PCS. • 0 = Normal operation • 1 = The PCS goes to the powerdown mode When the host clears this bit, the PCS resumes the normal operation.
10	*reserved*	R	0h	Always write 0, ignore on read.
9	RESTART_AN	RW	0h	Restart Auto-Negotiation. When the host writes this bit, the PCS initiates the auto-negotiation process. This bit is used to restart the auto-negotiation which is already initiated by setting AN_ENABLE. The PCS clears this bit after restarting the autonegotiation. Type: Self Clearing.
8	DUPLEX_MODE	RW	0h	Duplex Mode. This bit specifies the duplex mode of the PCS. • 0 = Half duplex • 1 = Full duplex If AN_ENABLE is set to 0, this bit determines the PHY link duplex mode. If AN_ENABLE is set to 1, then the PHY link duplex mode is independent of this bit (although the host can write any value) and is determined by the outcome of the Clause 37 auto-negotiation process. Note that only full duplex mode is supported.
7	*reserved*	R	0h	Always write 0, ignore on read.
6	SS6	RW	0h	Speed Selection (MSB). This bit, along with the SS13 bit of this register, indicates the speed. For more information, see description of the SS13 bit.
5:0	*reserved*	R	0h	ignore on read.

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7.4.1.202 SGMII_BASIC_STATUS register

Table 289. SGMII_BASIC_STATUS register - SGMII PCS register (address B001h)

Bit	Symbol	Access	Value	Description
15	ABL100T4	R	0h	100BASE-T4 Ability. Always returns 0 because this functionality is not supported.
14	FD100ABL	R	0h	100BASE-X Full-Duplex Ability. Always returns 0 because this functionality is not supported.
13	HD100ABL	R	0h	100BASE-X Half-Duplex Ability. Always returns 0 because this functionality is not supported.
12	FD10ABL	R	0h	10 Mbps Full-Duplex Ability. Always returns 0 because this functionality is not supported.
11	HD10ABL	R	0h	10 Mbps Half-Duplex Ability. Always returns 0 because this functionality is not supported.
10	FD100T	R	0h	100BASE-T2 Full-Duplex Ability. Always returns 0 because this functionality is not supported.
9	HD100T	R	0h	100BASE-T2 Half-Duplex Ability. Always returns 0 because this functionality is not supported.
1	*reserved*	R	0h	Ignore on read.
7	UN_DIR_ABL	R	0h	Unidirectional Ability. • 0 = The PCS is able to transmit frames only when the device has determined the valid link • 1 = The PCS is able to transmit frames irrespective of whether device has determined the valid link or not
6	MF_PRE_SUP	R	0h	MF Preamble Suppression. • 0 = The PCS accepts the MDIO frames with preamble suppressed • 1 = The PCS does not accept the MDIO frames with preamble suppressed Always returns 0.
5	AN_CMPL	R	0h	Auto-negotiation Complete. • 0 = The AN process is not completed • 1 = The AN process is completed When this bit is set to 1, the contents of the AN MMD Advertizement, AN MMD Link partner Ability, and AN MMD Expansion registers are valid. This bit returns 0 if AN_ENABLE is set to 0.
4	RF	R	0h	Remote Fault. When set to 1, this bit indicates that the receive link of the link partner is down. This bit is set based on the auto-negotiated (1000BASE-X auto-negotiation) information from the link partner. • 0 = The PCS did not detect a remote fault • 1 = The PCS detected a remote fault Note that this bit is only valid when using 1000BASE-X auto-negotiation (PCS_MODE=0x0) because remote fault information is part of the exchanged base page. When using SGMII autonegotiation (PCS_MODE=0x2) this bit must be ignored since remote fault information is not exchanged.

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Table 289. SGMII_BASIC_STATUS register - SGMII PCS register (address B001h)*...continued*

Bit	Symbol	Access	Value	Description
				Type: Latch High.
3	AN_ABL	R	1h	Auto-negotiation Ability. Always returns 1 because the PCS is able to perform auto-negotiation.
2	LINK_STS	R	0h	Link Status. When set to 1, it indicates that the Rx link is up. If the link goes down, it is latched until the host performs the read operation to this register. • 0 = Link Down • 1 = Link Up Type: Latch Low.
1	*reserved*	R	0h	Ignore on read.
0	EXT_REG_CAP	R	1h	Extended Register Capability. • 0 = Extended Register capability does not exist • 1 = Extended Register capability exists Always returns 1.

7.4.1.203 SGMII_SR_MII_DEV_ID1 register

Table 290. SGMII_SR_MII_DEV_ID1 register - SGMII PCS register (address B002h)

Bit	Symbol	Access	Value	Description
15:0	VS_MII_DEV_OUI_3_18	R	0h	Organizationally Unique Identifier[3:18]. This field contains Bits[18:3] of 24-bit OUI of device manufacturer.

7.4.1.204 SGMII_SR_MII_DEV_ID2 register

Table 291. SGMII_SR_MII_DEV_ID2 register - SGMII PCS register (address B003h)

Bit	Symbol	Access	Value	Description
15:10	VS_MMD_DEV_OUI_19_24	R	0h	Organizationally Unique Identifier[19:24]. This field contains Bits[24:19] of 24-bit OUI of device manufacturer.
9:4	VS_MMD_DEV_MMN_5_0	R	0h	Model Number. This field contains the 6-bit Model number of the device.
3:0	VS_MMD_DEV_RN_3_0	R	0h	Revision Number. This field contains the 4-bit Revision number of the device.

7.4.1.205 SGMII_AUTONEG_ADVERTISE register

Table 292. SGMII_AUTONEG_ADVERTISE register - SGMII PCS register (address B004h)

Bit	Symbol	Access	Value	Description
15	NP	R	0h	Next Page / SGMII Link Status. Always returns 0 when using 1000BASE-X auto-negotiation (PCS_MODE=0x0) because the Next Page feature is not supported.

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Table 292. SGMII_AUTONEG_ADVERTISE register - SGMII PCS register (address B004h)*...continued*

Bit	Symbol	Access	Value	Description
				When using SGMII autonegotiation (PCS_MODE=0x2) Next Page information is not exchanged. Instead, in SGMII PHY mode (TX_CONFIG=1), this field is read-only and indicates the PHY's Link Status that is communicated (advertised) to the SGMII link partner, in line with the definition of the tx_config_Reg in the SGMII specification. The value that is advertised depends on the setting of PHY_MODE_CTRL and bit 4 (SGMII_LINK_STS) of SGMII_PCS_CONFIG2.
14	*reserved*	R	0h	Always write 0, ignore on read.
13:12	RF	RW	0h	Remote Fault / SGMII Duplex Mode. This field indicates the fault signaling of the local device to be communicated (advertised) to the link partner. • 00 = No Error • 01 = Offline • 10 = Link Failure • 11 = Auto-negotiation Error Note that this field is only valid when using 1000BASE-X auto-negotiation (PCS_MODE=0x0). When using SGMII autonegotiation (PCS_MODE=0x2) remote fault information is not exchanged. Instead, in SGMII PHY mode (TX_CONFIG=1), bit 12 is read-only and indicates the duplex mode of operation that is communicated (advertised) to the SGMII link partner, in line with the definition of the tx_config_Reg in the SGMII specification. The value that is advertised depends on the setting of PHY_MODE_CTRL and bit 5 (FD) of SGMII_AUTONEG_ADVERTISE. • 0 = Half Duplex • 1 = Full Duplex
11:9	*reserved*	R	0h	Always write 0, ignore on read.
8:7	PAUSE	RW	0h	Pause Ability. This field indicates the Pause ability of the local device to be communicated (advertised) to the link partner. Software can program suitable values based on the capability of the MAC. • 00 = No Pause • 01 = Asymmetric Pause towards the link partner • 10 = Symmetric Pause • 11 = Symmetric Pause and Asymmetric Pause towards the local device Note that this field is only valid when using 1000BASE-X auto-negotiation (PCS_MODE=0x0). When using SGMII autonegotiation (PCS_MODE=0x2) Pause Ability information is not exchanged.
6	HD	RW	0h	Half Duplex. When this bit is set, it indicates that the device can operate in the half-duplex mode. ***Note:*** *Only valid when using 1000BASE-X auto-negotiation (PCS_MODE=0x0). When using SGMII autonegotiation (PCS_MODE=0x2) duplex information is not exchanged and this field must be ignored.*

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Table 292. SGMII_AUTONEG_ADVERTISE register - SGMII PCS register (address B004h)*...continued*

Bit	Symbol	Access	Value	Description
				Half-duplex operation is not supported at chip level.
5	FD	RW	1h	Full Duplex. When this bit is set, it indicates that the device can operate in the full-duplex mode. ***Note:*** *Only valid when using 1000BASE-X auto-negotiation (PCS_MODE=0x0). When using SGMII autonegotiation (PCS_MODE=0x2) duplex information is not exchanged and this field must be ignored.*
4:0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.206 SGMII_AUTONEG_LP_ABILITY register

Table 293. SGMII_AUTONEG_LP_ABILITY register - SGMII PCS register (address B005h)

Bit	Symbol	Access	Value	Description
15	LP_NP	R	0h	Link Partner Next Page / Link Partner Link Status. This bit indicates that the link partner can handle Next Page. To exchange information through Next Page, both devices (local and remote) should have the capability to handle Next Page. The PCS does not support Next Page (see SGMII_AUTONEG_ADVERTISE.NP) therefore, the Next Page exchange does not happen. ***Note:*** *Only valid when using 1000BASE-X auto-negotiation (PCS_MODE=0x0). When using SGMII autonegotiation (PCS_MODE=0x2) Next Page information is not exchanged and this field must be ignored. In SGMII auto-negotiation this field is used to indicate the Link Status of the SGMII link partner, in line with the definition of the tx_config_Reg in the SGMII specification.*
14	LP_ACK	R	0h	Link Partner Acknowledge. ACK bit from the Link Partner. This bit indicates that the link partner has successfully received the page sent by the local device.
13:12	LP_RF	R	0h	Link Partner Remote Fault / Link Partner Duplex Mode. This field indicates the fault signaling of the link partner. • 00 = No Error • 01 = Offline • 10 = Link Failure • 11 = Auto-negotiation Error ***Note:*** *Only valid when using 1000BASE-X auto-negotiation (PCS_MODE=0x0). When using SGMII autonegotiation (PCS_MODE=0x2) remote fault information is not exchanged and this field must be ignored. In SGMII auto-negotiation bit 12 of this field is used to indicate the duplex mode of operation of the SGMII link partner, in line with the definition of the tx_config_Reg in the SGMII specification.*
11:9	*reserved*	R	0h	Ignore on read.
8:7	LP_PAUSE	R	0h	Link Partner Pause Ability. This field indicates the Pause ability of the link partner. • 00 = No Pause

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Table 293. SGMII_AUTONEG_LP_ABILITY register - SGMII PCS register (address B005h)*...continued*

Bit	Symbol	Access	Value	Description
				• 01 = Asymmetric Pause towards the link partner • 10 = Symmetric Pause • 11 = Both Symmetric Pause and Asymmetric Pause towards the local device ***Note:*** *This field is only valid when using 1000BASE-X auto-negotiation (PCS_MODE=0x0).* When using SGMII autonegotiation (PCS_MODE=0x2) Pause Ability information is not exchanged.
6	LP_FD	R	0h	Link Partner Half Duplex. When this bit is set, it indicates that the link partner is capable of operating in the halfduplex mode. ***Note:*** *Only valid when using 1000BASE-X auto-negotiation (PCS_MODE=0x0). When using SGMII autonegotiation (PCS_MODE=0x2) duplex information is not exchanged and this field must be ignored.*
5	LP_HD	R	0h	Link Partner Full Duplex. When this bit is set, it indicates that the link partner is capable of operating in the full-duplex mode. ***Note:*** *Only valid when using 1000BASE-X auto-negotiation (PCS_MODE=0x0). When using SGMII autonegotiation (PCS_MODE=0x2) duplex information is not exchanged and this field must be ignored.*
4:0	*reserved*	R	0h	Ignore on read.

7.4.1.207 SGMII_AUTONEG_EXPANSION register

Table 294. SGMII_AUTONEG_EXPANSION register - SGMII PCS register (address B006h)

Bit	Symbol	Access	Value	Description
15:3	*reserved*	R	0h	Ignore on read.
2	LD_NP_ABL	R	0h	Local Device Next Page Able. • 0 = The local device does not have the next page ability • 1 = The local device has the next page ability The PCS always returns this bit as 0 because it does not support Next Page.
1	PG_RCVD	R	0h	Page Received. This bit indicates that the local device received a page from the link partner. • 0 = The local device did not receive a new page • 1 = The local device received a new page Type: Latch High.
0	*reserved*	R	0h	Ignore on read.

7.4.1.208 SGMII_EXTENDED_STATUS register

Table 295. SGMII_EXTENDED_STATUS register - SGMII PCS register (address B00Fh)

Bit	Symbol	Access	Value	Description
15	CAP_1G_X_FD	R	1h	1000BASE-X Full-Duplex Capable.

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Table 295. SGMII_EXTENDED_STATUS register - SGMII PCS register (address B00Fh)*...continued*

Bit	Symbol	Access	Value	Description
				• 0 = Not 1000BASE-X full-duplex capable • 1 = 1000BASE-X full-duplex capable The PCS always returns this bit as 1 because it does support this feature.
14	CAP_1G_X_HD	R	1h	1000BASE-X Half-Duplex Capable. • 0 = Not 1000BASE-X half-duplex capable • 1 = 1000BASE-X half-duplex capable The PCS always returns this bit as 1 because in principle it does have the capability to support a half-duplex MAC. However, it does not indicate actual half-duplex operation, this is not supported at chip level.
13	CAP_1G_T_FD	R	0h	1000BASE-T Full-Duplex Capable. • 0 = Not 1000BASE-T full-duplex capable • 1 = 1000BASE-T full-duplex capable The PCS always returns this bit as 0 because it does not support this feature.
12	CAP_1G_T_HD	R	0h	1000BASE-T Half-Duplex Capable. • 0 = Not 1000BASE-T half-duplex capable • 1 = 1000BASE-T half-duplex capable The PCS always returns this bit as 0 because it does not support this feature.
11:0	*reserved*	R	0h	Ignore on read.

7.4.1.209 SGMII_ADVANCED_CONTROL register

Table 296. SGMII_ADVANCED_CONTROL register - Advanced control (address B010h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0	always write 0, ignore on read.
14	CONFIG_ENABLE	RW		SGMII PHY configuration register write protection
			0*	SGMII PHY configuration registers cannot be changed
			1	SGMII PHY configuration registers can be changed
13:2	*reserved*	R	000h	always write 0, ignore on read.
1	GOTO_STANDBY	RW		goto standby mode The transition to STANDY mode needs 100 µs delay and no SMI transaction should be issued towards SGMII
			0*	no action
			1	go to standby
0	START_OPERATION	RW		start operation by exiting standby
			0*	no action
			1	exit standby

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7.4.1.210 SGMII_ADVANCED_STATUS register

Table 297. SGMII_ADVANCED_STATUS register - Advanced status (address B011h)

Bit	Symbol	Access	Value	Description
15:3	*reserved*	R	0h	Ignore on read.
2	LINK_STATUS	R	0h	This bit represents the actual link_status value of the SGMII PHY.
1:0	*reserved*	R	0h	Ignore on read.

7.4.1.211 SGMII_ADVANCED_CONFIG register

Table 298. SGMII_ADVANCED_CONFIG register - Advanced configuration (address B013h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15:1	*reserved*	R	0h	Always write 0, ignore on read.
0	AUTO	RW		Advanced configuration
			0*	PHY pauses in STANDBY at startup, until the start_operate bit is set.
			1	PHY autonomously proceeds to Normal mode for both startup and wake-up.

7.4.1.212 SGMII_IRQ_SOURCE register

Table 299. SGMII_IRQ_SOURCE register - SGMII interrupt source (address B014h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15	TOPDOWN_RESET	RW		Top-down SGMII reset due to global reset event. All configuration settings return to defaults. This bit is W1 C (Cleared on W1).
			0*	no topdown reset since last clear
			1	topdown reset since last clear
14:5	*reserved*	R	0h	Always write 0, ignore on read.
4	FIFO_OVF	RW		SGMII PCS RX FIFO overflow. This bit is W1C (cleared on W1).
			0*	no RX FIFO overflow since last clear
			1	RX FIFO overflow since last clear
3	FIFO_UNF	RW		SGMII PCS RX FIFO underflow. This bit is W1C (Cleared on W1).
			0*	no RX FIFO underflow since last clear
			1	RX FIFO underflow since last clear
2	AUTONEG_EVENT	RW		First read B001 h.5 to check Auto-Negotiation has completed. If so, the contents of the AN MMD Advertisement, AN MMD Link partner Ability, and AN MMD Expansion registers are valid. This bit is W1C (Cleared on W1).
			0*	no Auto-Negotiation process (re)start or completion event

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Table 299. SGMII_IRQ_SOURCE register - SGMII interrupt source (address B014h)*...continued*

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
			1	Auto-Negotiation process has been (re)started or completed
1	RX_LINK_STATUS_ EVENT	RW		The actual SGMII link status is available in register B011h.2. This bit is W1C (Cleared on W1).
			0*	This link did not change since last clear
			1	This link status changed since last clear
0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.213 SGMII_IRQ_ENABLE register

Table 300. SGMII_IRQ_ENABLE register - SGMII interrupt enables (address B015h)

Bit	Symbol	Access	Value	Description
15	TOPDOWN_ RESET_ENABLE	RW	0h	If set, the respective IRQ is enabled This setting is preserved during deep sleep.
14:5	*reserved*	RW	0h	Always write 0, ignore on read.
4	FIFO_OVF_ ENABLE	RW	0h	If set, the respective IRQ is enabled This setting is preserved during deep sleep.
3	FIFO_UNF_ ENABLE	RW	0h	If set, the respective IRQ is enabled This setting is preserved during deep sleep.
2	AUTONEG_ EVENT_ENABLE	RW	0h	If set, the respective IRQ is enabled This setting is preserved during deep sleep.
1	RX_LINK_STATUS_ EVENT_ENABLE	RW	0h	If set, the respective IRQ is enabled This setting is preserved during deep sleep.
0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.214 SGMII_IRQ_MSTATUS register

Table 301. SGMII_IRQ_MSTATUS register - SGMII interrupt mask status. This is AND of IRQ_SOURCE and IRQ_ENABLE. (address B016h)

Bit	Symbol	Access	Value	Description
15	TOPDOWN_ RESET_MSTS	R	0h	This is AND of IRQ_SOURCE and IRQ_ENABLE related to TOPDOWN_RESET _IRQ.
14:5	*reserved*	R	0h	Ignore on read.
4	FIFO_OVF_MSTS	R	0h	This is AND of IRQ_SOURCE and IRQ_ENABLE related to FIFO_OVF IRQ.
3	FIFO_UNF_MSTS	R	0h	This is AND of IRQ_SOURCE and IRQ_ENABLE related to FIFO_UNF IRQ.
2	AUTONEG_ EVENT_MSTS	R	0h	This is AND of IRQ_SOURCE and IRQ_ENABLE related to AUTONEG_EVENTIRQ.
1	RX_LINK_STATUS_ EVENT_MSTS	R	0h	This is AND of IRQ_SOURCE and IRQ_ENABLE related to RX_LINK_STATUS_EVENT IRQ.
0	*reserved*	R	0h	Ignore on read.

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7.4.1.215 SGMII_LED_TRIGGER0 register

Table 302. SGMII_LED_TRIGGER0 register - SGMII LED trigger 0 (address B040h)
Registers SGMII_LED_TRIGGER1 (B041h) to 3 (B043h) have identical entries

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	0h	If set, the trigger is enabled
14:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	FUNCTION	RW	0h	Selectable LED functions: 0 = Reserved 1 = Link_status 2 = Role: MAC(0) or PHY(1) 3 = PHY active (AFE enabled) 4 = Frame reception (RX_DV = true) 5 = Symbol error 6 = Frame transmission (TX_EN=true) 7 = Frame activity (RX_DV=true | TX_EN=true)

7.4.1.216 SGMII_PCS_CONTROL register

Table 303. SGMII_PCS_CONTROL register - SGMII PCS register (address B070h)

Bit	Symbol	Access	Value	Description
15	VR_RST	RW	0h	Vendor-Specific Soft Reset. When the host sets this bit, this triggers the vendor-specific software reset process in which all internal blocks are reset, except the APB interface block and PCS registers. When this bit is set, it does reset the PMA & PMD registers. This bit is self-cleared after 32 clock cycles of the APB bus clock. Type: Self Clearing.
14	R2TLBE	RW	0h	PCS Rx to Tx Loopback Enable. This bit controls the PCS loopback path from the GMII Rx to the GMII Tx at the GMII interface. • 0 = Loopback path is disabled • 1 = Loopback path is enabled
13	*reserved*	RW	0h	always write 0; ignore on read.
12	CL37_BP	R	0h	Enable Clause 37 AN in Backplane Configuration. The PCS always returns this bit as 0 because it does not support this feature.
11:10	*reserved*	R	0h	ignore on read.
9	MAC_AUTO_SW	RW	0h	Automatic Speed Mode Change after SGMII CL37 AN Completion. This bit controls the automatic change of the PCS speed and duplex mode settings after completion of SGMII Clause 37 auto-negotiation. • 0 = The PCS will not automatically change the speed/duplex mode and operate at the speed/duplex mode according to the values programmed to SR MII MMD Control Register (SGMII_BASIC_CONTROL). After the completion of CL37 AN, the application has to read the negotiated Speed/Duplex Mode from the VR MII MMD AN Interrupt and Status Register (SGMII_AUTONEG_

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Table 303. SGMII_PCS_CONTROL register - SGMII PCS register (address B070h)*...continued*

Bit	Symbol	Access	Value	Description
				STATUS) and then program the SR MII MMD Control Register (SGMII_BASIC_CONTROL) appropriately. • 1 = The PCS automatically switches to the negotiated SGMII speed/duplex mode settings, after the completion of Clause 37 auto-negotiation. This mode is valid only when PCS is configured as MAC-side SGMII (TX_CONFIG=0 and PCS_MODE set to the SGMII) and should be set only when auto-negotiation is enabled (AN_ENABLE=1). ***Note:*** *Only PCS speed/duplex mode settings are impacted and the application must still adjust other chip level speed/duplex mode settings appropriately.*
8	INIT	RW	0h	Datapath Initialization Control. This bit can be set to flush/initialize the various FIFOs implemented in the PCS. This is a self-clearing bit. After writing a 1 to this bit, the application should poll this bit continuously and proceed to any other operation only after reading this bit as 0. ***Note:*** *When this bit is programmed to 1, the RXFIFO_OVF/RXFIFO_UNF bits of VR MII MMD Digital Status Register (SGMII_PCS_STATUS) can get set incorrectly. Hence, these register bits must be read so that they get cleared.* Type: Self Clearing.
7	MSK_RD_ERR	R	0h	Mask Running Disparity Error. The PCS always returns this bit as 0 because it does not support this feature.
6	PRE_EMP	RW	0h	Pre-emption Packet Enable. This bit must be set to 1 to allow the PCS to properly receive/transmit pre-emption packets in SGMII 10M/100M Modes.
4:5	*reserved*	R	0h	always write 0, ignore on read.
4	DTXLANED_0	RW	0h	Disable Tx Lane of the PMD. This bit should always be set to 0 as the PCS does not support this feature.
3	*reserved*	RW	0h	always write 0; ignore on read.
2	EN_2_5G_MODE	RW	0h	Enable 2.5G SGMII Mode. This bit should be set to 1 to enable 2.5G SGMII mode of operation. • 0 = 2.5G SGMII mode is disabled and 1000M/100M/10M operation is supported • 1 = 2.5G SGMII mode is enabled and only 2500M operation is supported ***Note:*** *This bit must be set to 0 to allow SGMII to operate in 10/100/1000M.*
1	BYP_PWRUP	R	0h	Reserved
0	PHY_MODE_CTRL	RW	1h	PHY mode AN advertisement control. This bit controls the Clause 37 auto-negotiation advertisement when operating in SGMII PHY mode. • 0 = During SGMII autonegotiation the PCS advertises the values programmed to SGMII_LINK_STS of VR MII MMD AN Control Register (SGMII_PCS_CONFIG2), SS13 and

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Table 303. SGMII_PCS_CONTROL register - SGMII PCS register (address B070h)*...continued*

Bit	Symbol	Access	Value	Description
				SS6 of SR MII MMD Control Register (SGMII_BASIC_ CONTROL) and FD of SR MII MMD AN Advertisement Register (SGMII_AUTONEG_ADVERTISE). • 1 = During SGMII autonegotiation the PCS advertises the values of internal hardware input ports, controlled by chip level settings, in the SR MII MMD AN Advertisement Register (SGMII_AUTONEG_ADVERTISE). ***Note:*** *This bit should be set only when the PCS is configured as PHY-side SGMII i.e. TX_CONFIG=1.*

7.4.1.217 SGMII_PCS_STATUS register

Table 304. SGMII_PCS_STATUS register - SGMII PCS register (address B071h)

Bit	Symbol	Access	Value	Description
15:7	*reserved*	R	0h	ignore on read
6	RXFIFO_OVF	R	0h	Rx FIFO Overflow. This bit indicates overflow of the clock rate compensation FIFO. • 0 = Normal operation • 1 = FIFO overflow Type: Latch High.
5	RXFIFO_UNDF	R	0h	Rx FIFO Underflow. This bit indicates underflow of the clock rate compensation FIFO. • 0 = Normal operation • 1 = FIFO underflow Type: Latch High.
4:0	*reserved*	R	0h	ignore on read.

7.4.1.218 SGMII_PCS_CONFIG1 register

Table 305. SGMII_PCS_CONFIG1 register - SGMII PCS register (address B073h)

Bit	Symbol	Access	Value	Description
15:5	*reserved*	R	0h	Always write 0, ignore on read.
4	TX_POL_INV_0	RW	1h	Tx Polarity Invert. When set the data polarity on the Tx differential lines is reversed/inverted. Can be used to swap TXP and TXN. • 0 = Do not invert data polarity on Tx differential lines • 1 = Invert data polarity on Tx differential lines
3:1	*reserved*	R	0h	Always write 0, ignore on read.
0	RX_POL_INV_0	RW	0h	Rx Polarity Invert. When set the data received from Rx serial line is inverted. Can be used to swap RXP and RXN. • 0 = Do not invert data polarity of data received from Rx differential lines • 1 = Invert data polarity of data received from Rx differential lines

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7.4.1.219 SGMII_PCS_CONFIG2 register

Table 306. SGMII_PCS_CONFIG2 register - SGMII PCS register (address B074h)

Bit	Symbol	Access	Value	Description
15:9	*reserved*	R	0h	Always write 0, ignore on read.
8	MII_CTRL	RW	0h	SGMII MII Control. This bit controls the width of the internal MII interface when operating at SGMII speed modes of 10 Mbps or 100 Mbps. • 0 = 4-bit MII • 1 = 8-bit MII This bit must always be set to 0 for proper operation of the internal MII interface.
7:5	*reserved*	R	0h	Always write 0, ignore on read.
4	SGMII_LINK_STS	RW	1h	SGMII Link Status. This bit is used in Bit 15 of the tx_config_Reg during SGMII Clause 37 auto-negotiation when operating in SGMII PHY-side. • 0 = Link Down • 1 = Link Up This mode is relevant only when PCS is configured as PHY-side SGMII (TX_CONFIG=1 and PCS_MODE set to the SGMII) and when auto-negotiation is enabled (AN_ENABLE=1).
3	TX_CONFIG	RW	1h	SGMII Transmit Configuration Register select. This bit controls the tx_config_Reg values to be used during the Clause 37 auto-negotiation in SGMII mode. • 0 = Configures the PCS as MAC-side SGMII and uses MAC-to-PHY tx_config_Reg values for CL37 AN • 1 = Configures the PCS as PHY-side SGMII and uses PHY-to-MAC tx_config_Reg values for CL37 AN This mode is relevant only when PCS is configured as SGMII (PCS_MODE set to the SGMII) and when auto-negotiation is enabled (AN_ENABLE=1).
2:1	PCS_MODE	RW	0h	PCS Mode. This field controls the PCS auto-negotiation (and operating) mode. • 00 = 1000BASE-X mode (Clause 37 autonegotiation with tx_config_Reg according to IEEE802.3 Clause 37 and 1000Base-X) • 01 = Reserved • 10 = SGMII mode (Clause 37 auto-negotiation with tx_config_Reg according to the SGMII specification) • 11 = Reserved
0	MII_AN_INTR_EN	RW	1h	Clause 37 AN Complete Interrupt Enable. This bit controls the generation of an interrupt when Clause 37 auto-negotiation is completed. • 0 = Disables the generation of Clause 37 auto-negotiation complete interrupt output • 1 = Enables the generation of Clause 37 auto-negotiation complete interrupt output

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7.4.1.220 SGMII_AUTONEG_STATUS register

Table 307. SGMII_AUTONEG_STATUS register - SGMII PCS register (address B075h)

Bit	Symbol	Access	Value	Description
15:7	*reserved*	R	0h	ignore on read.
6	LP_CK_STP	R	0h	Link Partner EEE Clock Stop Capability. The PCS always returns this bit as 0 because it does not support this feature.
5	LP_EEE_CAP	R	0h	Link Partner EEE Capability. The PCS always returns this bit as 0 because it does not support this feature.
4:1	CL37_ANSGM_STS	R	0h	Clause 37 AN SGMII Status. This field indicates the status received from remote link after the SGMII autonegotiation is complete. • CL37_ANSGM_STS[0] - 0 = Half Duplex • CL37_ANSGM_STS[0] - 1 = Full Duplex • CL37_ANSGM_STS[2:1] - 00 = 10 Mbps speed link • CL37_ANSGM_STS[2:1] - 01 = 100 Mbps speed link • CL37_ANSGM_STS[2:1] - 10 = 1000 Mbps speed link • CL37_ANSGM_STS[2:1] - 11 = Reserved • CL37_ANSGM_STS[3] - 0 = Link is Down • CL37_ANSGM_STS[3] - 1 = Link is Up This field is valid only when PCS is configured as SGMII (PCS_MODE set to the SGMII) and the autonegotiation is complete.
0	CL37_ANCMPLT_INTR	RW	0h	Clause 37 AN Complete Interrupt. The PCS sets this bit when Clause 37 autonegotiation is complete. The host must clear this bit by writing 0 to it. Type: Self Set / Write Clear.

7.4.1.221 SGMII_LINK_DROP_COUNTER register

Table 308. SGMII_LINK_DROP_COUNTER register - Link drop counter (address B0A1h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	0h	If set, the trigger is enabled; 0 resets the counter
14:6	*reserved*	R	0h	Always write 0, ignore on read.
5:0	LINK_STATUS_DROPS	R	0h	Number of link drop events. This register tracks how many times the non-latched link_status bit has dropped. The value is reset when this register is disabled, during a power-cycle, or when inhibit is de-asserted, and on SGMII reset. The value saturates at max value and is not reset by disabling SGMII, link retraining, or sleep cycles as long as the core supply remains valid.

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7.5 Shared

7.5.1 Register Description

Table 309. Register overview: Shared

Address	Name	Access	Reset	Description
2h	PHY_IDENTIFIER0	R/W	1Bh	PHY Identifier 0 register
3h	PHY_IDENTIFIER1	R/W	B013h	PHY Identifier 1 register
5h	DEVICES_IN_PACKAGE1	R/W	Bh	Devices in package 1 register
6h	DEVICES_IN_PACKAGE2	R/W	4000h	Devices in package 2 register
Eh	PACKAGE_IDENTIFIER0	R/W	1Bh	Package identifier 0 register
Fh	PACKAGE_IDENTIFIER1	R/W	B010h	Package identifier 1 register
8021h	COMPOSITE_STATUS	R/W	3000h	Composite SGMII and Ethernet PHY status. Only applicable for TJA1104B/TJA1104D
8040h	PORT_CONTROL	R/W	0h	Port control
8046h	PORT_ABILITIES	R/W	A/C: 868h B/D: 808h	Port abilities
8048h	PORT_FUNC_ENABLES	R/W	2004h	Port functionality enable register
8070h	PORT_LEVEL_IRQ_SOURCES	R/W	0h	Port-level IRQ sources
8072h	PORT_LEVEL_IRQ_ENABLES	R/W	0h	Port-level IRQ enables
8074h	PORT_LEVEL_IRQ_MSTATUS	R/W	0h	Port-level IRQ masked status
8078h	PORT_FUNC_IRQ_STATUS	R/W	0h	Port-level IRQ status of sub-functions
807Ah	PORT_FUNC_IRQ_ENABLES	R/W	B00Ch	Port-level IRQ enables of sub-functions
807Ch	PORT_FUNC_IRQ_MSTATUS	R/W	0h	Port-level IRQ masked status of sub-functions
8080h	EPHY_CAT_IRQ_SOURCES	R/W	0h	Ethernet PHY category IRQ sources
8081h	EPHY_CAT_IRQ_ENABLES	R/W	555h	Ethernet PHY category enables
8082h	EPHY_CAT_IRQ_MSTATUS	R/W	0h	Ethernet PHY category status register
80A0h	EPHY_FUNCTIONAL_IRQ_SOURCES	R/W	0h	Ethernet PHY functional IRQ sources
80A1h	EPHY_FUNCTIONAL_IRQ_ENABLES	R/W	0h	Ethernet PHY functional IRQ enables
80A2h	EPHY_FUNCTIONAL_IRQ_MSTATUS	R/W	0h	Ethernet PHY functional IRQ masked status
80C0h	EPHY_CONFIG_REGXS_IRQ_SOURCES	R/W	0h	Ethernet PHY configuration register access IRQs sources
80C1h	EPHY_CONFIG_REGXS_IRQ_ENABLES	R/W	0h	Ethernet PHY configuration register access IRQs enables

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Table 309. Register overview: Shared*...continued*

Address	Name	Access	Reset	Description
80C2h	EPHY_CONFIG_REGXS_IRQ_MSTATUS	R/W	0h	Ethernet PHY configuration register access IRQs masked status
80D0h	EPHY_DIAGNOSTIC_IRQ_SOURCES	R/W	0h	Ethernet PHY diagnostic IRQ sources
80D1h	EPHY_DIAGNOSTIC_IRQ_ENABLES	R/W	0h	Ethernet PHY diagnostic IRQ enables
80D2h	EPHY_DIAGNOSTIC_IRQ_MSTATUS	R/W	0h	Ethernet PHY diagnostic masked IRQ masked status
8100h	PHY_CONTROL	R/W	0h	PHY control register
8102h	PHY_STATUS	R/W	0h	PHY Status register
8104h	PHY_LATCHED_STATUS	R/W	0h	PHY latched status register
8106h	PHY_ADDITIONAL_ABILITIES	R/W	6h	Additional PHY Abilities
8108h	PHY_CONFIG	R/W	0h	PHY configuration register. This register can only be changed if CONFIG_ENABLE or SUPER_CONFIG_ENABLE are set to 1.
810Ch	PHY_STATE	R/W	0h	PHY state register
810Eh	PHY_PARAMETERS	R/W	0h	PHY parameters register. This register can only be changed if CONFIG_ENABLE or SUPER_CONFIG_ENABLE are set to 1.
8180h	WAKE_SLEEP_CONTROL	R/W	0h	Wake sleep control register
8181h	WAKE_SLEEP_STATUS	R/W	0h	Wake sleep status register
8182h	WAKE_SLEEP_CONFIG	R/W	8032h	Wake sleep configuration register. This register can only be changed if CONFIG_ENABLE or SUPER_CONFIG_ENABLE are set to 1.
8184h	WAKE_SLEEP_PARAMETERS	R/W	6333h	Wake sleep parameters register. This register can only be changed if CONFIG_ENABLE or SUPER_CONFIG_ENABLE are set to 1.
8190h	EPHY_LED_TRIGGER0	R/W	8000h	Ethernet PHY LED trigger output 0 register
8191h	EPHY_LED_TRIGGER1	R/W	8004h	Ethernet PHY LED trigger output 1 register, see EPHY_LED_TRIGGER0
8192h	EPHY_LED_TRIGGER2	R/W	8006h	Ethernet PHY LED trigger output 2 register, see EPHY_LED_TRIGGER0
8193h	EPHY_LED_TRIGGER3	R/W	0000h	Ethernet PHY LED trigger output 3 register, see EPHY_LED_TRIGGER0
8319h	TXC_STATUS	R/W	0h	TXC status register
831Ah	RXC_STATUS	R/W	0h	RXC status register
831Bh	SGMII_LATENT_STATUS	R/W	0h	SGMII latent fault status register
831Ch	EPHY_LATENT_STATUS	R/W	0h	Ethernet PHY latent fault status register
831Dh	SGMII_CLK_STATUS	R/W	0h	SGMII clock status register

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Table 309. Register overview: Shared*...continued*

Address	Name	Access	Reset	Description
831Eh	EPHY_CLK_STATUS	R/W	0h	Ethernet PHY clock status
8320h	SIGNAL_QUALITY	R/W	70h	Signal quality
8324h	MSE	R/W	0h	Mean-square error register
8325h	MAX_MSE	R/W	0h	Max mean-square error register
8330h	CABLE_TEST	R/W	0h	Cable test register
8340h	LINK_TRAINING_TIMER	R/W	0h	Link training timer register
8341h	LOC_RCVR_STATUS_ TIMER	R/W	0h	Local receiver status timer register
8342h	REM_RCVR_STATUS_ TIMER	R/W	0h	Remote receiver status timer register
8343h	slave_SILENT_TIMER	R/W	0h	slave silent timer register
8350h	SYMBOL_ERROR_ COUNTER	R/W	0h	Symbol error counter register
8352h	ERROR_COUNTER_MISC	R/W	8000h	Error counter miscellaneous register
8353h	LINK_LOSSES_AND_ FAILURES	R/W	0h	Link losses and failures counter register
83F0h	PHY_COMPLIANCE_TEST	R/W	0h	PHY compliance test control register

7.5.1.1 PHY_IDENTIFIER0 register

Table 310. PHY_IDENTIFIER0 register - PHY Identifier 0 register (address 2h)

Bit	Symbol	Access	Value	Description
15:0	OUI_BITS_3_TO_18	R	1Bh	OUI bits [3:18] OUI = 00:60:37

7.5.1.2 PHY_IDENTIFIER1 register

Table 311. PHY_IDENTIFIER1 register - PHY Identifier 1 register (address 3h)

Bit	Symbol	Access	Value	Description
15:10	OUI_BITS_19_TO_ 24	R	2Ch	OUI bits [19:24] OUI = 00:60:37
9:4	MODEL	R	1h	6-bit PHY model number
3:0	REVISION	R	3h	4-bit PHY revision number

7.5.1.3 DEVICES_IN_PACKAGE1 register

Table 312. DEVICES_IN_PACKAGE1 register - Devices in package 1 register (address 5h)

Bit	Symbol	Access	Value	Description
15:8	*reserved*	R	0h	Ignore on read.
7	AN_PRESENT	R	0h	If set, Auto-Negotiation present in package

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Table 312. DEVICES_IN_PACKAGE1 register - Devices in package 1 register (address 5h)*...continued*

Bit	Symbol	Access	Value	Description
6	TC_PRESENT	R	0h	If set, TC present in package
5	DTE_XS_PRESENT	R	0h	If set, DTE XS present in package
4	PHY_XS_PRESENT	R	0h	If set, PHY XS present in package
3	PCS_PRESENT	R	1h	If set, PCS present in package
2	WIS_PRESENT	R	0h	If set, WIS present in package
1	PMD_PMA_ PRESENT	R	1h	If set, PMA/PMD present in package
0	CL22_PRESENT	R	1h	If set, Clause 22 registers present in package

7.5.1.4 DEVICES_IN_PACKAGE2 register

Table 313. DEVICES_IN_PACKAGE2 register - Devices in package 2 register (address 6h)

Bit	Symbol	Access	Value	Description
15	MMD31_PRESENT	R	A/B: 1h C/D: 0h	If set, Vendor specific device 2 present in package
14	MMD30_PRESENT	R	1h	If set, Vendor specific device 1 present in package
13	CL22_EXT_ PRESENT	R	0h	If set, Clause 22 extension present in package
12:0	*reserved*	R	0h	Ignore on read.

7.5.1.5 PACKAGE_IDENTIFIER0 register

Table 314. PACKAGE_IDENTIFIER0 register - Package identifier 0 register (address Eh)

Bit	Symbol	Access	Value	Description
15:0	OUI_BITS_3_TO_18	R	1Bh	OUI bits [3:18] OUI = 00:60:37

7.5.1.6 PACKAGE_IDENTIFIER1 register

Table 315. PACKAGE_IDENTIFIER1 register - Package identifier 1 register (address Fh)

Bit	Symbol	Access	Value	Description
15:10	OUI_BITS_19_TO_ 24	R	2Ch	OUI bits [19:24] OUI = 00:60:37
9:4	MODEL	R	1h	6-bit Package model number
3:0	REVISION	R	0h	4-bit Package revision number

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7.5.1.7 COMPOSITE_STATUS register

Table 316. COMPOSITE_STATUS register - Composite SGMII and Ethernet PHY status. Only applicable for TJA1104B/TJA1104D (address 8021h)

Bit	Symbol	Access	Value	Description
15	AN_LINK_STATUS	R	0h	If the device operates in SGMII-MAC mode, this bit indicates the received LINK_STATUS according to TX_CONFIG_REG[15].
14:12	LINK_STATUS_SELECT	R/W	3h	Selects the source for TX_CONFIG_REG[15] used in the SGMII auto negotiation process. This interface must operate in SGMII-PHY mode 000 = Reserved 001 = SGMII RX Link Status 010 = ePHY Link Availability 011 = ePHY Link Available & SGMII RX Link Status 1xx = Reserved This setting is preserved during deep sleep.
11:0	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.

7.5.1.8 PORT_CONTROL register

Table 317. PORT_CONTROL register - Port control (address 8040h)

Bit	Symbol	Access	Value	Description
15	RESET	R/W	0h	If set, reset all functionality of this PORT. This bit is self-clearing.
14	CONFIG_ENABLE	R/W	0h	If set, configuration registers of this PORT (see PORT_FUNC_ENABLES) can be changed. Otherwise, the configuration registers are write-protected.
13:0	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.

7.5.1.9 PORT_ABILITIES register

Table 318. PORT_ABILITIES register - Port abilities (address 8046h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Ignore on read.
11	BIST_ABILITY	R	1h	If set, this port supports BIST
10:7	*reserved*	R	0h	Ignore on read.
6	MAC_ADPTR_ABILITY	R	A/B = 1h C/D = 0h	If set, this port has MAC support
5	MACSEC_ABILITY	R	A/B = 1h C/D = 0h	If set, this port supports MACsec
4	*reserved*	R	0h	Ignore on read.
3	PTP_ABILITY	R	1h	If set, this port supports PTP
2:1	*reserved*	R	0h	Ignore on read.

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Table 318. PORT_ABILITIES register - Port abilities (address 8046h)*...continued*

Bit	Symbol	Access	Value	Description
0	PORT_ABILITY_XTD	R	0h	Extended port abilities

7.5.1.10 PORT_FUNC_ENABLES register

Table 319. PORT_FUNC_ENABLES register - Port functionality enable register (address 8048h)

Bit	Symbol	Access	Value	Description
15:14	*reserved*	R	0h	Always write 0, ignore on read.
13	MII_ENABLE	R/W	1h	If deasserted, the xMII interface (MII, RMII, RGMII, SGMII) is detached from the datapath. Device outputs become high-Z and inputs are disabled. This setting is configuration protected.
12	*reserved*	R	0h	Always write 0, ignore on read.
11	BIST_ENABLE	R/W	0h	If set, BIST functionality is enabled for this PORT This setting is configuration protected.
10:7	*reserved*	R	0h	Always write 0, ignore on read.
6[1]	MAC_ADPTR_ENABLE	R/W	0h	If set, the MAC adapter is enabled This setting is preserved during deep sleep.
5[1]	MACSEC_ENABLE	R/W	0h	If set, the MACsec subsystem is enabled This setting is preserved during deep sleep.
4	*reserved*	R	0h	Always write 0, ignore on read.
3	PTP_ENABLE	R/W	0h	If set, PTP functionality is enabled for this PORT. This setting is configuration protected.
2	EPHY_ENABLE	R/W	1h	If set, Ethernet PHY of this PORT is enabled. This setting is configuration protected.
1	*reserved*	R	0h	Always write 0, ignore on read.
0	TEST_ENABLE	R/W	0h	By default, entering test modes is restricted and guarded by respective TEST_ENABLE bits. This bit enables test mode selection of all sub-functions (ePHY, BIST and xMII) of a port. If this bit is not set, selecting a test mode has no effect, unless the TEST_ENABLE is set on global level or a sub-function specific TEST_ENABLE is set. If this bit is set, a test mode of all sub-functions can be entered. Note, the TEST_ENABLEs through the hierarchy (global, port, port sub-functions) are ORed. This setting is configuration protected. This setting is preserved during deep sleep.

[1] Reserved for C/D.
Always write 0, ignore on read

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7.5.1.11 PORT_LEVEL_IRQ_SOURCES register

Table 320. PORT_LEVEL_IRQ_SOURCES register - Port-level IRQ sources (address 8070h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14	EPHY_CLK_WARN	R/W	0h	If set, the Ethernet PHY clock warning is present, see EPHY_CLK_STATUS Write one to clear. This bit is latched-high.
13	SGMII_CLK_WARN	R/W	0h	If set, the SGMII clock warning is present, see SGMII_CLK_STATUS Write one to clear. This bit is latched-high.
12	EPHY_LF	R/W	0h	If set, the Ethernet latent fault is present, see EPHY_LATENT_STATUS Write one to clear. This bit is latched-high.
11	SGMII_LF	R/W	0h	If set, the SGMII latent fault is present, see SGMII_LATENT_STATUS Write one to clear. This bit is latched-high.
10	TXC_CLK_WARN	R/W	0h	If set, TXC clock warning is present, see TXC_STATUS register Write one to clear. This bit is latched-high.
9	RXC_CLK_WARN	R/W	0h	If set, RXC clock warning is present, see RXC_STATUS register. Write one to clear. This bit is latched-high.
8:0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.12 PORT_LEVEL_IRQ_ENABLES register

Table 321. PORT_LEVEL_IRQ_ENABLES register - Port-level IRQ enables (address 8072h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14	EPHY_CLK_WARN	R/W	0h	Enable for ePHY CLOCK_WARN interrupt This setting is preserved during deep sleep.
13	SGMII_CLK_WARN	R/W	0h	Enable for SGMII Clock Warn Interrupt This setting is preserved during deep sleep.
12	EPHY_LF	R/W	0h	Enable for ephy Link Latent Fault This setting is preserved during deep sleep.
11	SGMII_LF	R/W	0h	Enable for SGMII Link Latent Fault This setting is preserved during deep sleep.
10	TXC_CLK_WARN	R/W	0h	Enable for TXC Clock warn Interrupt This setting is preserved during deep sleep.

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Table 321. PORT_LEVEL_IRQ_ENABLES register - Port-level IRQ enables (address 8072h)*...continued*

Bit	Symbol	Access	Value	Description
9	RXC_CLK_WARN	R/W	0h	Enable for RXC Clock warn interrupt This setting is preserved during deep sleep.
8:0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.13 PORT_LEVEL_IRQ_MSTATUS register

Table 322. PORT_LEVEL_IRQ_MSTATUS register - Port-level IRQ masked status (address 8074h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Ignore on read.
14	EPHY_CLK_WARN	R	0h	Masked-status of Ethernet PHY CLOCK_WARN interrupt
13	SGMII_CLK_WARN	R	0h	Masked Status of SGMII Clock Warn Interrupt
12	EPHY_LF	R	0h	Masked status of ePHY link Latent fault
11	SGMII_LF	R	0h	Masked Status of SGMII Link Latent fault
10	TXC_CLK_WARN	R	0h	Masked status of TXC Clock warn Fault
9	RXC_CLK_WARN	R	0h	Masked status of RXC Clock warn fault
8:0	*reserved*	R	0h	Ignore on read.

7.5.1.14 PORT_FUNC_IRQ_STATUS register

Table 323. PORT_FUNC_IRQ_STATUS register - Port-level IRQ status of sub-functions (address 8078h)

Bit	Symbol	Access	Value	Description
15	PORT_LEVEL_IRQ	R	0h	Status bit indicating an interrupt at PORT-level from functionality that doesn't belong to one specific sub-function.
14	*reserved*	R	0h	Ignore on read.
13	SERDES_MII_IRQ	R	0h	Status bit indicating an interrupt from the SERDES-MII (SGMII) function of this port when it is configured to use this MII type.
12	INFRA_IRQ	R	0h	Status bit indicating an interrupt from INFRAstructure functionality for this port.
11:7	*reserved*	R	0h	Ignore on read.
6[1]	MAC_ADPTR_IRQ	R	0h	Status bit indicating an interrupt from the MACsec adapter of this port.
5[1]	MACSEC_IRQ	R	0h	Status bit indicating an interrupt from the MACsec subsytem of this port.
4	*reserved*	R	0h	Ignore on read.
3	PTP_IRQ	R	0h	Status bit indicating an interrupt from the PTP function of this port.
2	EPHY_IRQ	R	0h	Status bit indicating an interrupt from the ePHY of this port.
1:0	*reserved*	R	0h	Ignore on read.

[1] Reserved for C/D.
Always write 0, ignore on read

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7.5.1.15 PORT_FUNC_IRQ_ENABLES register

Table 324. PORT_FUNC_IRQ_ENABLES register - Port-level IRQ enables of sub-functions (address 807Ah)

Bit	Symbol	Access	Value	Description
15	PORT_LEVEL_IRQ	R/W	1h	Enable for port level interrupt This setting is preserved during deep sleep.
14	*reserved*	R	0h	Always write 0, ignore on read.
13	SERDES_MII_IRQ	R/W	1h	Enable for SERDES-MII interrupt This setting is preserved during deep sleep.
12	INFRA_IRQ	R/W	1h	Enable for INFRA interrupt This setting is preserved during deep sleep.
11:7	*reserved*	R	0h	Always write 0, ignore on read.
6[1]	MAC_ADPTR_IRQ	R/W	0h	Enable for MACsec adapter IRQ. This setting is preserved during deep sleep.
5[1]	MACSEC_IRQ	R/W	0h	Enable for MACsec subsystem IRQ. This setting is preserved during deep sleep.
4	*reserved*	R	0h	Always write 0, ignore on read.
3	PTP_IRQ	R/W	1h	Enable for PTP interrupt This setting is preserved during deep sleep.
2	EPHY_IRQ	R/W	1h	Enable for Ethernet PHY interrupt This setting is preserved during deep sleep.
1:0	*reserved*	R	0h	Always write 0, ignore on read.

[1] Reserved for C/D.
Always write 0, ignore on read

7.5.1.16 PORT_FUNC_IRQ_MSTATUS register

Table 325. PORT_FUNC_IRQ_MSTATUS register - Port-level IRQ masked status of sub-functions (address 807Ch)

Bit	Symbol	Access	Value	Description
15	PORT_LEVEL_IRQ	R	0h	Masked status of PORT_LEVEL interrupt
14	*reserved*	R	0h	Ignore on read.
13	SERDES_MII_IRQ	R	0h	Masked status of SERDES-MII interrupt
12	INFRA_IRQ	R	0h	Masked status of INFRA interrupt
11:7	*reserved*	R	0h	Ignore on read.
6[1]	MAC_ADPTR_IRQ	R	0h	Status bit indicating an interrupt from the MACsec adapter of this port. Masked status of MACsec adapter IRQ.
5[1]	MACSEC_IRQ	R	0h	Status bit indicating an interrupt from the MACsec subsystem of this port. Masked status of MACsec subsystem IRQ.
4	*reserved*	R	0h	Ignore on read.
3	PTP_IRQ	R	0h	Masked status of PTP interrupt
2	EPHY_IRQ	R	0h	Masked status of ePHY interrupt

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Table 325. PORT_FUNC_IRQ_MSTATUS register - Port-level IRQ masked status of sub-functions (address 807Ch)*...continued*

Bit	Symbol	Access	Value	Description
1:0	*reserved*	R	0h	Ignore on read.

[1] Reserved for C/D.
Always write 0, ignore on read

7.5.1.17 EPHY_CAT_IRQ_SOURCES register

Table 326. EPHY_CAT_IRQ_SOURCES register - Ethernet PHY category IRQ sources (address 8080h)

Bit	Symbol	Access	Value	Description
15:11	*reserved*	R	0h	Ignore on read.
10	EPHY_ DIAGNOSTIC_IRQ	R	0h	If set, pending interrupts present in 'Diagnosis and Test interrupts' source register
9	*reserved*	R	0h	Ignore on read.
8	EPHY_CONFIG_ REGXS_IRQ	R	0h	If set, pending interrupts present in 'Config and Register Access' interrupts source register
7:5	*reserved*	R	0h	Ignore on read.
4	EPHY_ FUNCTIONAL_IRQ	R	0h	If set, pending interrupts present in 'Functional baseline' source register
3:0	*reserved*	R	0h	Ignore on read.

7.5.1.18 EPHY_CAT_IRQ_ENABLES register

Table 327. EPHY_CAT_IRQ_ENABLES register - Ethernet PHY category enables (address 8081h)

Bit	Symbol	Access	Value	Description
15:11	*reserved*	R	0h	Always write 0, ignore on read.
10	EPHY_ DIAGNOSTIC_IRQ	R/W	1h	Enable EPHY_DIAGNOSTIC_IRQ category
9	*reserved*	R	0h	Always write 0, ignore on read.
8	EPHY_CONFIG_ REGXS_IRQ	R/W	1h	Enable EPHY_CONFIG_REGXS_IRQ category
7:5	*reserved*	R	0h	Always write 0, ignore on read.
4	EPHY_ FUNCTIONAL_IRQ	R/W	1h	Enable EPHY_FUNCTIONAL_IRQ category
3:0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.19 EPHY_CAT_IRQ_MSTATUS register

Table 328. EPHY_CAT_IRQ_MSTATUS register - Ethernet PHY category status register (address 8082h)

Bit	Symbol	Access	Value	Description
15:11	*reserved*	R	0h	Ignore on read.
10	EPHY_ DIAGNOSTIC_IRQ	R	0h	Masked status of EPHY_DIAGNOSTIC_IRQ category

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Table 328. EPHY_CAT_IRQ_MSTATUS register - Ethernet PHY category status register (address 8082h)*...continued*

Bit	Symbol	Access	Value	Description
9	*reserved*	R	0h	Ignore on read.
8	EPHY_CONFIG_ REGXS_IRQ	R	0h	Masked status of EPHY_CONFIG_REGXS_IRQ category
7:5	*reserved*	R	0h	Ignore on read.
4	EPHY_ FUNCTIONAL_IRQ	R	0h	Masked status of EPHY_FUNCTIONAL_IRQ category
3:0	*reserved*	R	0h	Ignore on read.

7.5.1.20 EPHY_FUNCTIONAL_IRQ_SOURCES register

Table 329. EPHY_FUNCTIONAL_IRQ_SOURCES register - Ethernet PHY functional IRQ sources (address 80A0h)

Bit	Symbol	Access	Value	Description
15	POLARITY_ CONFLICT	R/W	0h	If set, unresolved polarity conflict Write one to clear. This bit is latched-high.
14	TRAINING_ FAILURE	R/W	0h	If set, failure occurred during training. This including but is not necessarily restricted to expiration of the max_wait_timer. Write one to clear. This bit is latched-high.
13	TRANSMIT_ FAILURE	R/W	0h	If set, attempt to transmit data without a link available Write one to clear. This bit is latched-high.
12:5	*reserved*	R	0h	Always write 0, ignore on read.
4	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
3	WAKE_SLEEP_ EVENT	R/W	0h	If set, the PHY's Wake-Sleep status has changed Write one to clear. This bit is latched-high.
2	LINK_AVAILABLE_ EVENT	R/W	0h	If set, LINK_AVAILABILITY has changed Write one to clear. This bit is latched-high.
1	LINK_STATUS_ EVENT	R/W	0h	If set, LINK_STATUS has changed Write one to clear. This bit is latched-high.
0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.21 EPHY_FUNCTIONAL_IRQ_ENABLES register

Table 330. EPHY_FUNCTIONAL_IRQ_ENABLES register - Ethernet PHY functional IRQ enables (address 80A1h)

Bit	Symbol	Access	Value	Description
15	POLARITY_ CONFLICT	R/W	0h	If set, POLARITY_CONFLICT IRQ is enabled This setting is preserved during deep sleep.

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Table 330. EPHY_FUNCTIONAL_IRQ_ENABLES register - Ethernet PHY functional IRQ enables (address 80A1h)*...continued*

Bit	Symbol	Access	Value	Description
14	TRAINING_ FAILURE	R/W	0h	If set, TRAINING_FAILURE IRQ is enabled This setting is preserved during deep sleep.
13	TRANSMIT_ FAILURE	R/W	0h	If set, TRANSMIT_FAILURE IRQ is enabled This setting is preserved during deep sleep.
12:5	*reserved*	R	0h	Always write 0, ignore on read.
4	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
3	WAKE_SLEEP_ EVENT	R/W	0h	If set, WAKE_SLEEP_EVENT IRQ is enabled This setting is preserved during deep sleep.
2	LINK_AVAILABLE_ EVENT	R/W	0h	If set, LINK_AVAILABLE_EVENT IRQ is enabled This setting is preserved during deep sleep.
1	LINK_STATUS_ EVENT	R/W	0h	If set, LINK_STATUS_EVENT IRQ is enabled This setting is preserved during deep sleep.
0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.22 EPHY_FUNCTIONAL_IRQ_MSTATUS register

Table 331. EPHY_FUNCTIONAL_IRQ_MSTATUS register - Ethernet PHY functional IRQ masked status (address 80A2h)

Bit	Symbol	Access	Value	Description
15	POLARITY_ CONFLICT	R	0h	Masked status of POLARITY_CONFLICT
14	TRAINING_ FAILURE	R	0h	Masked status of TRAINING_FAILURE
13	TRANSMIT_ FAILURE	R	0h	Masked status of TRANSMIT_FAILURE
12:5	*reserved*	R	0h	Ignore on read.
4	*reserved*	R	0h	Ignore on read.
3	WAKE_SLEEP_ EVENT	R	0h	Masked status of WAKE_SLEEP_EVENT
2	LINK_AVAILABLE_ EVENT	R	0h	Masked status of LINK_AVAILABLE_EVENT
1	LINK_STATUS_ EVENT	R	0h	Masked status of LINK_STATUS_EVENT
0	*reserved*	R	0h	Ignore on read.

7.5.1.23 EPHY_CONFIG_REGXS_IRQ_SOURCES register

Table 332. EPHY_CONFIG_REGXS_IRQ_SOURCES register - Ethernet PHY configuration register access IRQs sources (address 80C0h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.

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Table 332. EPHY_CONFIG_REGXS_IRQ_SOURCES register - Ethernet PHY configuration register access IRQs sources (address 80C0h)*...continued*

Bit	Symbol	Access	Value	Description
14:13	*reserved*	R	0h	Always write 0, ignore on read.
12	RECONFIG_ DENIED	R/W	0h	If set, Denied config attempt to PHY register Write one to clear. This bit is latched-high.
11	INIT_ERROR	R/W	0h	If set, PHY initialization at (re)boot or restart has not completed successfully Write one to clear. This bit is latched-high.
10:2	*reserved*	R	0h	Always write 0, ignore on read.
1	SMI_ERROR	R/W	0h	If set, this bit flags invalid SMI transactions to this PHY's register space, like writing to read-only registers and reading or writing non-existent registers.[1] Write one to clear. This bit is latched-high.
0	*reserved*	R	0h	Always write 0, ignore on read.

[1] SMI_ERROR gets set when reading clause 22 always accessible register or with a subsequent SMI access after a wrong clause 22 op code. Ignore SMI_ ERROR in those cases.

7.5.1.24 EPHY_CONFIG_REGXS_IRQ_ENABLES register

Table 333. EPHY_CONFIG_REGXS_IRQ_ENABLES register - Ethernet PHY configuration register access IRQs enables (address 80C1h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
14:13	*reserved*	R	0h	Always write 0, ignore on read.
12	RECONFIG_ DENIED	R/W	0h	If set, RECONFIG_DENIED_EN IRQ is enabled This setting is preserved during deep sleep.
11	INIT_ERROR	R/W	0h	If set, INIT_ERROR_EN IRQ is enabled This setting is preserved during deep sleep.
10:2	*reserved*	R	0h	Always write 0, ignore on read.
1	SMI_ERROR	R/W	0h	If set, SMI_ERROR IRQ is enabled This setting is preserved during deep sleep.
0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.25 EPHY_CONFIG_REGXS_IRQ_MSTATUS register

Table 334. EPHY_CONFIG_REGXS_IRQ_MSTATUS register - Ethernet PHY configuration register access IRQs masked status (address 80C2h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Ignore on read.
14:13	*reserved*	R	0h	Ignore on read.

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Table 334. EPHY_CONFIG_REGXS_IRQ_MSTATUS register - Ethernet PHY configuration register access IRQs masked status (address 80C2h)*...continued*

Bit	Symbol	Access	Value	Description
12	RECONFIG_ DENIED	R	0h	Masked status of RECONFIG_DENIED IRQ
11	INIT_ERROR	R	0h	Masked status of INIT_ERROR IRQ
10:2	*reserved*	R	0h	Ignore on read.
1	SMI_ERROR	R	0h	Masked status of SMI_ERROR IRQ
0	*reserved*	R	0h	Ignore on read.

7.5.1.26 EPHY_DIAGNOSTIC_IRQ_SOURCES register

Table 335. EPHY_DIAGNOSTIC_IRQ_SOURCES register - Ethernet PHY diagnostic IRQ sources (address 80D0h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Always write 0, ignore on read.
1	SQI_WARN	R/W	0h	If set, SQI dipped below the selected warning level. Write one to clear. This bit is latched-high.
0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.27 EPHY_DIAGNOSTIC_IRQ_ENABLES register

Table 336. EPHY_DIAGNOSTIC_IRQ_ENABLES register - Ethernet PHY diagnostic IRQ enables (address 80D1h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Always write 0, ignore on read.
1	SQI_WARN	R/W	0h	If set, SQI_WARN IRQ is enabled This setting is preserved during deep sleep.
0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.28 EPHY_DIAGNOSTIC_IRQ_MSTATUS register

Table 337. EPHY_DIAGNOSTIC_IRQ_MSTATUS register - Ethernet PHY diagnostic masked IRQ masked status (address 80D2h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Ignore on read.
1	SQI_WARN	R	0h	Masked status of SQI_WARN IRQ
0	*reserved*	R	0h	Ignore on read.

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7.5.1.29 PHY_CONTROL register

Table 338. PHY_CONTROL register - PHY control register (address 8100h)

Bit	Symbol	Access	Value	Description
15	RESET	R/W	0h	If set, the PHY is reset This bit is self-clearing.
14	CONFIG_ENABLE	R/W	0h	If set, configuration registers of this PHY can be changed. Otherwise access is restricted. This setting is preserved during deep sleep.
13:2	*reserved*	R	0h	Always write 0, ignore on read.
1	GOTO_STANDBY	R/W	0h	If set, the PHY is directed to go to STANDBY. This bit is self clearing. On read it returns 1, if the PHY is still in transition. This bit is self-clearing.
0	START_ OPERATION	R/W	0h	If set, exit standby state and start PHY operation. This bit is self clearing. On read it returns 1, if the PHY is still in transition. This bit is self-clearing.

7.5.1.30 PHY_STATUS register

Table 339. PHY_STATUS register - PHY Status register (address 8102h)

Bit	Symbol	Access	Value	Description
15:8	*reserved*	R	0h	Ignore on read.
7	DETECTED_ POLARITY	R	0h	If set, Polarity inversed with respect to pin definitions, polarity correction required
6	LINK_AVAILABLE	R	0h	If set, Link is available for use by higher OSI layers
5	SENDN_OR_DATA	R	0h	If set, PHY in SEND_IDLE_OR_DATA state
4	REM_RCVR_ STATUS	R	0h	If set, the remote receiver status is set
3	SCRAMBLER_ STATUS	R	0h	If set, the scrambler is synchronized
2	LINK_STATUS	R	0h	If set, the Link status from Link Monitor FSM is set
1	LOC_RCVR_ STATUS	R	0h	If set, the PHY is synchronized to the received signal and symbols values can be robustly detected
0	*reserved*	R	0h	If set, Additional status bits in register 8103h Ignore on read.

7.5.1.31 PHY_LATCHED_STATUS register

Table 340. PHY_LATCHED_STATUS register - PHY latched status register (address 8104h)

Bit	Symbol	Access	Value	Description
15:3	*reserved*	R	0h	Always write 0, ignore on read.
2	REMOTE_FAILURE	R/W	0h	If set, a link failure caused by remote event. Write one to clear. This bit is latched-high.

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Table 340. PHY_LATCHED_STATUS register - PHY latched status register (address 8104h)*...continued*

Bit	Symbol	Access	Value	Description
1	EXPIRED_MAX_WAIT	R/W	0h	If set, MAX_WAIT_TIMER has expired since last clear. Write one to clear. This bit is latched-high.
0	*reserved*	R	0h	If set, Additional latched status in register 8105h Always write 0, ignore on read.

7.5.1.32 PHY_ADDITIONAL_ABILITIES register

Table 341. PHY_ADDITIONAL_ABILITIES register - Additional PHY Abilities (address 8106h)

Bit	Symbol	Access	Value	Description
15:3	*reserved*	R	0h	Ignore on read.
2	TC10_ABILITY	R	1h	If set, PHY is able to support TC10 Wake-Sleep
1	BASIC_SLEEP_ABILITY	R	1h	If set, PHY is able to support SLEEP state
0	*reserved*	R	0h	If set, Additional abilities in register 8107h Ignore on read.

7.5.1.33 PHY_CONFIG register

Table 342. PHY_CONFIG register - PHY configuration register. This register can only be changed if CONFIG_ENABLE or SUPER_CONFIG_ENABLE are set to 1. (address 8108h)

Bit	Symbol	Access	Value	Description
15:6	*reserved*	R	0h	Always write 0, ignore on read.
5	STATUS_FILTER_DISABLE	R/W	0h	Burst errors robustness is enhanced by filtering signals that drive status bits and the state machine. This avoids burst error-triggered retraining when the PHY is still locked and will likely auto-recover quickly. This bit allows to turn off that filtering. Disabling is strongly discouraged This setting is configuration protected.
4	JUMBO_ENABLE	R/W	0h	If set, Enable support for jumbo packets This setting is configuration protected. This setting is preserved during deep sleep.
3	POLARITY_SWAP	R/W	0h	Polarity swap The bits [POLARITY_SWAP, POLARITY_CORRECT_DISABLE] encode the following behavior: 00 = Polarity correction, default pin polarity 01 = No polarity correction, default pin polarity 10 = Polarity correction plus forced polarity swap 11 = Forced polarity swap, no polarity correction This setting is configuration protected. This setting is preserved during deep sleep.
2	POLARITY_CORRECT_DISABLE	R/W	0h	If set, automatic polarity correction is disabled and no link established in normal mode if input polarity is swapped. This setting is configuration protected.

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Table 342. PHY_CONFIG register - PHY configuration register. This register can only be changed if CONFIG_ENABLE or SUPER_CONFIG_ENABLE are set to 1. (address 8108h)*...continued*

Bit	Symbol	Access	Value	Description
				This setting is preserved during deep sleep.
1	TEST_ENABLE	R/W	0h	This bit guards against accidental test mode selection of the Ethernet PHY. If this bit is not set, selecting a test mode has no effect, unless TEST_ENABLE was set on port or super level. Test modes do also include loopbacks, cable test, and other standardized or proprietary diagnostic features which are available to the customer but interfere with normal operation. This setting is configuration protected. This setting is preserved during deep sleep.
0	AUTO_OPERATION	R/W	0h	If set, PHY autonomously proceeds to NORMAL mode for both start-up and wake-up. The default configuration is latched from pin strapping. This setting is configuration protected.

7.5.1.34 PHY_STATE register

Table 343. PHY_STATE register - PHY state register (address 810Ch)

Bit	Symbol	Access	Value	Description
15	BUSY	R	0h	If set, PHY busy transitioning to next state
14	TEST_OR_ DIAGNOSIS	R	0h	If set, the PHY is not functionally operational but in a test or diagnosis state
13:12	*reserved*	R	0h	Ignore on read.
11:8	TRANSMIT_STATE	R	0h	Transmit state. The PHY reports DISABLED when in PMA test modes.
7:4	*reserved*	R	0h	Ignore on read.
3:0	BASIC_STATE	R	0h	Basic state. 0000 = DISABLED 0001 = STANDBY 0010 = FAILSAFE/Reserved 0011 = SLEEP 1010 = Reserved 1001 = SLAVE_SILENT 1000 = TRAINING 1100 = SEND_IDLE (state) 1101 = SEND_IDLE_OR_DATA 1110 = Reserved 1111 = SEND_IDLE_OR_DATA_SLEEP_CHECK 1011 = SLEEP_SILENT 0100 = Reserved xx11 = SLEEP-related

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7.5.1.35 PHY_PARAMETERS register

Table 344. PHY_PARAMETERS register - PHY parameters register. This register can only be changed if CONFIG_ENABLE or SUPER_CONFIG_ENABLE are set to 1. (address 810Eh)

Bit	Symbol	Access	Value	Description
15:3	*reserved*	R	0h	Always write 0, ignore on read.
2:0	LINK_AVAILABLE_HYST_TIME	R/W	0h	Time hysteresis on link availability assertion 000 = 0 ms 001 = 250 µs 010 = 500 µs 011 = 1 ms 100 = 2 ms 101 = 4 ms 110 = 8 ms 111 = 16 ms This setting is configuration protected. This setting is preserved during deep sleep.

7.5.1.36 WAKE_SLEEP_CONTROL register

Table 345. WAKE_SLEEP_CONTROL register - Wake sleep control register (address 8180h)

Bit	Symbol	Access	Value	Description
15	WAKEUP_PHY_REQUEST	R/W	0h	If set, wake-up request to this PHY This bit is self-clearing.
14:3	*reserved*	R	0h	Always write 0, ignore on read.
2	SLEEP_REJECT	R/W	0h	If set, reject sleep request This bit is self-clearing.
1	SLEEP_ACCEPT	R/W	0h	If set, accept sleep request received on MDI This bit is self-clearing.
0	SLEEP_REQUEST	R/W	0h	If set, start the handshake to bring this PHY and it link partner into sleep state This bit is self-clearing.

7.5.1.37 WAKE_SLEEP_STATUS register

Table 346. WAKE_SLEEP_STATUS register - Wake sleep status register (address 8181h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
14	WU_IO_RECEIVED	R/W	0h	If set, Wake-up request on WAKE_IN_OUT pin received. Write one to clear. This bit is latched-high.
13	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
12	WU_PHY_RECEIVED	R/W	0h	If set, Wake-up request received by WU_PHY bit Write one to clear. This bit is latched-high.

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Table 346. WAKE_SLEEP_STATUS register - Wake sleep status register (address 8181h)*...continued*

Bit	Symbol	Access	Value	Description
11	WUP_RECEIVED	R/W	0h	If set, Wake-Up Pulse received Write one to clear. This bit is latched-high.
10	WUR_RECEIVED	R/W	0h	If set, Wake-Up Relay received Write one to clear. This bit is latched-high.
9	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
8:4	*reserved*	R	0h	Always write 0, ignore on read.
3	AUTO_SLEEP_EVENT	R/W	0h	If set, Auto sleep has been triggered. This bis is cleared automatically, when the PHY wakes up. Write one to clear. This bit is latched-high.
2	SLEEP_FAILED	R/W	0h	If set, Failed to enter Sleep mode. This bis is cleared automatically, when the PHY wakes up. Write one to clear. This bit is latched-high.
1	LPS_RECEIVED	R/W	0h	If set, Sleep request received. This bis is cleared automatically, when the PHY wakes up. Write one to clear. This bit is latched-high.
0	SLEEP_STATUS	R	0h	If set, PHY has entered Sleep state

7.5.1.38 WAKE_SLEEP_CONFIG register

Table 347. WAKE_SLEEP_CONFIG register - Wake sleep configuration register. This register can only be changed if CONFIG_ENABLE or SUPER_CONFIG_ENABLE are set to 1. (address 8182h)

Bit	Symbol	Access	Value	Description
15:14	FUNCTION_SELECT	R/W	2h	Wakeup sleep function selection register 00 = No Wake-Sleep functionality 01 = Basic Wake-Sleep functionality 10 = TC10 Wake-Sleep functionality 11 = Reserved This setting is configuration protected. This setting is preserved during deep sleep.
13:12	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
11	WU_IO_ENABLE	R/W	0h	If set, wake-up the PHY on WAKE_IN_OUT pin assertion This setting is configuration protected. This setting is preserved during deep sleep.
10	WU_SMI_ENABLE	R/W	0h	If this bit is set to one, this PHY wakes up when the system wake-up WU_SMI bit is asserted in register ALWAYS_ACCESSIBLE. This setting is configuration protected. This setting is preserved during deep sleep.

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Table 347. WAKE_SLEEP_CONFIG register - Wake sleep configuration register. This register can only be changed if CONFIG_ENABLE or SUPER_CONFIG_ENABLE are set to 1. (address 8182h)*...continued*

Bit	Symbol	Access	Value	Description
9	FWD_WU_PHY_TO_WU_IO	R/W	0h	If set, the PHY forwards a wake request by WAKE_REQUEST_PHY control bit to WAKE_IN_OUT This setting is configuration protected. This setting is preserved during deep sleep.
8	FWD_WUPWUR_TO_WU_IO	R/W	0h	If set, a WUP/WUR requests received on MDI by this PHY is forwarded to the device WAKE_IN_OUT pin This setting is configuration protected. This setting is preserved during deep sleep.
7:6	*reserved*	R	0h	Always write 0, ignore on read.
5	SLEEP_SILENT_CHECK	R/W	1h	If set, sleep responder waits for the initiator to become silent before entering SLEEP_SILENT. This setting is configuration protected. This setting is preserved during deep sleep.
4	WUR_IN_SEND_LPS	R/W	1h	If this bit is set to zero, a wake-up request during SEND_LPS will never be serviced directly but be stored and serviced after the sleep request procedure completes, successful or not. If this bit is set to one, a wake-up request occurring during SEND_LPS before either PHY has become SILENT, will cause a WUR to be transmitted. Note that this WUR will imply discontinuation of the LPS sequence. This WUR provides fastest forwarding of the wake-up event and stops the sleep request if possible. If the sleep process wasn't stopped by the WUR, the pending wake-up will be serviced after the sleep request process completes. This setting is configuration protected. This setting is preserved during deep sleep.
3	AUTO_REJECT	R/W	0h	If set, a sleep request is automatically rejected on SLEEP_ACK_TIMER expiration. Otherwise, sleep requests are auto-accepted by the responder on SLEEP_ACK_TIMER expiration (TC10) This setting is configuration protected. This setting is preserved during deep sleep.
2	AUTO_SLEEP_ON_IDLE	R/W	0h	If set, long idle autonomous sleep request enabled This setting is configuration protected. This setting is preserved during deep sleep.
1	AUTO_SLEEP_ON_SILENCE	R/W	1h	If set, a slave automatically returns to SLEEP after wake-up if no signal is received from the master This setting is configuration protected. This setting is preserved during deep sleep.
0	NO_AUTO_ON_WAKE	R/W	0h	If set, the PHY pauses in "active STANDBY" after wake-up until the START_OPERATION bit is set. Otherwise, the PHY autonomously proceeds to normal mode after wake-up. This setting is configuration protected. This setting is preserved during deep sleep.

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7.5.1.39 WAKE_SLEEP_PARAMETERS register

Table 348. WAKE_SLEEP_PARAMETERS register - Wake sleep parameters register. This register can only be changed if CONFIG_ENABLE or SUPER_CONFIG_ENABLE are set to 1. (address 8184h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14:12	SLEEP_ACK_ TIMER	R/W	6h	Timer for auto accept (or reject) sleep request: 0 = 0 µs, 1 = 250 µs, 2 = 500 µs, 3 = 1 ms, 4 = 2 ms, 5 = 4 ms, 6 = 8 ms, 7 = 12 ms This setting is configuration protected. This setting is preserved during deep sleep.
11	*reserved*	R	0h	Always write 0, ignore on read.
10:8	AUTO_SLEEP_ IDLE_TIMER	R/W	3h	Auto sleep idle timer 0 = 125 ms, 1 = 250 ms, 2 = 500 ms, 3 = 1s, 4 = 2 s, 5 = 4 s, 6 = 8 s, 7 = 16 s This setting is configuration protected. This setting is preserved during deep sleep.
7	*reserved*	R	0h	Always write 0, ignore on read.
6:4	AUTO_SLEEP_ SILENCE_TIMER	R/W	3h	Auto sleep silence timer 0 = 125 ms, 1 = 250 ms, 2 = 500 ms, 3 = 1 s, 4 = 2 s, 5 = 4 s, 6 = 8 s, 7 = 16 s This setting is configuration protected. This setting is preserved during deep sleep.
3	*reserved*	R	0h	Always write 0, ignore on read.
2:0	MIN_SILENT_ TIMER	R/W	3h	Minimum duration of SLEEP_SILENT state: 0 = 125 µs, 1 = 250 µs, 2 = 500 µs, 3 = 1 ms, 4 = 2 ms, 5 = 4 ms, 6 = 8 ms, 7 = 16 ms This setting is configuration protected. This setting is preserved during deep sleep.

7.5.1.40 EPHY_LED_TRIGGER0 register

Table 349. EPHY_LED_TRIGGER0 register - Ethernet PHY LED trigger output 0 register (address 8190h)

Bit	Symbol	Access	Value	Description
15	ENABLE	R/W	1h	If set, trigger is enabled
14:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	FUNCTION	R/W	0h	LED trigger function selection 0 = Link_available 1 = Link_status 2 = Role: master(1) or slave(0) 3 = EPHY active (AFE enabled) 4 = Frame reception (RX_DV = true) 5 = Symbol error 6 = Frame transmission (TX_EN = true) 7 = Frame activity (RX_DV=true | TX_EN = true)

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7.5.1.41 EPHY_LED_TRIGGER1 register

Table 350. EPHY_LED_TRIGGER1 register - Ethernet PHY LED trigger output 1 register, see EPHY_LED_TRIGGER0 (address 8191h)

Bit	Symbol	Access	Value	Description
15	ENABLE	R/W	1h	If set, trigger is enabled
14:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	FUNCTION	R/W	4h	LED trigger function selection, see EPHY_LED_TRIGGER0

7.5.1.42 EPHY_LED_TRIGGER2 register

Table 351. EPHY_LED_TRIGGER2 register - Ethernet PHY LED trigger output 2 register, see EPHY_LED_TRIGGER0 (address 8192h)

Bit	Symbol	Access	Value	Description
15	ENABLE	R/W	1h	If set, trigger is enabled
14:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	FUNCTION	R/W	6h	LED trigger function selection, see EPHY_LED_TRIGGER0

7.5.1.43 EPHY_LED_TRIGGER3 register

Table 352. EPHY_LED_TRIGGER3 register - Ethernet PHY LED trigger output 3 register, see EPHY_LED_TRIGGER0 (address 8193h)

Bit	Symbol	Access	Value	Description
15	ENABLE	R/W	0h	If set, trigger is enabled
14:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	FUNCTION	R/W	0h	LED trigger function selection, see EPHY_LED_TRIGGER0

7.5.1.44 TXC_STATUS register

Table 353. TXC_STATUS register - TXC status register (address 8319h)

Bit	Symbol	Access	Value	Description
15	HIGH_FQ_EVENT	R/W	0h	If set, Ratio between the TXC and FRO frequencies was too high. Write one to clear. This bit is latched-high.
14	LOW_FQ_EVENT	R/W	0h	If set, Ratio between the TXC and FRO frequencies was too low. Write one to clear. This bit is latched-high.
13	HIGH_FQ_NOW	R	0h	If set, Ratio of TXC and FRO frequencies is too high
12	LOW_FQ_NOW	R	0h	If set, Ratio of TXC and FRO frequencies is too low
11	CLK_PRESENT	R	0h	If set, TXC PRESENT
10:0	*reserved*	R	0h	Always write 0, ignore on read.

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7.5.1.45 RXC_STATUS register

Table 354. RXC_STATUS register - RXC status register (address 831Ah)

Bit	Symbol	Access	Value	Description
15	HIGH_FQ_EVENT	R/W	0h	If set, Ratio between the RXC and FRO frequencies was too high. Write one to clear. This bit is latched-high.
14	LOW_FQ_EVENT	R/W	0h	If set, Ratio between the RXC and FRO frequencies was too low. Write one to clear. This bit is latched-high.
13	HIGH_FQ_NOW	R	0h	If set, Ratio of RXC and FRO frequencies is too high
12	LOW_FQ_NOW	R	0h	If set, Ratio of RXC and FRO frequencies is too low
11	CLK_PRESENT	R	0h	If set, RXC PRESENT
10:0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.46 SGMII_LATENT_STATUS register

Table 355. SGMII_LATENT_STATUS register - SGMII latent fault status register (address 831Bh)

Bit	Symbol	Access	Value	Description
15	TXPLL_STATUS	R/W	0h	SGMII TX PLL latent fault event detected at startup. Write one to clear. This bit is latched-high.
14	RXPLL_STATUS	R/W	0h	SGMII RX PLL latent fault event detected at startup. Write one to clear. This bit is latched-high.
13:11	*reserved*	R	0h	Always write 0, ignore on read.
10	LINK_STATUS	R/W	0h	SGMII LINK latent fault detected at startup. Write one to clear. This bit is latched-high.
9:0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.47 EPHY_LATENT_STATUS register

Table 356. EPHY_LATENT_STATUS register - Ethernet PHY latent fault status register (address 831Ch)

Bit	Symbol	Access	Value	Description
15	PLL_STATUS	R/W	0h	EPHY PLL Latent Fault. Write one to clear. This bit is latched-high.
14	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
13	LOC_RCVR_STATUS	R/W	0h	EPHY LOC RCVR latent fault detected at startup. Write one to clear. This bit is latched-high.

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Table 356. EPHY_LATENT_STATUS register - Ethernet PHY latent fault status register (address 831Ch)*...continued*

Bit	Symbol	Access	Value	Description
12	REM_RCVR_ STATUS	R/W	0h	EPHY REM RCVR latent fault detected at startup. Write one to clear. This bit is latched-high.
11	SCR_STATUS	R/W	0h	EPHY SCR latent fault detected at startup. Write one to clear. This bit is latched-high.
10	LINK_STATUS	R/W	0h	EPHY LINK latent fault detected at startup. Write one to clear. This bit is latched-high.
9:0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.48 SGMII_CLK_STATUS register

Table 357. SGMII_CLK_STATUS register - SGMII clock status register (address 831Dh)

Bit	Symbol	Access	Value	Description
15	HIGH_FQ_EVENT	R/W	0h	If set, Ratio of TCLK and REFCLK frequencies has been too high. Write one to clear. This bit is latched-high.
14	LOW_FQ_EVENT	R/W	0h	If set, Ratio of TCLK and REFCLK frequencies has been too low. Write one to clear. This bit is latched-high.
13	HIGH_FQ_NOW	R	0h	If set, Ratio of TCLK and REFCLK frequencies is too high
12	LOW_FQ_NOW	R	0h	If set, Ratio of TCLK and REFCLK frequencies is too low
11	CLK_PRESENT	R	0h	If set, SGMII CLK PRESENT
10:0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.49 EPHY_CLK_STATUS register

Table 358. EPHY_CLK_STATUS register - Ethernet PHY clock status (address 831Eh)

Bit	Symbol	Access	Value	Description
15	HIGH_FQ_EVENT	R/W	0h	If set, Ratio of TCLK and REFCLK frequencies has been too high. Write one to clear. This bit is latched-high.
14	LOW_FQ_EVENT	R/W	0h	If set, Ratio of TCLK and REFCLK frequencies has been too low. Write one to clear. This bit is latched-high.
13	HIGH_FQ_NOW	R	0h	If set, Ratio of TCLK and REFCLK frequencies is too high
12	LOW_FQ_NOW	R	0h	If set, Ratio of TCLK and REFCLK frequencies is too low

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Table 358. EPHY_CLK_STATUS register - Ethernet PHY clock status (address 831Eh)*...continued*

Bit	Symbol	Access	Value	Description
11	CLK_PRESENT	R	0h	If set, ePHY CLK PRESENT
10:0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.50 SIGNAL_QUALITY register

Table 359. SIGNAL_QUALITY register - Signal quality (address 8320h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R/W	0h	always write 0, 0s, all 0, 0000, ignore on read.
14	VALID	R	0h	If set, the SQI value is valid
13:11	*reserved*	R	0h	Always write 0, ignore on read.
10:8	SQI_WARN_LIMIT	R/W	0h	SQI warning threshold. 000 = no warn limit 001 = class A SQI limit 010 = class B SQI limit 011 = class C SQI limit 100 = class D SQI limit 101 = class E SQI limit 110 = class F SQI limit 111 = class G SQI limit Exceeding the warn limit (SQI value is below the programmed value) will fire an SQI_WARN_IRQ Exceeding the warn limit (SQI value is below the programmed value) will fire an SQI_WARN_IRQ This setting is preserved during deep sleep.
7	*reserved*	R	0h	Always write 0, ignore on read.
6:4	SQI_WORST	R	7h	Lowest SQI level since last cleared. Reading sets the value to the current SQI level. This bit is latched-low.
3	*reserved*	R	0h	Always write 0, ignore on read.
2:0	SQI	R	0h	Current SQI level

7.5.1.51 MSE register

Table 360. MSE register - Mean-square error register (address 8324h)

Bit	Symbol	Access	Value	Description
15	ENABLE	R/W	0h	If set, the MSE is calculated. Otherwise, MSE reports zeros. This setting is preserved during deep sleep.
14:12	*reserved*	R	0h	Always write 0, ignore on read.
11:4	MSE	R	0h	Mean-square value of slicer error
3:0	*reserved*	R	0h	Always write 0, ignore on read.

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7.5.1.52 MAX_MSE register

Table 361. MAX_MSE register - Max mean-square error register (address 8325h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Ignore on read.
11:4	MAX_MSE	R	0h	The MAX_MSE tracks the peak MSE value. This field is cleared on read. MSE values while the PHY is in SEND_IDLE_OR_DATA. During all other PHY states this register value will be preserved. This flag is cleared on read. This bit is latched-high.
3:0	*reserved*	R	0h	Ignore on read.

7.5.1.53 CABLE_TEST register

Table 362. CABLE_TEST register - Cable test register (address 8330h)

Bit	Symbol	Access	Value	Description
15	ENABLE	R/W	0h	If set, Cable test functionality enabled
14	START	R/W	0h	Start This bit is self-clearing.
13	VALID	R	0h	If set, test results are available and valid
12:3	*reserved*	R	0h	Always write 0, ignore on read.
2:0	FAULT_TYPE	R	0h	000 = No cabling fault detected 001 = Shorted pair 010 = Open pair 011, 100, 101, 110 = Reserved 111 = Not able to detect

7.5.1.54 LINK_TRAINING_TIMER register

Table 363. LINK_TRAINING_TIMER register - Link training timer register (address 8340h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Ignore on read.
11:4	TIME_MS	R	0h	This value returns the time in ms and saturates at 251ms. If invalid, 255 is returned. The measurement is started when the PHY initially enters SLAVE_SILENT and stopped when LINK_AVAILABILITY=1.
3:1	TIME_SUBMS	R	0h	Fractional part in multiples of 125us
0	*reserved*	R	0h	Ignore on read.

7.5.1.55 LOC_RCVR_STATUS_TIMER register

Table 364. LOC_RCVR_STATUS_TIMER register - Local receiver status timer register (address 8341h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Ignore on read.

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Table 364. LOC_RCVR_STATUS_TIMER register - Local receiver status timer register (address 8341h) *...continued*

Bit	Symbol	Access	Value	Description
11:4	TIME_MS	R	0h	This value returns the time in ms and saturates at 251ms. If invalid, 255 is returned. The measurement is started when the PHY initially enters SLAVE_SILENT and stopped when LOC_RCVR_STATUS=1.
3:1	TIME_SUBMS	R	0h	Fractional part in multiples of 125us
0	*reserved*	R	0h	Ignore on read.

7.5.1.56 REM_RCVR_STATUS_TIMER register

Table 365. REM_RCVR_STATUS_TIMER register - Remote receiver status timer register (address 8342h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Ignore on read.
11:4	TIME_MS	R	0h	This value returns the time in ms and saturates at 251ms. If invalid, 255 is returned. The measurement is started when the PHY initially enters SLAVE_SILENT and stopped when REM_RCVR_STATUS=1.
3:1	TIME_SUBMS	R	0h	Fractional part in multiples of 125us
0	*reserved*	R	0h	Ignore on read.

7.5.1.57 SLAVE_SILENT_TIMER register

Table 366. SLAVE_SILENT_TIMER register - slave silent timer register (address 8343h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Ignore on read.
11:4	TIME_MS	R	0h	This value returns the time in ms and saturates at 251ms. The measurement is started when the PHY initially enters SLAVE_SILENT and ends when the PHY transitions to TRAINING.
3:1	TIME_SUBMS	R	0h	Fractional part in multiples of 125us
0	*reserved*	R	0h	Ignore on read.

7.5.1.58 SYMBOL_ERROR_COUNTER register

Table 367. SYMBOL_ERROR_COUNTER register - Symbol error counter register (address 8350h)

Bit	Symbol	Access	Value	Description
15:0	SYMBOL_ERRORS	R	0h	This register counts symbol errors when LINK_AVAILABLE=1 and ERROR_COUNTER_MISC[COUNTERS_ENABLE]=1. When reaching its maximum value, this register does not saturate, but automatically rolls-over to zero. The value does not get reset by link retraining, but does not increment while LINK_AVAILABLE=0. The counter is reset when the counter is disabled (COUNTER_ENABLE=0), during a power-cycle, when inhibit

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Table 367. SYMBOL_ERROR_COUNTER register - Symbol error counter register (address 8350h)

Bit	Symbol	Access	Value	Description
				is de-asserted, on PHY_CONTROL[RESET] or MMD1.PMA_CONTROL1[RESET]

7.5.1.59 ERROR_COUNTER_MISC register

Table 368. ERROR_COUNTER_MISC register - Error counter miscellaneous register (address 8352h)

Bit	Symbol	Access	Value	Description
15	COUNTER_ENABLE	R/W	1h	If set, the SYM_ERRORS, LINK_AVAILABLE_DROPS, LINK_STATUS_DROPS, LINK_FAILURES and LINK_LOSSES counters are enabled. This setting is preserved during deep sleep.
14	*reserved*	R	0h	Always write 0, ignore on read.
13:8	LINK_STATUS_DROPS	R	0h	This register counts how many times the non-latched LINK_STATUS bit has dropped unintentionally. The value saturates at max value. The counter is reset when the counter is disabled (COUNTER_ENABLE=0), during a power-cycle, when inhibit is de-asserted, on PHY_CONTROL[RESET] or MMD1.PMA_CONTROL1[RESET]
7:6	*reserved*	R	0h	Always write 0, ignore on read.
5:0	LINK_AVAILABLE_DROPS	R	0h	This register counts the number of times that LINK_AVAILBILITY has dropped unintentionally. The value saturates at max value. The counter is reset when the counter is disabled (COUNTER_ENABLE=0), during a power-cycle, when inhibit is de-asserted, on PHY_CONTROL[RESET] or MMD1.PMA_CONTROL1[RESET]

7.5.1.60 LINK_LOSSES_AND_FAILURES register

Table 369. LINK_LOSSES_AND_FAILURES register - Link losses and failures counter register (address 8353h)

Bit	Symbol	Access	Value	Description
15:10	LINK_LOSSES	R	0h	A link loss is counted when the LOC_RCVR_STATUS becomes 0 and the PHY will go through a full training.
9:0	LINK_FAILURES	R	0h	If COUNTER_ENABLED is set, this counter counts the number of link failures that do not cause a link drop. This includes SSD and ESD errors, as well as transmitting SEND_I when the REM_RCVR_STATUS=0 and LOC_RCVR_STATUS=1. Note that symbol errors in an idle sequence are covered by the symbol error counter and not counted as link failure. The value saturates at max value. The counter is reset when the counter is disabled (COUNTER_ENABLE=0), during a power-cycle, when inhibit is de-asserted, on PHY_CONTROL[RESET] or MMD1.PMA_CONTROL1[RESET]

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7.5.1.61 PHY_COMPLIANCE_TEST register

Table 370. PHY_COMPLIANCE_TEST register - PHY compliance test control register (address 83F0h)

Bit	Symbol	Access	Value	Description
15:1	*reserved*	R	0h	Always write 0, ignore on read.
0	TX_TCLK_EN	R/W	0h	If enabled, the PHY provides the transmit clock for jitter test on an external pin. This setting is configuration protected.

8 Thermal characteristics

Table 371. Thermal characteristics

Value determined for free convection conditions on a JEDEC 2S2P board[1].

Symbol	Parameter	Conditions	Typ	Unit
$R_{th(j-a)}$	thermal resistance from junction to ambient		38	K/W
$R_{th(j-c)}$	thermal resistance from junction to case	[2]	14	K/W
Ψ_{j-top}	thermal characterization parameter from junction to top of package		8	K/W

[1] According to JEDEC JESD51-2, JESD51-5 and JESD51-7 at natural convection on 2s2p board. Board with two inner copper layers (thickness: 35 µm) and thermal via array under the exposed pad connected to the first inner copper layer (thickness: 70 µm)

[2] Case temperature refers to the center of the heatsink at the bottom of the package.

9 Limiting values

Table 372. Limiting values

In accordance with the Absolute Maximum Rating System (IEC 60134); all voltages are defined with respect to ground unless otherwise specified.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_x	Voltage on pin x	on pins MDC, INT_N, RST_N, INH, WAKE_IN_OUT, GPIO0, GPIO1, RX_ER, RX_CTL, RXDx, RXC, TX_ER, TXC, TX_CTL, TXDx, MDIO, GPIOx	-0.5	—	4	V
		on pins XI, XO	-0.5	—	1.6	V
		on pins TRX_P, TRX_M	-0.5	—	4	V
		on pins SIN, SIP	-0.5	—	1.6	V
		on pins SOP, SON	-0.5	—	1.6	V
		on pin VDDIO	-0.5	—	4	V
		on pin VDD_CORE	-0.5	—	1.6	V
		on pin VREGA_OUT	-0.5	—	4	V
		on pin VDDA_AO	-0.5	—	4	V
		on pin VREGD_IN, VREGA_IN	-0.5	—	4	V

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Table 372. Limiting values...*continued*

In accordance with the Absolute Maximum Rating System (IEC 60134); all voltages are defined with respect to ground unless otherwise specified.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
		on pins VDDA_TRX	-0.5	—	4	V
		on pins VDDA_SI, VDDA_SO	-0.5	—	4	V
V_{trt}	transient voltage	on on pins TRX_P, TRX_M [1]				
		pulse 1	-100	—	—	V
		pulse 2a	—	—	+75	V
		pulse 3a	-150	—	—	V
		pulse 3b	—	—	100	V
V_{ESD}	electrostatic discharge voltage	IEC 61000-4-2; 150 pF, 330 Ω [2]				
		on pins TRX_P, TRX_M to GND	-6.0	—	+6.0	kV
V_{ESD}	electrostatic discharge voltage	Human Body Model (HBM)				
		on any pin[3]	-2	—	+2	kV
		on pins TRX_P, TRX_M to GND[4]	-4	—	+4	kV
V_{ESD}	electrostatic discharge voltage	Charge Device Model (CDM)				
		on all pins[5]	-500	—	+500	V
T_{vj}	virtual junction temperature	[6]	-40	—	+150	°C
T_{stg}	storage temperature	[7]	-55	—	+150	°C
T_{amb}	ambient temperature	[8]	-40	—	+125	°C

[1] Verified by an external test house according to IEC TS62228 Section 4.2.4; parameters for standardisation defined in ISO7637 Part 2
[2] Verified by an external test house according to IEC TS62228 Section 4.3
[3] According to AEC-Q100-002
[4] Pins stressed to GND, emulating the application circuits (Section 13). HBM pulse as specified in AECQ100-002 used.
[5] According to AEC-Q100-011
[6] In accordance with IEC 60747-1. An alternative definition of virtual junction temperature is: T_{vj} = Tamb + P × Rth(vj-a), where Rth(vj-a) is the thermal resistance from virtual junction to ambient of the device (see Table 371) and P is the customer-application-controlled power dissipation. The rating for T_{vj} limits the allowable combinations of power dissipation (P) and ambient temperature (Tamb).
[7] T_{stg} in application according to IEC61360-4. For component transport and storage conditions, see instead IEC61760-2.
[8] Specification for virtual junction temperature T_{vj} is leading, i.e., maximum ambient temperature of 125 °C can only be sustained if self-heating through power dissipation is ≤ 25 °C.

10 Static characteristics

Table 373. Static characteristics

T_{vj} = -40 °C to + 150 °C; all voltages are defined with respect to ground unless otherwise specified; positive currents flow into the IC. [1]

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
Power supply						
V_{DD}	supply voltage	on pin VDDIO, 1.8 V signaling	1.65	1.8	1.95	V
		on pin VDDIO, 2.5 V signaling	2.25	2.5	2.75	V
		on pin VDDIO, 3.3 V signaling	3.00	3.30	3.6	V

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Table 373. Static characteristics...*continued*

T_{vj} = -40 °C to + 150 °C; all voltages are defined with respect to ground unless otherwise specified; positive currents flow into the IC. [1]

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
		on pin VDD_CORE if VREGD is bypassed and VREGD_IN is connected to VDD_CORE.	1.045	1.1	1.155	V
		on pin VDDA_AO, 3.3 V operation [2]	3.135	3.3	3.465	V
		on pin VDDA_AO, 2.5 V operation	2.375	2.5	2.625	V
		on pin VREGD_IN, if integrated VREGD LDO is enabled	3.135	3.3	3.465	V
		on pin VREGA_IN, if integrated VREGA LDO is enabled	3.135	3.3	3.465	V
		on pins VDDA_TRX if VREGA is bypassed and VREGA_IN is connected to VDDA_TRX.	2.375	2.5	2.625	V
		on pins VDDA_SI, VDDA_SO	1.045	1.1	1.155	V
Current consumption (single supply mode)[3][4]						
$I_{DD(RMS)}$	RMS supply current	on pin VREGA_IN; mode DEVICE_OPERATIONAL [5]	—	A/C = 18 B/D = 22	A/C = 24 B/D = 27	mA
		on pin VREGA_IN; mode DEVICE_STANDBY [5]	—	1	1	mA
		on pin VREGD_IN; variants B and D; mode DEVICE_OPERATIONAL 100BASE-T1 LEADER mode; 100 BASE-T1 link-up and SGMII link-up; MACsec in bypass	—	52	62	mA
		on pin VREGD_IN; variants B and D; mode DEVICE_OPERATIONAL 100BASE-T1 LEADER mode; 100 BASE-T1 link-up and SGMII link-up; MACsec operational 100 % load RX/TX	—	58	72	mA
		on pin VREGD_IN; variants B and D; mode DEVICE_OPERATIONAL 100BASE-T1 FOLLOWER mode; 100 BASE-T1 link-up and SGMII link-up; MACsec in bypass	—	52	62	mA
		on pin VREGD_IN; variants B and D; mode DEVICE_OPERATIONAL 100BASE-T1 FOLLOWER mode; 100 BASE-T1 link-up and SGMII link-up; MACsec operational 100 % load RX/TX	—	58	72	mA
		on pin VREGD_IN; variants B and D;	—	32	40	mA

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Table 373. Static characteristics...*continued*

T_{vj} = -40 °C to + 150 °C; all voltages are defined with respect to ground unless otherwise specified; positive currents flow into the IC. [1]

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
		mode DEVICE_STANDBY; SGMII link-up				
		on pin VREGD_IN; variants B and D mode DEVICE_SLEEP; SGMII link-down	—	11	18	mA
		on pin VREGD_IN; variants A and C; mode DEVICE_OPERATIONAL 100BASE-T1 LEADER mode; 100 BASE-T1 link-up; MACsec in bypass	—	19	25	mA
		on pin VREGD_IN; variants A and C; mode DEVICE_OPERATIONAL 100BASE-T1 LEADER mode; 100 BASE-T1 link-up; MACsec operational 100 % load RX/TX	—	26	35	mA
		on pin VREGD_IN; variants A and C; mode DEVICE_OPERATIONAL 100BASE-T1 FOLLOWER mode; 100 BASE-T1 link-up; MACsec in bypass	—	19	29	mA
		on pin VREGD_IN; variants A and C; mode DEVICE_OPERATIONAL 100BASE-T1 FOLLOWER mode; 100 BASE-T1 link-up; MACsec operational 100 % load RX/TX	—	28	38	mA
		on pin VREGD_IN; variants A and C; mode DEVICE_STANDBY	—	10	15	mA
		on pin VREGD_IN; variants A and C; mode DEVICE_SLEEP [6]	—	10	15	mA
Current consumption (individual supply mode)[3][4]						
$I_{DD(RMS)}$	RMS supply current	on pins VDDA_TRX; DEVICE_OPERATIONAL	—	17	19	mA
		on pin VDDA_TRX; mode DEVICE_STANDBY	—	0.25	0.35	mA
		on pin VDDA_TRX; mode DEVICE_SLEEP	—	0.25	0.35	mA
		on pin VDDA_SI; SGMII link-up; (variants B and D only)	—	—	25	mA
		on pin VDDA_SO; SGMII link-up (variants B and D only)	—	—	15	mA
		on pin VDD_CORE mode DEVICE_OPERATIONAL 100BASE-T1 LEADER mode; 100 BASE-T1 link-up; MACsec in bypass	—	22	30	mA
		on pin VDD_CORE	—	29	38	mA

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Table 373. Static characteristics...*continued*

T_{vj} = -40 °C to + 150 °C; all voltages are defined with respect to ground unless otherwise specified; positive currents flow into the IC. [1]

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
		mode DEVICE_OPERATIONAL 100BASE-T1 LEADER mode; 100 BASE-T1 link-up; MACsec operational 100 % load RX/TX				
		on pin VDD_CORE mode DEVICE_OPERATIONAL 100BASE-T1 FOLLOWER mode; 100 BASE-T1 link-up; MACsec in bypass	—	22	30	mA
		on pin VDD_CORE mode DEVICE_OPERATIONAL 100BASE-T1 FOLLOWER mode; 100 BASE-T1 link-up; MACsec operational 100 % load RX/TX	—	29	38	mA
		on pin VDD_CORE mode DEVICE_STANDBY	—	10	20	mA
		on pin VDD_CORE; mode DEVICE_SLEEP	—	10	20	mA
		on pin VREGD_IN [7]	—	—	0	mA
Current consumption (IO)[3][4]						
$I_{DD(RMS)}$	RMS supply current	on pins VDDIO; 3.3V VDDIO [8]				
		MII mode; frame size 1500 bytes; default IO settings; payload pattern 0xA5 [9]	—	—	30	mA
		RMII mode; frame size 1500 bytes; default IO settings; payload pattern 0xCC [9]	—	—	18	mA
		RGMII mode; frame size 1500 bytes; default IO settings; payload pattern 0xA5 [9]	—	—	24	mA
		MII mode; frame size 128 bytes; PRBS pattern; default IO settings; 25°C ambient [9]	—	20	—	mA
		RMII mode; frame size 128 bytes; PRBS pattern; default IO settings; 25°C ambient [9]	—	3	—	mA
		RGMII mode; frame size 128 bytes; PRBS pattern; default IO settings; 25°C ambient [9]	—	12	—	mA
Current consumption (other pins)[3]						
$I_{DD(RMS)}$	RMS supply current	on pin VDDA_AO	—	—	1	mA
I_{sleep}	sleep current	on pin VDDA_AO; device mode DEVICE_DEEP_SLEEP [10]	—	20	55	uA
Integrated LDO (VREGD)						

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Table 373. Static characteristics...*continued*

T_{vj} = -40 °C to + 150 °C; all voltages are defined with respect to ground unless otherwise specified; positive currents flow into the IC. [1]

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_O	output voltage	on pin VDD_CORE	1.078	1.1	1.137	V
I_O	output current	on pin VDD_CORE	—	50	65	mA
$V_{th(act)}$	regulator activation threshold	on pin VREGD_IN [3] Below the activation threshold, the device expects a supply connected directly to VDD_CORE	2.26	—	—	V
I_{PD}	power-down current	LDO disabled [3]	1	—	30	uA
Integrated LDO (VREGA)						
V_O	output voltage	on pin VDDA_OUT	2.437	2.5	2.59	V
I_O	output current	on pin VDDA_OUT	—	—	25	mA
$V_{th(act)}$	regulator activation threshold	on pin VREGA_IN [3] Below the activation threshold, the device expects a supply connected directly to VDDA_TRX	3.0	—	—	V
Digital I/O pins[11]						
V_{IH}	HIGH-level input voltage	3.3, 2.5, 1.8 V signaling [12]	0.7 × V_{DDIO}	—	—	V
V_{IL}	LOW-level input voltage	3.3, 2.5, 1.8 V signaling	—	—	0.3 × V_{DDIO}	V
V_{OH}	HIGH-level output voltage	3.3 V signaling I_{OH} = -2 mA	V_{DDIO} - 0.4	—	—	V
		2.5 V signaling I_{OH} = -2 mA	V_{DDIO} - 0.4	—	—	V
		1.8 V signals I_{OH} = -2 mA	V_{DDIO} - 0.4	—	—	V
V_{OL}	LOW-level output voltage	3.3 V signaling I_{OL} = 2 mA	—	—	0.4	V
		2.5 V signaling I_{OL} = 2 mA	—	—	0.4	V
		1.8 V signaling I_{OL} = 2 mA	—	—	0.4	V
C_i	input capacitance	[3]	—	—	5.0	pF
V_{hys}	input hysteresis voltage	[3]	0.1 × V_{DDIO}	—	—	V
I_{OSH}	HIGH-level short-circuit output current	[3]	—	—	144	mA
I_{OSL}	LOW-level short-circuit output current	[3]	—	—	144	mA
I_{IL}	HIGH-level input current	V_{IL}= $V_{DD(IO)}$ internal pull-up or pull-down disabled	—	—	10	uA

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Table 373. Static characteristics...*continued*

T_{vj} = -40 °C to + 150 °C; all voltages are defined with respect to ground unless otherwise specified; positive currents flow into the IC. [1]

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
I_{IH}	LOW-level input current	V_{IL}=0 V internal pull-up or pull-down disabled	-10	—	—	uA
R_{pd}	pull-down resistance		40	50	62	kΩ
R_{pu}	pull-up resistance		40	50	62	kΩ
R	output impedance	Default register settings. VDDIO = 3.3 V [3]	—	50	—	Ω
		Default register settings. VDDIO = 2.5 V [3]	—	50	—	Ω
		Default register settings. VDDIO = 1.8 V [3]	—	50	—	Ω
SGMII[3]						
$\|V_{o(dif)}\|$	differential output voltage (absolute value)	R_L = 100 Ω	150	—	400	mV
$R_{term(se)}$	single-ended termination resistance		—	50	—	Ω
$V_{ring(o)}$	ringing voltage		—	—	0.1 × Vo(dif)	V
$Z_{o(se)}$	single-ended output impedance		40	—	140	Ω
$\Delta Z_{o(se)}$	single-ended output impedance mismatch		—	—	10	%
$\|\Delta V_{o(dif)}\|$	differential output voltage difference	change in differential output voltage between complementary output states R_L = 100 Ω	—	—	25	mV
ΔV_{cm}	common-mode voltage difference	change in output offset voltage between complementary output states R_L = 100 Ω	—	—	25	mV
$V_{th(i)dif}$	differential input threshold voltage		-50	—	50	mV
$Z_{i(dif)RX}$	receiver differential input impedance		80	100	120	Ω
100BASE-T1[3]						
$\|V_{o(dif)}\|$	differential output voltage (absolute value)	load 100 Ω meassured at MDI	—	—	2.2	V
R_{term}	termination resistance		—	100	—	Ω
Oscillator[3]						
C_i	input capacitance	on pin XI	—	2	—	pF
C_L	load capacitance	[13]	—	8	—	pF
$V_{i(p-p)}$	input voltage	on pin XI for externally connected AC-coupled clock.	0.8	—	1.21	V_{pp}

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Table 373. Static characteristics...*continued*

T_{vj} = -40 °C to + 150 °C; all voltages are defined with respect to ground unless otherwise specified; positive currents flow into the IC. [1]

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
		C_{AC}=100 pF				
Voltage monitors						
V_{uvd}	undervoltage detection voltage	on pin VDD_CORE	0.99	—	—	V
		on pins VDDA_TRX	2.25	—	—	V
V_{uvr}	undervoltage recovery voltage	on pins VDDA_TRX	—	—	2.38	V
V_{ovd}	overvoltage detection voltage	on pin VDD_CORE	—	—	1.21	V
		on pins VDDA_TRX	—	—	2.75	V
		on pin VREGA_IN	—	—	3.63	V
		on pin VREGD_IN	—	—	3.63	V
V_{ovr}	overvoltage recovery voltage	on pins VDDA_TRX	2.63	—	—	V
		on pin VREGA_IN	3.47	—	—	V
		on pin VREGD_IN	3.47	—	—	V

[1] All parameters are guaranteed over the junction temperature range by design. Factory testing uses correlated test conditions to cover the specified temperature and power supply voltage ranges.
[2] Note that INH and WAKE_IN_OUT logic levels are according to VDDA_AO
[3] Not measured in production, guaranteed by design.
[4] Typical values for 25 °C and typical operating voltages
[5] VDDA_TRX is supplied through VREGA_OUT
[6] Device in sleep mode, but VREGD_IN is still supplied.
[7] When supplied from externally, pin VREGD_IN is externally connected to VDD_CORE, VDDA_SI and VDDA_SO.
[8] If VDDIO is supplied with a lower voltage, the current is guaranteed to be less than this value.
[9] Characterized on a 18.5 cm FR4 trace and 10k pull-down on RXD2.
[10] For max value at an ambient temperature of 85 °C, for typical value at 25°C; supply on pins other than VDDA_AO is expected to be removed in DEVICE_DEEP_SLEEP.
[11] Some pins are input-only or output-only. In this case only the relevant information applies
[12] Pins RST_N, WAKE_IN_OUT and INH are V_{DDA_AO} referenced, see pinning list for details.
[13] Value of external load capacitors depend on the selected crystal. Consult the TJA1104 Appnote for details.

11 Dynamic characteristics

Table 374. Dynamic characteristics

T_{vj} = -40 °C to + 150 °C; all voltages are defined with respect to ground unless otherwise specified. Not measured in production, guaranteed by design.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
Digital I/Os (JTAG, GPIOs)[1]						
t_r	rise time	15 cm PCB transmission line: 50 Ω; 5 pF far-end load. 20% - 80%				
		1.8 V, default IO settings	1.3	—	3.9	ns
		2.5 V, default IO settings	1.5	—	4.26	ns
		3.3 V, default IO settings	2.2	—	4.7	ns
t_f	fall time	15 cm PCB transmission line: 50 Ω; 5 pF far-end load.				

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Table 374. Dynamic characteristics...*continued*

T_{vj} = -40 °C to + 150 °C; all voltages are defined with respect to ground unless otherwise specified. Not measured in production, guaranteed by design.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
		20% - 80%				
		1.8 V, default IO settings	1.3	—	3.9	ns
		2.5 V, default IO settings	1.5	—	4.26	ns
		3.3 V, default IO settings	2.2	—	4.7	ns
Digital I/Os (MDIO, MII, RMII, RGMII)[1]						
t_r	rise time	15 cm PCB transmission line: 50 Ω; 5 pF far-end load. 20% - 80%				
		1.8 V, default IO settings	0.39	—	2	ns
		2.5 V, default IO settings	0.92	—	3.3	ns
		3.3 V, default IO settings	1.2	—	3.6	ns
t_f	fall time	15 cm PCB transmission line: 50 Ω; 5 pF far-end load. 20% - 80%				
		1.8 V, default IO settings	0.39	—	2	ns
		2.5 V, default IO settings	0.92	—	3.3	ns
		3.3 V, default IO settings	1.2	—	3.6	ns
MII transmit timing						
T_{clk}	clock period	on pin TX_CLK	—	40	—	ns
δ	duty cycle	on pin TX_CLK	35	—	65	%
t_{WH}	pulse width HIGH	on pin TX_CLK	—	20	—	ns
t_{WL}	pulse width LOW	on pin TX_CLK	—	20	—	ns
t_{su}	set-up time	from pins TXD[3:0], TX_ER, TX_EN to TX_CLK				
		MII mode	10	—	—	ns
		revMII mode	10	—	—	ns
t_h	hold time	from TX_CLK to TXD[3:0], TX_ER, TX_EN				
		MII mode	0	—	—	ns
		revMII mode	10	—	—	ns
MII receive timing						
T_{clk}	clock period	on pin RX_CLK	—	40	—	ns
δ	duty cycle	on pin RX_CLK	35	—	65	%
t_{WH}	pulse width HIGH	on pin RX_CLK	—	20	—	ns
t_{WL}	pulse width LOW	on pin RX_CLK	—	20	—	ns
t_d	delay time	from RX_CLK to RXD[3:0], RX_ER, RX_DV				

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Table 374. Dynamic characteristics*...continued*

T_{vj} = -40 °C to + 150 °C; all voltages are defined with respect to ground unless otherwise specified. Not measured in production, guaranteed by design.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
		25 pF load default IO settings 15 pF load MII_DATA_IO_CONFIG[ALTERNATE_CONFIG]=1				
		MII	10	—	25	ns
		revMII	0	—	25	ns
RMII transmit/receive timing						
T_{clk}	clock period	on pin REF_CLK	—	20	—	ns
δ	duty cycle	on pin REF_CLK	35	—	65	%
t_{WH}	pulse width HIGH	on pin REF_CLK	—	10	—	ns
t_{WL}	pulse width LOW	on pin REF_CLK	—	10	—	ns
t_{su}	set-up time	from pins TXD[1:0], TX_EN, TX_ER to REF_CLK	4	—	—	ns
t_h	hold time	from REF_CLK to TXD[1:0], TX_EN, TX_ER	2	—	—	ns
t_d	delay time	from REF_CLK to RXD[1:0], CRSDV, RX_ER 25 pF load default IO settings 15 pF load MII_DATA_IO_CONFIG[ALTERNATE_CONFIG]=1	2	—	10	ns
RGMII transmit/receive timing						
T_{clk}	clock period	on pin TXC	—	40	—	ns
δ	duty cycle	on pin TXC	40	50	60	%
$t_{sk(o)TX}$	transmitter output skew time	RGMII mode, from pin RXC to RXD[3:0], RX_CTL [2] 25 pF load default IO settings 15 pF load MII_DATA_IO_CONFIG[ALTERNATE_CONFIG]=1	-500	—	500	ps
$t_{sk(i)RX}$	receiver input skew time	RGMII mode, from pin TXC to TXD[3:0], TX_CTL	1	—	2.6	ns
$t_{su(o)TX}$	transmitter output setup time	RGMII-ID mode, from pins RXD[3:0], RX_CTL to RXC [3] 25 pF load default IO settings 15 pF load MII_DATA_IO_CONFIG[ALTERNATE_CONFIG]=1	1.2	—	—	ns
$t_{h(o)TX}$	transmitter output hold time	RGMII-ID mode, from pin RXC to RXD[3:0], RX_CTL 25 pF load default IO settings 15 pF load MII_DATA_IO_CONFIG[ALTERNATE_CONFIG]=1	1.2	—	—	ns
$t_{su(o)RX}$	receiver output setup time	RGMII-ID mode, from pins TXD[3:0], TX_CTL to TXC	1	—	—	ns

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Table 374. Dynamic characteristics...*continued*

T_{vj} = -40 °C to + 150 °C; all voltages are defined with respect to ground unless otherwise specified. Not measured in production, guaranteed by design.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$t_{h(o)RX}$	receiver output hold time	RGMII-ID mode, from pin TXC to TXD[3:0], TX_CTL	1	—	—	ns
SGMII						
t_r	rise time	20% - 80%	40	—	200	ps
t_f	fall time	20% - 80%	40	—	200	ps
t_{jit}	jitter time	TIE (Time Interval Error): RMS value	—	—	10	ps
MDIO timing						
T_{clk}	clock period	on pin MDC	50	—	1000	ns
δ	duty cycle	on pin MDC	—	50	—	%
t_{su}	set-up time	from pin MDIO to MDC	10	—	—	ns
t_h	hold time	from pin MDC to MDIO	10	—	—	ns
t_d	delay time	from pin MDC to MDIO	0	—	15	ns
JTAG						
f_{clk}	clock frequency	on pin TCK	0.1	—	25	MHz
δ	duty cycle	on pin TCK	40	50	60	%
t_{su}	set-up time		8	—	—	ns
t_h	hold time		25	—	—	ns
t_{ov}	output valid time		—	—	20	ns
CLK_OUT[4]						
f_{clk}	clock frequency	on pins CLK_OUT	—	25	—	MHz
Δf	frequency deviation	on pins CLK_OUT [5]	-100	—	100	ppm
MDI						
$t_{d,tx}$[6]	transmit latency	Jumbo frames disabled				
		MII from TX_EN to SSD on MDI	145	—	207	ns
		revMII from TX_EN to SSD on MDI	266	—	366	ns
		RMII from TX_EN to SSD on MDI	385	—	486	ns
		RGMII from TX_EN to SSD on MDI	425	—	526	ns
		SGMII /S/ to SSD on MDI	1000	—	1120	ns
$t_{d,rx}$[7]	receive latency	SSD to RX_DV, MII	651	—	808	ns
		SSD to RX_DV, revMII	791	—	987	ns
		SSD to DV_CRS, RMII	831	—	1047	ns
		SSD to TX_EN on RGMII	651	—	807	ns
		SSD to /S/ on SGMII	980	—	1200	ns
Voltage monitor						

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Table 374. Dynamic characteristics...*continued*

T_{vj} = -40 °C to + 150 °C; all voltages are defined with respect to ground unless otherwise specified. Not measured in production, guaranteed by design.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$t_{uvd(det)}$	undervoltage detection time	from crossing detection level to IRQ generation	—	—	5	ms
$t_{ovd(det)}$	overvoltage detection time	from crossing detection level to IRQ generation	—	—	5	ms
Power-up timing						
dV_{ru}/dt	rate of change of ramp-up voltage	supply ramp up rate; 20 % - 80 %				
		on pin VDD_AO	1	—	35	V/ms
		on pin VDD_CORE	1	—	35	V/ms
		on pin VDDIO	1	—	35	V/ms
		on pin VDDA_SI, VDDA_SO	1	—	35	V/ms
		on pin VREGA_IN	1	—	35	V/ms
		on pin VREGD_IN	1	—	35	V/ms
$t_{init(ps)}$	pin strapping initialization time	from deasserting RST_N pin to completion of pin strapping	—	—	3	ms
Oscillator						
f_{osc}	oscillator frequency		—	25	—	MHz
Δf	frequency deviation	over lifetime and temperature range [8]				
		MII, SGMII	-100	—	100	ppm
		RGMII, RMII [9]	-50	—	50	ppm
Oscillator Follower Mode (external clock)[10]						
δ	duty cycle	on pin XI	45	50	55	%
$t_{jit(i)rms}$	RMS input jitter time	on pin XI	—	—	tbd	ps
Reset						
$t_{det(rst)}$	reset detection time	on pin RST_N	5	—	—	us

[1] For reference only. Run simulations with the IBIS model and your custom board for accurate results.
[2] Device configured to operate in RGMII mode.
[3] Device configured to operate in RGMII-ID mode, delay is in the transmitter.
[4] CLK_OUT is disabled after device reset
[5] With 100 ppm crystal or clock source connected to XI pin.
[6] Latency is provided for non-MACsec variants. For MACsec Variants, MACsec latency should be added from Table 26
[7] Latency is provided for non-MACsec variants. For MACsec Variants, MACsec latency should be added from Table 27
[8] The frequency deviation depends on the selected crystal
[9] Device can tolerate a 100 ppm crystal in RGMII/RMII mode, but clock output ppm specifications are not met.
[10] Consult the TJA1104 Application Note for more information on clocking.

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12 Timing diagrams

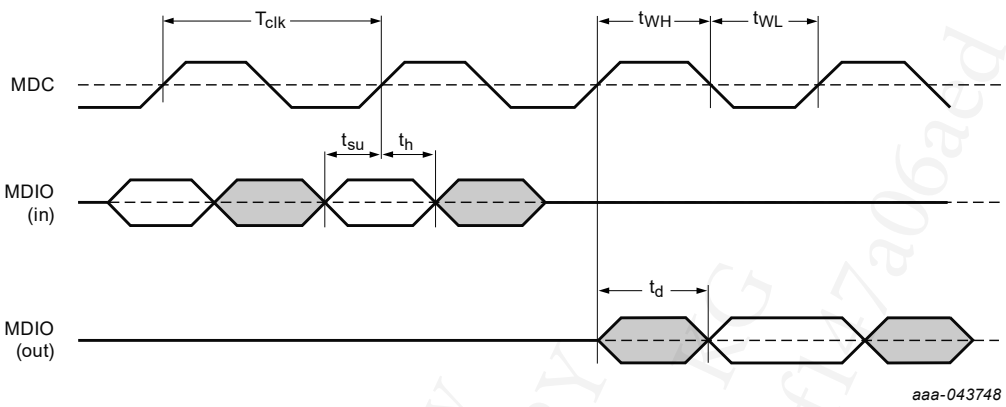


Figure 47. MDIO timing diagram

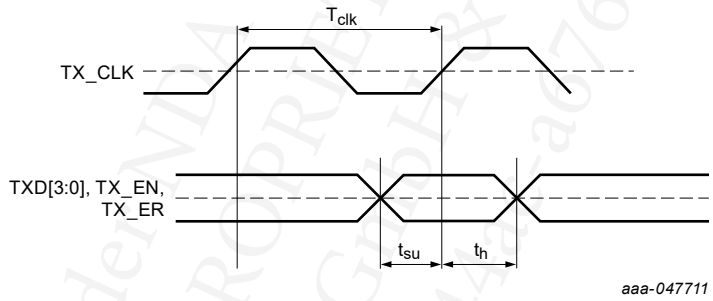


Figure 48. MII transmit timing diagram

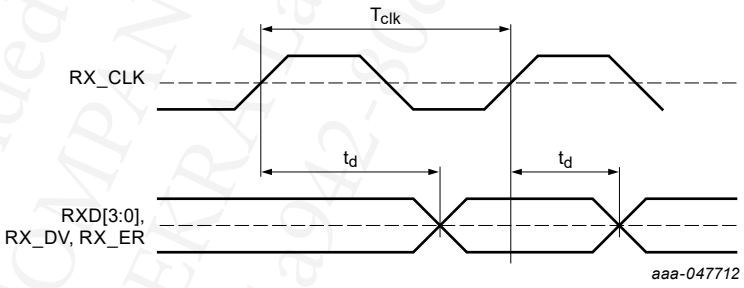


Figure 49. MII receive timing diagram

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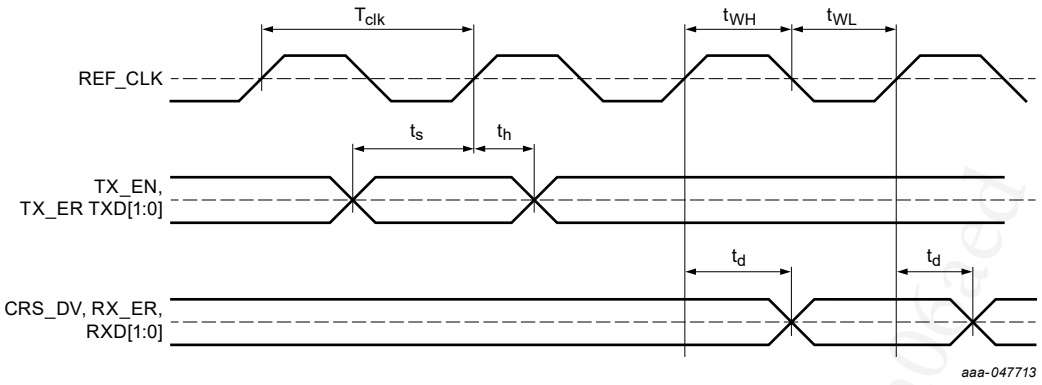


Figure 50. RMII transmit/receive timing diagram

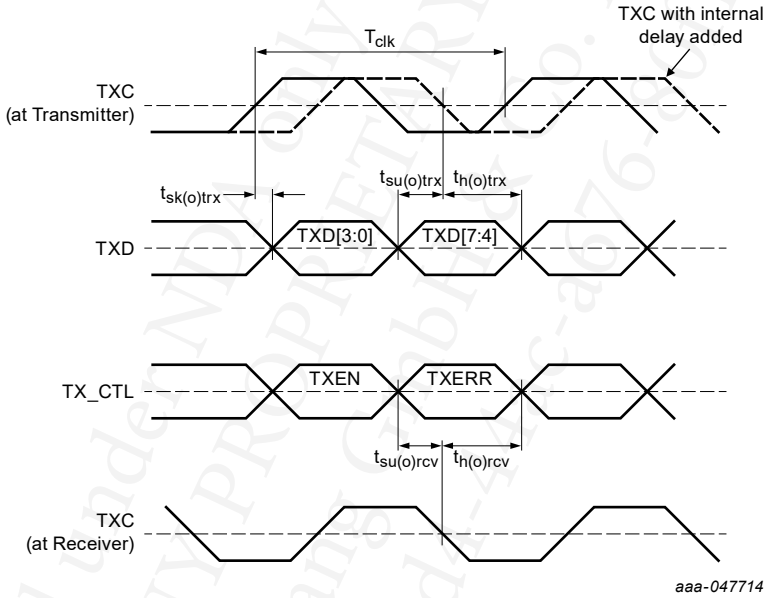


Figure 51. RGMII transmit timing diagram

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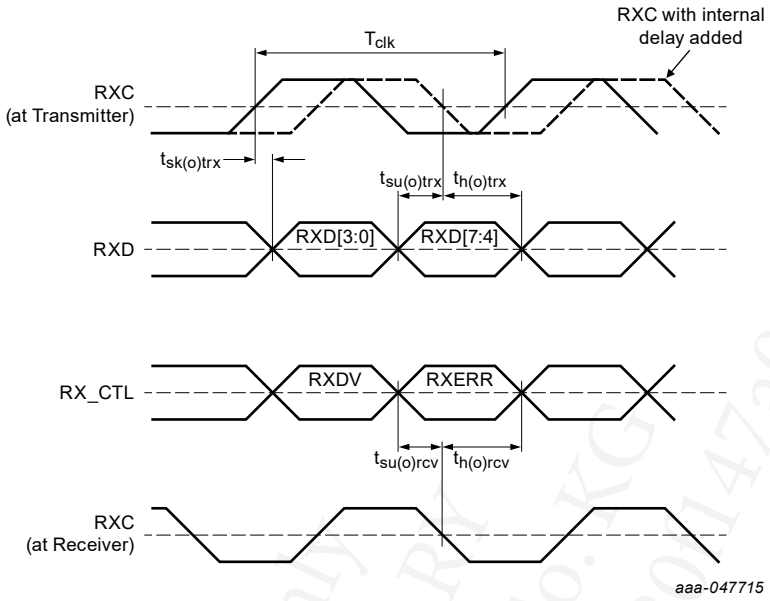


Figure 52. RGMII receive timing diagram

13 Application diagrams

The schematics in Figure 53 to Figure 56 show application examples. See application note AN13662 for further information.

13.1 Supply

The TJA1104 offers several supply options including single supply from a 3.3 V or 2.5 V source as well as individual supplies. Furthermore the supply options for the TJA1104A/C and TJA1104B/D variants are slightly different. The optimal supply configuration depends on the application, however, in general single-supply operation reduces BOM costs and PCB space while individual supplies allow lower power consumption. The following sections highlight the most common options.

13.1.1 Single supply

The diagrams Figure 53 and Figure 54 show the 3.3 V single supply use-case for TJA1104A/TJA1104C and TJA1104B/TJA1104D.

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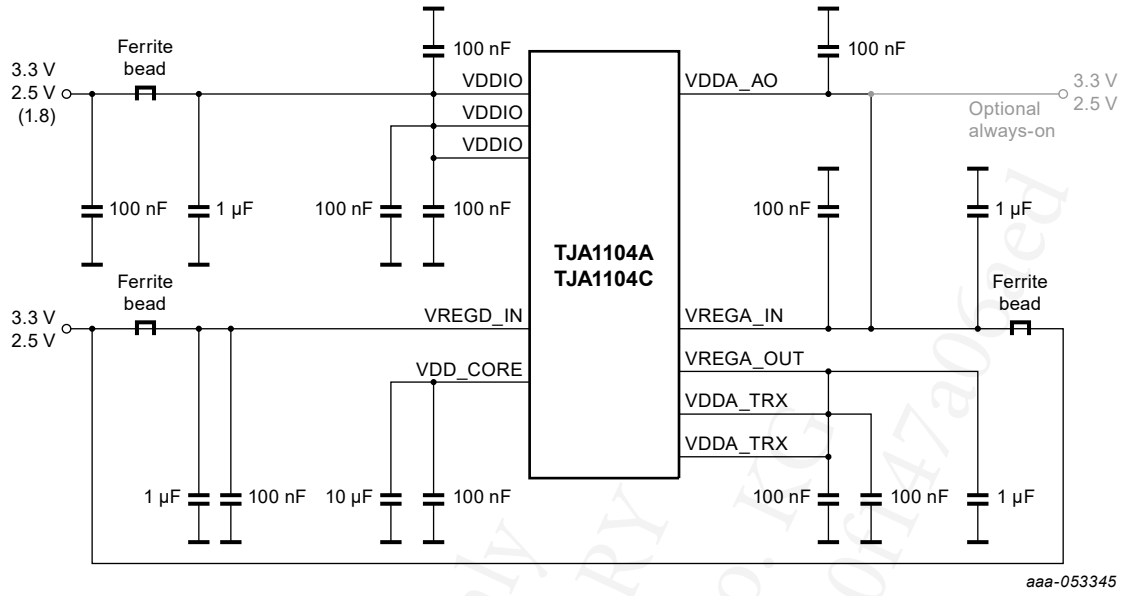


Figure 53. 3.3 V single supply circuit diagram TJA1104A/TJA1104C

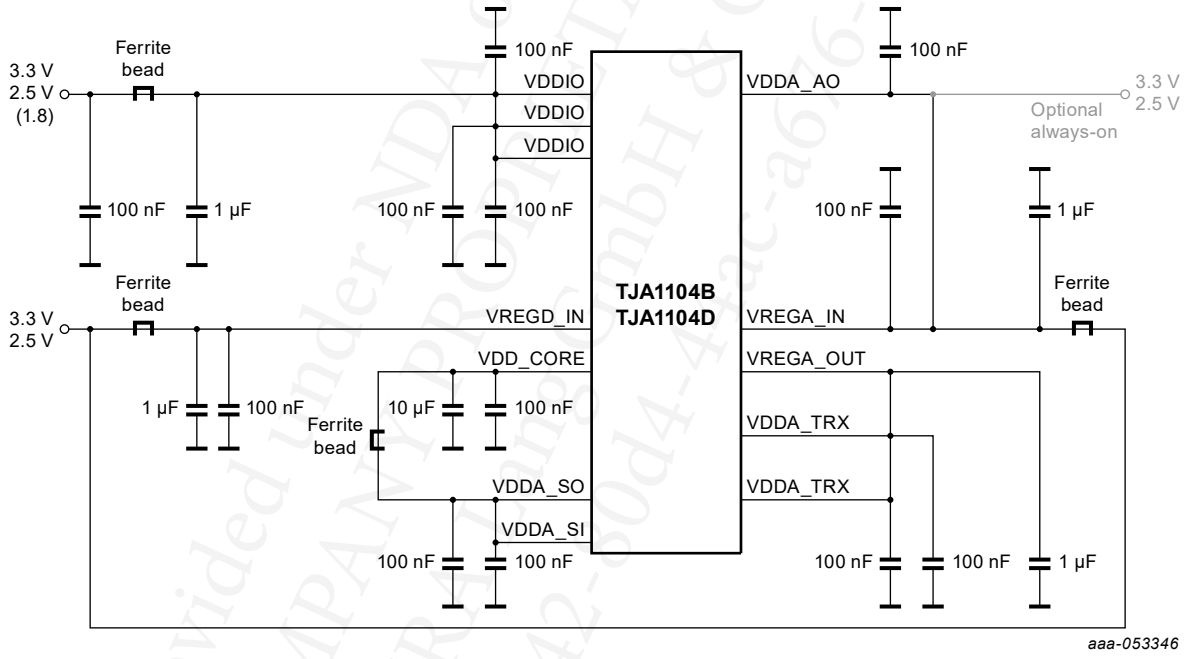


Figure 54. 3.3 V single supply circuit diagram TJA1104B/TJA1104D

13.1.2 Individual supply

The diagrams Figure 55 and Figure 56 show the individual supply use-case for TJA1104A/TJA1104C and TJA1104B/TJA1104D. Pins VREGD_IN and VDD_CORE are shorted together and connected to an external 1.1 V source. Pins VREGA_IN, VREGA_OUT and VDDA_TRX are shorted together and connected to an external 2.5 source. For TJA1104B/TJA1104D the pins VREGD_IN, VDD_CORE, VDDA_SO and VDDA_SI must be supplied from the same 1.1 V supply source.

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It is possible to supply 1.1 V directly as shown Figure 55 and Figure 56 but use the internal regulator of the TJA1104 to generate the supply for VDDA_TRX as shown in Section 13.1.1. This approach is recommended when a 1.1 V supply is already available but 2.5 V is not.

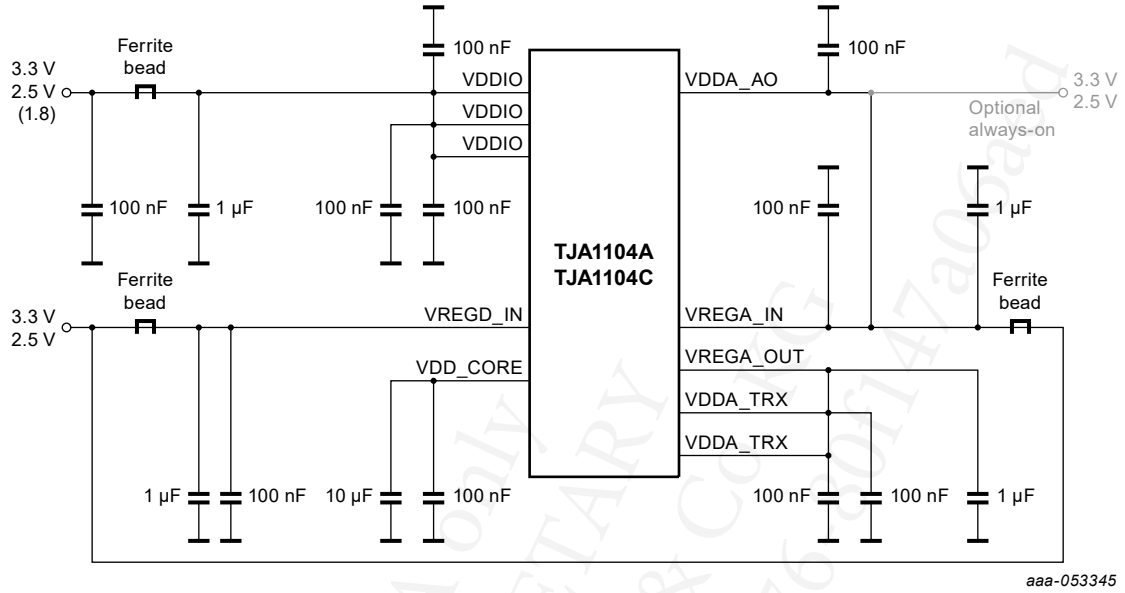


Figure 55. Individual supply circuit diagram TJA1104A/TJA1104C

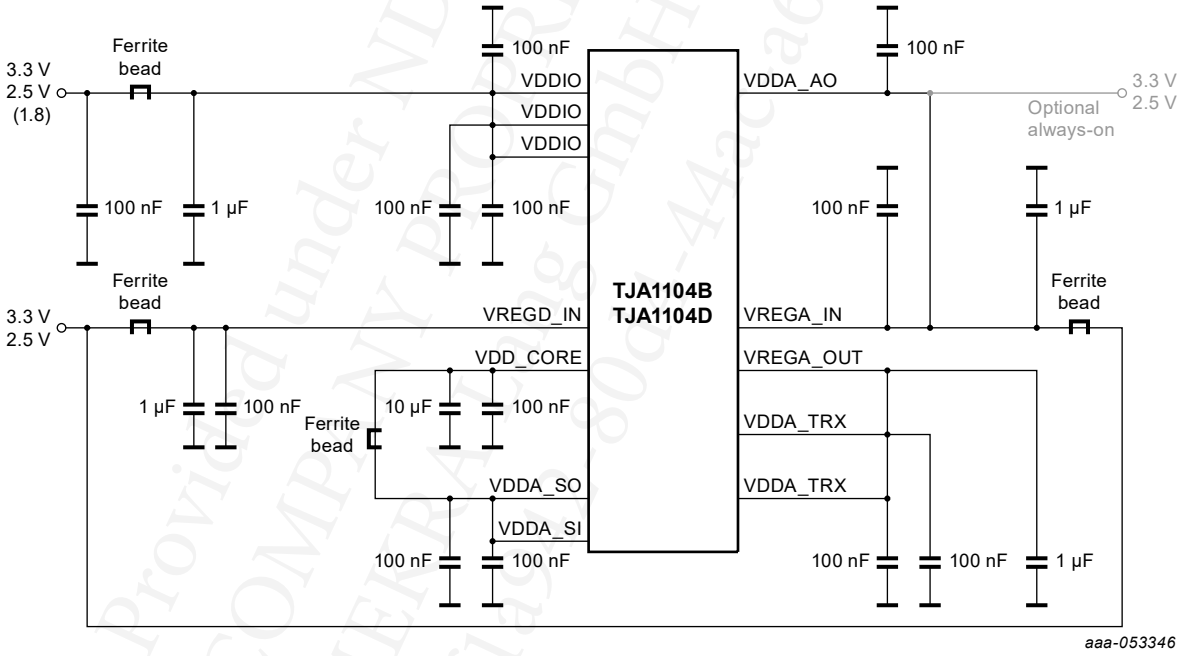


Figure 56. Individual supply circuit diagram TJA1104B/TJA1104D

13.2 Wakeup-Sleep

In case the wakeup/sleep feature is used, VDDA_AO must be supplied by a dedicated standby supply. The VDDA_AO supply of the device can operate on 2.5 V or 3.3 V, however commonly 3.3 V is used. Consult TJA1104 Application Notes for further information. The pins INH, WAKE_IN_OUT and RST_N are in the VDDA_AO domain. Thus any interfacing to these pins must be compatible with the level of VDDA_AO.

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13.3 Reset

The RST_N pin is part of the VDDA_AO voltage domain. This makes sure that the device is not automatically going through a reset sequence after a deep-sleep during which only VDDA_AO is present. In case VDDA_AO and VDDIO are different supplies it must be ensured that RST_N is driven with voltage levels according to VDDA_AO.

For example, if VDDIO is 1.8 V then all microcontroller pins operate in 1.8 V mode. In this case, RST_N cannot be connected directly to a microcontroller pin, because RST_N has thresholds relative to VDDA_AO and not VDDIO. See the application notes for examples.

Figure 57 shows two possibilities how a microcontroller and TJA1104 can be connected. Additional components such as pull-up resistors may be required. It is mandatory to keep the RST_N at defined levels during all phases of system startup, operation, sleep and wakeup.

Note: *Never assert RST_N during wake/sleep cycles. A device reset clears all wakeup/sleep related state. Attention when the microcontroller controls RST_N. Check that the GPIO behavior of the microcontroller during startup/reset is well defined. A glitch on the GPIO outputs can lead to a TJA1104 reset.*

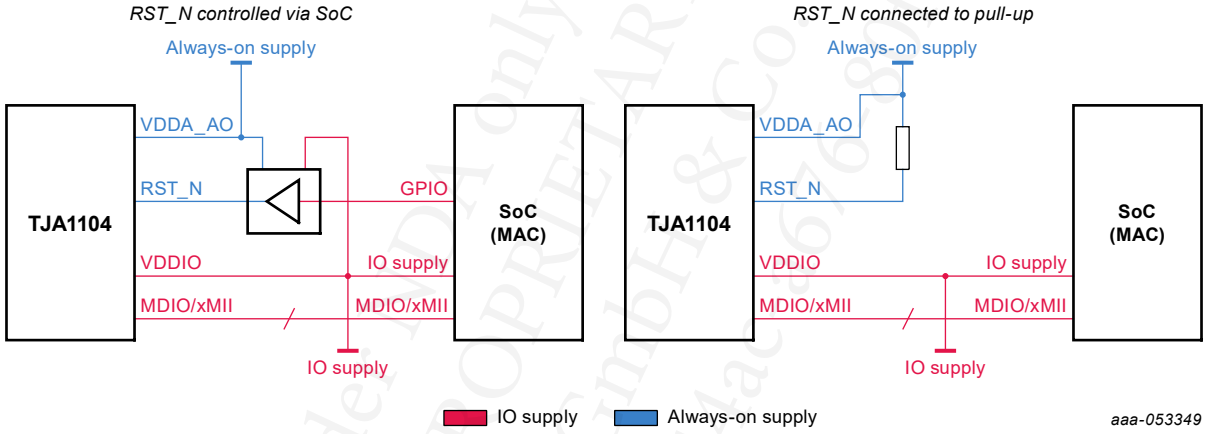


Figure 57. Driving the RST_N pin

14 Test information

Quality information

This product has been qualified in accordance with the Automotive Electronics Council (AEC) standard Q100 Rev-H - *Failure mechanism based stress test qualification for integrated circuits*, and is suitable for use in automotive applications.

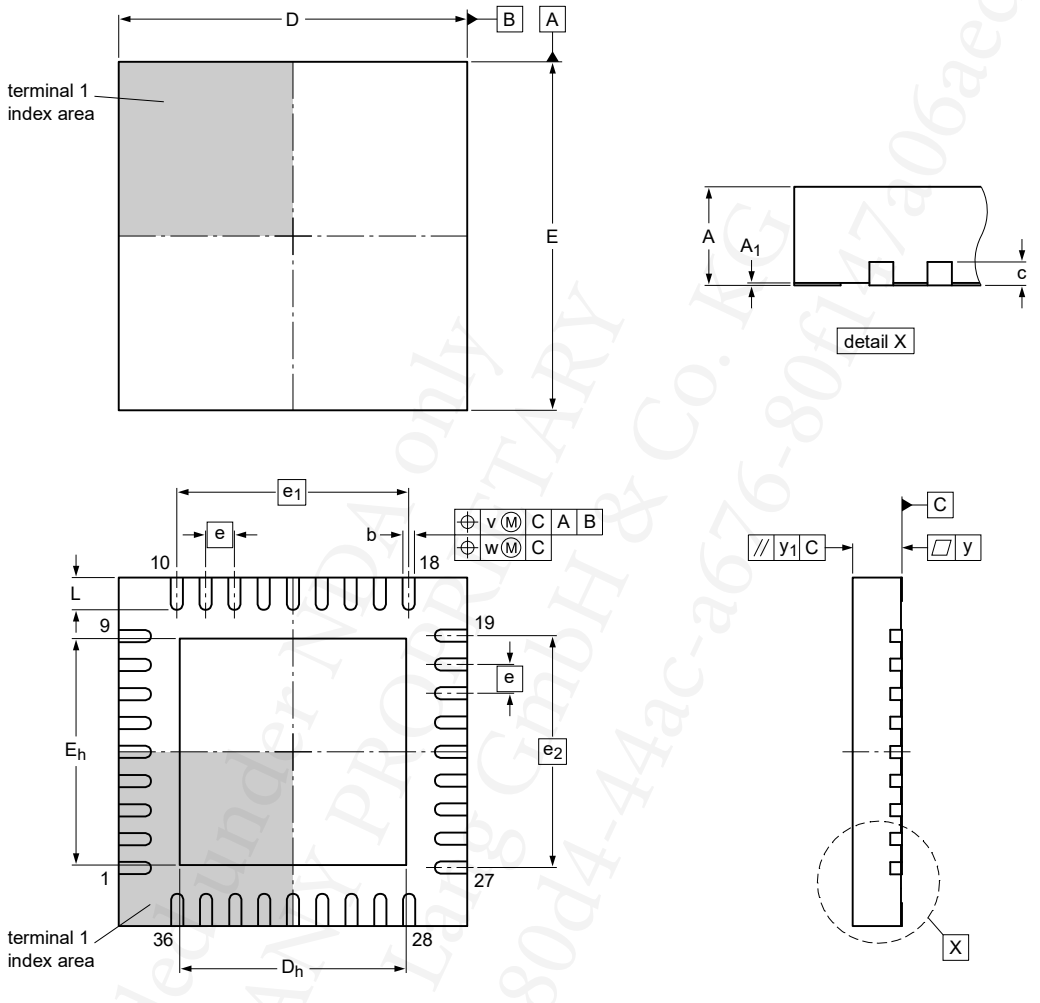
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15 Package outline

HVQFN36: plastic thermal enhanced very thin quad flat package; no leads; 36 terminals; body 6 x 6 x 0.85 mm **SOT1092-2**



Dimensions

Unit		$A^{(1)}$	A_1	b	c	$D^{(1)}$	D_h	$E^{(1)}$	E_h	e	e_1	e_2	L	v	w	y	y_1
mm	max	1.00	0.05	0.30		6.1	4.05	6.1	4.05				0.65				
	nom	0.85	0.02	0.21	0.2	6.0	3.90	6.0	3.90	0.5	4	4	0.55	0.1	0.05	0.05	0.1
	min	0.80	0.00	0.18		5.9	3.75	5.9	3.75				0.45				

Note
1. Plastic or metal protrusions of 0.075 mm maximum per side are not included.

sot1092-2_po

Outline version	References				European projection	Issue date
	IEC	JEDEC	JEITA			
SOT1092-2		---	---			~~09-02-23~~ 09-02-24

Figure 58. Package outline TJA1104HVQFN36

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16 Soldering

This text provides a very brief insight into a complex technology. A more in-depth account of soldering ICs can be found in Application Note *AN10365 "Surface mount reflow soldering description"*.

16.1 Introduction to soldering

Soldering is one of the most common methods through which packages are attached to Printed Circuit Boards (PCBs), to form electrical circuits. The soldered joint provides both the mechanical and the electrical connection. There is no single soldering method that is ideal for all IC packages. Wave soldering is often preferred when through-hole and Surface Mount Devices (SMDs) are mixed on one printed wiring board; however, it is not suitable for fine pitch SMDs. Reflow soldering is ideal for the small pitches and high densities that come with increased miniaturization.

16.2 Wave and reflow soldering

Wave soldering is a joining technology in which the joints are made by solder coming from a standing wave of liquid solder. The wave soldering process is suitable for the following:

- Through-hole components
- Leaded or leadless SMDs, which are glued to the surface of the printed circuit board

Not all SMDs can be wave soldered. Packages with solder balls, and some leadless packages which have solder lands underneath the body, cannot be wave soldered. Also, leaded SMDs with leads having a pitch smaller than ~0.6 mm cannot be wave soldered, due to an increased probability of bridging.

The reflow soldering process involves applying solder paste to a board, followed by component placement and exposure to a temperature profile. Leaded packages, packages with solder balls, and leadless packages are all reflow solderable.

Key characteristics in both wave and reflow soldering are:

- Board specifications, including the board finish, solder masks and vias
- Package footprints, including solder thieves and orientation
- The moisture sensitivity level of the packages
- Package placement
- Inspection and repair
- Lead-free soldering versus SnPb soldering

16.3 Wave soldering

Key characteristics in wave soldering are:

- Process issues, such as application of adhesive and flux, clinching of leads, board transport, the solder wave parameters, and the time during which components are exposed to the wave
- Solder bath specifications, including temperature and impurities

16.4 Reflow soldering

Key characteristics in reflow soldering are:

- Lead-free versus SnPb soldering; note that a lead-free reflow process usually leads to higher minimum peak temperatures (see Figure 1) than a SnPb process, thus reducing the process window
- Solder paste printing issues including smearing, release, and adjusting the process window for a mix of large and small components on one board

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- Reflow temperature profile; this profile includes preheat, reflow (in which the board is heated to the peak temperature) and cooling down. It is imperative that the peak temperature is high enough for the solder to make reliable solder joints (a solder paste characteristic). In addition, the peak temperature must be low enough that the packages and/or boards are not damaged. The peak temperature of the package depends on package thickness and volume and is classified in accordance with Table 1 and Table 2

Table 375. SnPb eutectic process (from J-STD-020D)

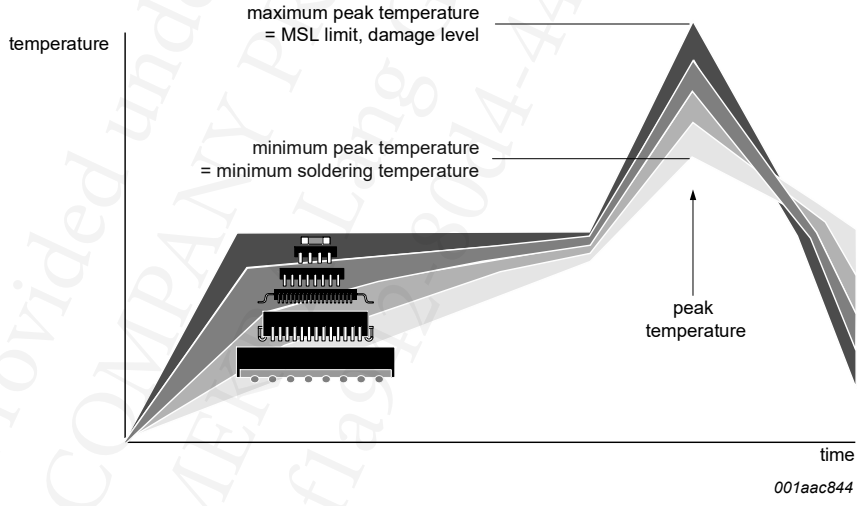
Package thickness (mm)	Package reflow temperature (°C)	
	Volume (mm³)	
	< 350	≥ 350
< 2.5	235	220
≥ 2.5	220	220

Table 376. Lead-free process (from J-STD-020D)

Package thickness (mm)	Package reflow temperature (°C)		
	Volume (mm³)		
	< 350	350 to 2000	> 2000
< 1.6	260	260	260
1.6 to 2.5	260	250	245
> 2.5	250	245	245

Moisture sensitivity precautions, as indicated on the packing, must be respected at all times.

Studies have shown that small packages reach higher temperatures during reflow soldering, see Figure 1.



MSL: Moisture Sensitivity Level

Figure 59. Temperature profiles for large and small components

For further information on temperature profiles, refer to Application Note *AN10365 "Surface mount reflow soldering description"*.

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17 Abbreviations

Table 377. List of Abbreviations

Abbreviation	Description
ADAS	advanced driver assistance system
AEC	automotive electronics council
AFE	analog front end
ASIL	automotive safety integrity level
BIST	built-in self-test
BSCAN	boundary scan
CDM	charged-device model
CMOS	complementary metal-oxide-semiconductor
ECU	electronic control unit
EMC	electromagnetic compatibility
ESD	electrostatic discharge, end of stream delimiter [3]
FUSA	functional safety (as per ISO 26262)
GPIO	general-purpose Input-output
HBM	human body model
HVQFN	enhanced very thin quad flat package
IEEE	institute of electrical and electronics engineers
IRQ	interrupt
IPG	Inter-packet gap
JTAG	joint test action group
LED	light emitting diode
LVCMOS	low-voltage CMOS
MAC	media access control
MACsec	media access control security
MDI	Media-dependant interface
MDIO	management data input/output
MII	media independent interface
MMD	MDIO manageable devices
MSE	mean square error
OUI	organizationally unique identifier
OV	over voltage
PCB	printed circuit board
PHY	physical layer
PPS	pulse-per-second
PTP	precision time protocol

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Table 377. List of Abbreviations...*continued*

Abbreviation	Description
RGMII	reduced gigabit media independent interface
RMII	reduced media independent interface
SC	secure channel
SCS	secure channel selection
SerDes	serializer / deserializer
SFD	start of frame delimiter
SSD	start of stream delimiter
SGMII	serial gigabit media independent interface
SQI	signal quality indicator
UV	under voltage
xMII	x media independent interface

18 References

[1] DIN EN 61000-4-2:2009-12, Electromagnetic compatibility (EMC) - Part 4-2: Testing and measurement techniques - Electrostatic discharge immunity test (IEC 61000-4-2:2008)

[2] IEC 62228-5:2021, Integrated circuits - EMC evaluation of transceivers - Part 5: Ethernet transceivers, 2021-04-26

[3] IEEE Std 802.3-2018, 31 August 2018

[4] RMII Specification 1.2, RMII Consortium, 20 March 1998

[5] RGMII Specification 2.0, Broadcom, HP, Marvell, 1 April 2002

[6] Serial-GMII Specification 1.8, Cisco Systems, 27 April 2005

[7] AN13662, TJA1104 Application Notes, NXP, http://www.docstore.nxp.com/

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[9] OPEN Sleep/Wake-up Specification, v2.0, OPEN Alliance, Feb 21, 2017

[10] TJA1104 safety manual, UM12054

[11] MACsec core user manual, UM12059

[12] ISO 21111-2:2020 Road vehicles — In-vehicle Ethernet — Part 2: Common physical entity requirements

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19 Revision history

Table 378. Revision history

Document ID	Release date	Description
TJA1104 v.1.0	20240417	• Initial release

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Document status[1][2]	Product status[3]	Definition
Objective [short] data sheet	Development	This document contains data from the objective specification for product development.
Preliminary [short] data sheet	Qualification	This document contains data from the preliminary specification.
Product [short] data sheet	Production	This document contains the product specification.

[1] Please consult the most recently issued document before initiating or completing a design.
[2] The term 'short data sheet' is explained in section "Definitions".
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